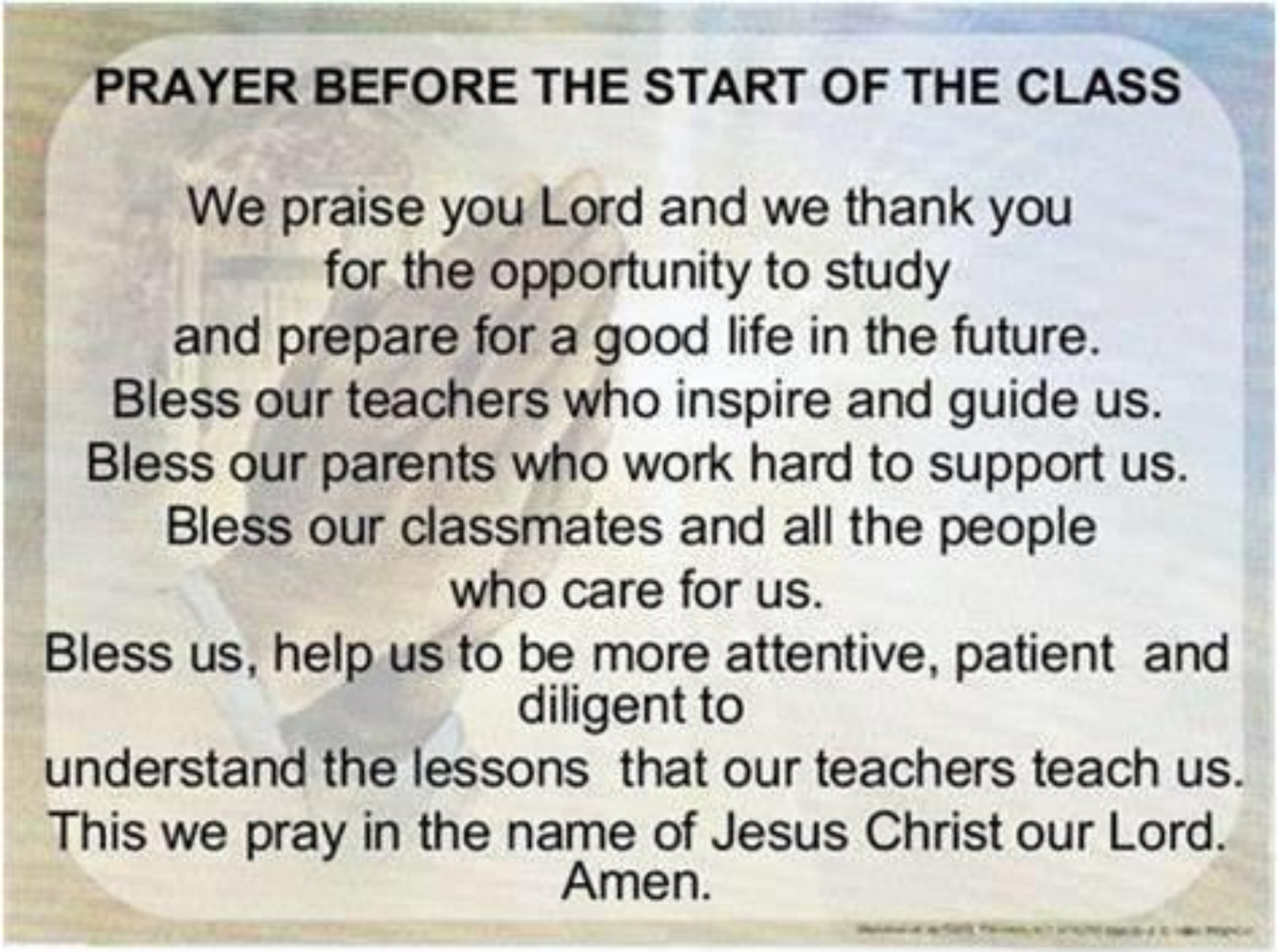




NERVOUS SYSTEM

ANATOMY AND PHYSIOLOGY (MC1)
COLLEGE OF NURSING
PRESIDENT RAMON MAGSAYSAY STATE UNIVERSITY


PRAYER BEFORE THE START OF THE CLASS
We praise you Lord and we thank you
for the opportunity to study
and prepare for a good life in the future.
Bless our teachers who inspire and guide us.
Bless our parents who work hard to support us.
Bless our classmates and all the people
who care for us.
Bless us, help us to be more attentive, patient and diligent to
understand the lessons that our teachers teach us.
This we pray in the name of Jesus Christ our Lord.
Amen.

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PREPARED BY: REIKA ATSUMI, MD

MARIA NOVIEL DOMINGUEZ, RN, MD

COLLEGE OF NURSING, PRESIDENT RAMON MAGSAYSAY STATE UNIVERSITY

ANATOMY AND PHYSIOLOGY

2020

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OBJECTIVES



At the end of the session, students are expected to:

1. Enumerate the **functions of the nervous system.**
2. Identify the **cells of the nervous system:**
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    2. Neurons
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OBJECTIVES

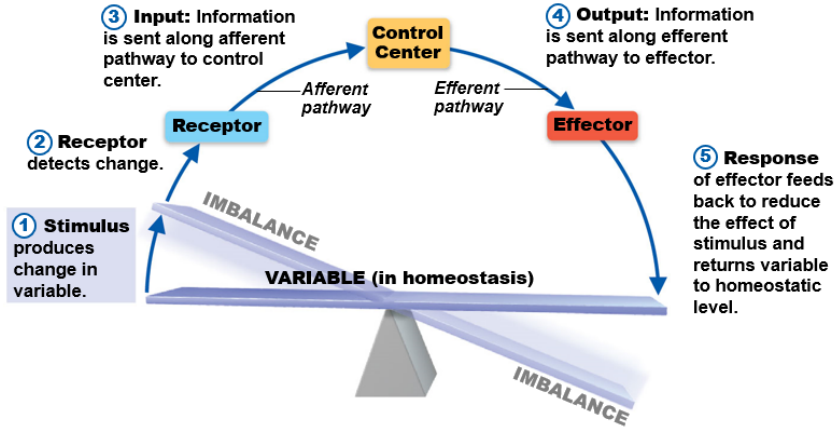
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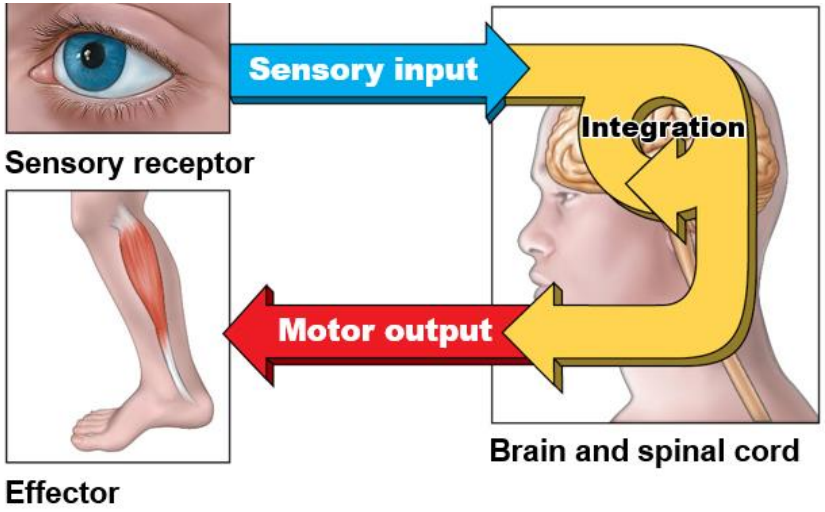
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FUNCTIONS OF THE NERVOUS SYSTEM

Functions of the Nervous System

1. **Sensory Input**
   - Gathering information
   - Sensory receptors detect internal and external changes (stimuli)
2. **Integration**
   - Processing and interpreting sensory input
   - Decides appropriate response
3. **Motor Output**
   - Initiates a response
   - Activates muscles (movement) or glands (secretion)





FUNCTIONS OF THE NERVOUS SYSTEM

4. **Homeostasis**
- Detects, interprets, and responds to internal and external changes

5. **Control of Muscles and Glands**
- Stimulates skeletal muscles for movement
- Regulates smooth muscle, cardiac muscle, and glandular activity

6. **Mental Activity**
- Involves consciousness, memory, and thought processes

FUNCTIONS OF THE NERVOUS SYSTEM
MENTAL ACTIVITY

CONSCIOUSNESS	MEMORY	THINKING
▪ Awareness of unique thoughts, feelings, sensations, and environment ▪ Subjective awareness of self and surroundings	▪ Processes of encoding, storage, and retrieval of past experiences or learnings	▪ Using the mind to consider, reason, or make judgments ▪ Involves rational thought and intelligence



OBJECTIVES

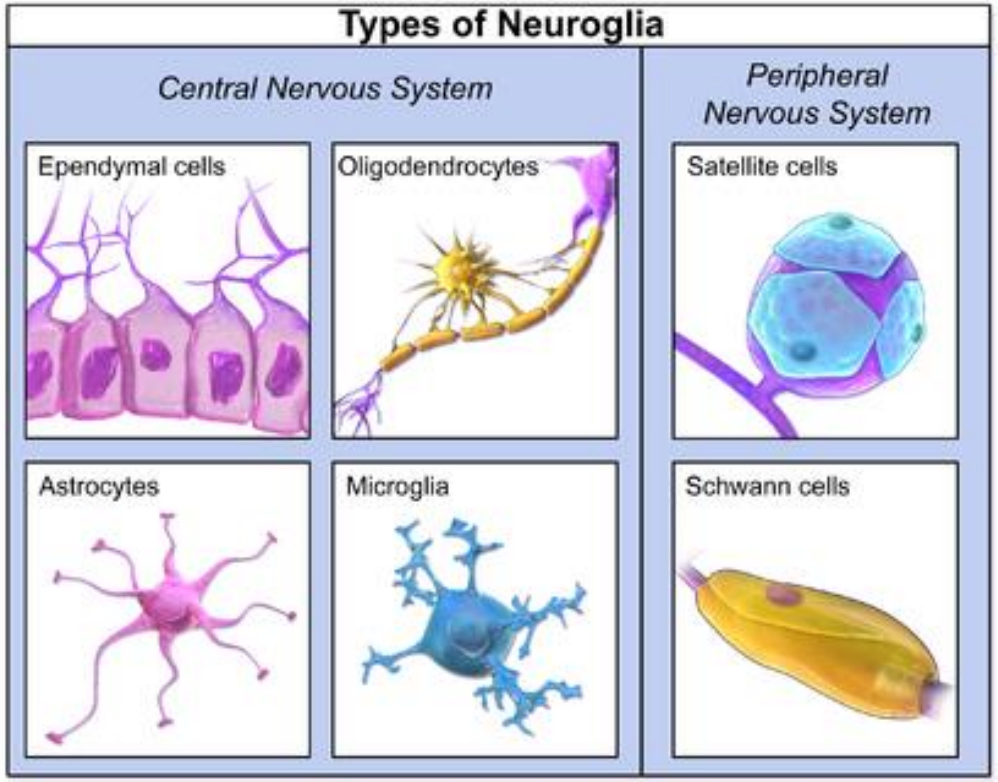
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NERVOUS TISSUE: STRUCTURE AND FUNCTION

Cells of Nervous Tissue

1. **Neuroglia (Glial Cells/Supporting Cells)**
   - Resemble neurons but are more numerous
   - Unable to conduct nerve impulses
   - Retain ability to divide
   - Functions: Support, insulate, and protect neurons
2. **Neurons**
   - The functional and structural units of the nervous system
   - Specialized to conduct electrical impulses



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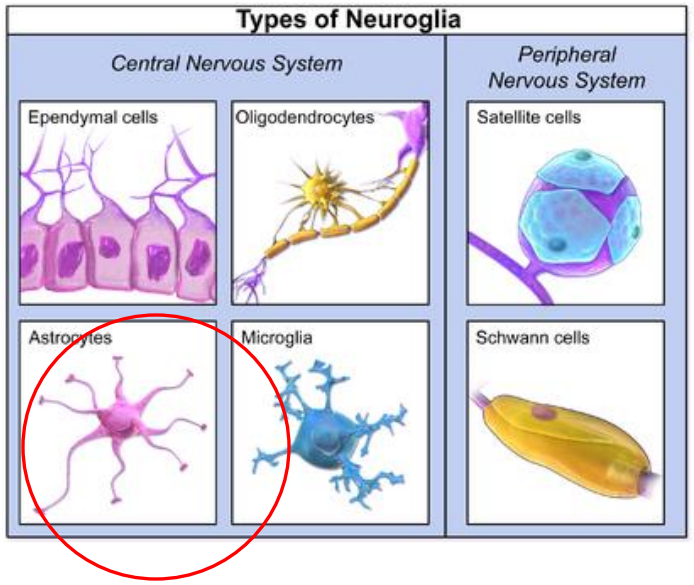
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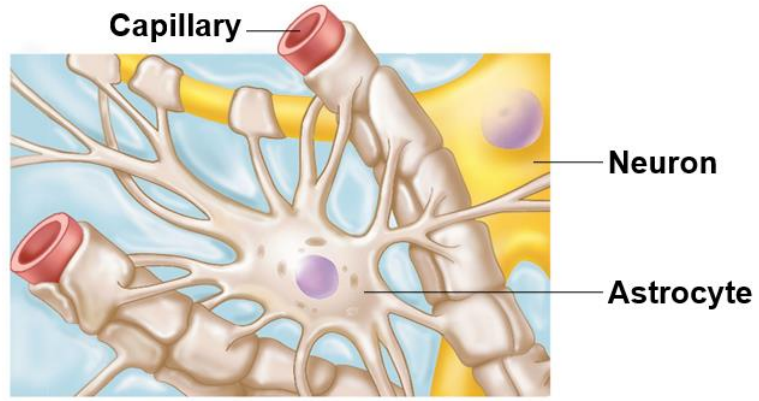
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NERVOUS TISSUE

CNS GLIAL CELLS

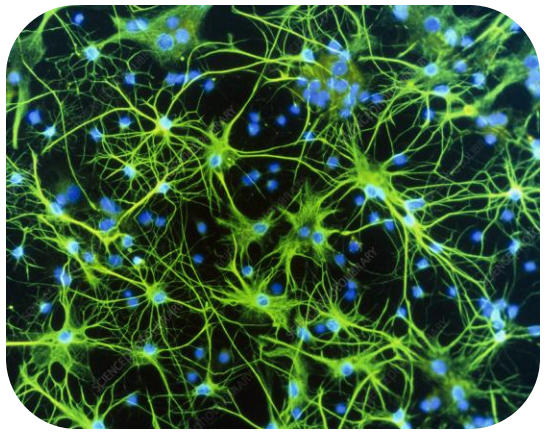


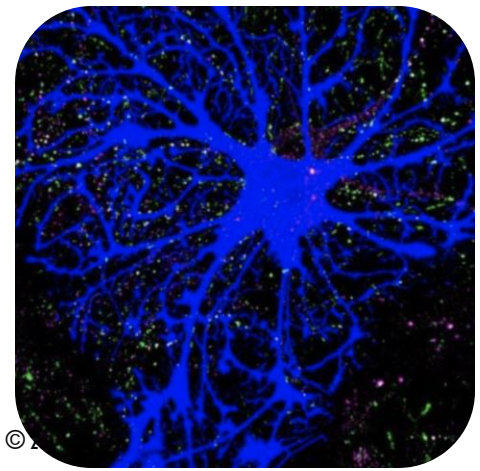


(a) Astrocytes are the most abundant and versatile neuroglia.

Astrocytes

- Most abundant glial cells in the CNS
- Star-shaped, with many processes
- Anchor neurons to blood capillaries
- Form part of the **blood-brain barrier** (control permeability and exchanges between blood and neurons)
- Protect neurons from harmful blood-borne substances
- Regulate the chemical environment of the brain

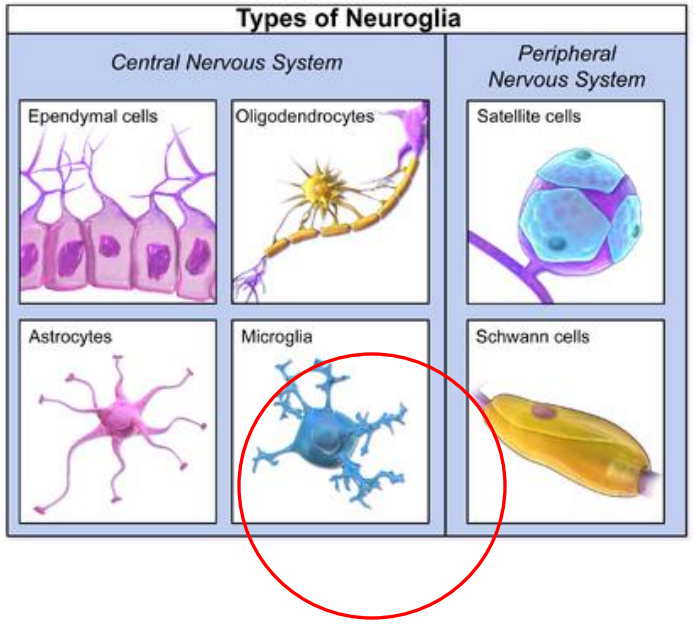


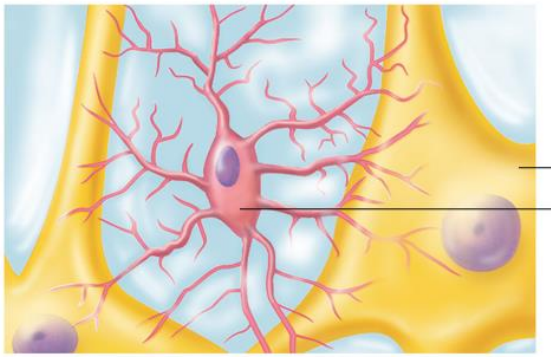


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NERVOUS TISSUE

CNS GLIAL CELLS

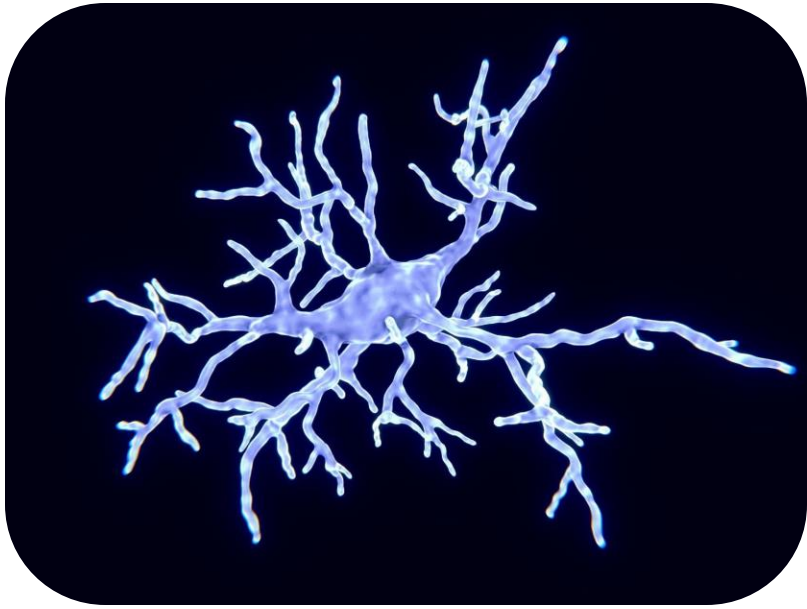




(b) Microglial cells are phagocytes that defend CNS cells.

Microglial Cells

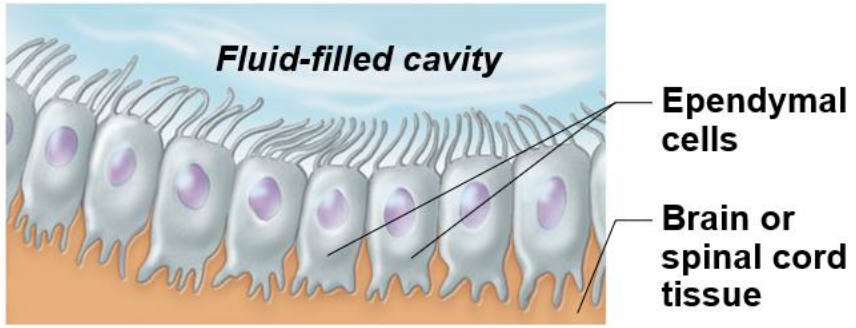
- Small, spiderlike phagocytes
- Monitor health of nearby neurons
- Dispose of debris, pathogens, and dead cells
- Become actively phagocytic during inflammation or injury



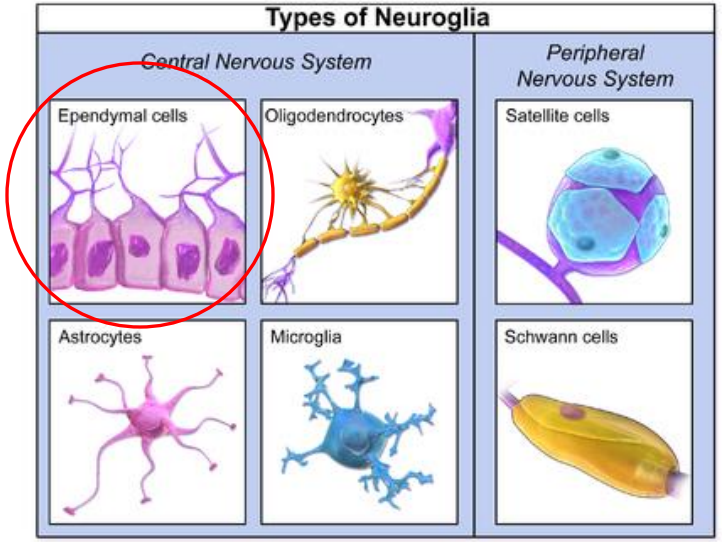
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NERVOUS TISSUE

CNS GLIAL CELLS

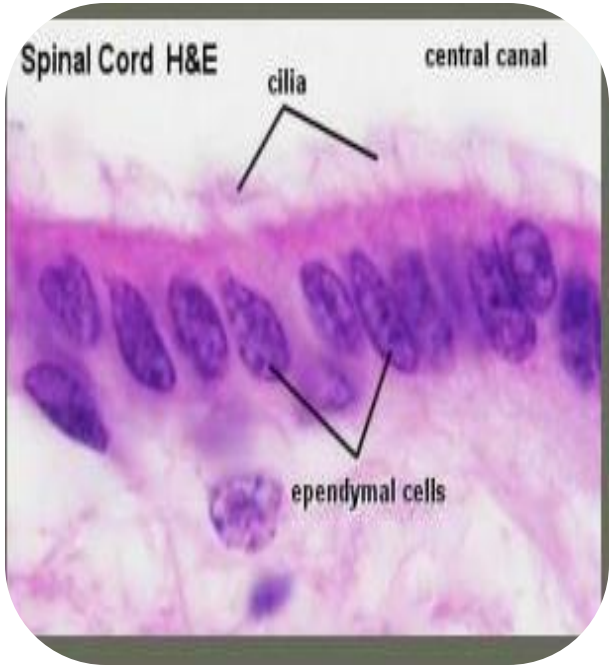


(c) Ependymal cells line cerebrospinal fluid–filled cavities.



Ependymal Cells

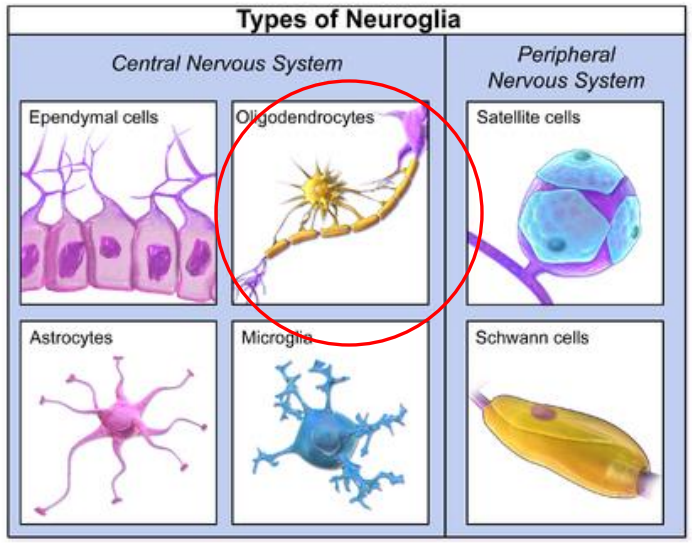
- Line the cavities of the brain and spinal cord
- Squamous to columnar epithelial-like cells
- Many are ciliated → assist in circulation of cerebrospinal fluid (CSF)
- Specialized ependymal cells contribute to CSF production in the choroid plexus



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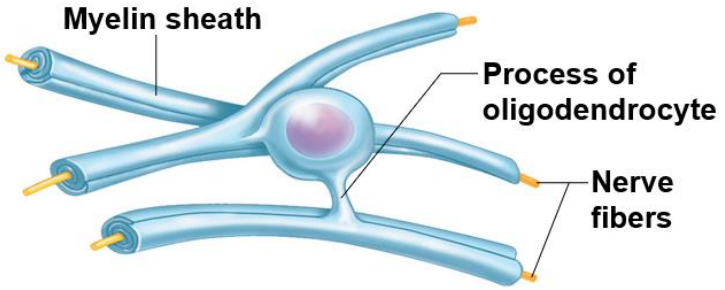
NERVOUS TISSUE

CNS GLIAL CELLS

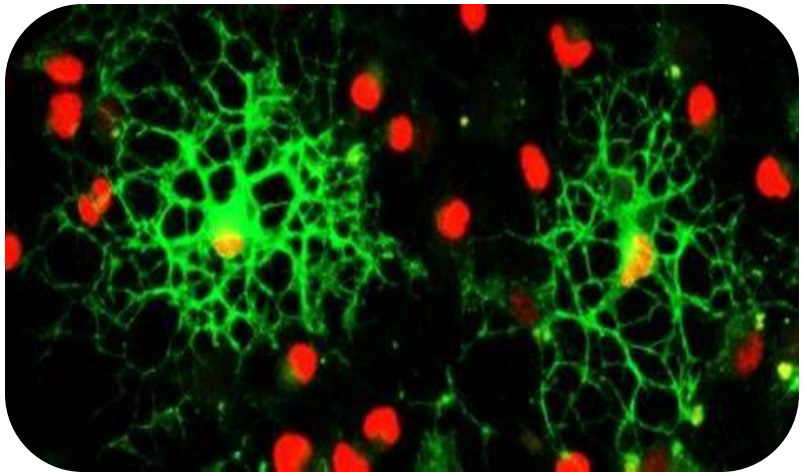


Oligodendrocytes

- Found in the central nervous system
- Extend processes that wrap around nerve fibers
- Form **myelin sheaths** in the CNS
- Function: Insulate axons and increase the speed of nerve impulse conduction



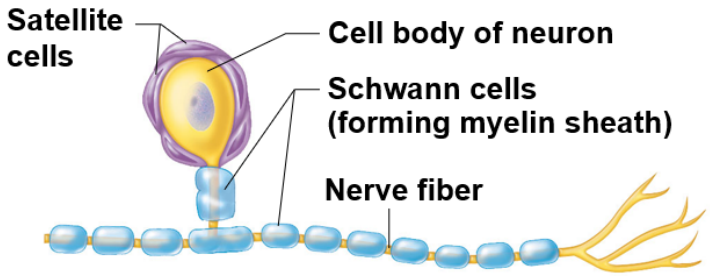
(d) Oligodendrocytes have processes that form myelin sheaths around CNS nerve fibers.



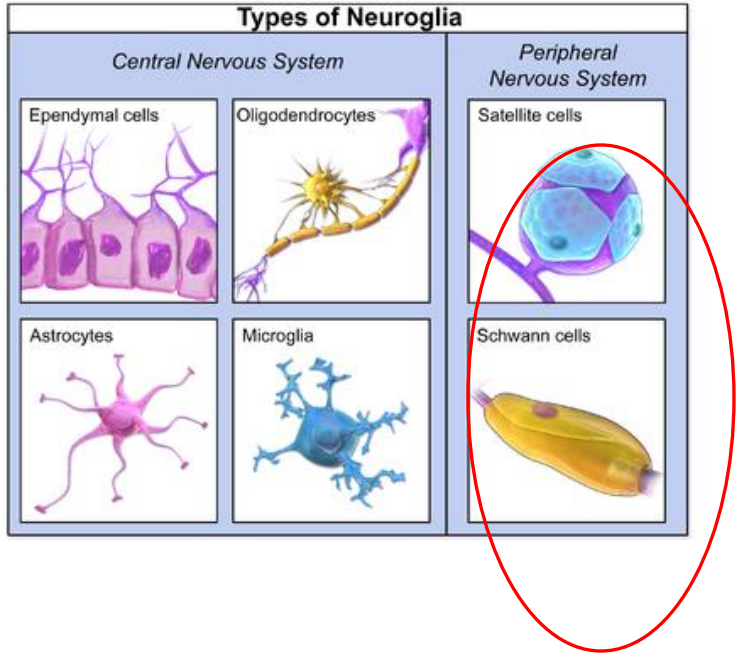
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NERVOUS TISSUE

PNS GLIAL CELLS

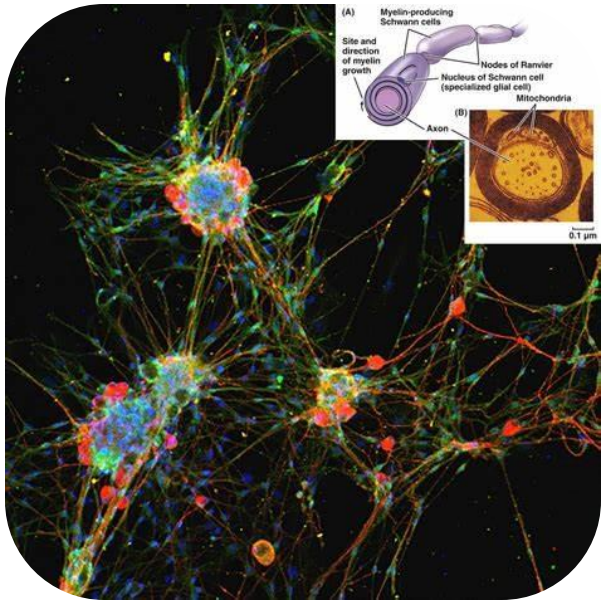


(e) Satellite cells and Schwann cells (which form myelin) surround neurons in the PNS.



Schwann Cells (Neurolemmocytes/ Neurolemma Cells)

- Form myelin sheath around axons in the PNS
- Function: Insulate axons and speed up nerve impulse conduction
- Each Schwann cell myelinates only one segment of an axon



Satellite Cells

- Surround and protect neuron cell bodies in ganglia
- Provide cushioning and regulate the chemical environment

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NERVOUS TISSUE

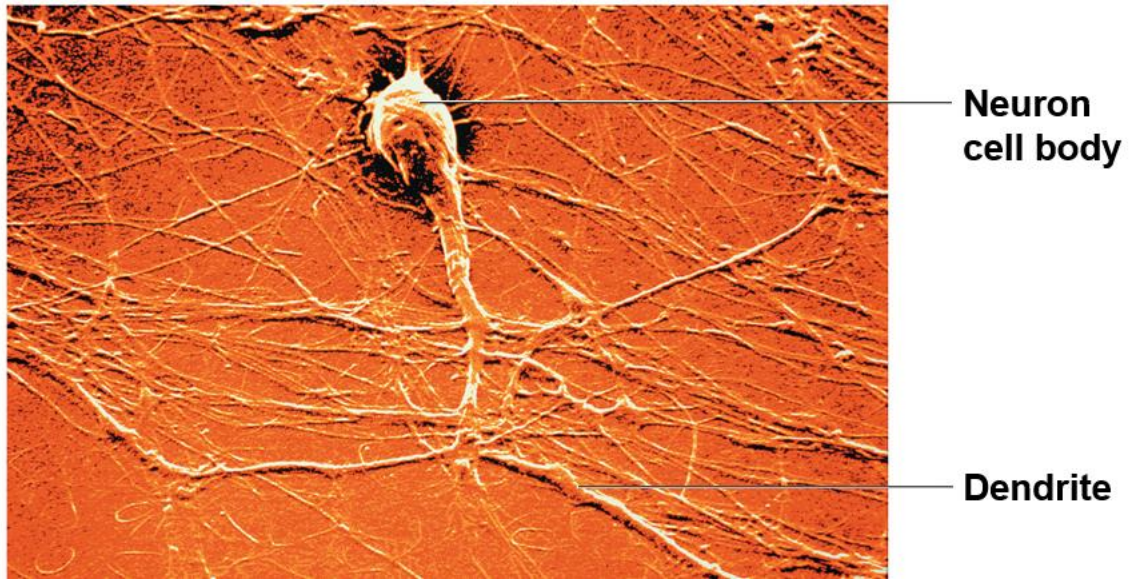
NEURONS

Neurons (Nerve Cells)

- Specialized cells that transmit **nerve impulses (action potentials)**
- Structural and functional units of the nervous system

Major Regions of a Neuron:

1. **Cell Body (Soma)**
    1. Contains the **nucleus** and **metabolic center** of the neuron
    2. Responsible for maintaining cell health and protein synthesis
2. **Processes**
    1. Extensions from the cell body: **dendrites** and **axons**
    2. **Dendrites:** Receive incoming signals
    3. **Axon:** Conducts impulses away from the cell body



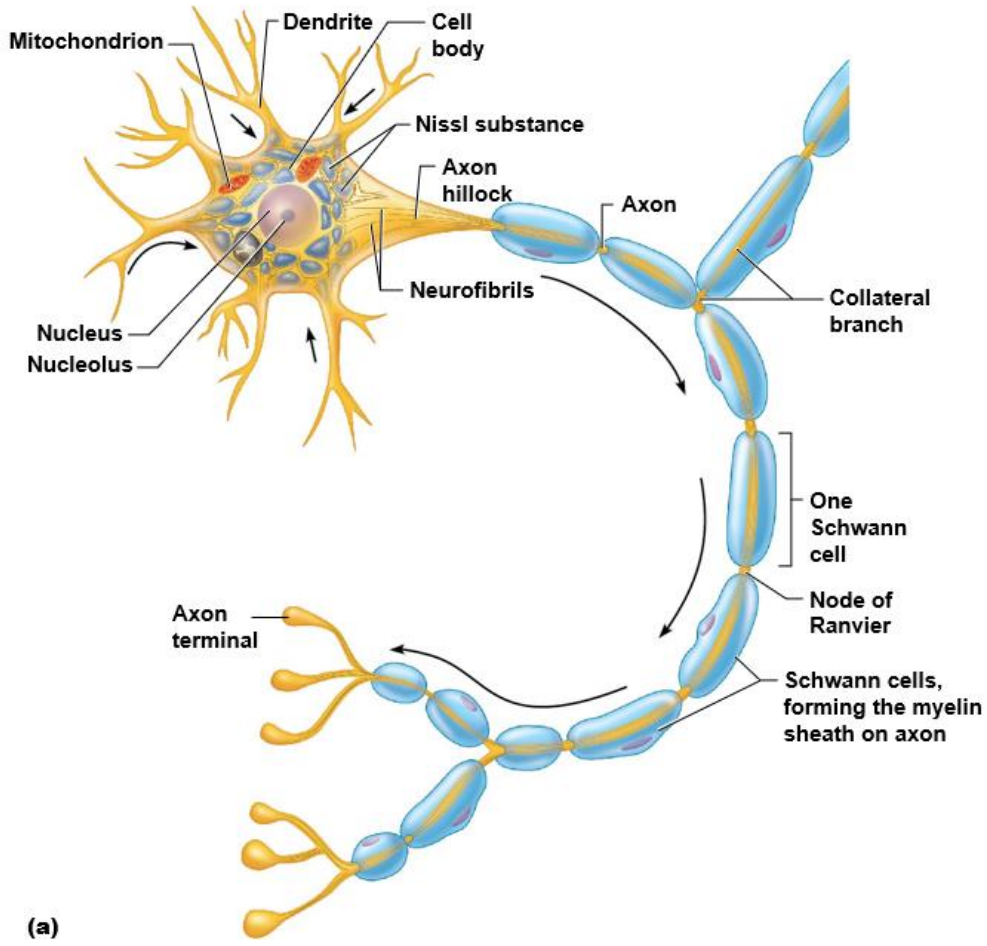
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NERVOUS TISSUE

NEURONS

Neuron Cell Body (Soma)

- **Metabolic center** of the neuron
- Contains **nucleus** with large nucleolus — directs protein synthesis
- **Nissl bodies**: Rough endoplasmic reticulum; site of protein synthesis
- **Neurofibrils**: Intermediate filaments that maintain cell shape and structural integrity

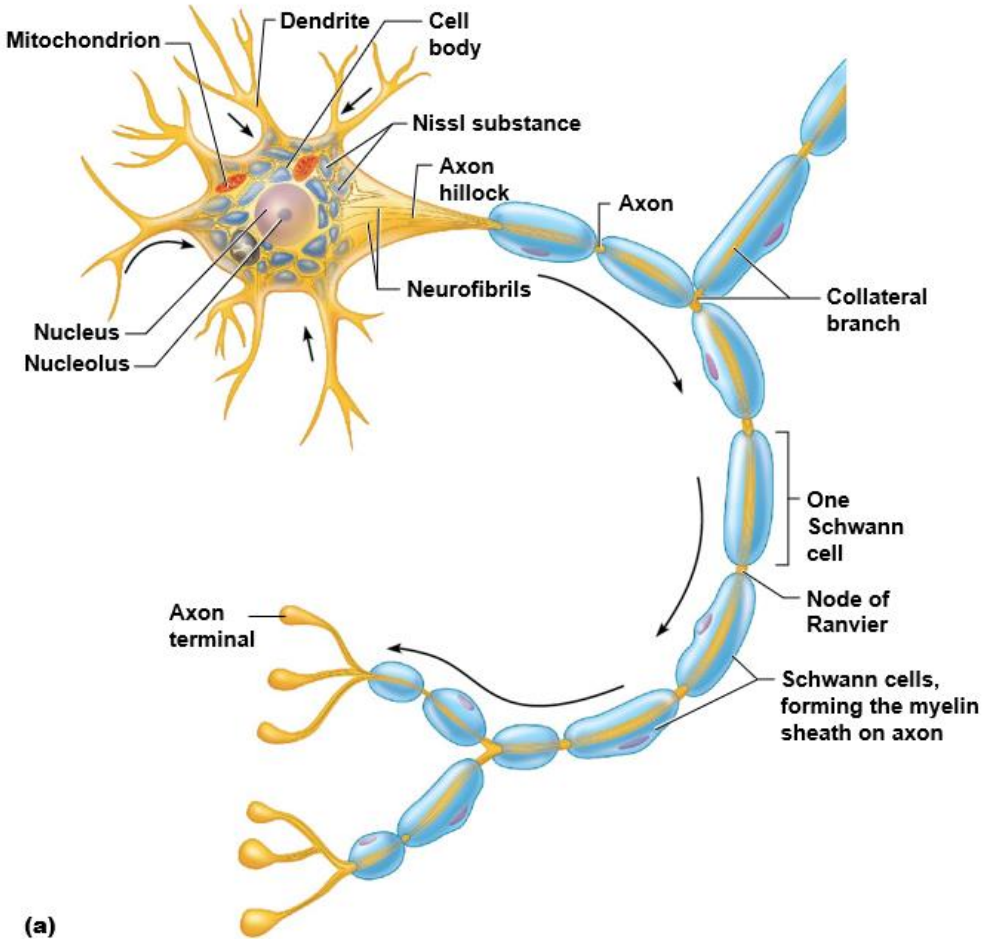


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NERVOUS TISSUE
NEURONS

Neuron Processes: Dendrites

- **Processes (fibers)** extend from the neuron's ce body
- **Dendrites:** short, highly branched cytoplasmic extensions
- Neurons may have **hundreds of dendrites**
- Function: **Conduct nerve impulses toward the body**

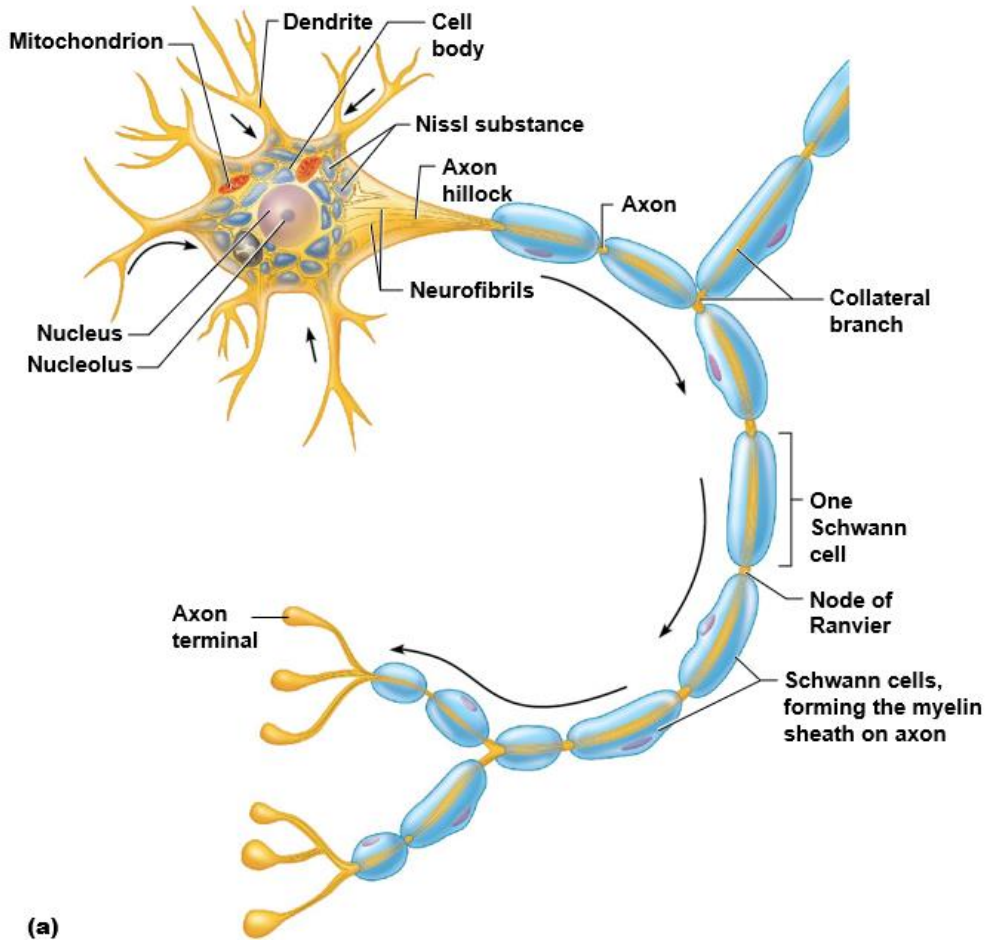


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NERVOUS TISSUE
NEURONS

Neuron Processes: Axons

- Each neuron has **only one axon**, arising from the **axon hillock**
- **Conducts nerve impulses away** from the cell b
- **Ends in axon terminals** containing vesicles fillec with neurotransmitters
- **Axon terminals** are separated from the next neuron by a **synaptic cleft**
- In the **PNS**, axons are **surrounded by Schwann cells** forming the **myelin sheath**



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NERVOUS TISSUE
NEURONS

Myelin Sheath

- **White, fatty material** covering axons
- **Protects and insulates** nerve fibers
- **Increases the speed** of nerve impulse transmission
- Formed by **Schwann cells** in the PNS and **oligodendrocytes** in the CNS

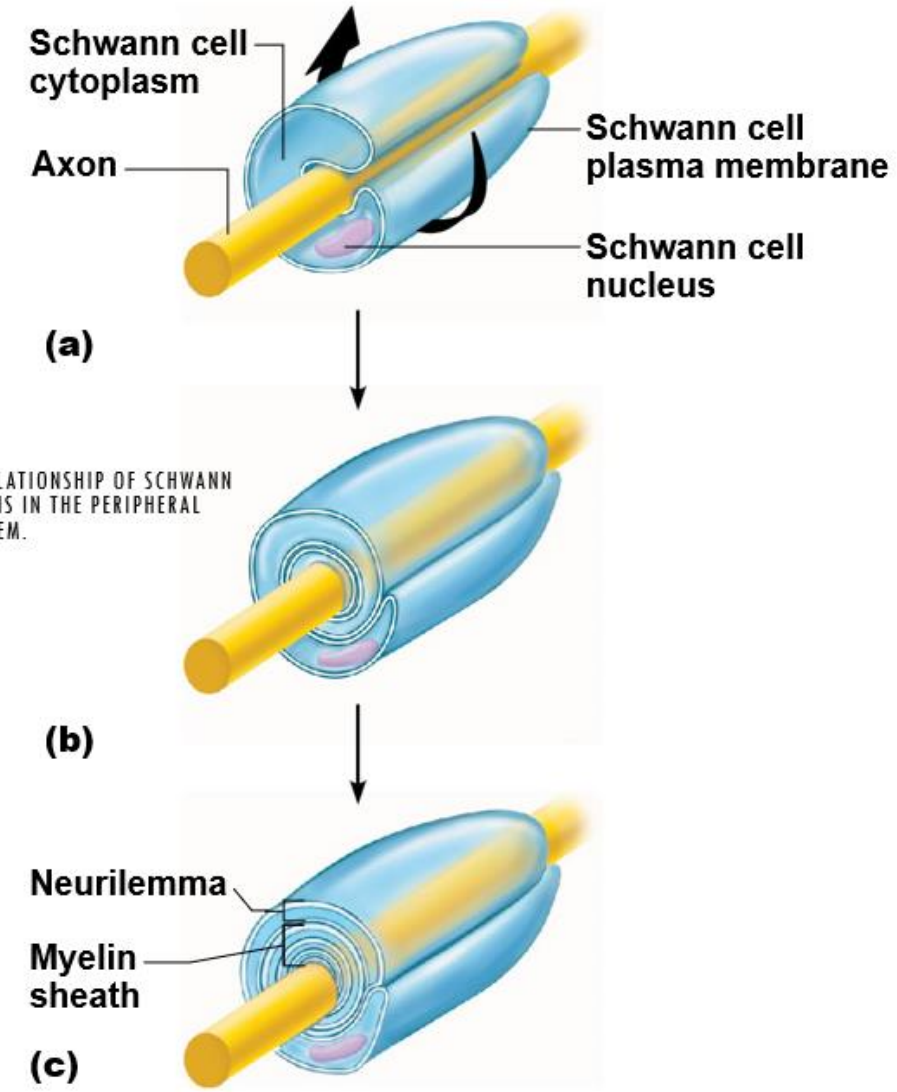


FIGURE 7.5 RELATIONSHIP OF SCHWANN CELLS TO AXONS IN THE PERIPHERAL NERVOUS SYSTEM.

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NERVOUS TISSUE

NEURONS

Myelin Sheaths and Associated Cells

- **Schwann cells (PNS):** Wrap axons in a *jelly roll–like fashion* to form the myelin sheath
- **Neurilemma:** Outer portion of the Schwann cell external to the myelin sheath; aids in axonal regeneration
- **Nodes of Ranvier:** Gaps between myelin segments along the axon; allow saltatory conduction
- **Oligodendrocytes (CNS):** Form myelin sheaths around multiple axons simultaneously
- **CNS axons lack a neurilemma,** hence limited regeneration

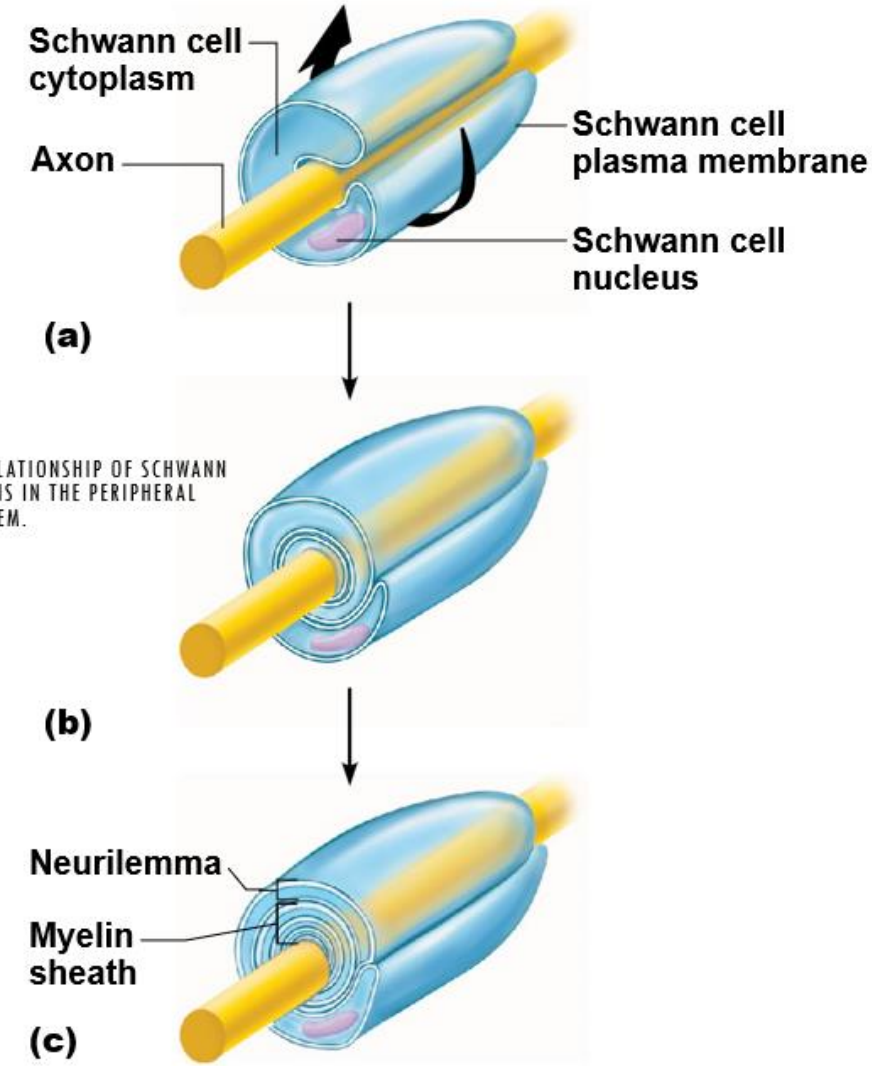


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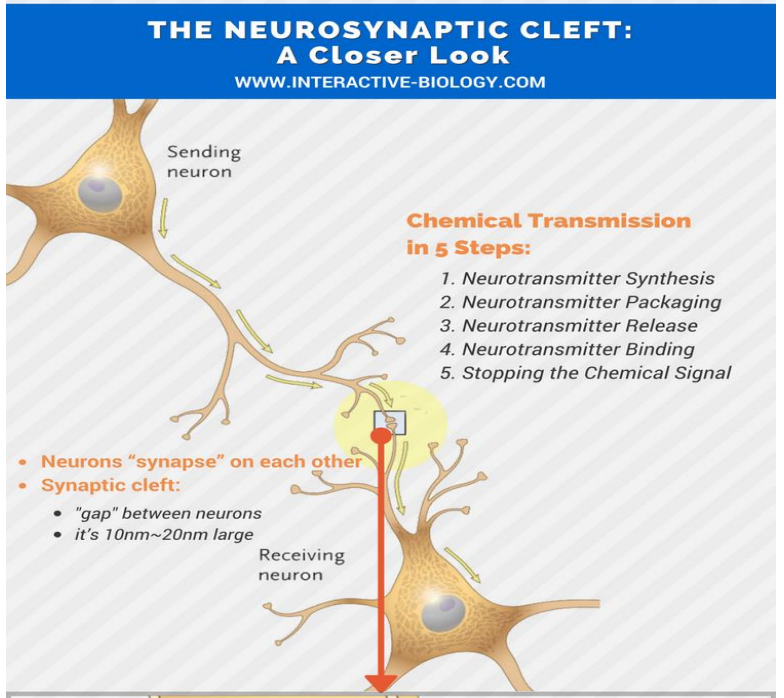
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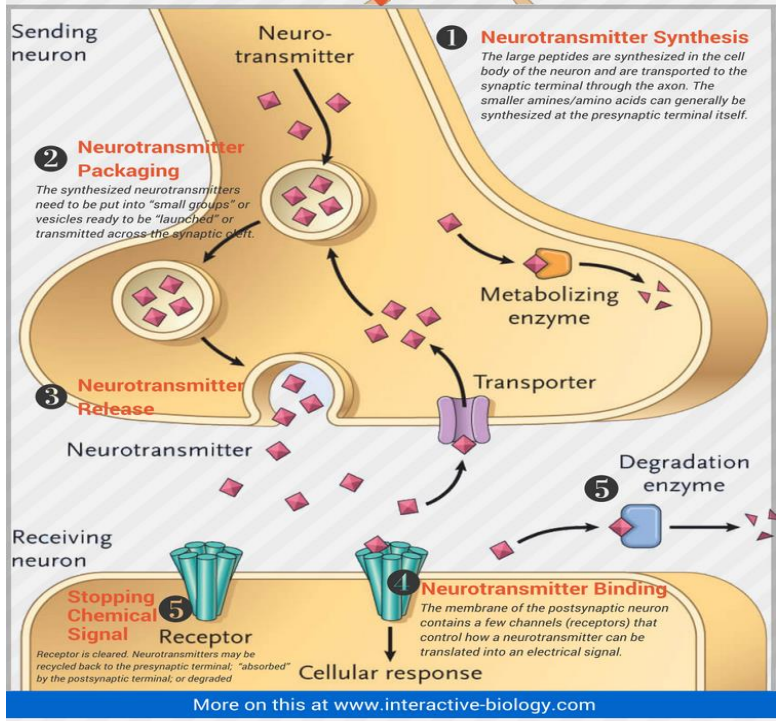
NERVOUS TISSUE

NEURONS

Synapse and Synaptic Cleft

- **Synapse:** Functional junction between neurons where a **nerve impulse is transmitted**
- **Synaptic cleft:** Tiny **gap** between the **axon terminal of one neuron** and the **dendrite or cell body of the next**
- Transmission occurs through **neurotransmitters** released from vesicles in the axon terminal
- Ensures **one-way communication** between neurons





NERVOUS TISSUE
KEY TERMINOLOGY

- **Nuclei:** Clusters of neuron cell bodies in the **CNS**
- **Ganglia:** Collections of neuron cell bodies in the **PNS**
- **Tracts:** Bundles of nerve fibers in the **CNS**
- **Nerves:** Bundles of nerve fibers in the **PNS**
- **White matter:** Regions with **myelinated fibers (tracts)**
- **Gray matter:** Regions with **unmyelinated fibers and cell bodies**

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NEURONS

FUNCTIONAL PROPERTIES OF NEURONS

- **Irritability**: Ability to **respond to a stimulus** and **convert it into a nerve impulse**
- **Conductivity:** Ability to **transmit the impulse** to other neurons, muscles, or glands
- These two properties enable **communication** and **coordination** within the nervous system

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NEURONS
CLASSIFICATION

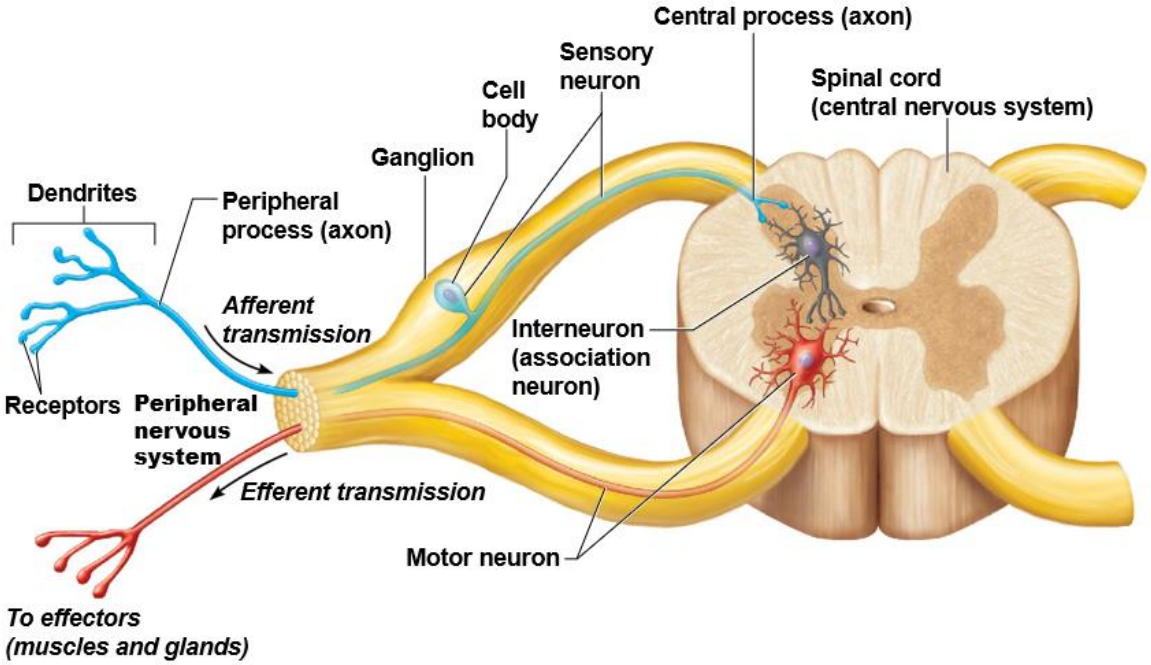
FUNCTIONAL	STRUCTURAL
Sensory (afferent) neurons	Multipolar neurons
Motor (efferent) neurons	Bipolar neurons
Interneurons (association neurons)	Unipolar neurons

NEURONS

FUNCTIONAL CLASSIFICATION

Sensory (Afferent) Neurons

- Transmit impulses **from sensory receptors to the central nervous system (CNS)**
- Responsible for detecting **changes in the internal and external environment**
- **Receptors include:**
  - ***Cutaneous sense organs*** – detect touch, pain, pressure, and temperature in the skin
  - ***Proprioceptors*** – detect body position, muscle tension, and movement in muscles and tendons



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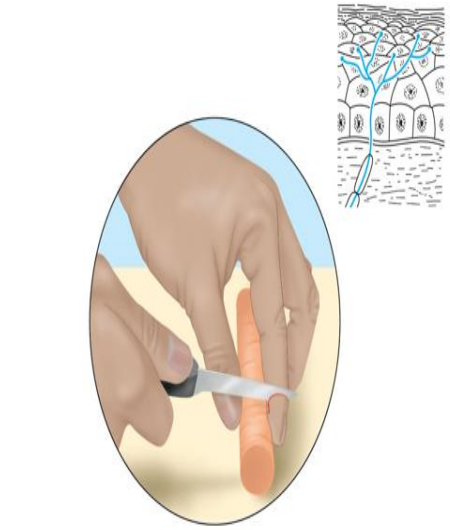
INTEGRATION TASK

Sensory Discrimination Task
Instructions:

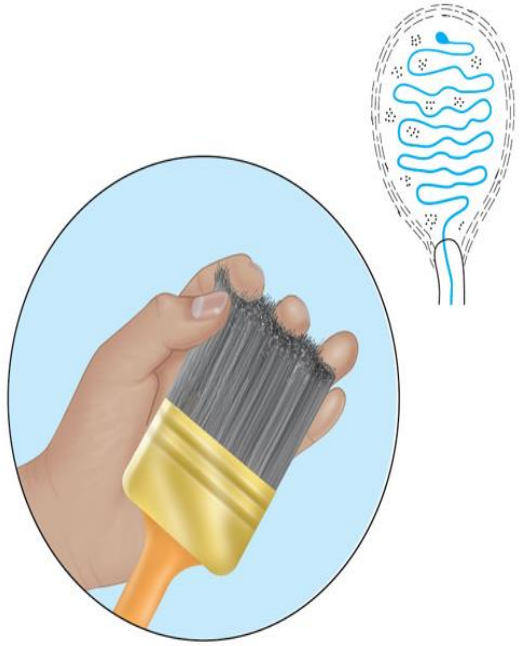
1. Close your eyes.
2. Using a **cotton bud,** gently touch your fingertip.
3. Using a **toothpick,** lightly prick your fingertip.
4. Identify which stimulus is **soft** and which is **sharp.**
5. Reflect on how your nervous system perceives and interprets these different sensations.



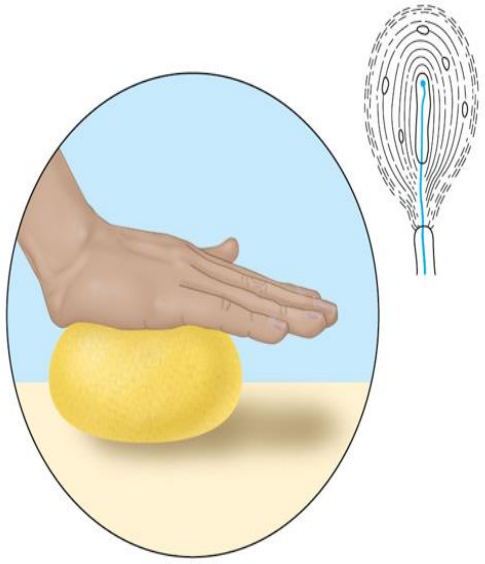
TYPES OF SENSORY RECEPTORS



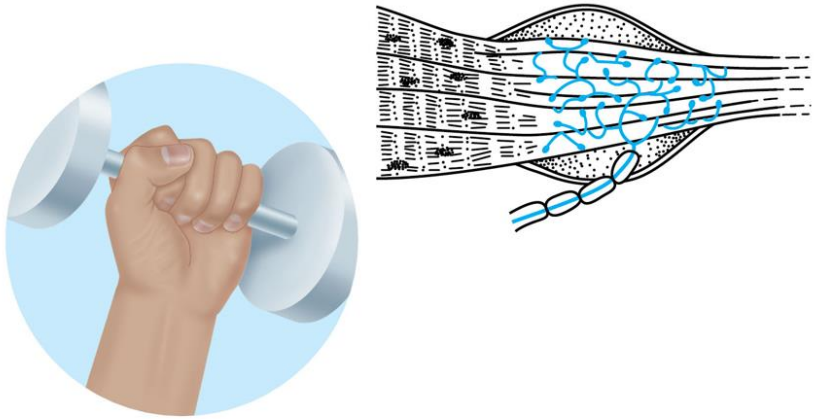
(a) Free nerve endings (pain and temperature receptors)



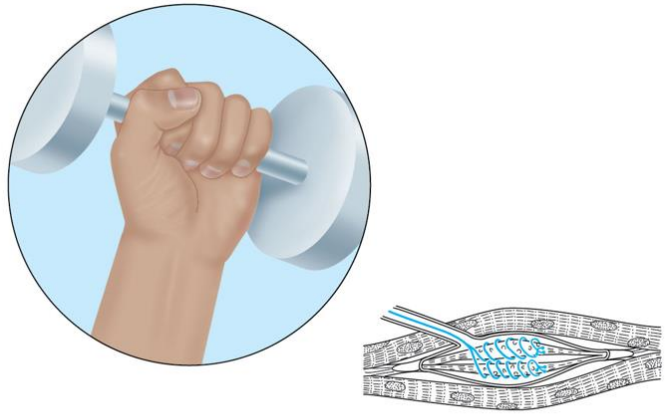
(b) Meissner's corpuscle (touch receptor)



(c) Lamellar corpuscle (deep pressure receptor)



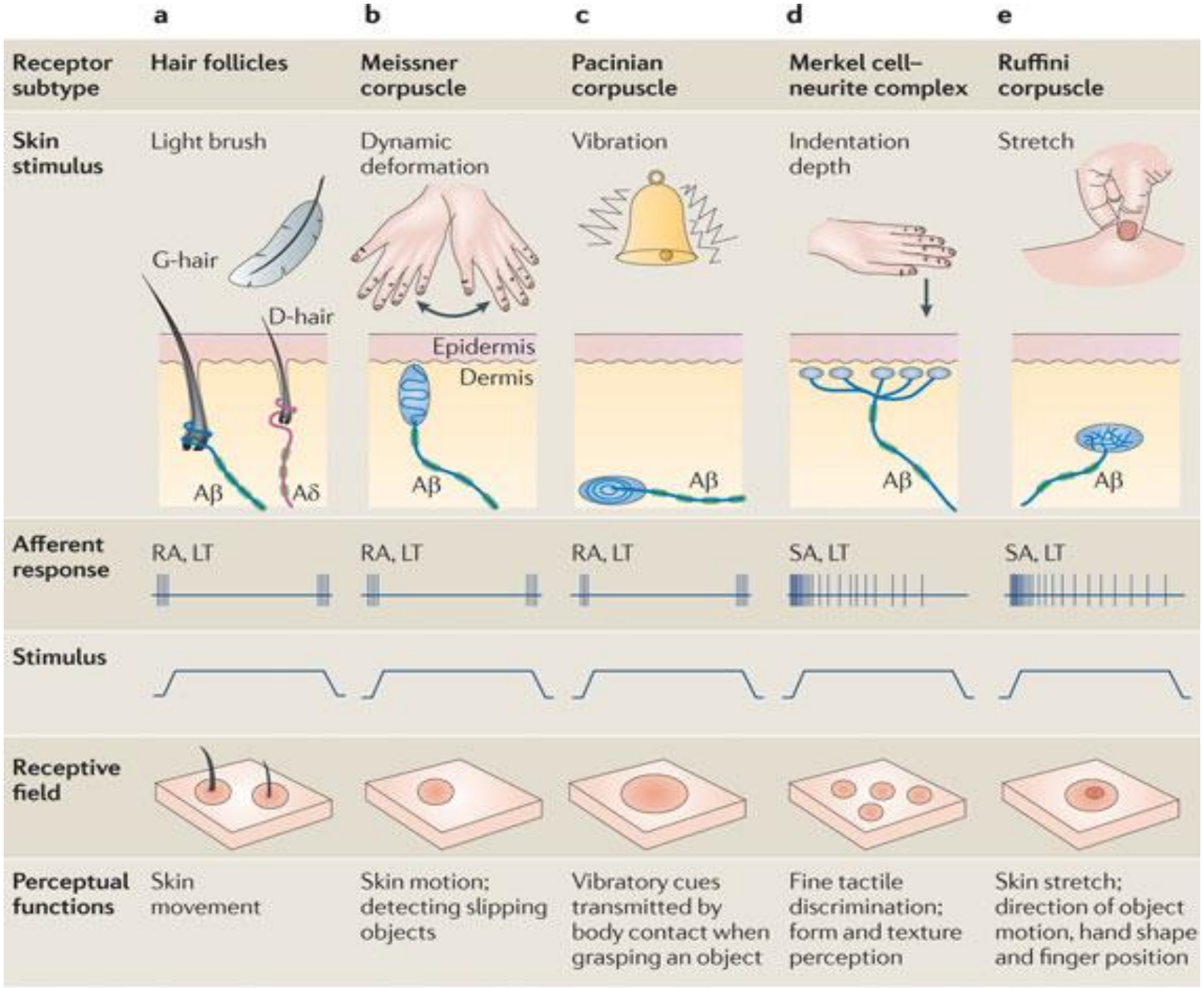
(d) Golgi tendon organ (proprioceptor)



(e) Muscle spindle (proprioceptor)

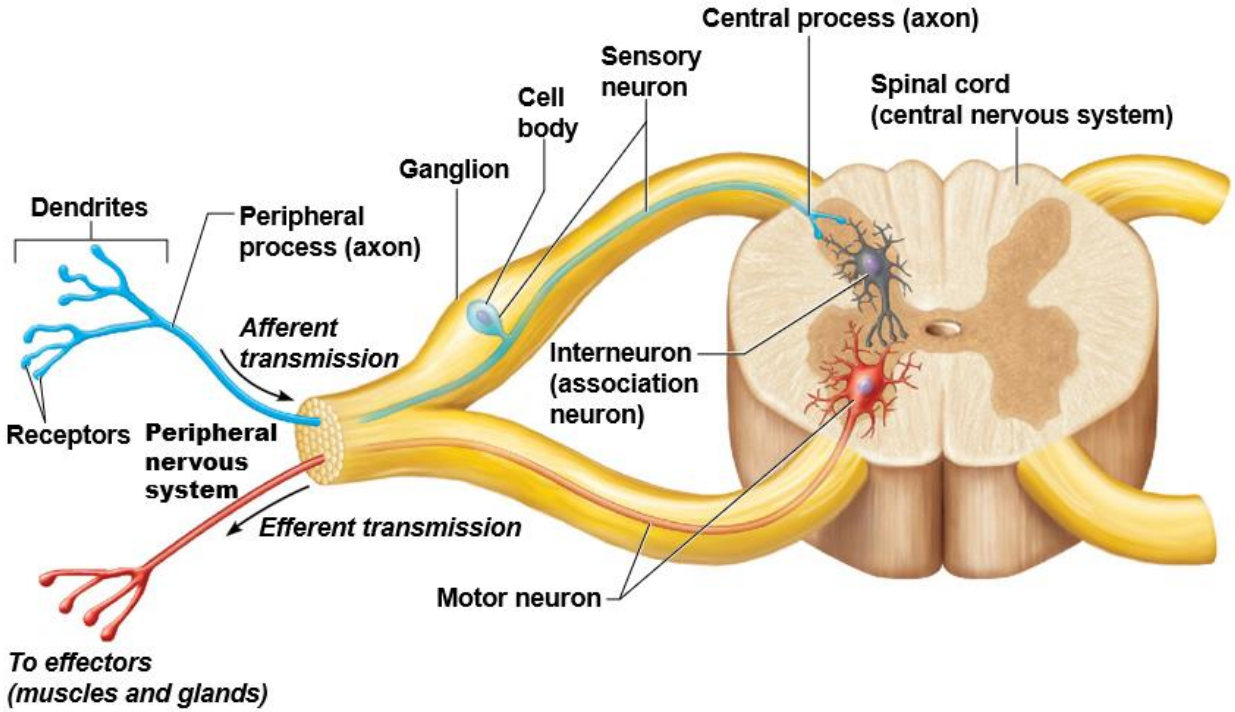
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TYPES OF SENSORY RECEPTORS



NEURONS

FUNCTIONAL CLASSIFICATION



1. **Motor (Efferent) Neurons**
   - Transmit impulses **from the CNS to effectors** (muscles and glands)
   - Responsible for initiating **movement** or **glandular secretion**
   - Cell bodies located in the **CNS**, axons extend to **PNS effectors**

2. **Interneurons (Association Neurons)**
   - Found **entirely within the CNS**
   - Connect **sensory (afferent)** and **motor (efferent)** neurons
   - Involved in **integration, reflexes,** and **decision-making**

NEURONS

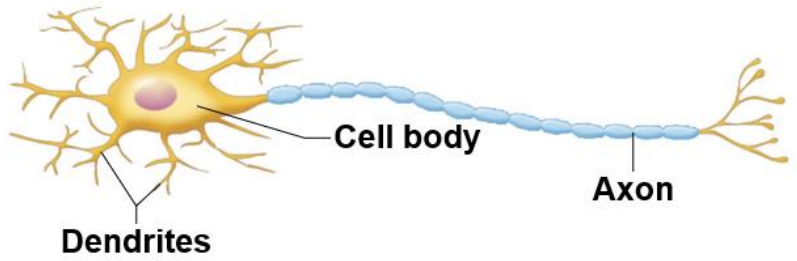
STRUCTURAL CLASSIFICATION

(Based on the number of processes extending from the cell body)

1. **Multipolar Neurons**
   - Many extensions from the cell body
   - One axon and multiple dendrites
   - Includes **all motor neurons** and **interneurons**
   - **Most common** structural type in the CNS

2. **Bipolar Neurons**
   - One axon and one dendrite
   - Found in **special sense organs** (e.g., retina of the eye, olfactory epithelium of the nose)

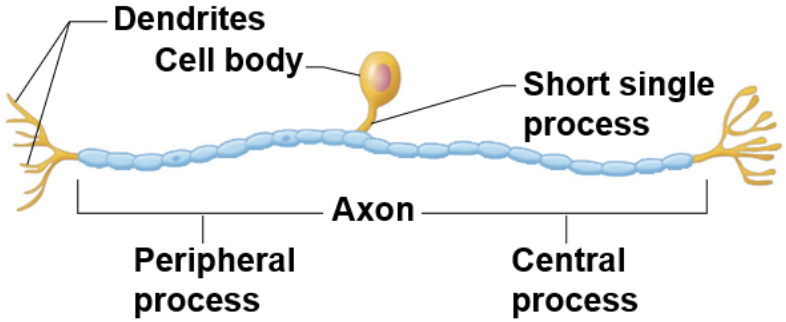
3. **Unipolar (Pseudounipolar) Neurons**
   - A single short process divides into two branches
   - Found in **sensory neurons** of the **PNS ganglia**
   - Conduct impulses **toward and away from** the cell body



(a) Multipolar neuron



(b) Bipolar neuron



(c) Unipolar neuron



OBJECTIVES

At the end of the session, students are expected to:

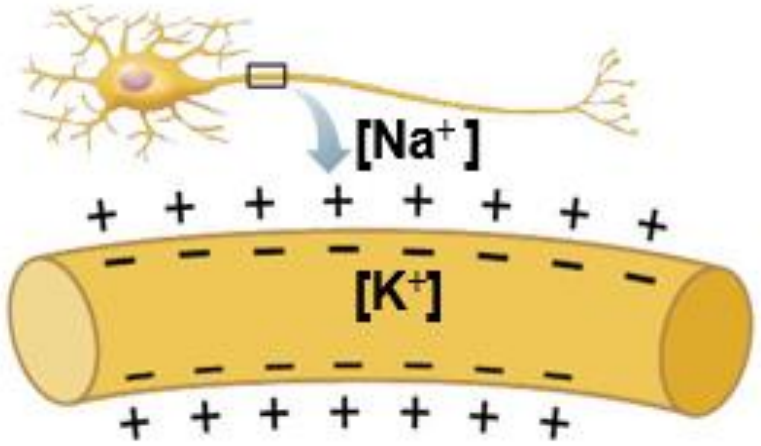
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            3. Enteric Nervous System
        2. Somatic Nervous System
        3. Sensory (Afferent) Nervous System
        4. Motor (Efferent) Nervous System

NEURONS

ACTION POTENTIAL

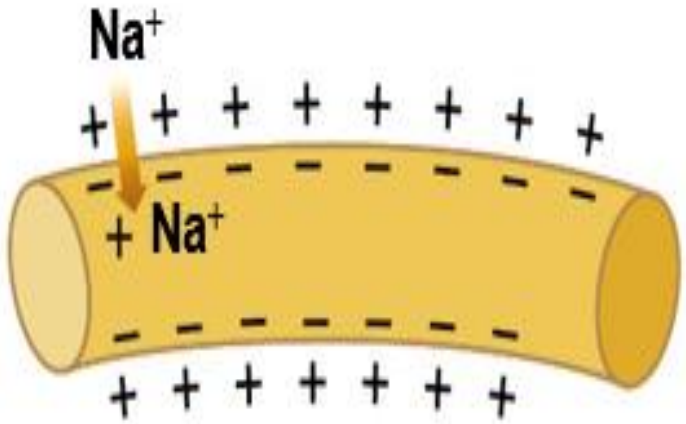
1. Resting Membrane is Polarized

- The **resting neuron** has a **polarized membrane**—a difference in electrical charge across the plasma membrane.
- The **inside of the neuron** is **negatively charged** relative to the **outside.**
- This resting potential is typically around **–70 mV**.
- Maintained by the **sodium-potassium (Na^+/K^+) pump** and **selective membrane permeability.**



NEURONS
ACTION POTENTIAL

2. Stimulus Initiates Local Depolarization

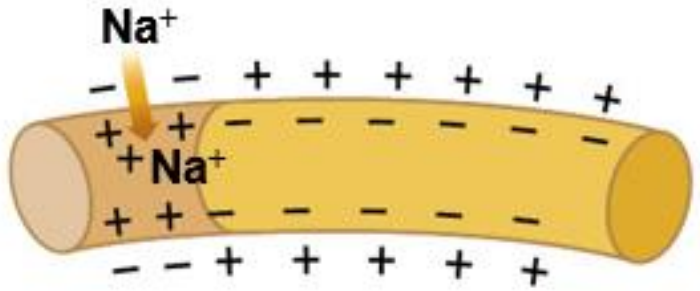


- A **stimulus** changes the **membrane permeability** to sodium (Na^+) ions.
- **Sodium channels open,** allowing Na^+ to diffuse **into** the neuron.
- The **influx of Na^+** reverses the membrane polarity — this is **depolarization.**
- A **graded potential** (localized depolarization) occurs when the inside becomes more positive.
- If the stimulus is **strong enough,** the **threshold potential** is reached, and an **action potential** is generated.

NEURONS

ACTION POTENTIAL

3. Depolarization and Generation of an Action Potential

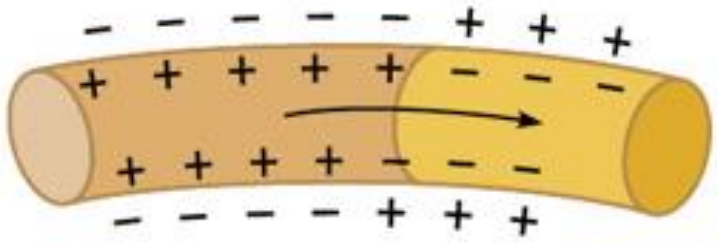


- At rest, the **neuronal membrane is polarized:**
  - **More positive ions outside** than inside.
  - **K^+** is the major intracellular cation.
  - **Na^+** is the major extracellular cation.
- A **strong stimulus** opens sodium channels, allowing **Na^+ influx**, reversing membrane polarity.
- When polarity is completely reversed, an **action potential** is generated.
- The **resting membrane potential** (~–70 mV) is temporarily lost during depolarization.

NEURONS
ACTION POTENTIAL

4. Propagation of the Action Potential

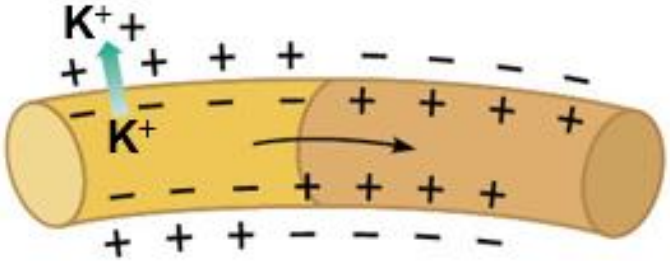
- If sufficient **Na^+ influx** occurs, the **action potential** is generated and **propagated** along the entire axon.
- The **all-or-none principle**: a nerve impulse **either occurs completely or not at all.**
- **Myelinated fibers** conduct impulses **faster** than unmyelinated ones due to **saltatory conduction** (impulse jumps from node to node).



NEURONS
ACTION POTENTIAL

5. Repolarization

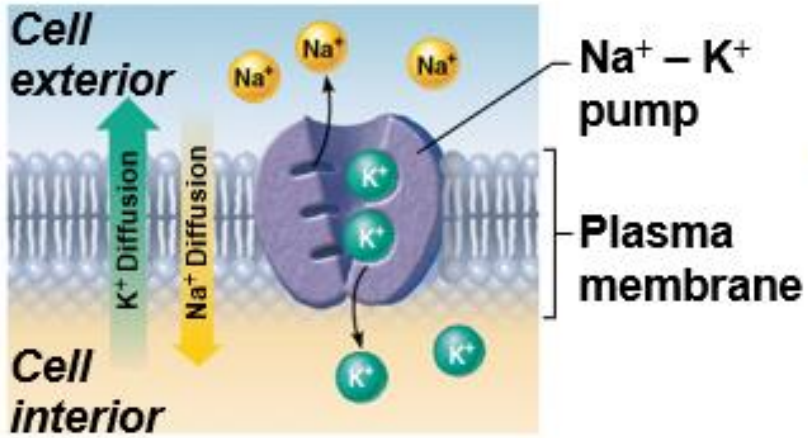
- Membrane becomes **impermeable to Na^+** and **permeable to K^+.**
- **K^+ ions diffuse out** of the neuron, restoring the **negative internal charge.**
- **Repolarization** re-establishes resting polarity:
  - Inside of membrane → **negative**
  - Outside of membrane → **positive**
- Prepares the neuron for the **next action potential.**



NEURONS
ACTION POTENTIAL

6. Restoration of Ionic Conditions (Na^+/K^+ Pump)

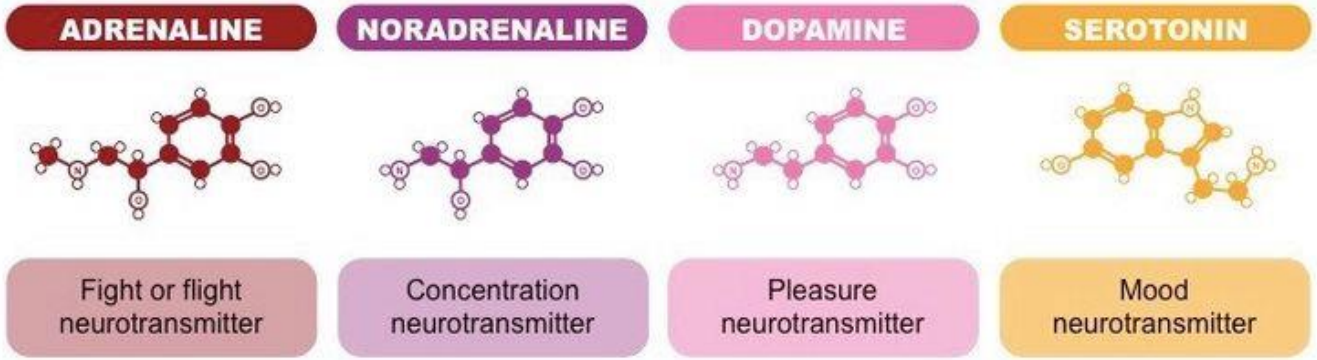
- After repolarization, ionic balance is restored by the **sodium–potassium pump (Na^+/K^+-ATPase)**.
- The pump uses **ATP** to:
  - **Eject 3 Na^+ ions** from the neuron.
  - **Return 2 K^+ ions** into the neuron.
- This re-establishes **resting ionic distribution** and **membrane potential**.
- The neuron remains in a **refractory period** and **cannot fire again** until repolarization is complete.



BIOFLIX: HOW SYNAPSES WORK

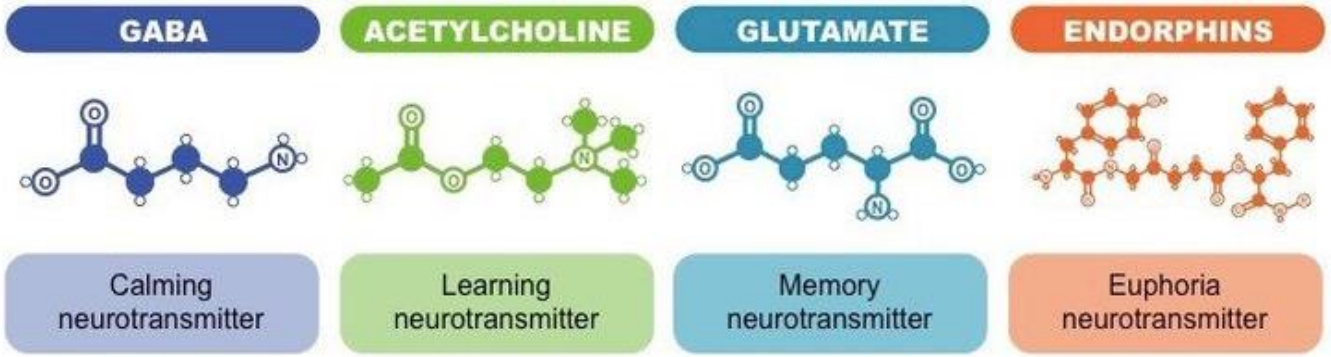


NEUROTRANSMITTERS



- **Adrenaline (Epinephrine):** Increases heart rate and prepares the body for "fight-or-flight."
- **Noradrenaline (Norepinephrine):** Involved in alertness, attention, and mood regulation.
- **Dopamine:** Regulates movement, motivation, and pleasure; imbalance linked to Parkinson's disease and schizophrenia.
- **Serotonin:** Affects mood, sleep, and appetite; low levels associated with depression.

NEUROTRANSMITTERS



- **GABA (Gamma-Aminobutyric Acid):** Primary inhibitory neurotransmitter; reduces neuronal excitability and anxiety.
- **Acetylcholine (ACh):** Involved in muscle contraction, learning, and memory; decreased levels seen in Alzheimer's disease.
- **Glutamate:** Main excitatory neurotransmitter; essential for learning and memory but excessive levels cause excitotoxicity.
- **Endorphins:** Natural painkillers that produce euphoria and reduce stress perception.



OBJECTIVES

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            3. Enteric Nervous System
        2. Somatic Nervous System
        3. Sensory (Afferent) Nervous System
        4. Motor (Efferent) Nervous System

REFLEXES

- **Reflexes** are **rapid, predictable, and involuntary** responses to stimuli.
- They occur through specific **neural pathways called reflex arcs.**
- **Types of Reflexes:**
  - **Somatic Reflexes:** Involve skeletal muscles (e.g., knee-jerk reflex).
  - **Autonomic Reflexes**: Involve smooth muscles, cardiac muscles, or glands (e.g., pupil constriction, heart rate regulation).

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REFLEXES

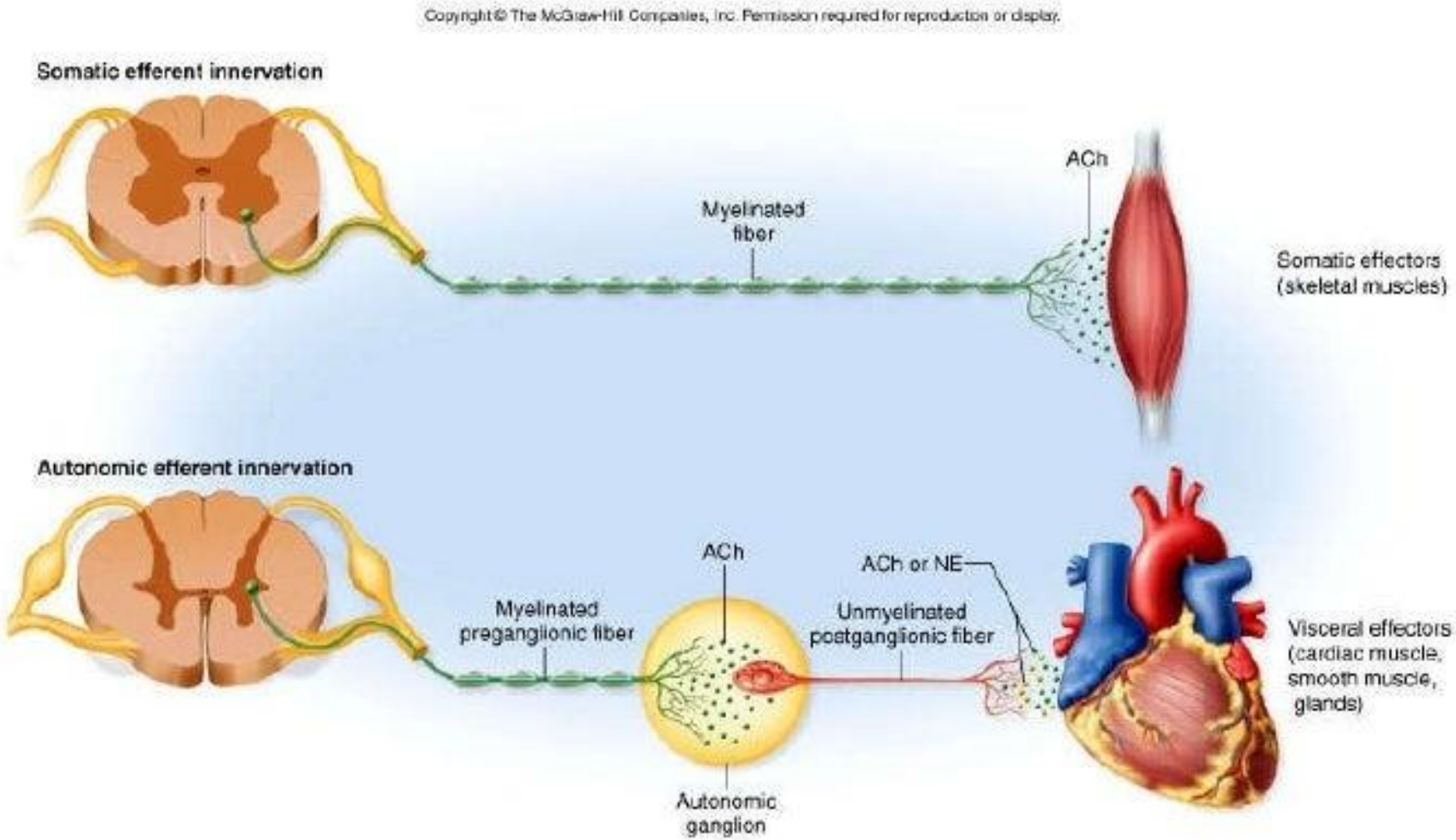
Somatic Reflexes

- Reflexes that stimulate **skeletal muscles.**
- Occur **involuntarily,** even though skeletal muscle is usually under voluntary control.
- **Example:** Pulling your hand away from a hot object (withdrawal reflex).

Autonomic Reflexes

- Regulate activities of **smooth muscles,** the **heart,** and **glands.**
- Maintain **homeostasis** in involuntary body functions.
- **Examples:** Regulation of blood pressure, heart rate, glandular secretions, and digestion.

Somatic vs. Autonomic

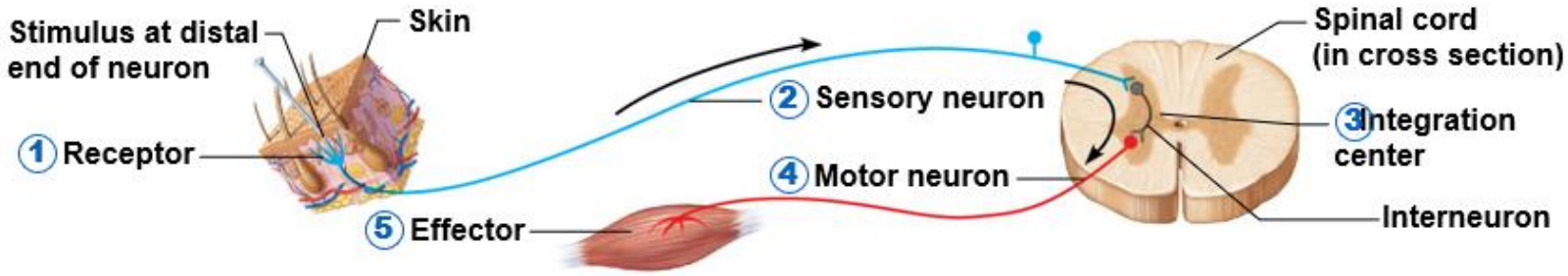


REFLEXES

REFLEX ARC

Five Elements of a Reflex Arc

1. **Sensory Receptor** – Detects or responds to a specific stimulus (e.g., pain, pressure, temperature).
2. **Sensory Neuron** – Transmits the nerve impulse from the receptor to the **integration center** (CNS).
3. **Integration Center (CNS)** – Processes incoming information and determines an appropriate response.
4. **Motor Neuron** – Conducts impulses from the integration center to the effector.
5. **Effector Organ** – The **muscle or gland** that carries out the response (e.g., muscle contraction or gland secretion).



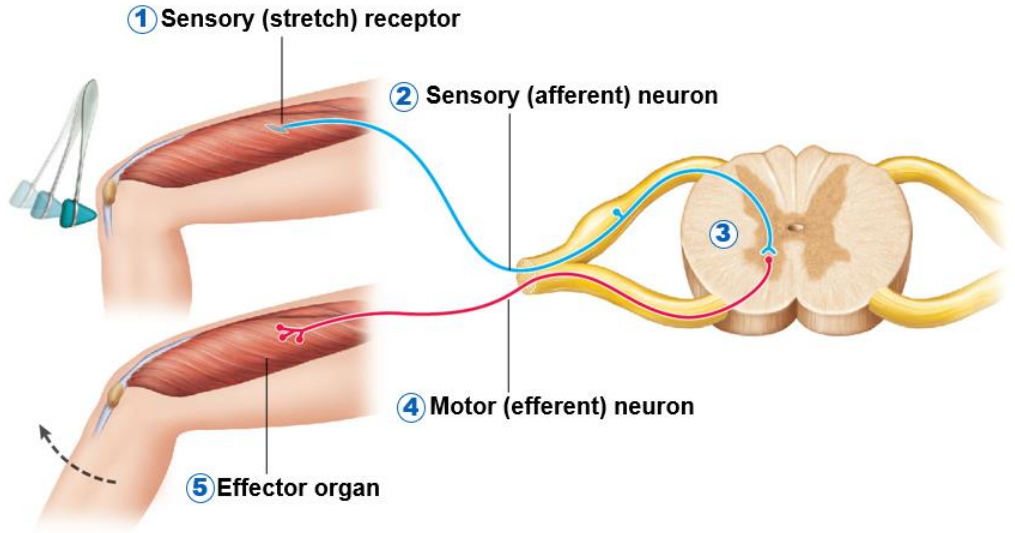
(a) Five basic elements of reflex arc

REFLEXES

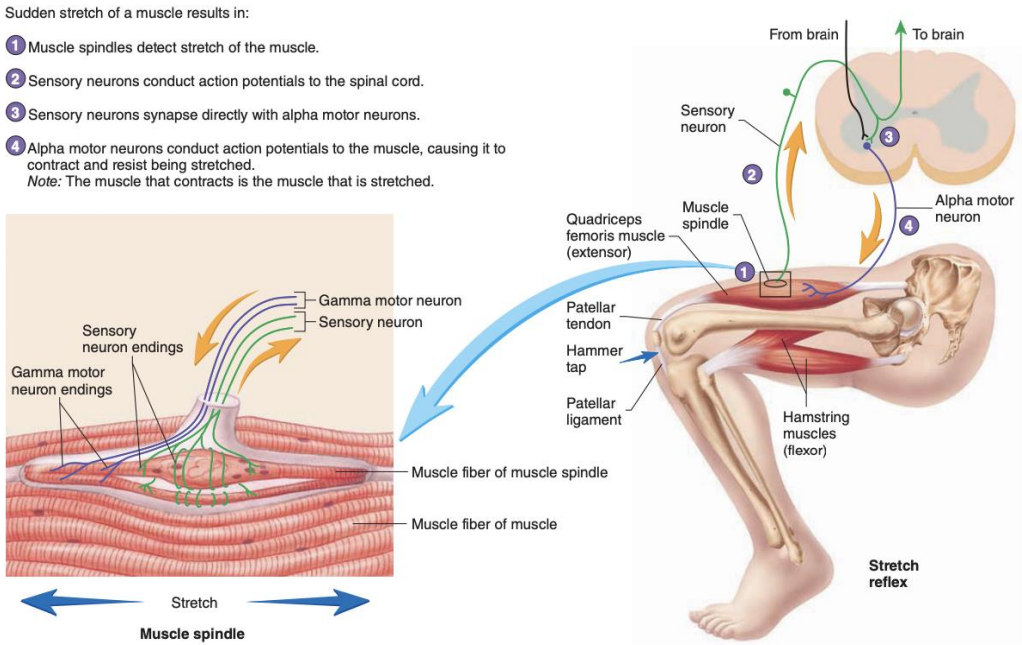
REFLEX ARC

Two-Neuron Reflex Arcs

- **Simplest reflex arc** type, also called **monosynaptic reflex.**
- Involves only **two neurons:**
  - **Sensory neuron –** carries the impulse from the receptor to the spinal cord.
  - **Motor neuron –** carries the impulse from the spinal cord to the effector.
- **Examples:**
  - **Patellar (knee-jerk) reflex**
  - **Achilles (ankle-jerk) reflex**



(b) Two-neuron reflex arc



PROCESS FIGURE 12.6 Stretch Reflex

REFLEXES

REFLEX ARC

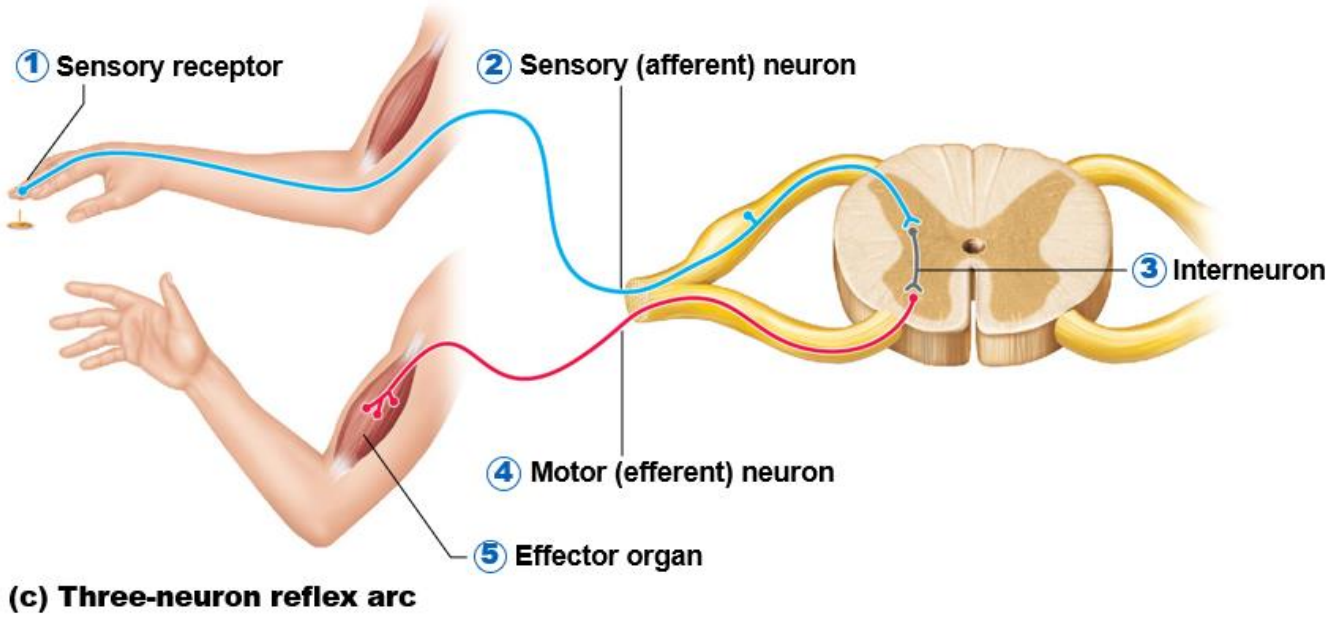
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- **Examples:**
  - **Patellar (knee-jerk) reflex**
  - **Achilles (ankle-jerk) reflex**



REFLEXES

REFLEX ARC



(c) Three-neuron reflex arc

Three-Neuron Reflex Arcs

- **Also called polysynaptic reflex arcs.**
- Consists of **five elements:**
  - **Receptor** – detects stimulus.
  - **Sensory neuron** – transmits signal to the CNS.
  - **Interneuron** – processes information in the CNS and coordinates response.
  - **Motor neuron** – carries impulse to the effector.
  - **Effector** – muscle or gland that performs the response.
- **Example: Flexor (withdrawal) reflex.**

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OBJECTIVES

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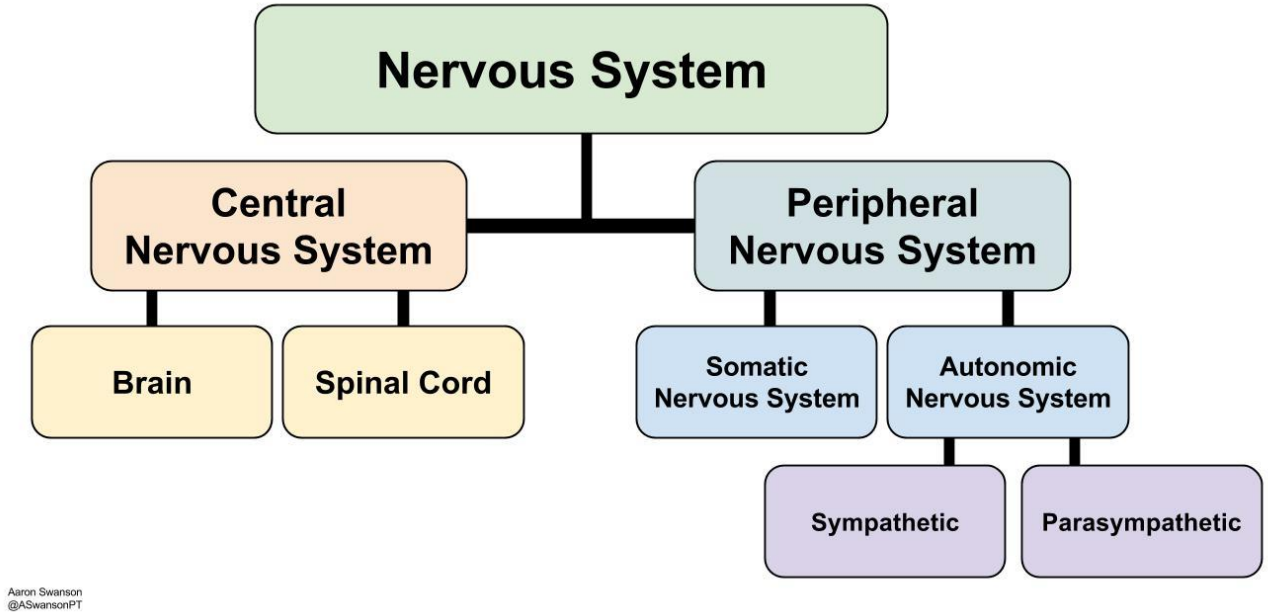
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ORGANIZATION OF THE NERVOUS SYSTEM

Nervous System Classifications

1. Structural Classification: Based on anatomical organization of the nervous system.

2. Functional Classification: Based on the activities or functions performed by different parts of the nervous system.



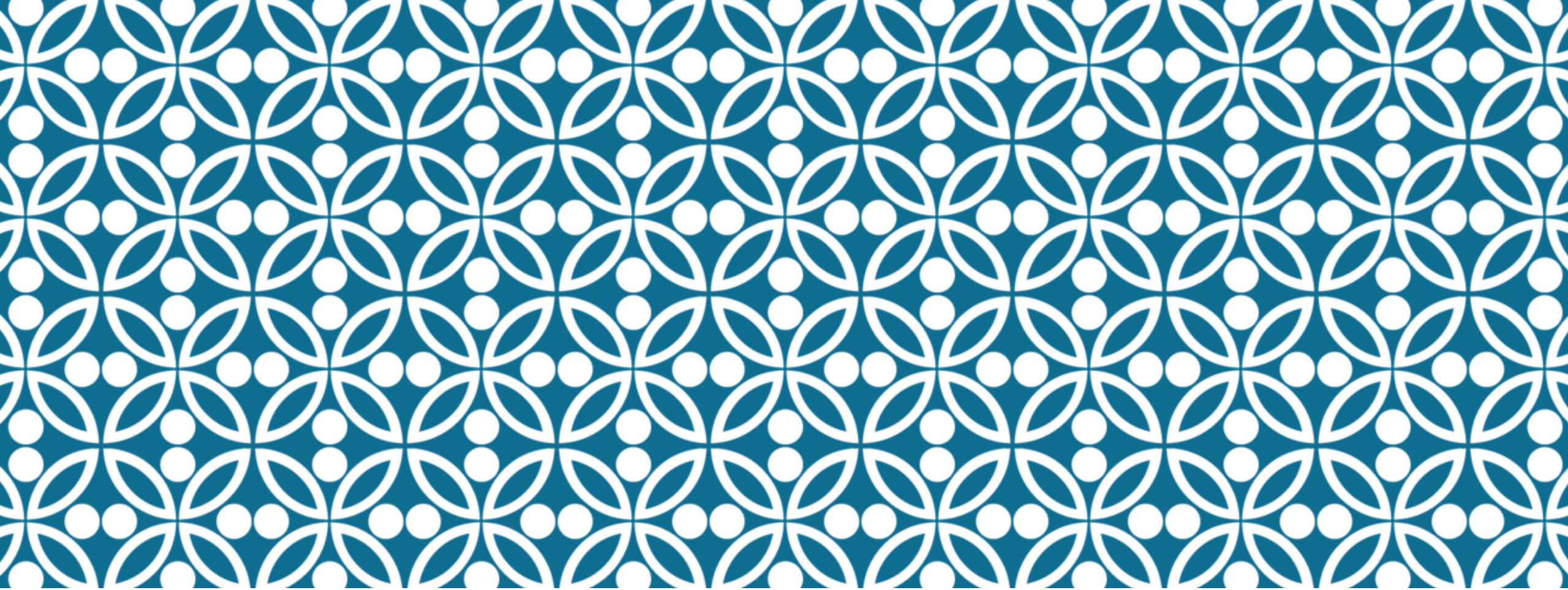
STRUCTURAL CLASSIFICATION

Central Nervous System (CNS)	Peripheral Nervous System (PNS)
Organs ❖ Brain ❖ Spinal cord	Nerves extending from the brain and spinal cord (located outside the brain and spinal cord) ❖ Spinal nerves—carry impulses to and from the spinal cord ❖ Cranial nerves—carry impulses to and from the brain
Function ✓ Integration; command center ✓ Interprets incoming sensory information ✓ Issues outgoing instructions	Functions ✓ Serve as communication lines among sensory organs, the brain and spinal cord, and glands or muscles

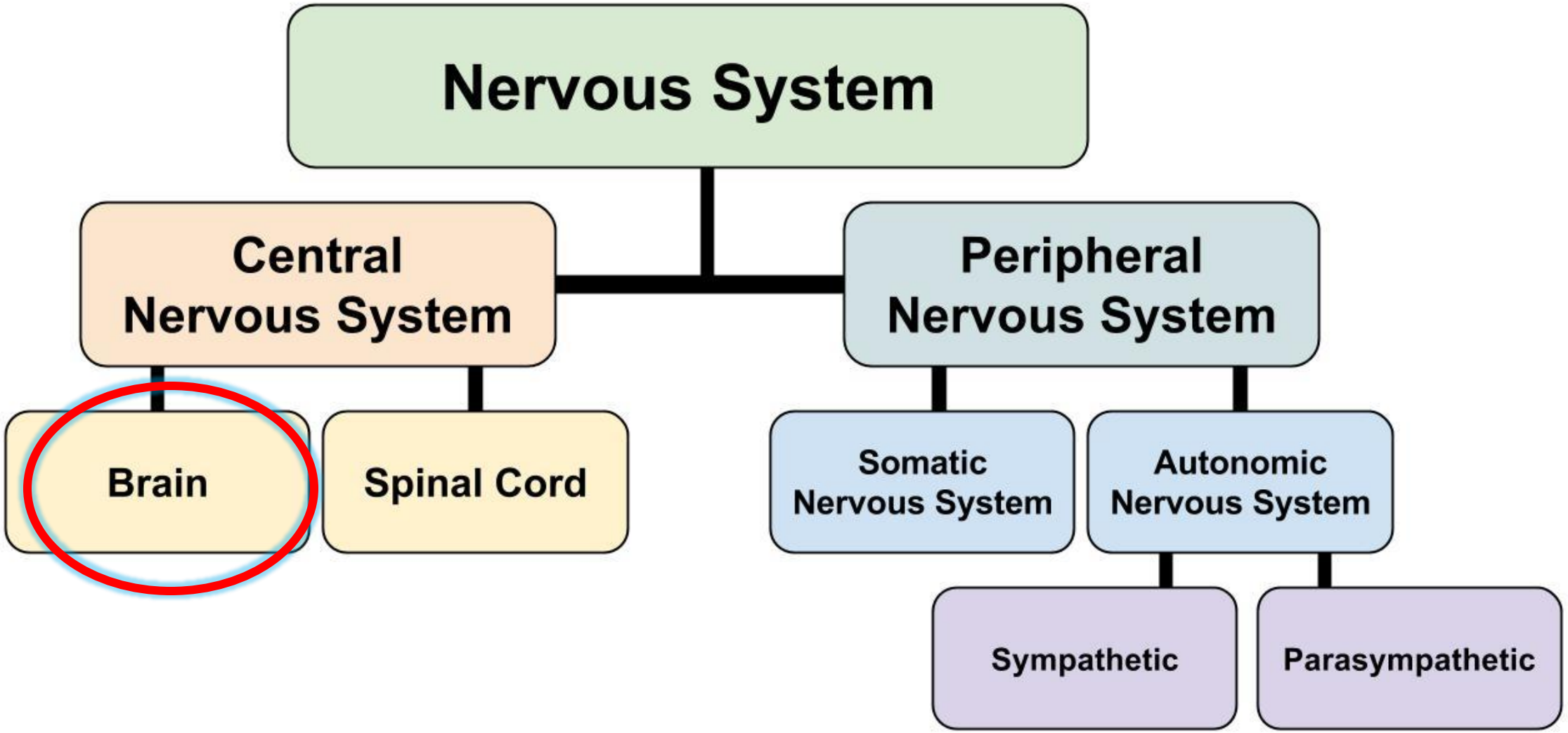
FUNCTIONAL CLASSIFICATION

Sensory (Afferent) Division	Motor (efferent) division
Nerve fibers that carry **information to the central nervous system**	Nerve fibers that carry impulses away from the central nervous system organs to effector organs (muscles and glands)
Two subdivisions ❖ Somatic sensory (afferent) fibers carry information from the skin, skeletal muscles, and joints ❖ Visceral sensory (afferent) fibers carry information from visceral organs	Two subdivisions ❖ Somatic nervous system = voluntary Consciously (voluntarily) controls skeletal muscles ❖ Autonomic nervous system = involuntary Automatically controls smooth and cardiac muscles and glands Further divided into the sympathetic and parasympathetic nervous systems

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CENTRAL NERVOUS SYSTEM



Aaron Swanson
@ASwansonPT

BRAIN
ANATOMY OF THE BRAIN

4 Major Brain Regions

1. Cerebral hemispheres/ Cerebrum
   - Cortex
   - White matter
   - Basal nuclei
2. Diencephalon
   - Thalamus
   - Hypothalamus
   - Epithalamus
3. Brainstem
   - Midbrain
   - Pons
   - Medulla oblongata
4. Cerebellum

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BRAIN

EMBRYONIC DEVELOPMENT OF THE HUMAN BRAIN

Three Primary Brain Vesicles (at 3–4 weeks

1. **Prosencephalon (Forebrain)**
2. **Mesencephalon (Midbrain)**
3. **Rhombencephalon (Hindbrain)**

Five Secondary Brain Vesicles (at 5 weeks):

- **Prosencephalon** → Telencephalon, Diencephalon
- **Mesencephalon** → remains as Midbrain
- **Rhombencephalon** → Metencephalon, Myelencephalon

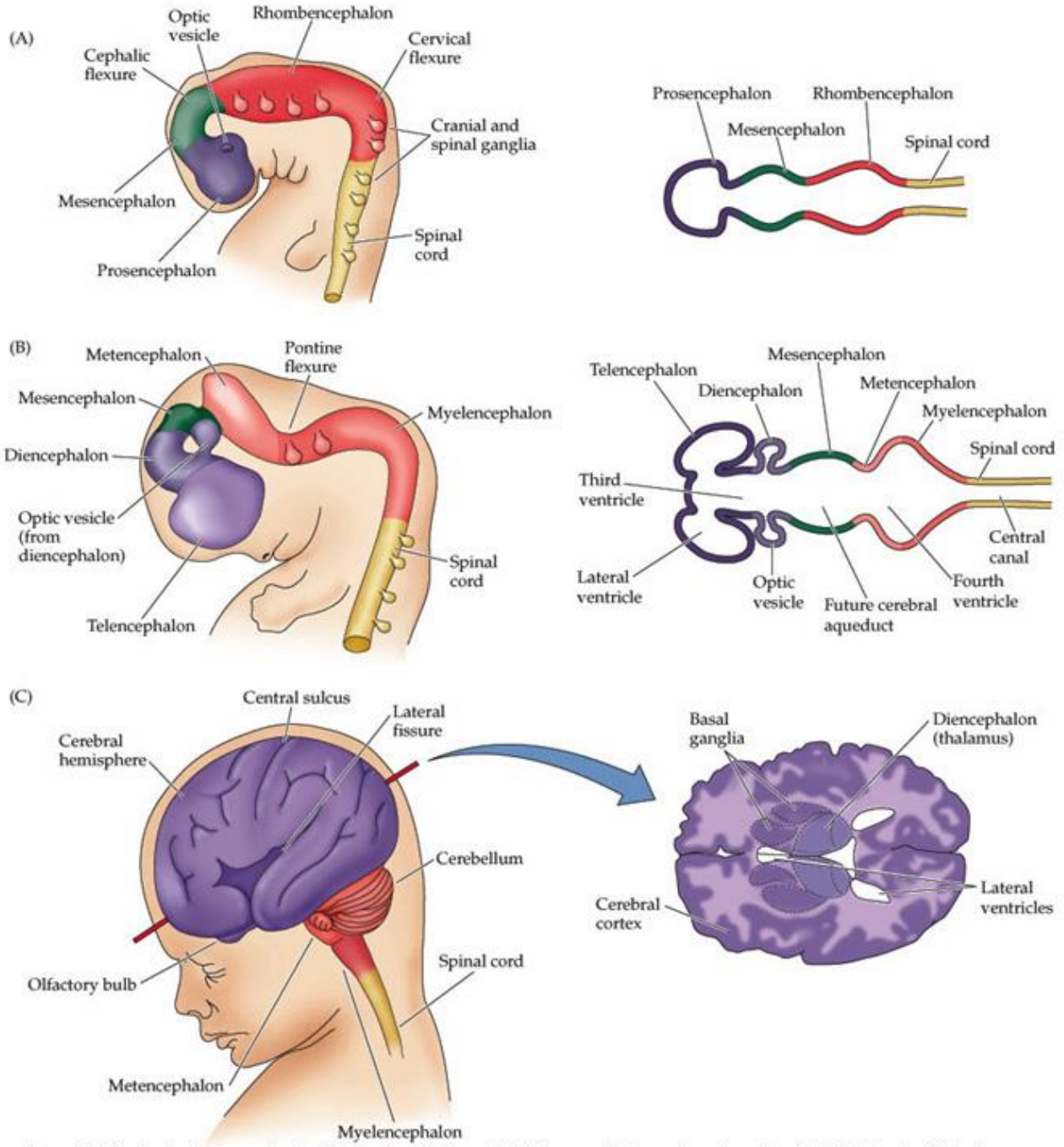
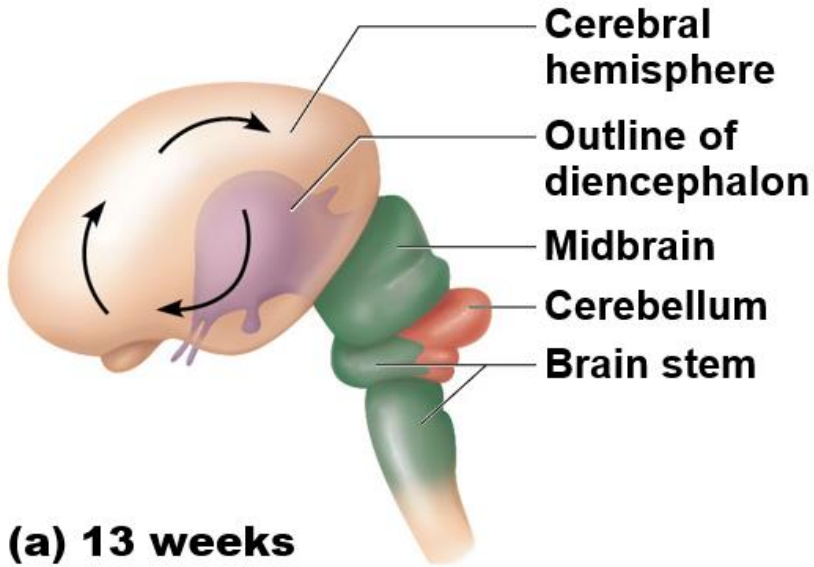


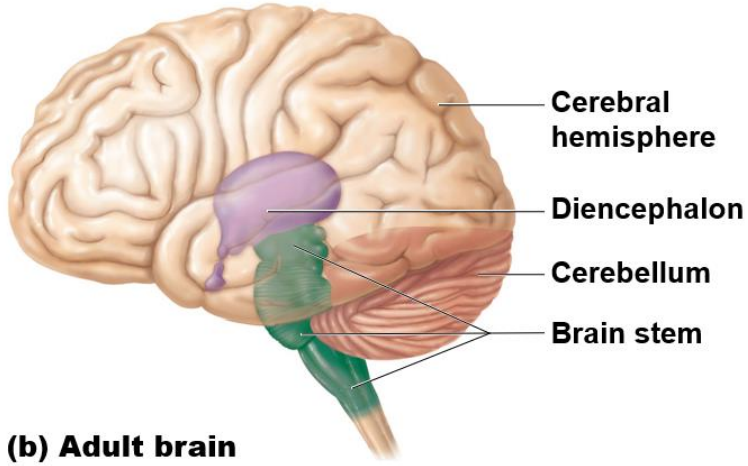
Figure 1.1. The developing human brain. Left, drawings (not to scale) of the neural tube as viewed from the side. Right, top two illustrations drawn as though the tube were stretched out to make it straight and the top were cut off; view is from the top. The bottom right illustration (C) shows the appearance of a section through the forebrain at the level indicating to the left (Figure 22.3 from Neuroscience, 5th Ed.).

BRAIN
DEVELOPMENT AND MAJOR REGIONS OF THE HUMAN BRAIN

At 13 weeks (embryonic brain):

- *Cerebral hemispheres* begin to enlarge
- *Diencephalon* outlined
- *Midbrain, cerebellum, and brain stem* form early structural divisions





BRAIN
EMBRYONIC DEVELOPMENT OF THE HUMAN BRAIN

Adult Brain Structures Derived:

- **Telencephalon** → Cerebral hemispheres
- **Diencephalon** → Thalamus, Hypothalamus, Epithalamus
- **Mesencephalon** → Midbrain
- **Metencephalon** → Pons, Cerebellum
- **Myelencephalon** → Medulla oblongata

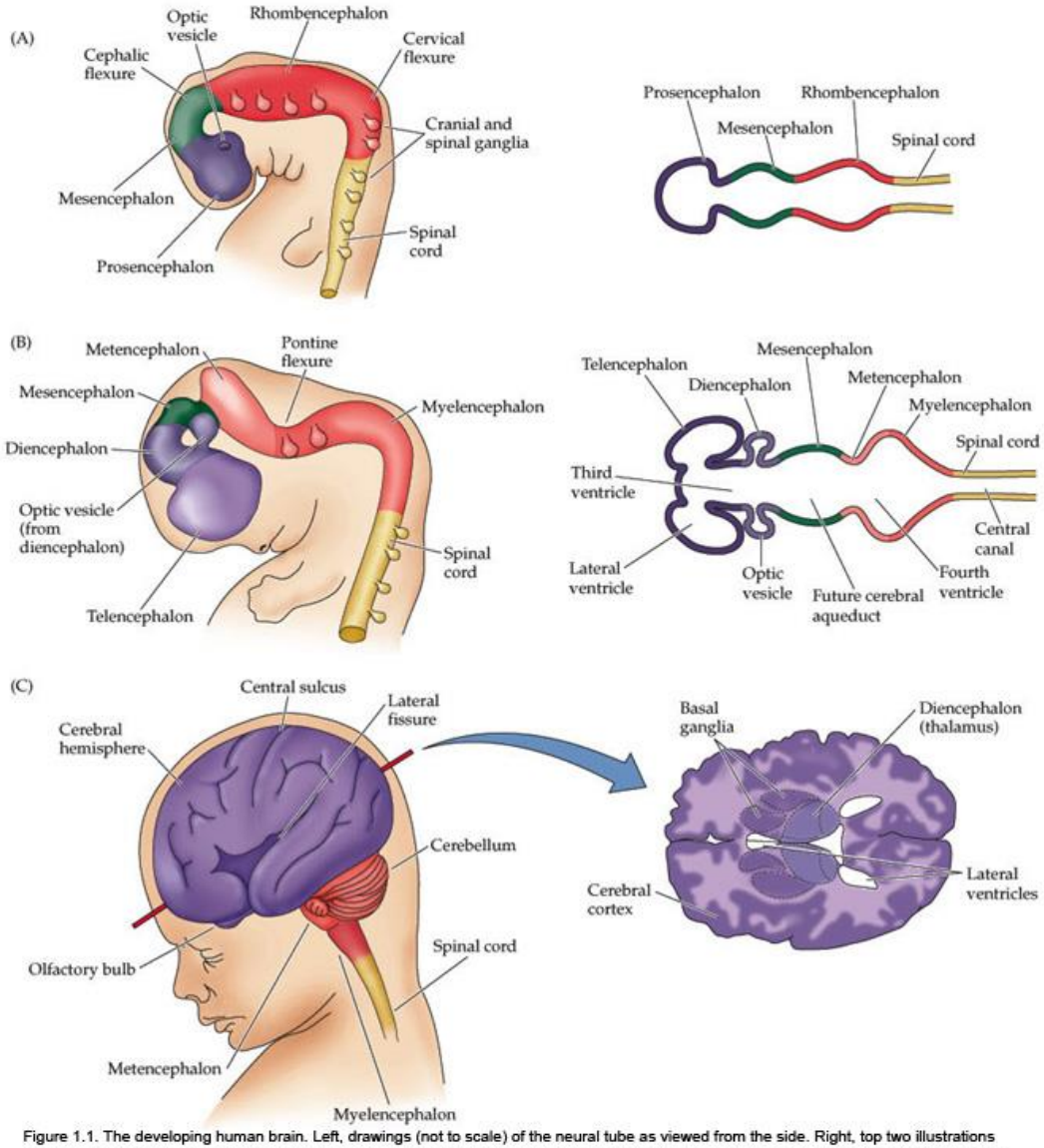


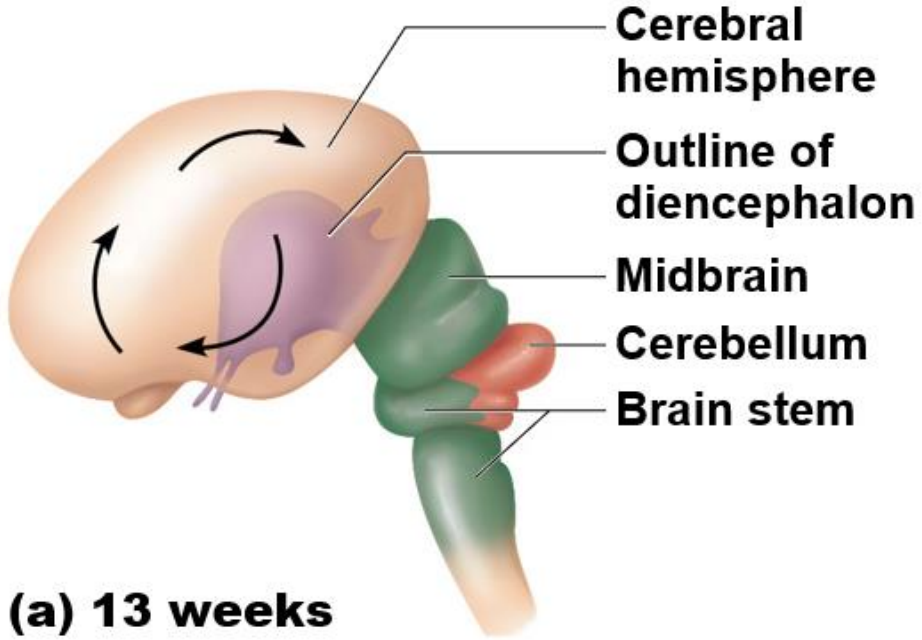
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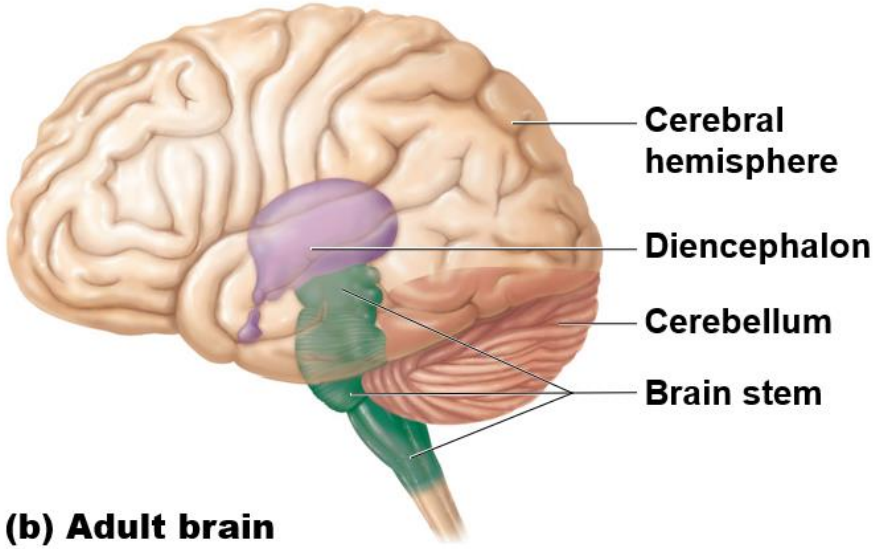
BRAIN

DEVELOPMENT AND MAJOR REGIONS OF THE HUMAN BRAIN

Adult brain:

- **Cerebral hemispheres** – largest part; responsible for higher cognitive functions
- **Diencephalon** – includes thalamus and hypothalamus; processes sensory and autonomic information
- **Cerebellum** – coordinates movement and balance
- **Brain stem** – connects brain to spinal cord; controls vital autonomic functions





BRAIN
ANATOMY OF THE BRAIN

4 Major Brain Regions

1. Cerebral hemispheres/ Cerebrum
   - Cortex
   - White matter
   - Basal nuclei

2. Diencephalon
   - Thalamus
   - Hypothalamus
   - Subthalamus
   - Epithalamus

3. Brainstem
   - Midbrain
   - Pons
   - Medulla oblongata

4. Cerebellum

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BRAIN

ANATOMY OF THE BRAIN

1. **Cerebral Hemispheres (Cerebrum)**
    1. **Cortex:** Outer gray matter responsible for conscious thought, perception, and voluntary movement.
    2. **White matter:** Myelinated fiber tracts connecting different brain regions.
    3. **Basal nuclei:** Deep gray matter involved in movement regulation and procedural learning.

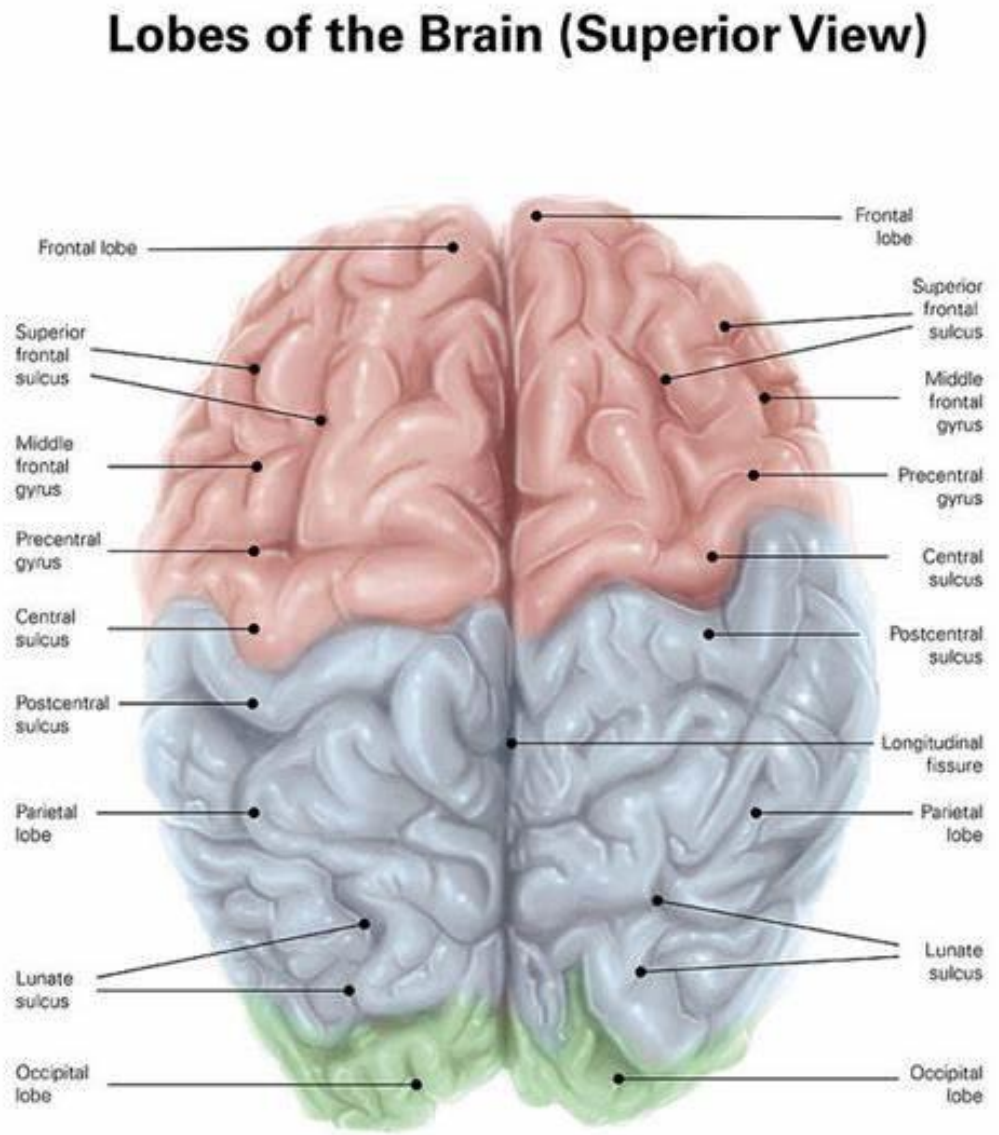
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BRAIN

CEREBRAL HEMISPHERES/CEREBRUM

Cerebral Hemispheres

- **Paired structures:** Left and right hemispheres.
- **Location:** Superior parts of the brain.
- **Weight:** Largest portion of the brain (~1200 g in females, ~1400 g in males).
- **Lobes:** Named after overlying cranial bones (frontal, parietal, temporal, occipital).
- **Brain size correlation:** Related to **body size**, not intelligence.



BRAIN

CEREBRAL HEMISPHERES/CEREBRUM

Functions of the Cerebral Hemispheres (Cerebrum)

- Controls **balance.**
- Provides **precise timing for skeletal muscle activity.**
- Coordinates **body movements.**

INTEGRATION TASK

Thumb Overlap Test

- **Activity:** Join hands as shown in the image.
- **Observation:** Note **which thumb overlaps** the other.



BRAIN

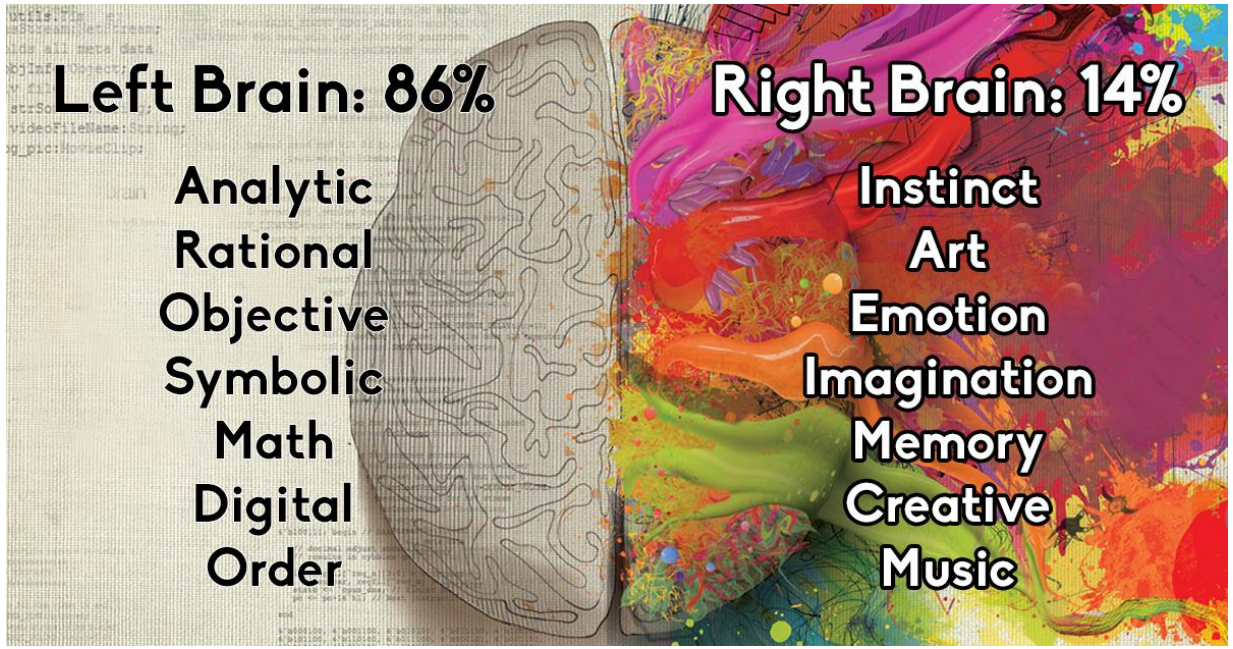
CEREBRAL HEMISPHERES/CEREBRUM

Right Hemisphere:

- Spatial perception
- Musical ability
- Artistic and visual-spatial skills

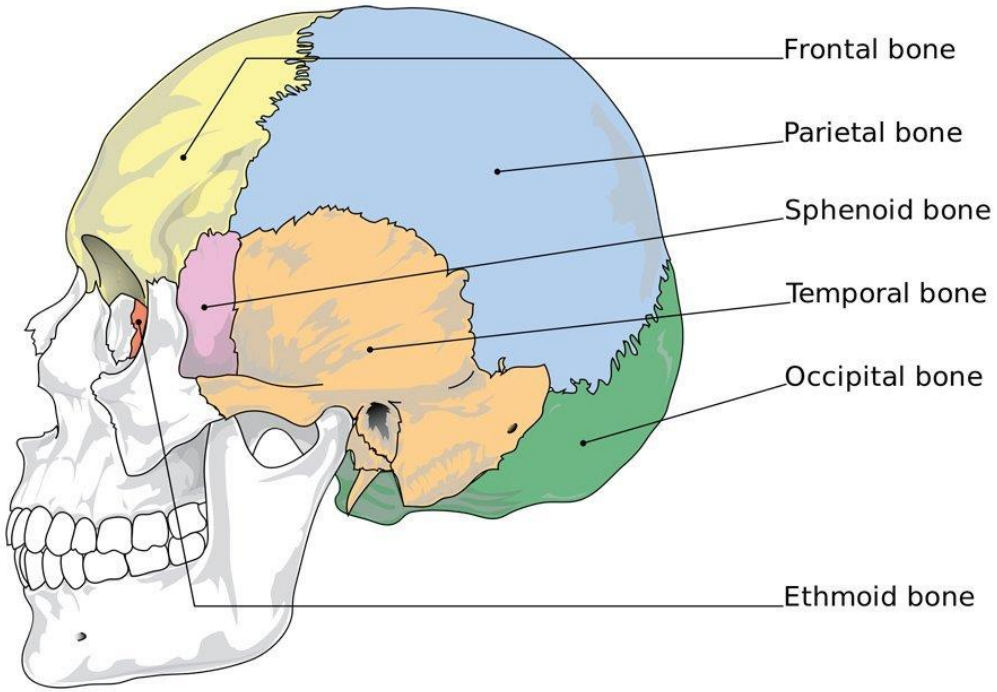
Left Hemisphere:

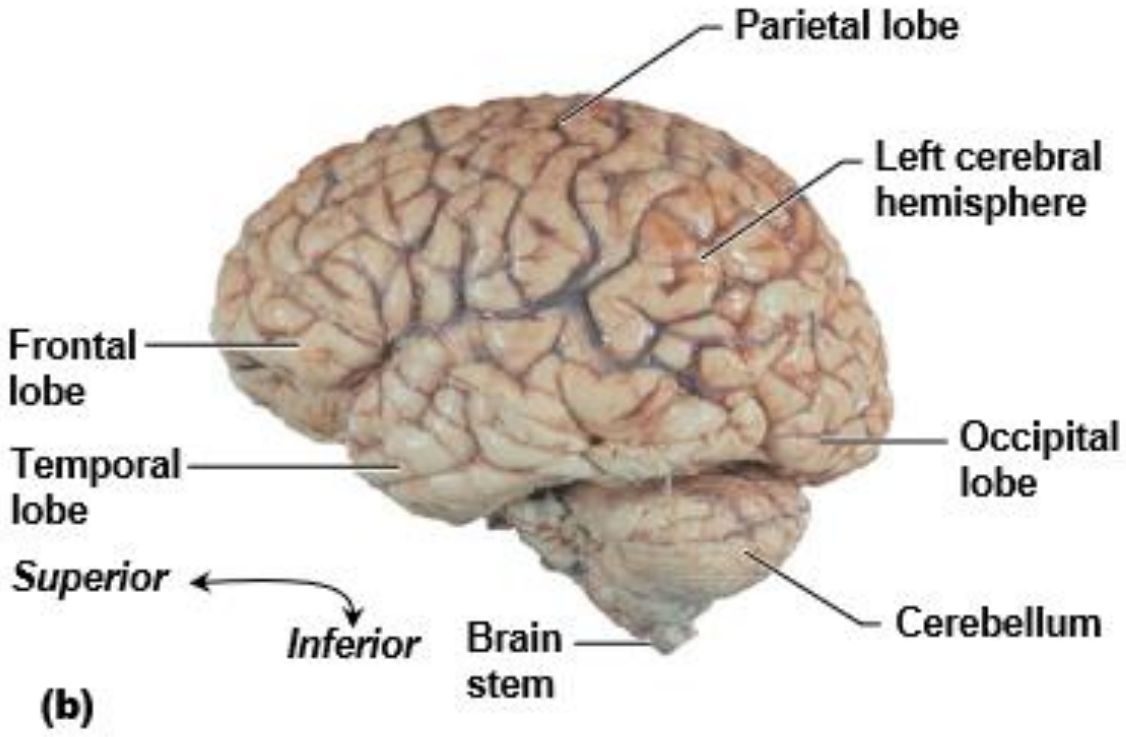
- Analytical thinking
- Language processing (speech and comprehension)
- Logical reasoning and problem-solving



*statistical data on the picture not verified

LEFT LATERAL VIEW OF THE BRAIN





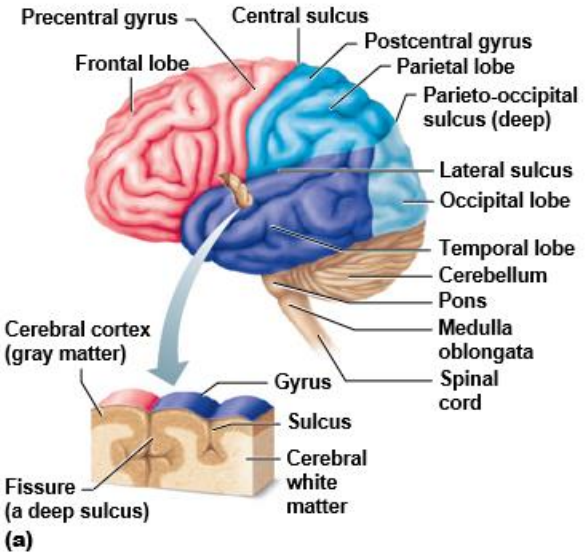
LEFT LATERAL VIEW OF THE BRAIN

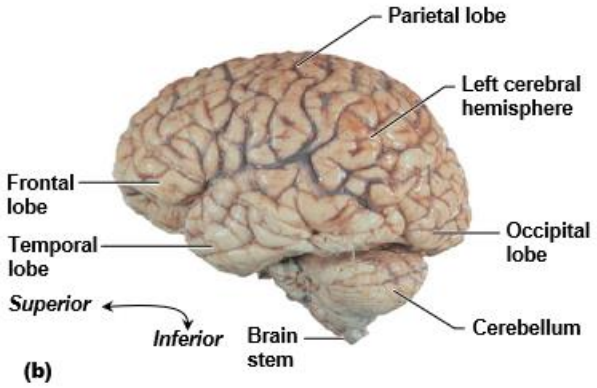
Surface Features of the Cerebral Hemisphere (Left Lateral View)

•**Gyri:** Ridges; numerous folds on the surface of each hemisphere.

•**Sulci:** Shallow grooves between gyri; divide each hemisphere into lobes.

•**Fissures:** Deep grooves; prominent landmarks separating major brain regions.





INTEGRATION TASK

Single-Leg Balance

- **Activity:**
  - Everyone **stand up.**
  - **Raise one leg** off the ground.
- **Observation:** Maintain balance and note stability.

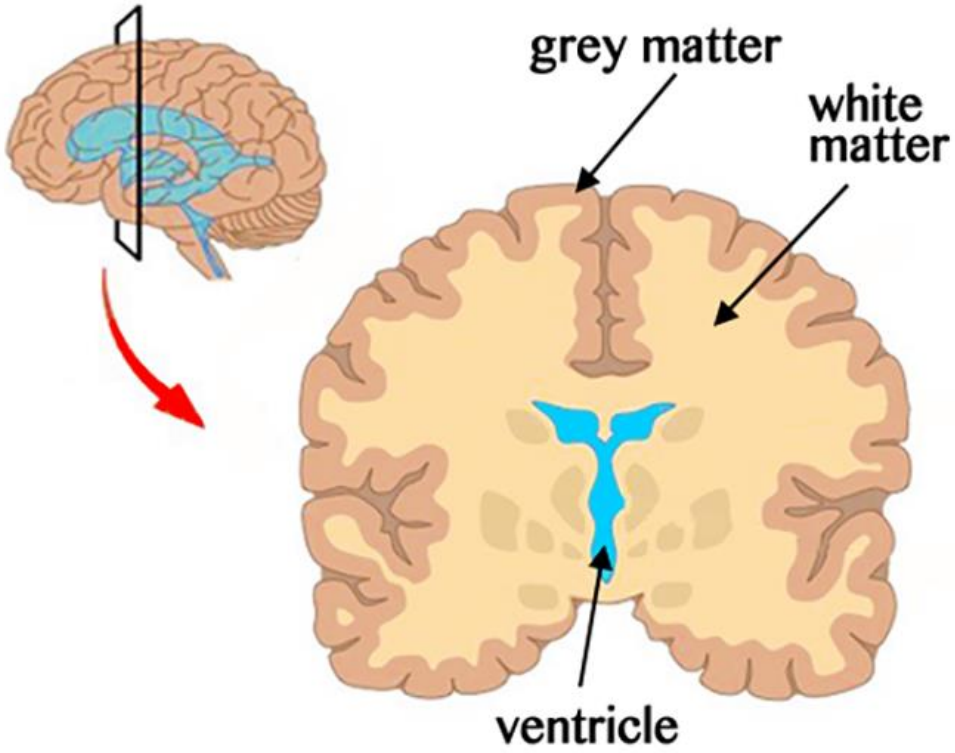


BRAIN

CEREBRAL HEMISPHERES/CEREBRUM

Three Main Regions of the Cerebral Hemispheres

1. **Gray Matter (Cerebral Cortex)** – outer layer composed mostly of neuron cell bodies; responsible for higher brain functions such as consciousness, sensation, and voluntary movement
2. **White Matter (Cerebral Medulla)** – deeper tissue composed of myelinated nerve fibers that connect various parts of the brain and spinal cord
3. **Basal Nuclei (Basal Ganglia)** – islands of gray matter deep within the white matter; help regulate voluntary motor activities and muscle tone

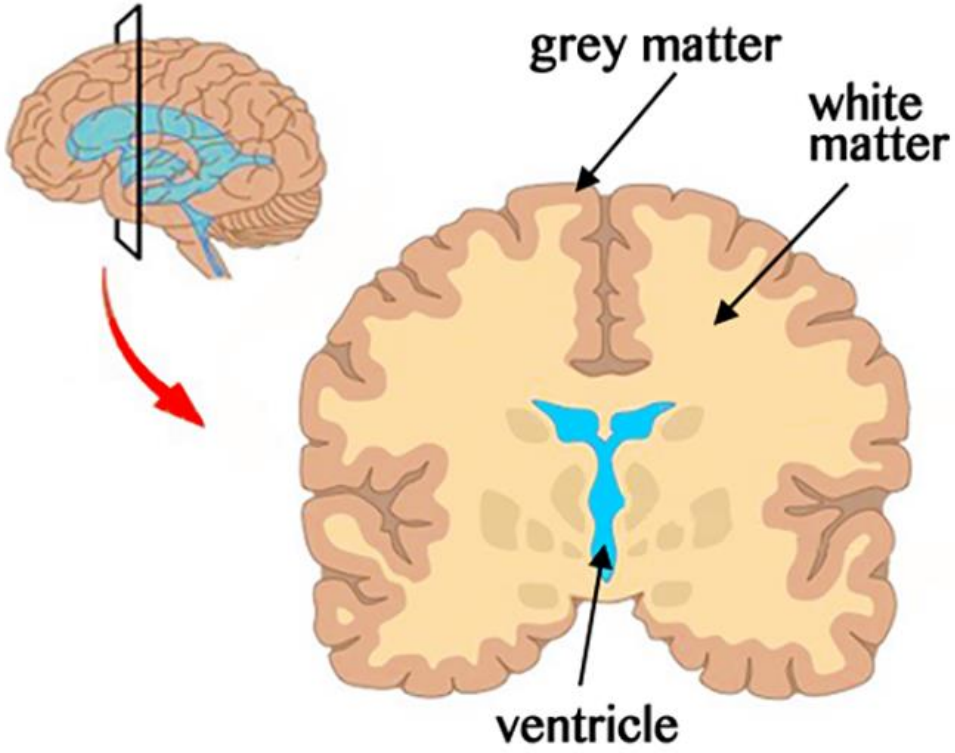


BRAIN
CEREBRAL HEMISPHERES/CEREBRUM

Three Main Regions of Cerebral Hemisphere

1. Gray Matter / Cerebral Cortex

- The **outer surface** of the cerebrum
- Contains various **types of neurons** (e.g., fusiform cells, stellate cells, pyramidal cells) named mainly by their shape
- **Thickness varies** across regions:
  - *Primitive areas* (2–3 layers)
  - *Advanced regions* (up to 6 layers)
- Responsible for higher brain functions—**sensation, voluntary movement, thought, reasoning, and memory**



INTEGRATION TASK



Sensory Processing and Pain Response

1. **Pinch your arm** as hard as you can.
2. **Apply ice** on the same area.

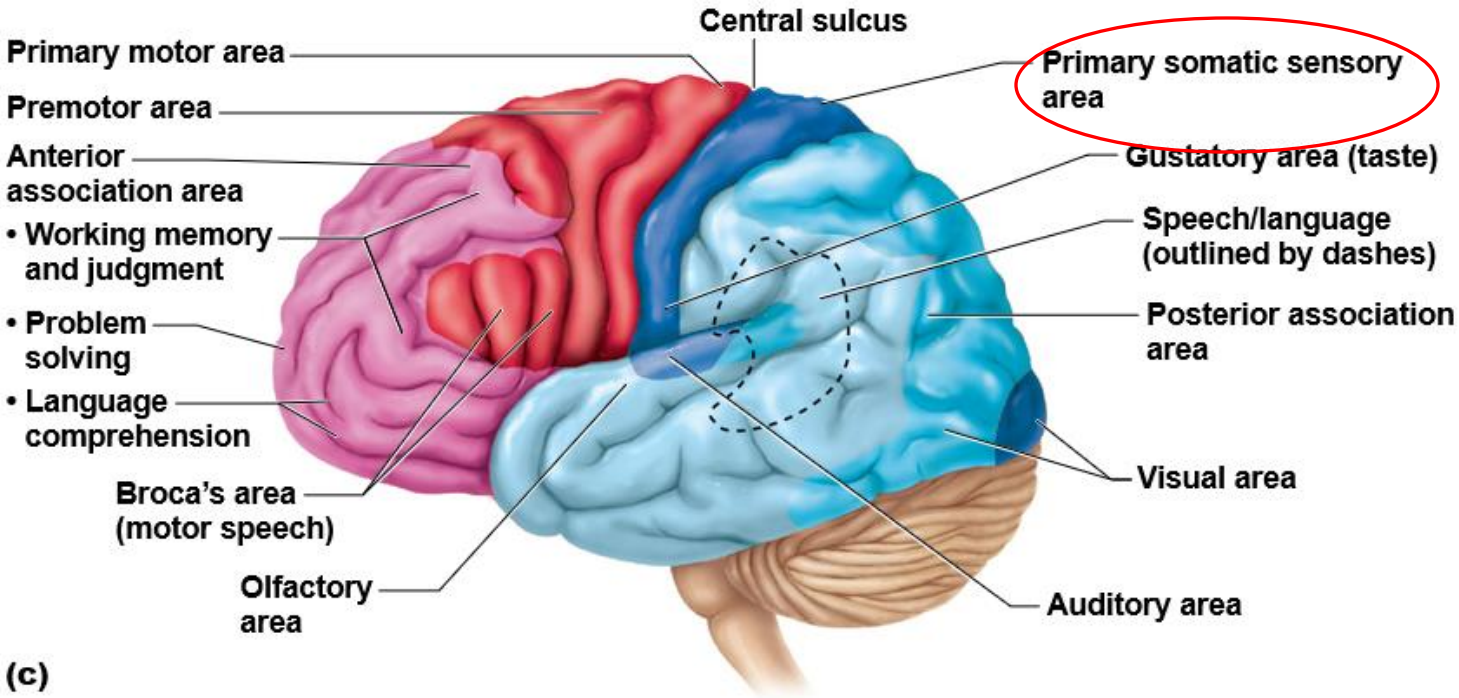


Observe the **difference in sensation** between the two stimuli.

BRAIN
CEREBRAL CORTEX

Primary Somatic Sensory Area

- **Location:** Parietal lobe, **posterior to the central sulcus**
- **Function:** Receives and interprets **sensory impulses** from the body's receptors
- **Stimuli detected:** Pain, temperature, light touch
- **Feature:** Organized as a **sensory homunculus**—a spatial map of body regions

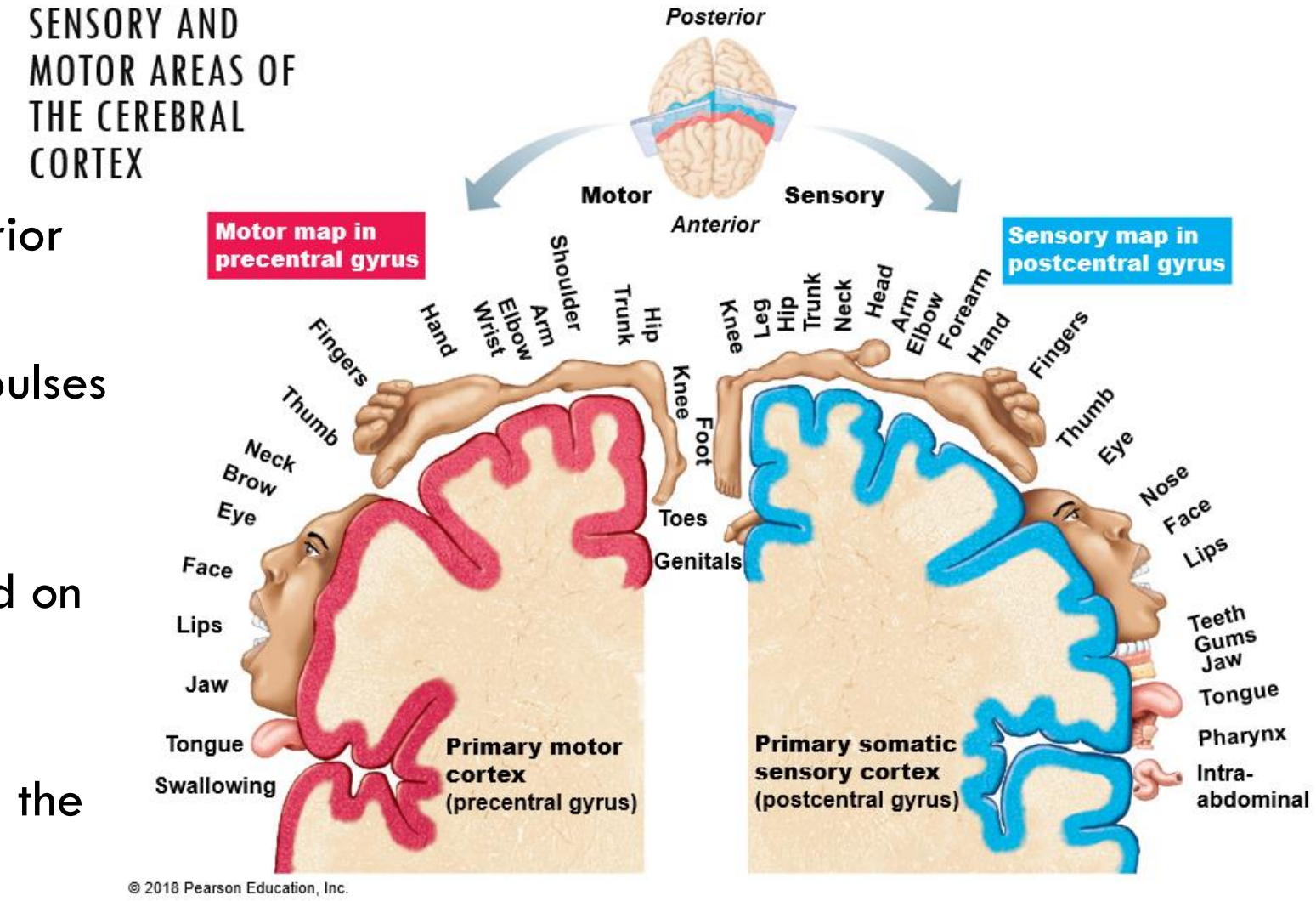


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BRAIN
CEREBRAL CORTEX

Primary Somatic Sensory Area

- Located in the **parietal lobe**, posterior to the central sulcus
- Receives and interprets sensory impulses from the body's sensory receptors
- **Sensory homunculus**: spatial representation of body parts based on sensory input density
- **Contralateral control**: left cortex receives input from the right side of the body, and vice versa



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INTEGRATION TASK

Motor Function and Voluntary Movement

1. **Pick up the heaviest object** on your table.
2. **Lift it above your head.**
3. **Put it back** on the table carefully.

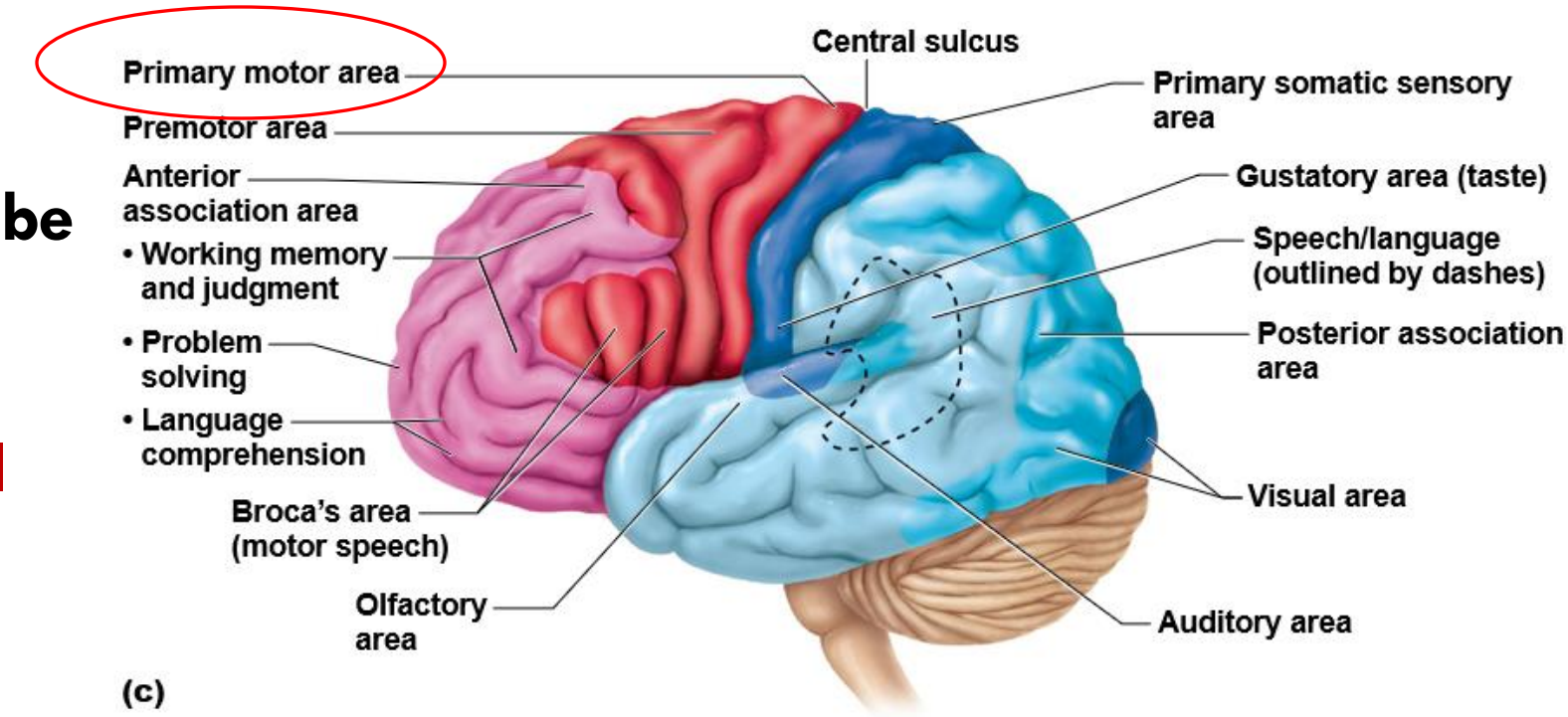
Observe how your brain and muscles coordinate during the activity.



BRAIN
CEREBRAL CORTEX

Primary Motor Area

- **Location:** Anterior to the central sulcus in the **frontal lobe** (precentral gyrus)
- **Function:** Allows **conscious control of voluntary skeletal muscle movements**
- **Feature:** Motor output is **contralateral**—right hemisphere controls left body side, and vice versa

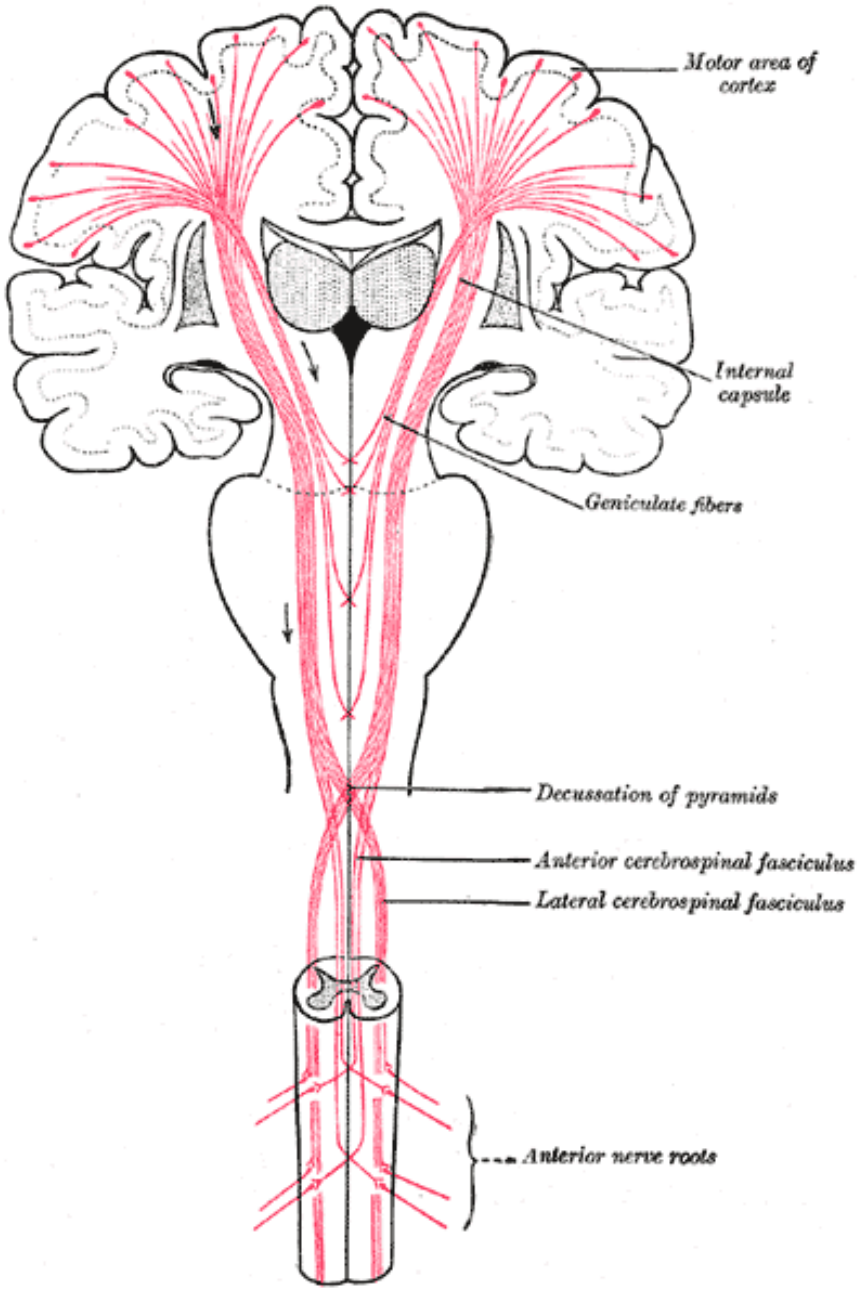


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BRAIN
CEREBRAL CORTEX

Pyramidal (Corticospinal) Tract and Motor Homunculus

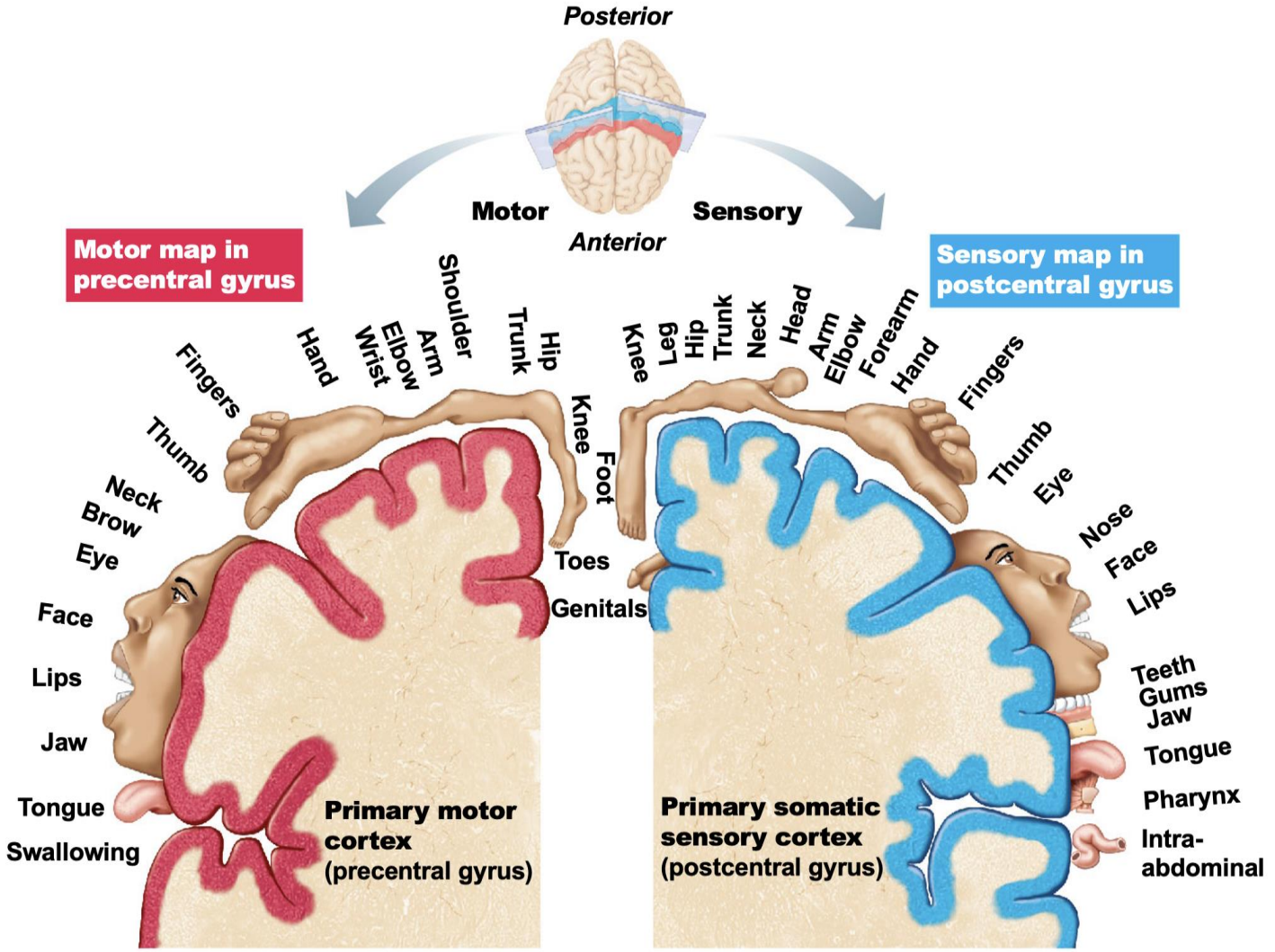
- **Motor neurons** in the primary motor cortex form the **pyramidal (corticospinal) tract,** which **descends to the spinal cord** to control voluntary movements.



SENSORY AND MOTOR AREAS OF THE CEREBRAL CORTEX

Pyramidal (Corticospinal) Tract and Motor Homunculus

- **Motor homunculus** is a **spatial map** of the body representing how much cortical area is devoted to control specific muscles.
- Body areas requiring **fine motor control** (e.g., hands, face) occupy **larger cortical regions.**



INTEGRATION TASK

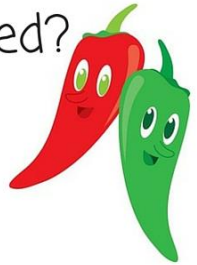
Repeat it 3 times quickly!

Observe how your **tongue and mouth muscles** coordinate to form precise sounds.

TONGUE TWISTERS

Peter Piper picked a peck of pickled peppers.
A peck of pickled peppers Peter Piper picked.
If Peter Piper picked a peck of pickled peppers,
Where's the peck of pickled peppers Peter Piper picked?



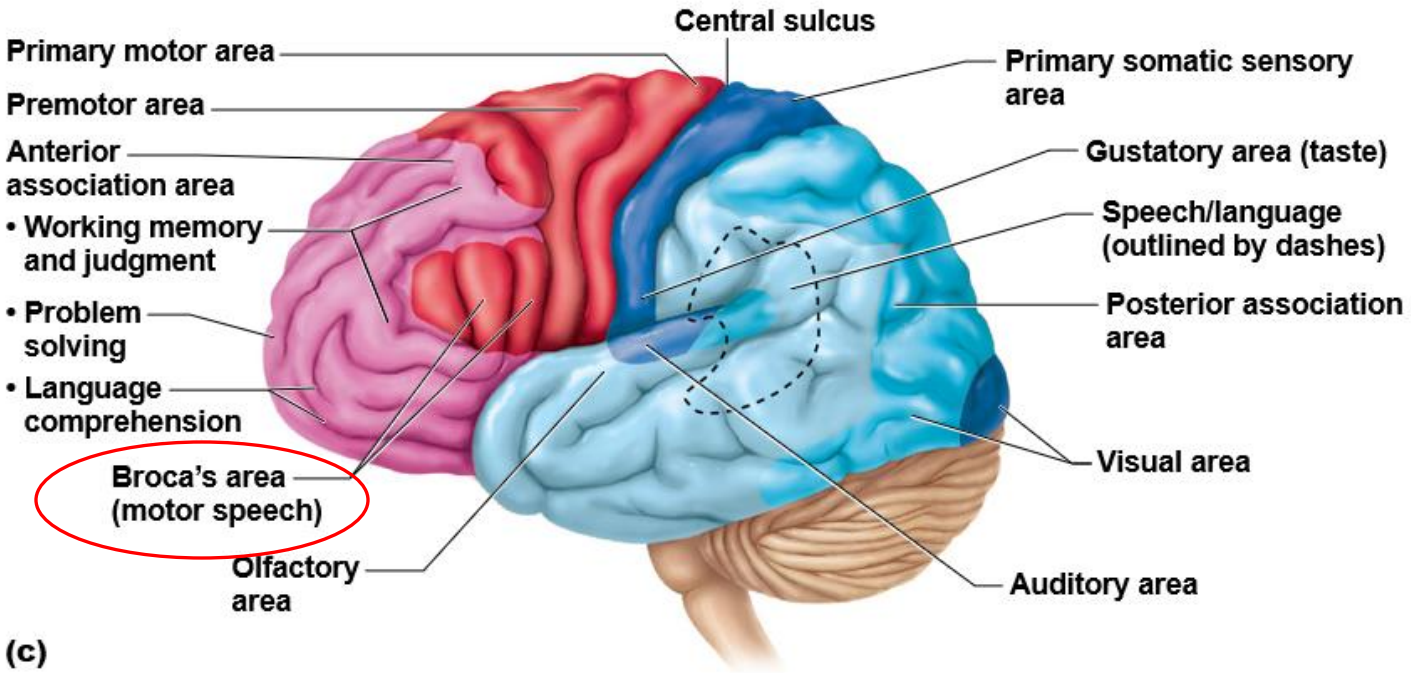


CookingWithLanguages.com

BRAIN
CEREBRAL CORTEX

Broca's Area (Motor Speech Area)

- Located in the **inferior frontal gyrus** of the **frontal lobe** (usually in the **left hemisphere**)
- Responsible for **speech production and articulation**
- Coordinates the **motor movements** required for speaking

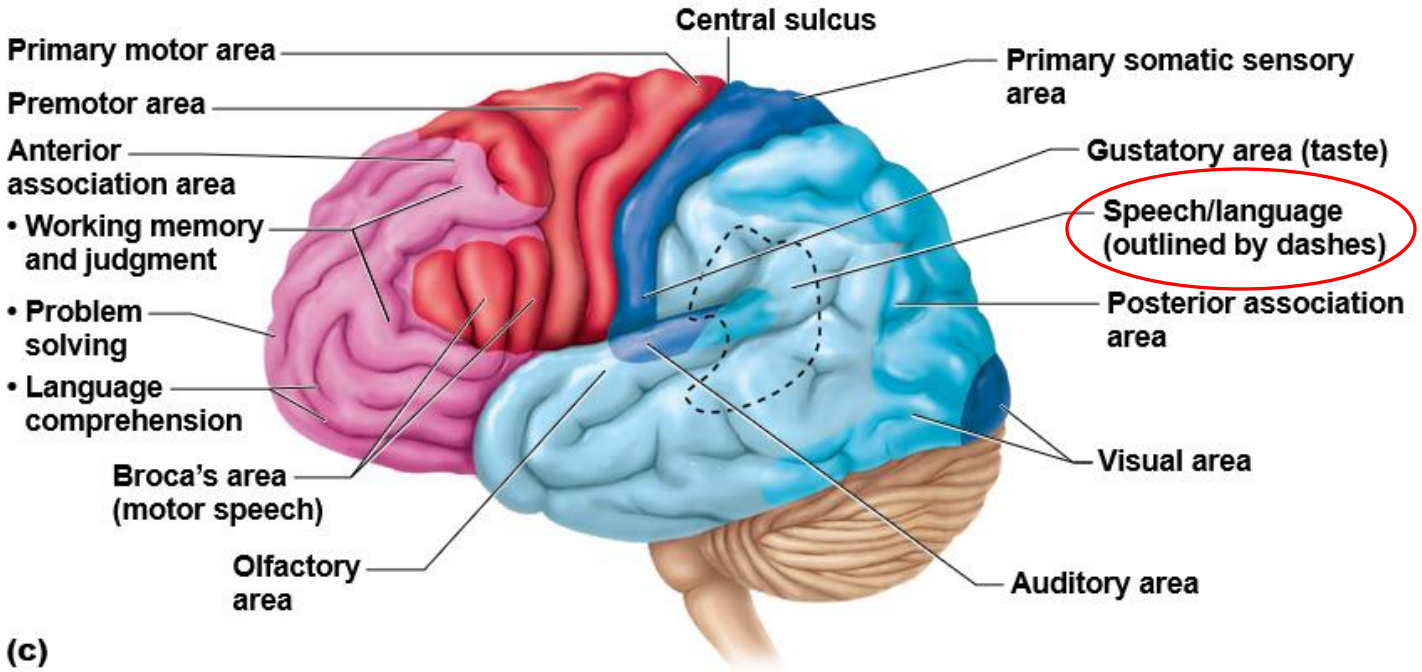


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BRAIN
CEREBRAL CORTEX

Wernicke's Area (Sensory Speech Area)

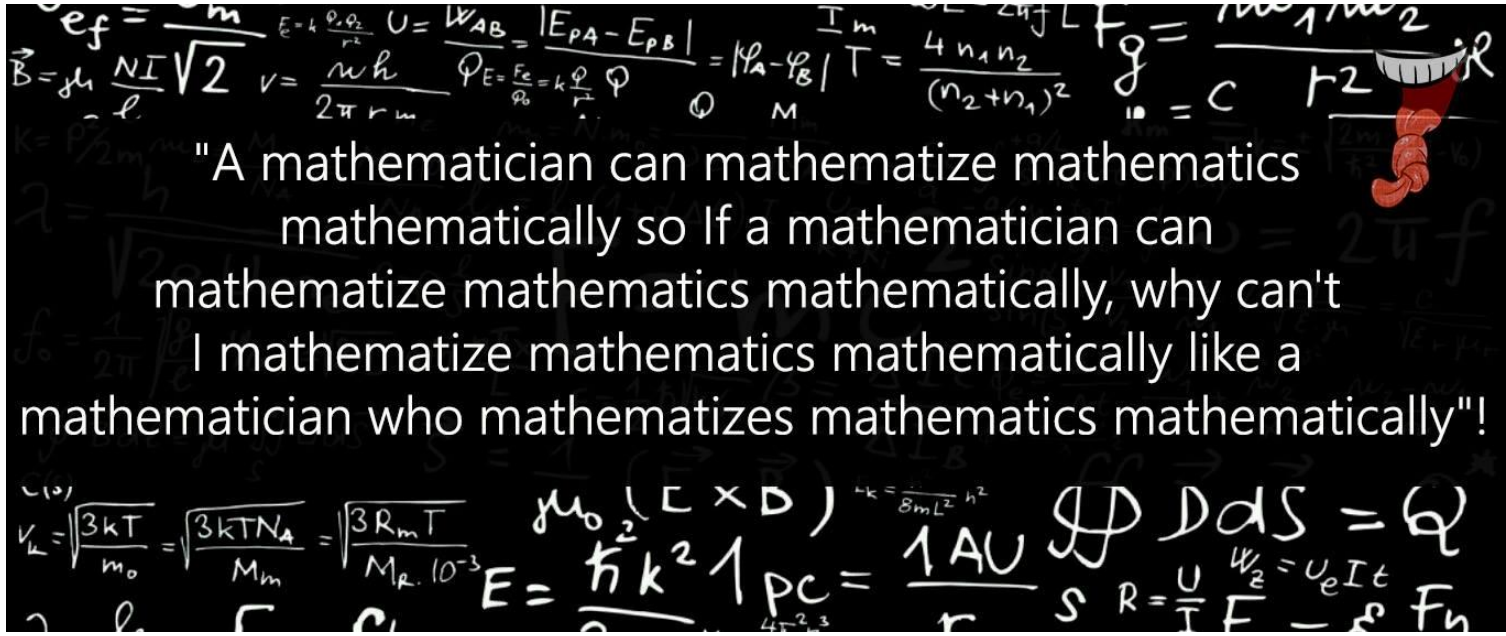
- Located in the **posterior part of the superior temporal gyrus** (usually in the **left hemisphere**)
- Responsible for **comprehension of written and spoken language**
- Works with **Broca's area** for coherent speech and language processing



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INTEGRATION TASK

Repeat this tongue twister clearly:



BRAIN

CEREBRAL CORTEX

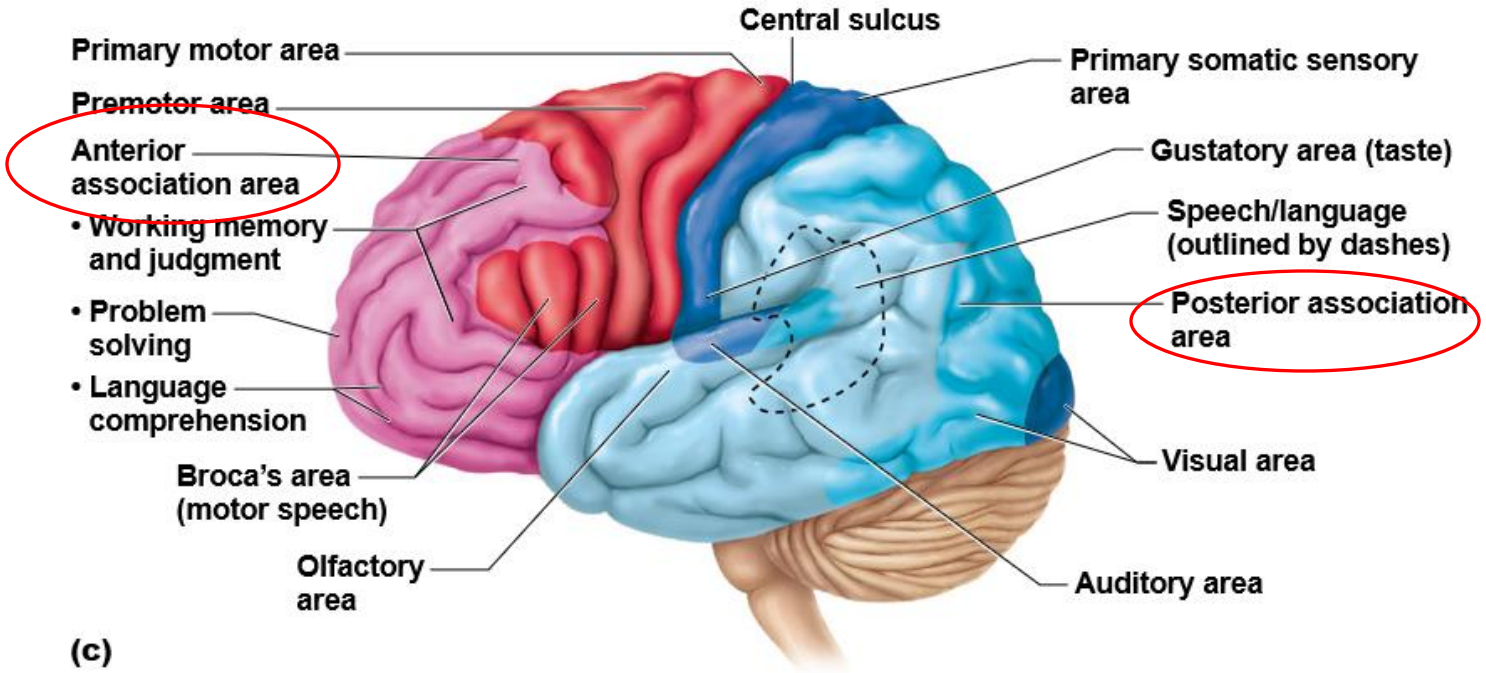
Other Specialized Areas of the Cerebral Cortex

1. Anterior Association Area (Frontal Lobe)

- Involved in **higher intellectual reasoning, judgment, personality,** and **complex learning**
- Responsible for **recognition of sensory inputs** and **planning of responses**

2. Posterior Association Area (Posterior Cortex)

- Region where **visual, auditory, and somatosensory association areas** meet
- Plays a key role in **pattern recognition, awareness of spatial relationships,** and **integrating sensory information** into a cohesive perception of the environment

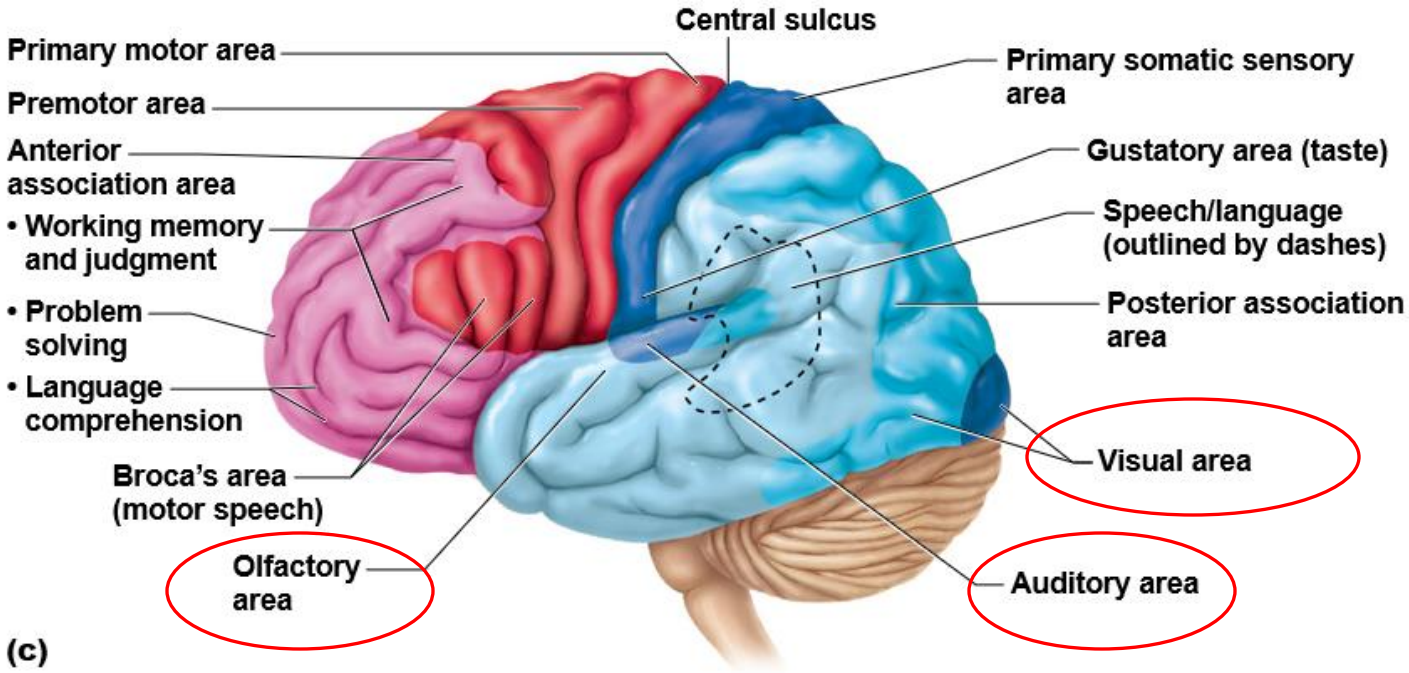


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BRAIN
CEREBRAL CORTEX

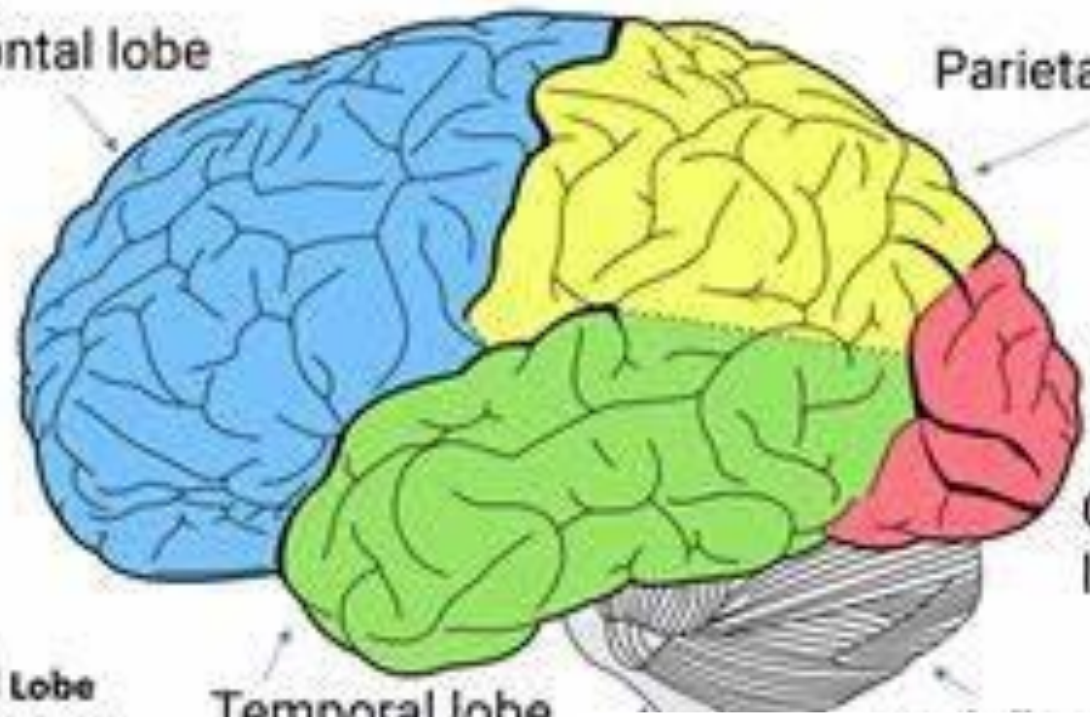
Other Specialized Areas of the Cerebral Cortex

- **Visual Area (Occipital Lobe):** Responsible for processing visual information such as shape, color, and motion.
- **Auditory Area (Temporal Lobe):** Interprets sound stimuli including pitch, rhythm, and volume.
- **Olfactory Area (Temporal Lobe):** Detects and processes smell stimuli from the nasal cavity.



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Frontal Lobe:
Reasoning, motor skills, higher level congition, expressive language.



Parietal Lobe
Processing tactile sensory information, such as pressure, pain and touch.

Occipital Lobe
Interpreting visual stimuli and information and the primary visual cortex.

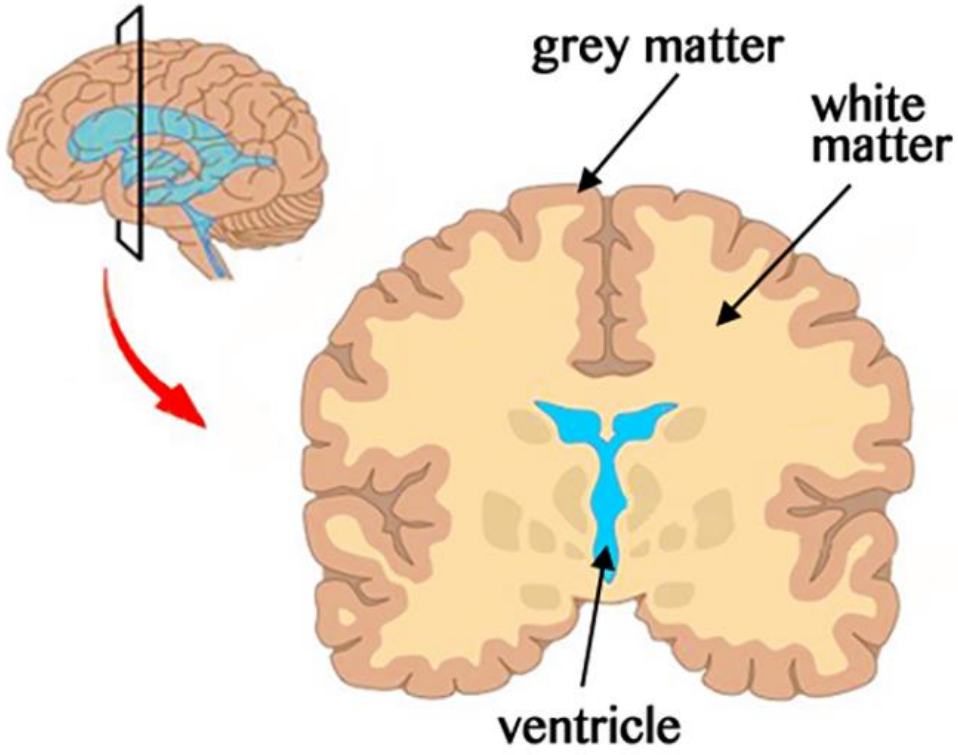
Temporal Lobe
Primary auditory cortex (for interpreting sounds and language), hippocampus (for memories)

BRAIN

CEREBRAL HEMISPHERES/CEREBRUM

Three main regions of cerebral hemisphere

1. Gray matter / Cerebral cortex
2. White matter / Cerebral medulla
3. Basal nuclei

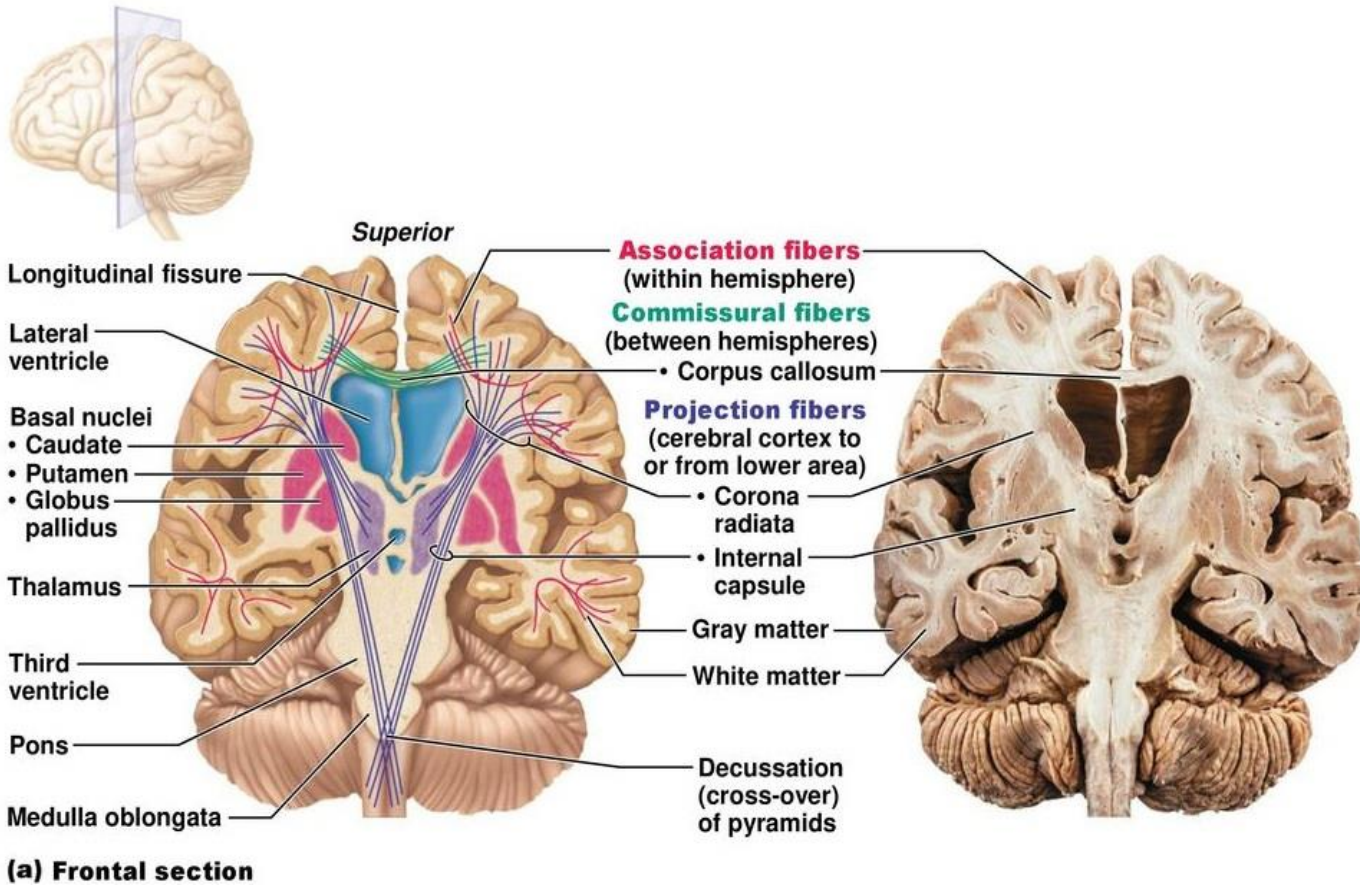


BRAIN

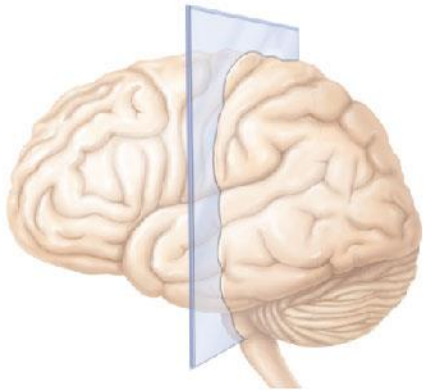
CEREBRAL WHITE MATTER

Composed of myelinated fiber tracts located deep to the cerebral cortex (gray matter).

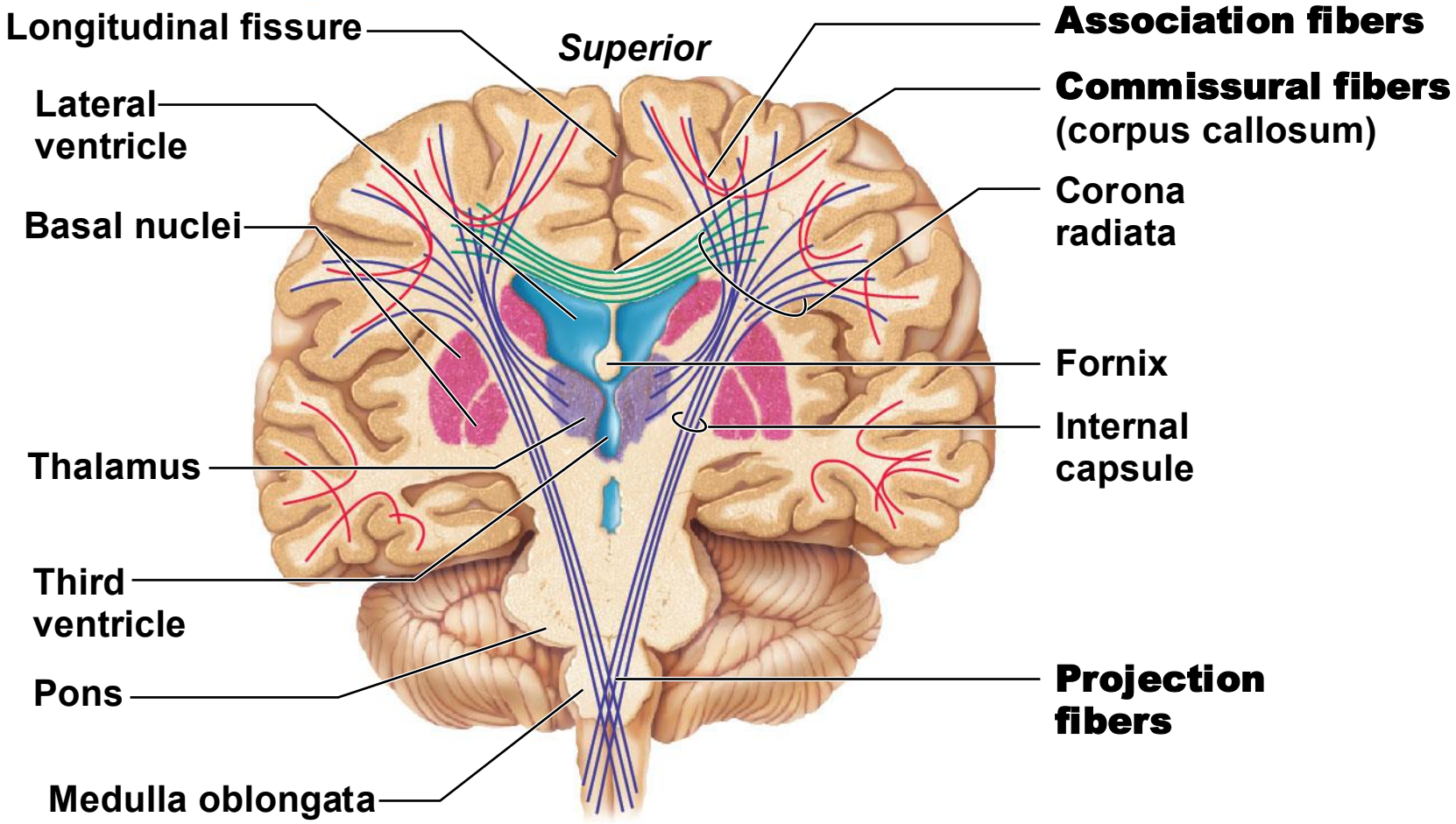
- **Corpus Callosum:** Major commissural tract connecting the left and right cerebral hemispheres.
- **Commissural Fibers:** Tracts that connect corresponding areas between hemispheres.
- **Association Fibers:** Connect regions within the same hemisphere.
- **Projection Fibers:** Link the cerebrum with lower brain regions and the spinal cord.



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FRONTAL SECTION (FACING POSTERIORLY) OF THE BRAIN SHOWING COMMISSURAL, ASSOCIATION, AND PROJECTION FIBERS RUNNING THROUGH THE CEREBRUM AND THE LOWER CNS.



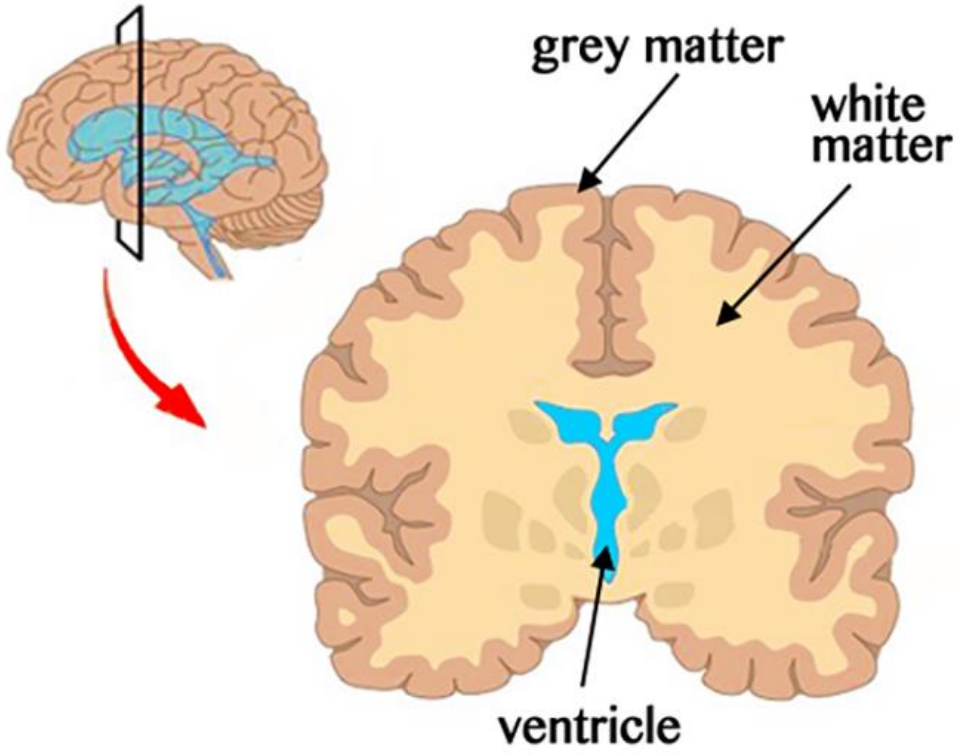
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BRAIN

CEREBRAL HEMISPHERES/CEREBRUM

Three main regions of cerebral hemisphere

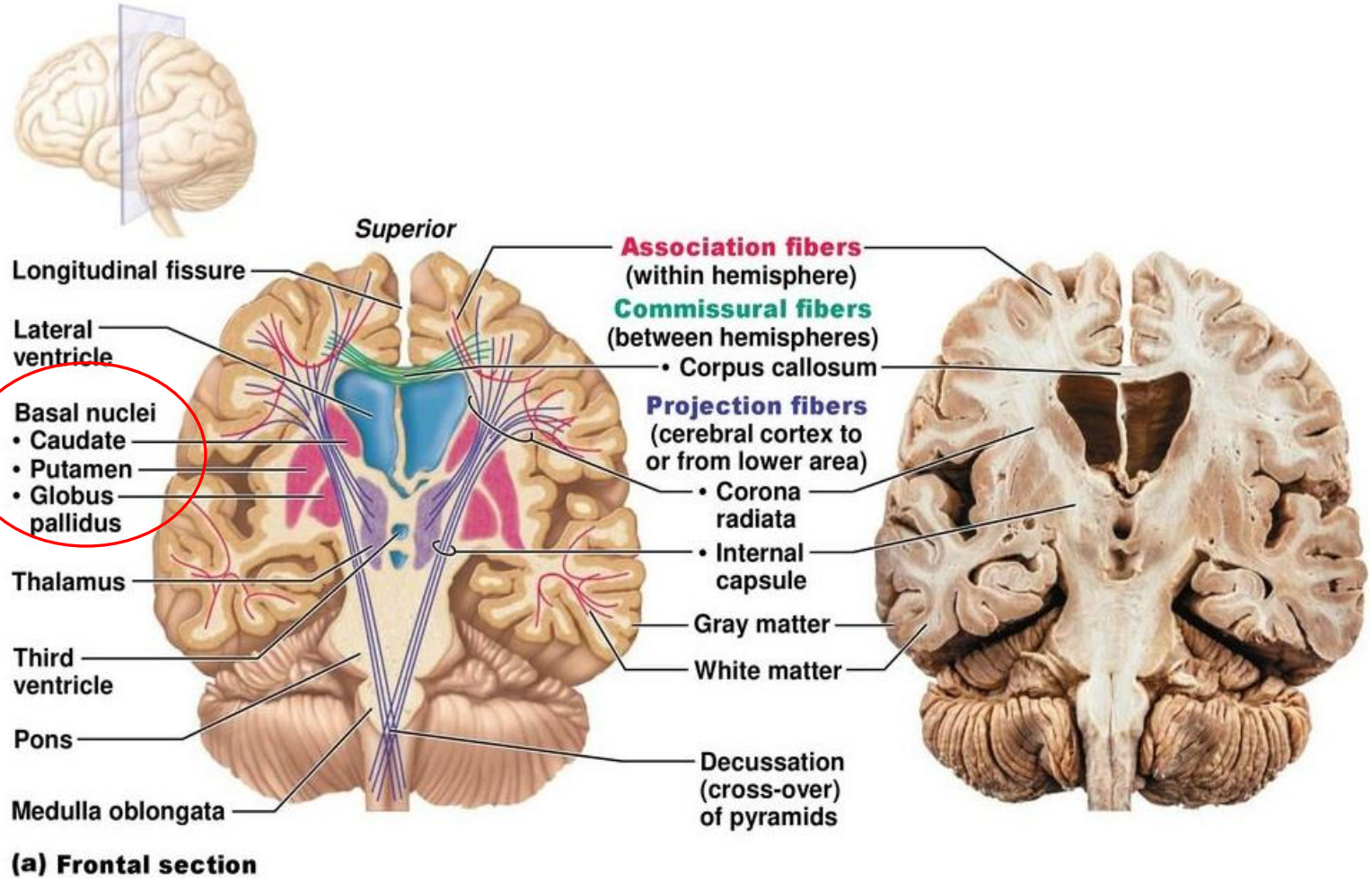
1. Gray matter / Cerebral cortex
2. White matter
3. Basal nuclei



BRAIN
BASAL NUCLEI

Basal Nuclei (Basal Ganglia)

- "Islands" of **gray matter** buried deep within the **white matter** of the cerebrum
- **Regulate voluntary motor activities** by modifying instructions sent to skeletal muscles from the **primary motor cortex**
- Responsible for **planning, organizing, and coordinating** motor movements and **maintaining posture**
- Major nuclei include: **caudate nucleus, putamen, and globus pallidus**

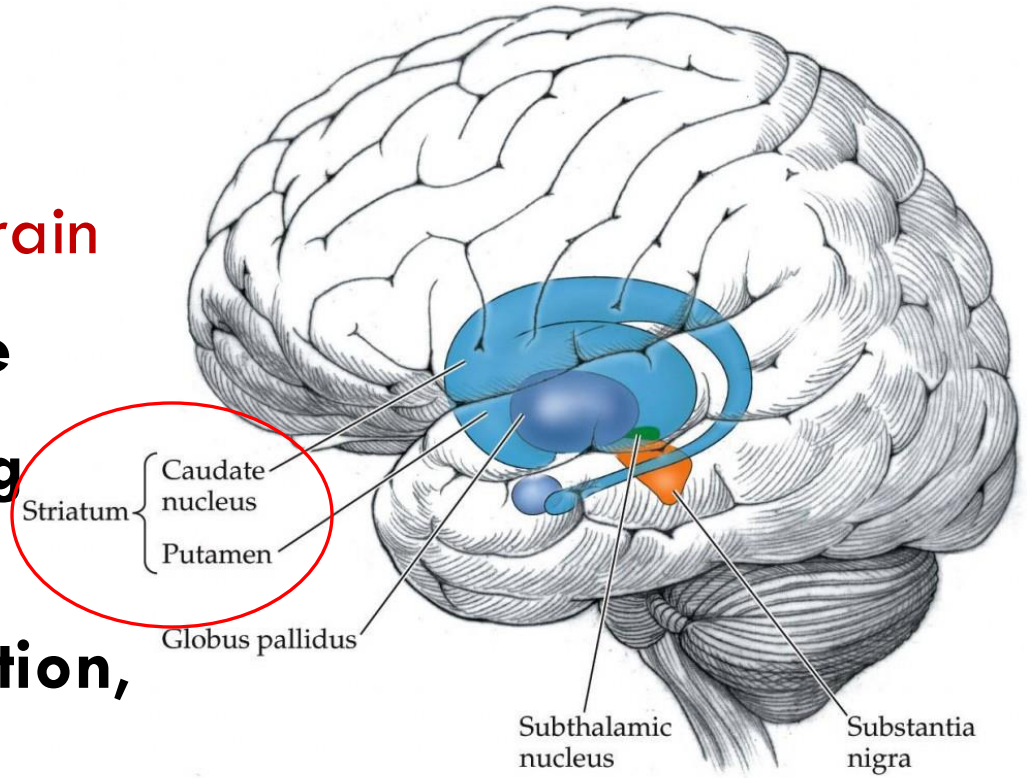


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BRAIN
BASAL NUCLEI: STRIATUM

Striatum

- Known as the **"reward system"** of the brain
- Produces **feelings of reward or pleasure**
- Coordinates **multiple aspects of thinking** related to **decision-making**
- Involved in **movement planning, motivation, reinforcement learning**, and **reward perception**
- Functions through the release of **dopamine** from the **substantia nigra**



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LIMBIC SYSTEM
GROUPED PARTS OF THE CEREBRUM AND DIENCEPHALON

Limbic System

A group of structures in the **cerebrum** and **diencephalon** forming a functional network involved in **emotion, memory, motivation, and survival behaviors.**

Functions:

- Controls **emotional behavior** and **motivational states**
- Integrates **memory and learning** with emotional experience
- Regulates **autonomic and endocrine responses** to emotion
- Influences **behavioral drives** (e.g., feeding, reproduction, survival instincts)

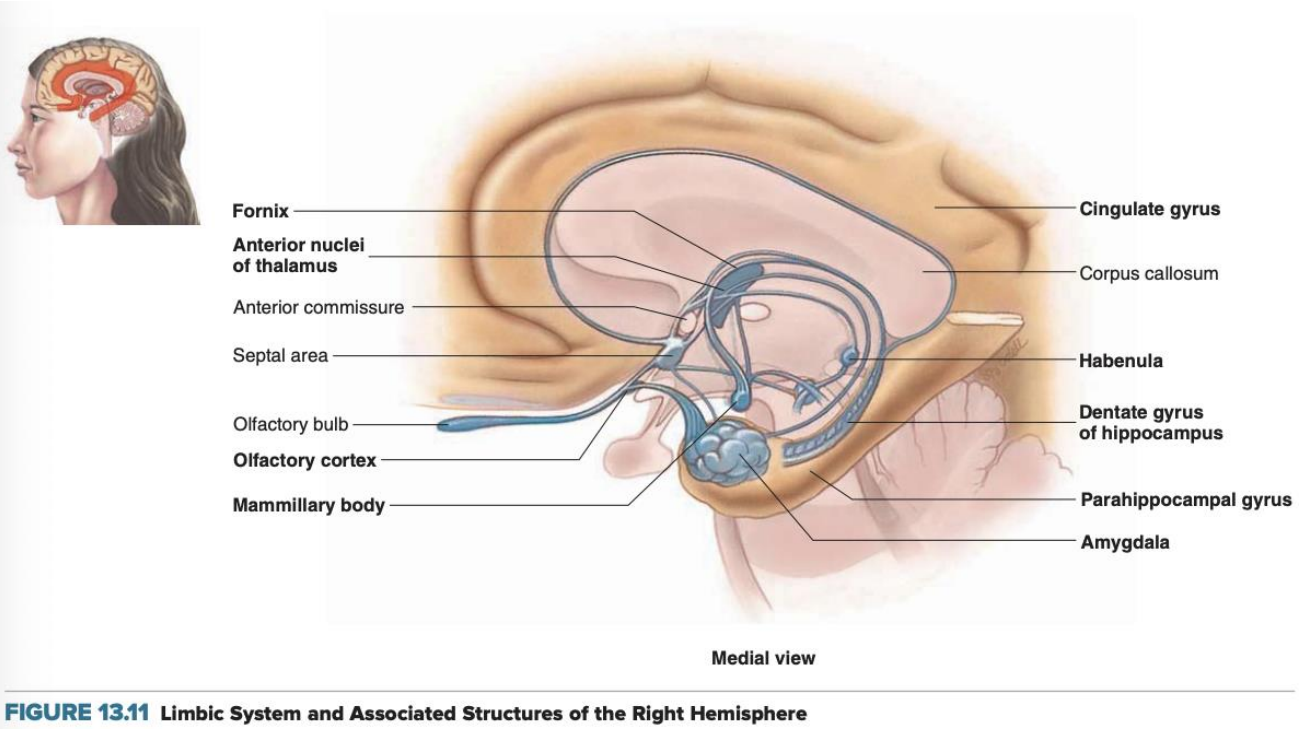


FIGURE 13.11 Limbic System and Associated Structures of the Right Hemisphere

LIMBIC SYSTEM
GROUPED PARTS OF THE CEREBRUM AND DIENCEPHALON

Major Components and Structures:

- **Cingulate gyrus** – Emotional processing and motivation.
- **Parahippocampal gyrus** – Memory encoding and retrieval.
- **Hippocampus & Dentate gyrus** – Long-term memory formation.
- **Amygdala** – Emotional learning, especially fear and aggression.
- **Hypothalamus (mammillary bodies)** – Integration of emotional and autonomic responses.
- **Thalamic anterior nuclei & Habenula** – Relays emotion-related sensory information.
- **Olfactory cortex** – Links smell with memory and emotion.
- **Fornix** – Fiber tract connecting the hippocampus to the mammillary bodies and thalamus.

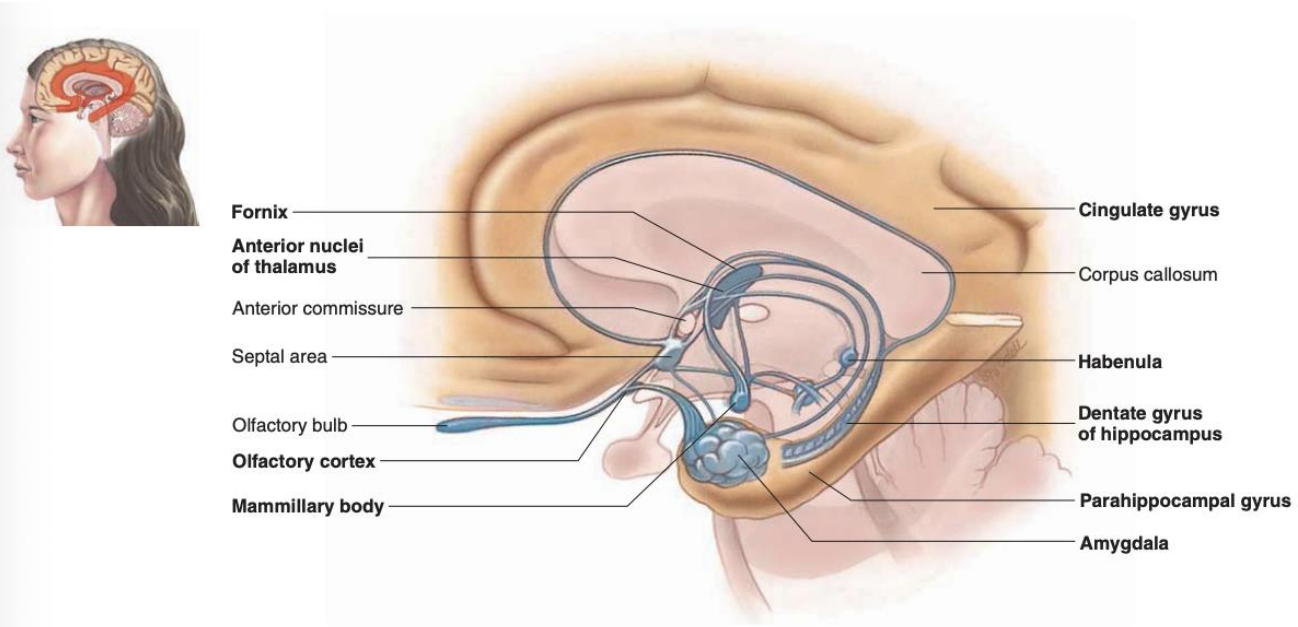


FIGURE 13.11 Limbic System and Associated Structures of the Right Hemisphere

BRAIN
ANATOMY OF THE BRAIN

4 Major Brain Regions

1. Cerebral hemispheres/ Cerebrum
   - Cortex
   - White matter
   - Basal nuclei

2. Diencephalon
   - Thalamus
   - Hypothalamus
   - Subthalamus
   - Epithalamus

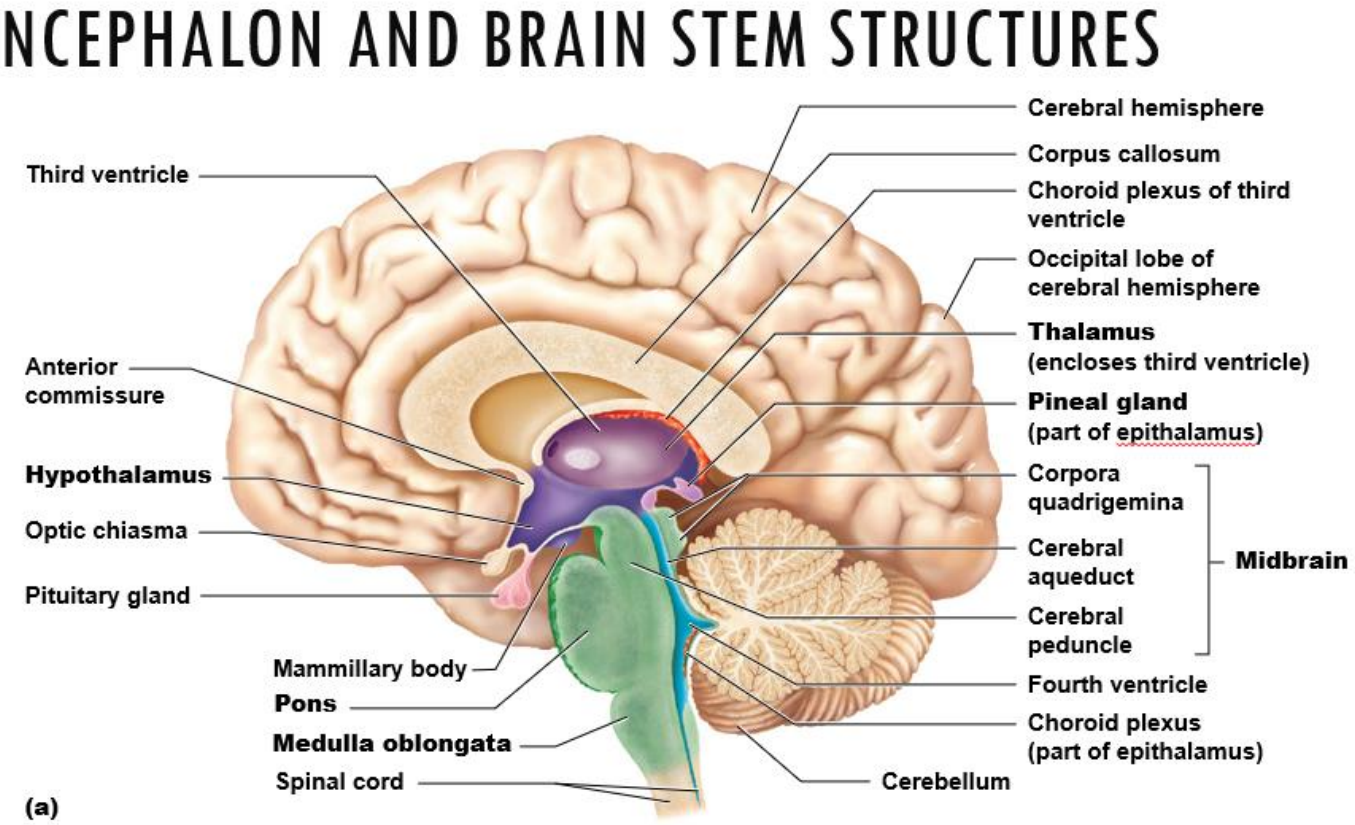
3. Brainstem
   - Midbrain
   - Pons
   - Medulla oblongata

4. Cerebellum

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BRAIN
DIENCEPHALON

- Sits on top of the brain stem
- Enclosed by the cerebral hemispheres
- Made of four structures
    1. Thalamus
    2. Hypothalamus
    3. Subthalamus
    4. Epithalamus



BRAIN
DIENCEPHALON

Located between the **cerebral hemispheres** and the **midbrain**, the diencephalon acts as a **central processing and relay center.**

1. Thalamus
- Serves as the **relay station** for **sensory information** traveling to the **cerebral cortex.**
- Sorts and edits sensory impulses (except smell).
- Plays a role in **awareness, pain perception,** and **consciousness.**

2. Hypothalamus
- Regulates **autonomic functions** (heart rate, blood pressure, digestion, temperature).
- Controls **endocrine activity** via the **pituitary gland.**
- Maintains **homeostasis** (balance in internal body conditions).
- Regulates **emotions, hunger, thirst,** and **sleep-wake cycles.**

3. Subthalamus
- Plays a critical role in **motor control and coordination**
- Modulates **voluntary motor activity** via connections with basal nuclei and midbrain structures

4. Epithalamus
- Contains the **pineal gland,** which secretes **melatonin.**
- Helps regulate **circadian rhythms** and the **sleep-wake cycle.**
- Involved in **emotional responses** to odors via connections with the limbic system.

INTEGRATION TASK

Question: Who is your first teacher in school?

Take a moment to recall their name, face, and your first classroom experience.



INTEGRATION TASK

Question: What is the name of your first pet?

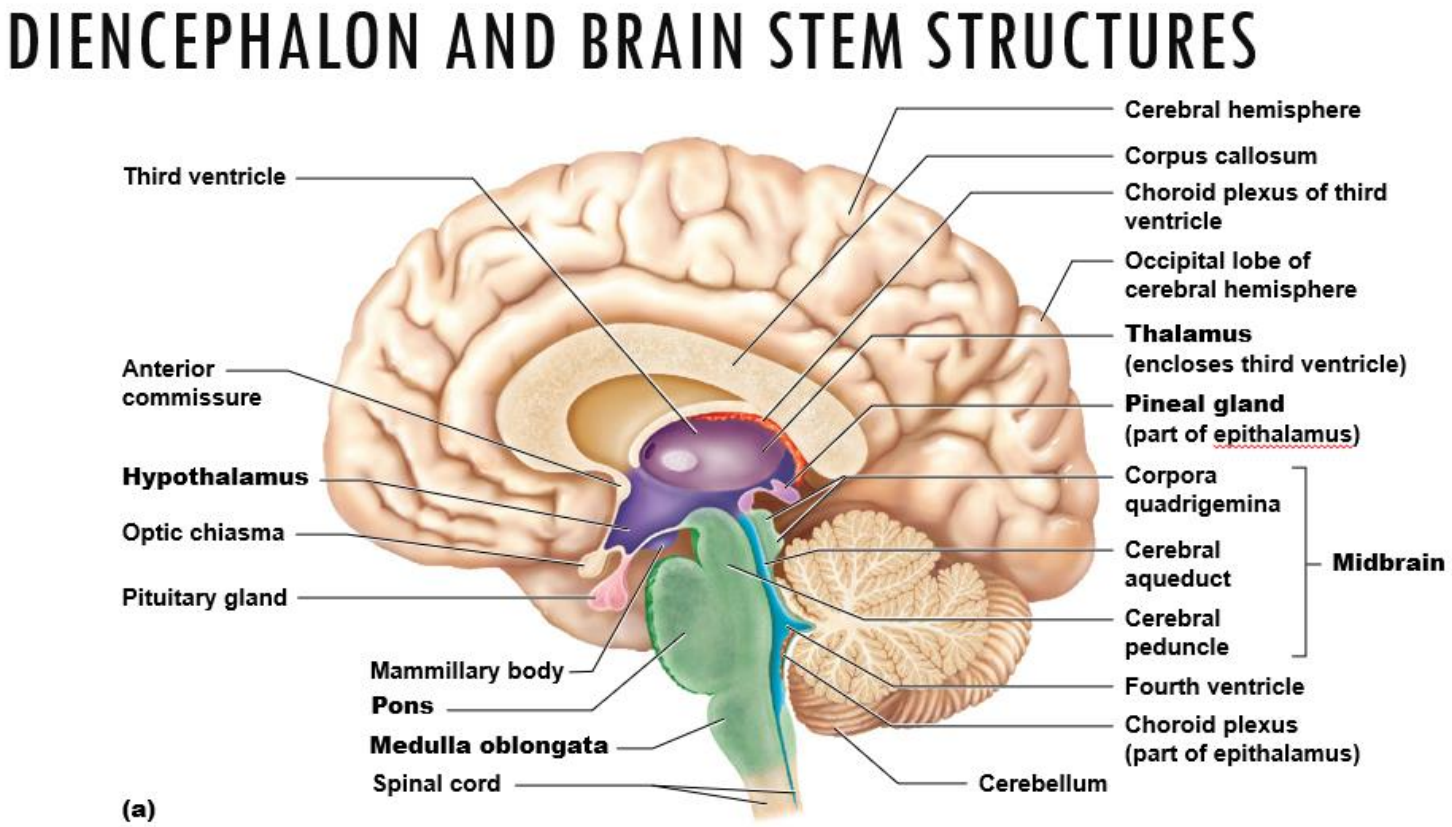
Pause for a moment. Recall its name, face, and how you felt spending time with it.



BRAIN
DIENCEPHALON

Main Component:

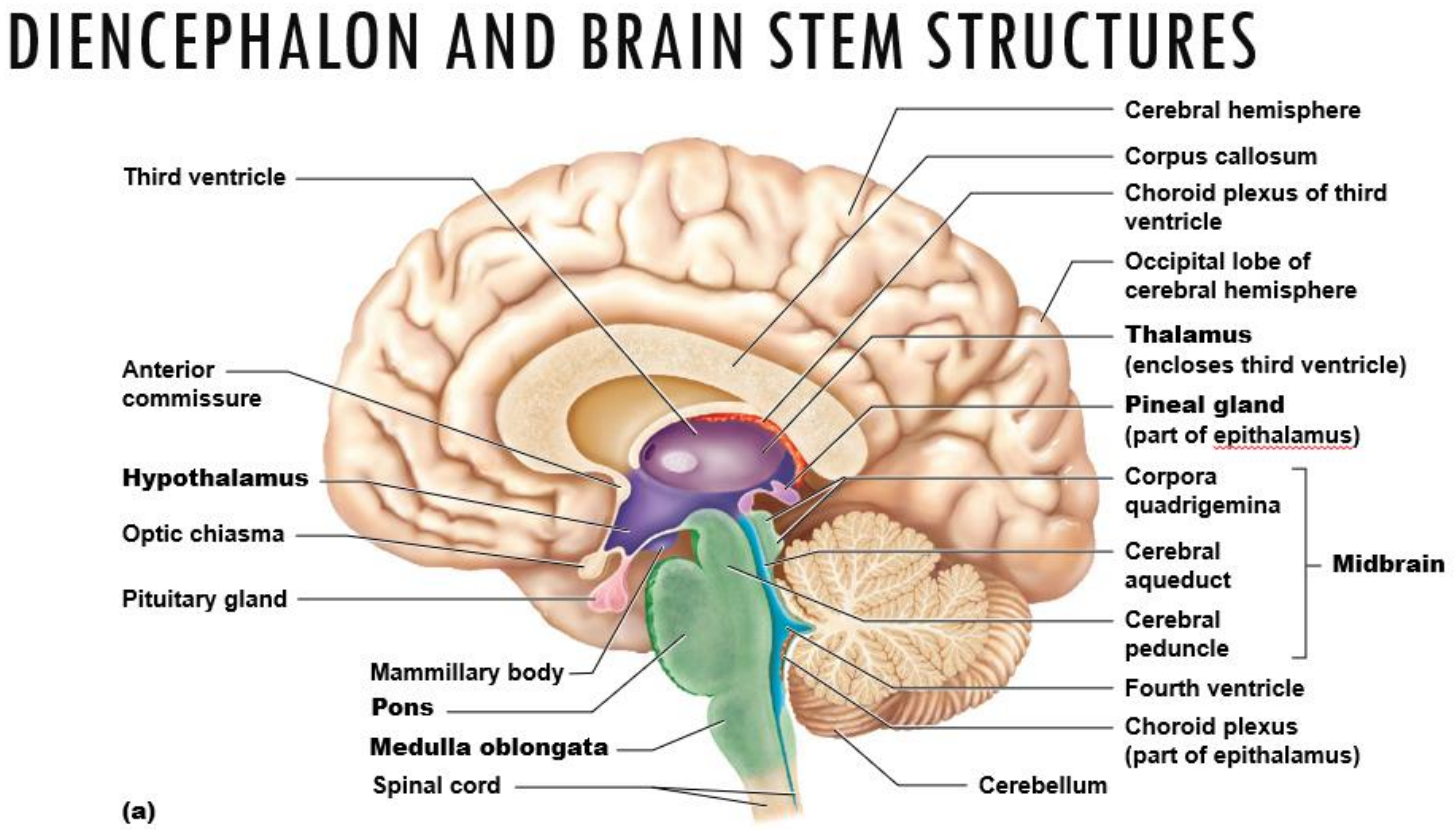
1. **Thalamus**
2. Hypothalamus
3. Subthalamus
4. Epithalamus



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BRAIN
DIENCEPHALON: THALAMUS

- Encloses the **third ventricle**
- Serves as the **main relay station** for sensory impulses traveling to the **cerebral cortex**
- **Involved in memory** processing and emotional functions
- **Transfers impulses** to the appropriate cortical areas for **localization and interpretation**



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INTEGRATION TASK

Check your temperature using thermometer.



INTEGRATION TASK

Drink 1 Glass of Water

- Take a short break and drink **one full glass of water.**
- Notice how your body feels afterward — refreshed, alert, and more focused.
- Remember: **Even mild dehydration can affect brain performance and mood.**



INTEGRATION TASK

Who Do You Love the Most?

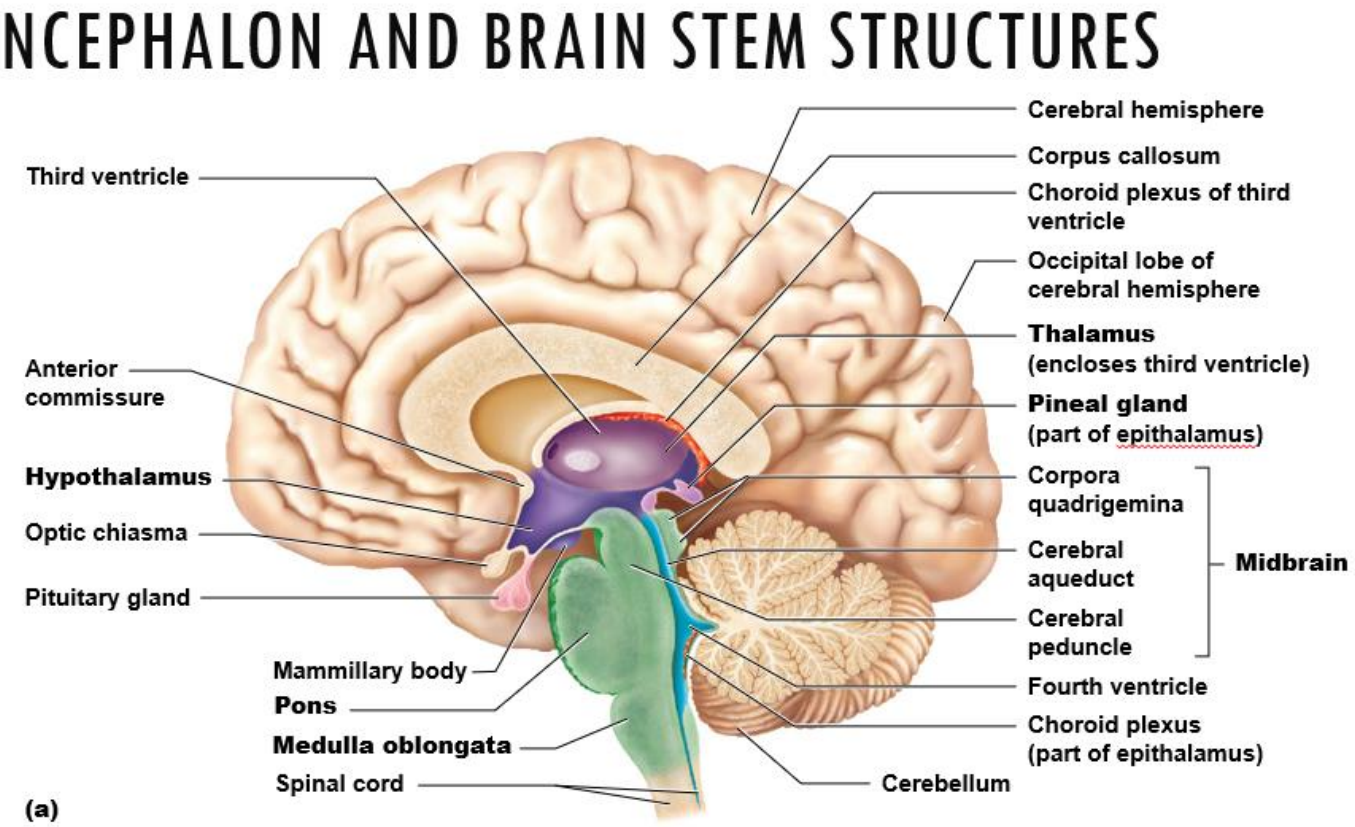
- Think about the person (or people) you love the most.
- Notice how simply recalling them affects your emotions, heart rate, or even breathing.
- This emotional response involves specific **brain regions** that process **memory, emotion, and reward**.



BRAIN
DIENCEPHALON

Main Component:

1. Thalamus
2. **Hypothalamus**
3. Subthalamus
4. Epithalamus



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BRAIN
DIENCEPHALON: HYPOTHALAMUS

- **Location:** Most inferior part of the diencephalon

Main Structures:

- **Mammillary bodies** – involved in olfactory reflexes, emotional responses to odors, and memory
- **Infundibulum** – connects the hypothalamus to the pituitary gland

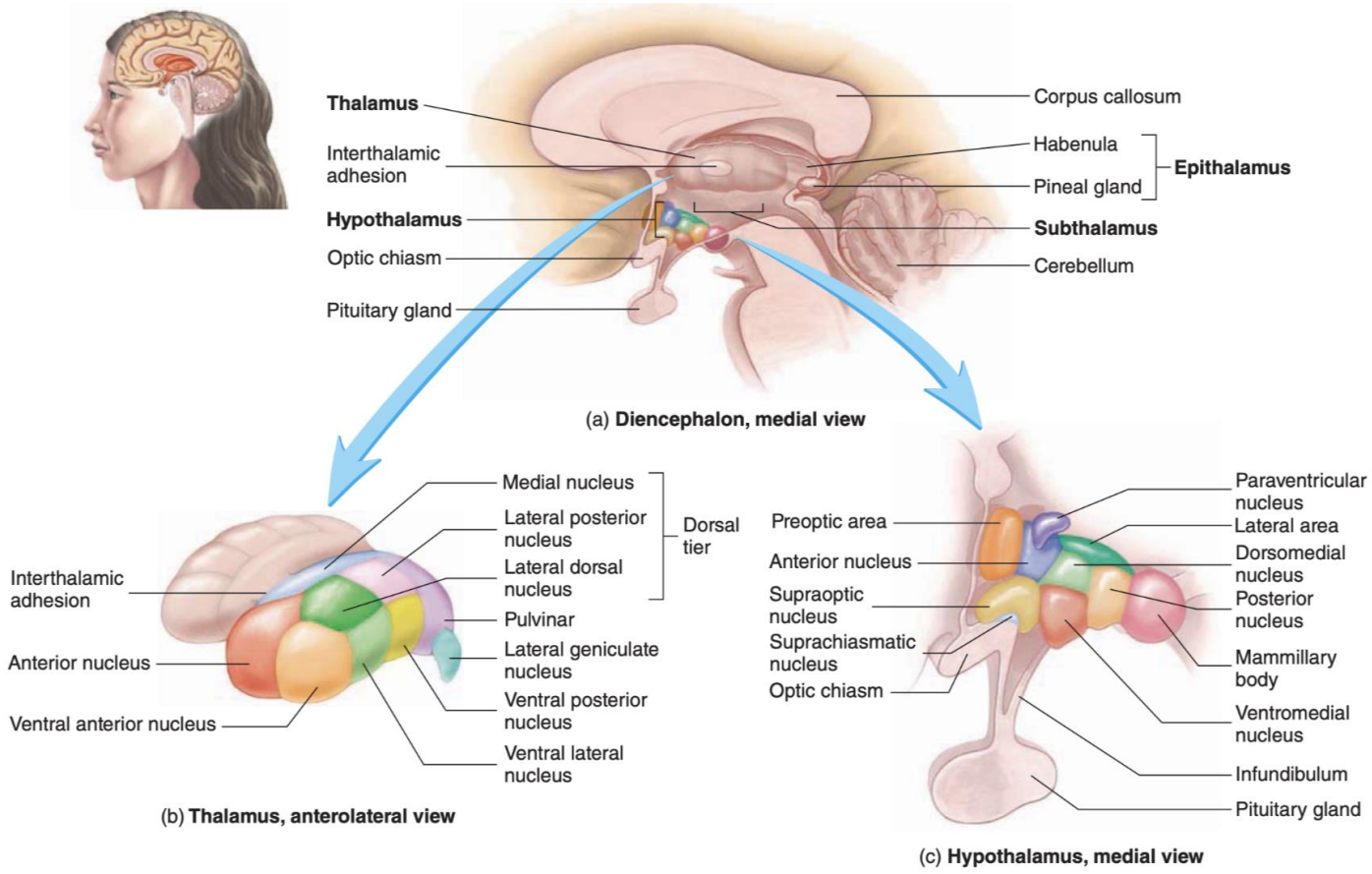


FIGURE 13.7 Diencephalon
(*a*) General overview of the right half of the diencephalon as seen in a median section. (*b*) Thalamus, showing the nuclei of the left half. (*c*) Hypothalamus, showing the nuclei and right half of the pituitary.

BRAIN
DIENCEPHALON: HYPOTHALAMUS

- **Major Functions:** Regulates the **pituitary gland** → controls endocrine functions (metabolism, reproduction, stress response, urine production)
- Maintains **homeostasis** by regulating:
  - Autonomic functions (heart rate, digestion)
  - Body temperature (sweating, shivering)
  - Thirst, hunger, and sex drive
  - Emotional behavior (rage, pleasure, fear)
  - Sleep-wake cycle (via reticular activating system)
- **Integrates sensory input** from internal organs, taste, smell, touch (nipples, genitalia), and visual cues

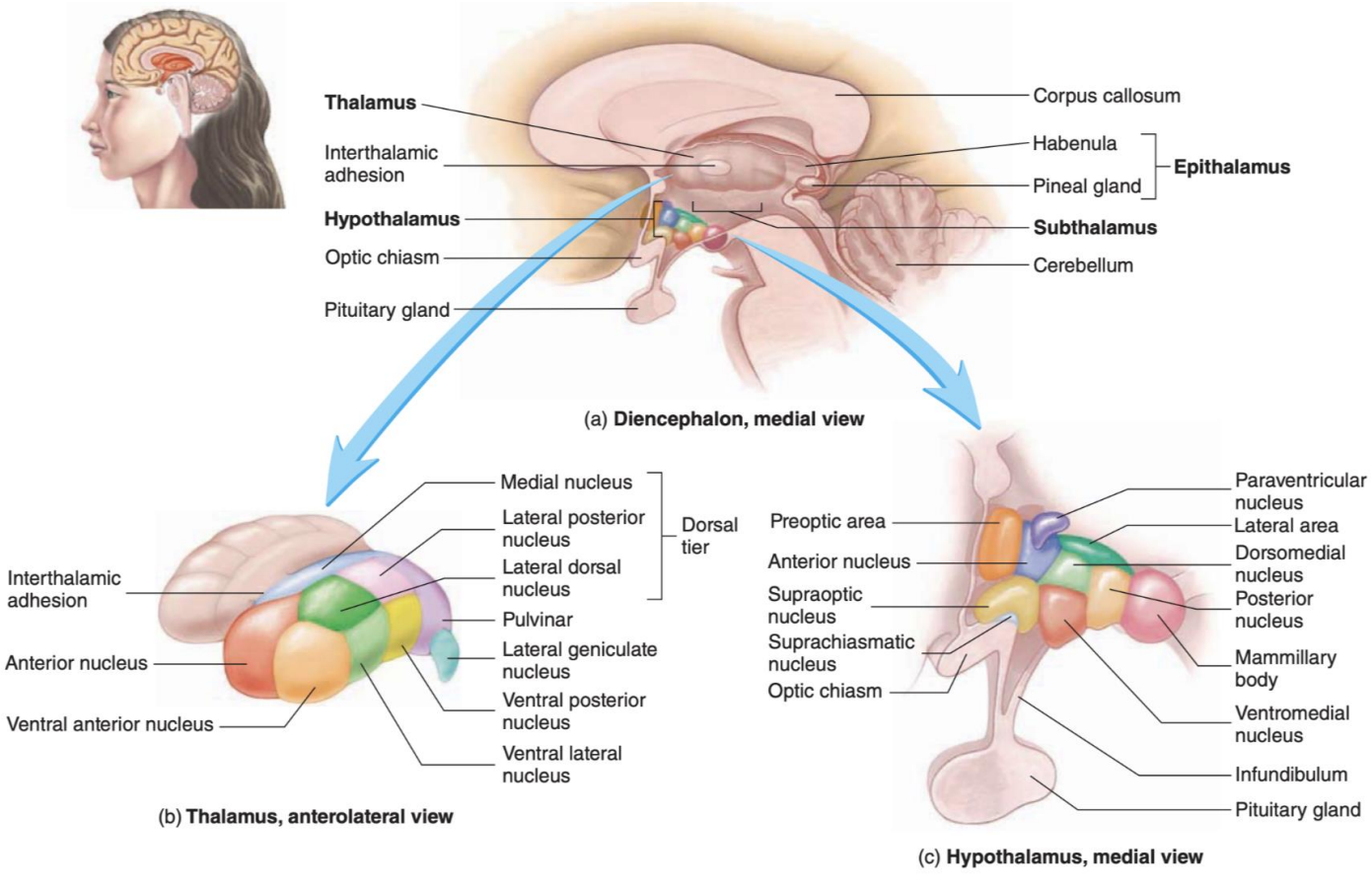


FIGURE 13.7 Diencephalon
(*a*) General overview of the right half of the diencephalon as seen in a median section. (*b*) Thalamus, showing the nuclei of the left half. (*c*) Hypothalamus, showing the nuclei and right half of the pituitary.

BRAIN

DIENCEPHALON: HYPOTHALAMUS

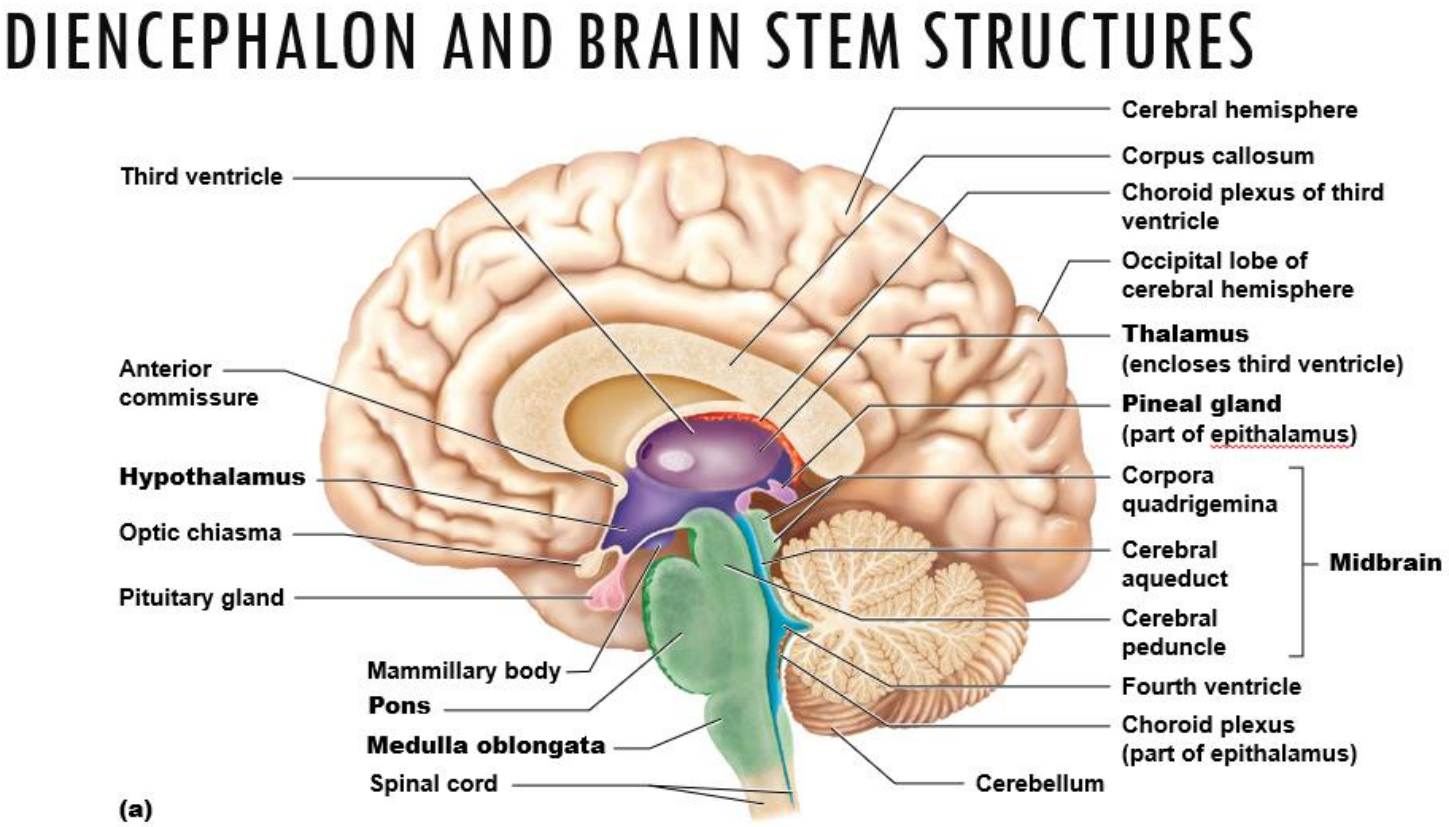
TABLE 13.3 Functions of the Hypothalamus

Function	Hypothalamic Nuclei	Description
Autonomic	Preoptic area and anterior nucleus (parasympathetic) Lateral area and posterior nucleus (sympathetic)	Helps control heart rate, urine release from the bladder, movement of food through the digestive tract, and blood vessel diameter
Endocrine	Paraventricular nucleus Supraoptic nucleus	Helps regulate pituitary gland secretions and influences metabolism, ion concentration, sexual development, and sexual functions; serves as the site of antidiuretic hormone and oxytocin production (see chapter 18)
Muscle control	Lateral area	Controls the muscles involved in swallowing; stimulates shivering
Regulation of body temperature	Preoptic area Anterior nucleus Posterior nucleus	Promotes heat loss when the hypothalamic temperature increases by increasing sweat production (anterior nucleus); promotes heat production when the hypothalamic temperature decreases by stimulating shivering (posterior nucleus); aspirin reduces fever by affecting the preoptic area
Regulation of food and water intake	Ventromedial nucleus Lateral area	Hunger center promotes eating; satiety center inhibits eating; thirst center promotes water intake
Emotions	Lateral area Medial area	Large range of emotional influences over body functions; directly involved in stress-related and psychosomatic illnesses and in feelings of fear and rage
Regulation of the sleep-wake cycle	Lateral area Suprachiasmatic nucleus	Coordinates responses to the sleep-wake cycle with the other areas of the brain (e.g., the reticular activating system); suprachiasmatic nucleus receives direct input from the eyes concerning light/dark cycles and is implicated in jet lag
Sexual development and behavior	Preoptic area Dorsomedial nucleus Ventromedial nucleus	Stimulates sexual development, sexual arousal, and sexual behavior; the preoptic area is larger in males than in females

BRAIN
DIENCEPHALON

Main Component:

1. Thalamus
2. Hypothalamus
3. **Subthalamus**
4. Epithalamus



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BRAIN
DIENCEPHALON: SUBTHALAMUS

- **Location:** Small region **inferior to the thalamus**

Main Structures:

- **Subthalamic nuclei** – functionally associated with the **basal nuclei**
- Contains **ascending and descending tracts**
- Portions of **red nucleus** and **substantia nigra** extend into this area

Major Functions:

1. Plays a critical role in **motor control and coordination**

2. Modulates **voluntary motor activity** via connections with basal nuclei and midbrain structures

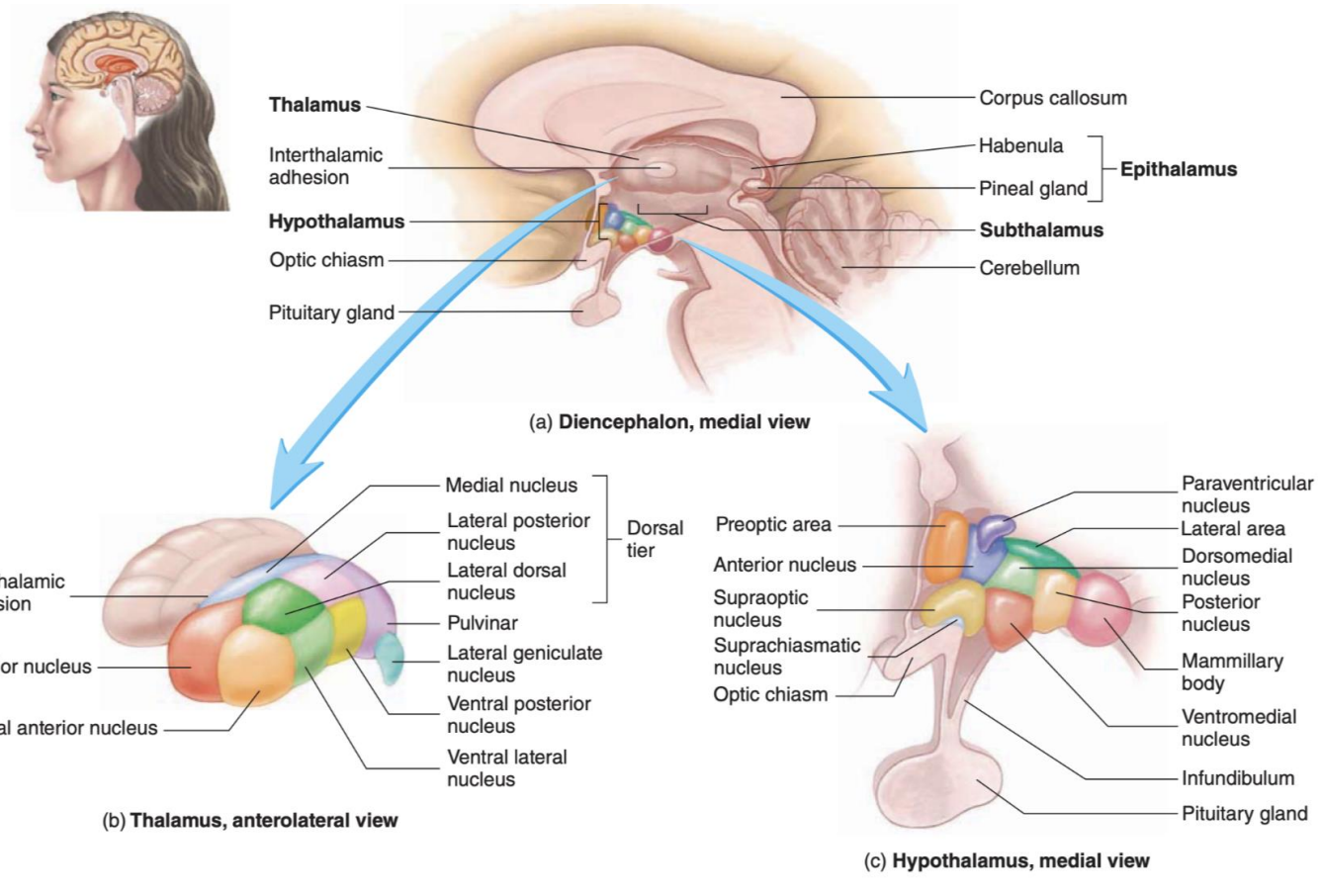
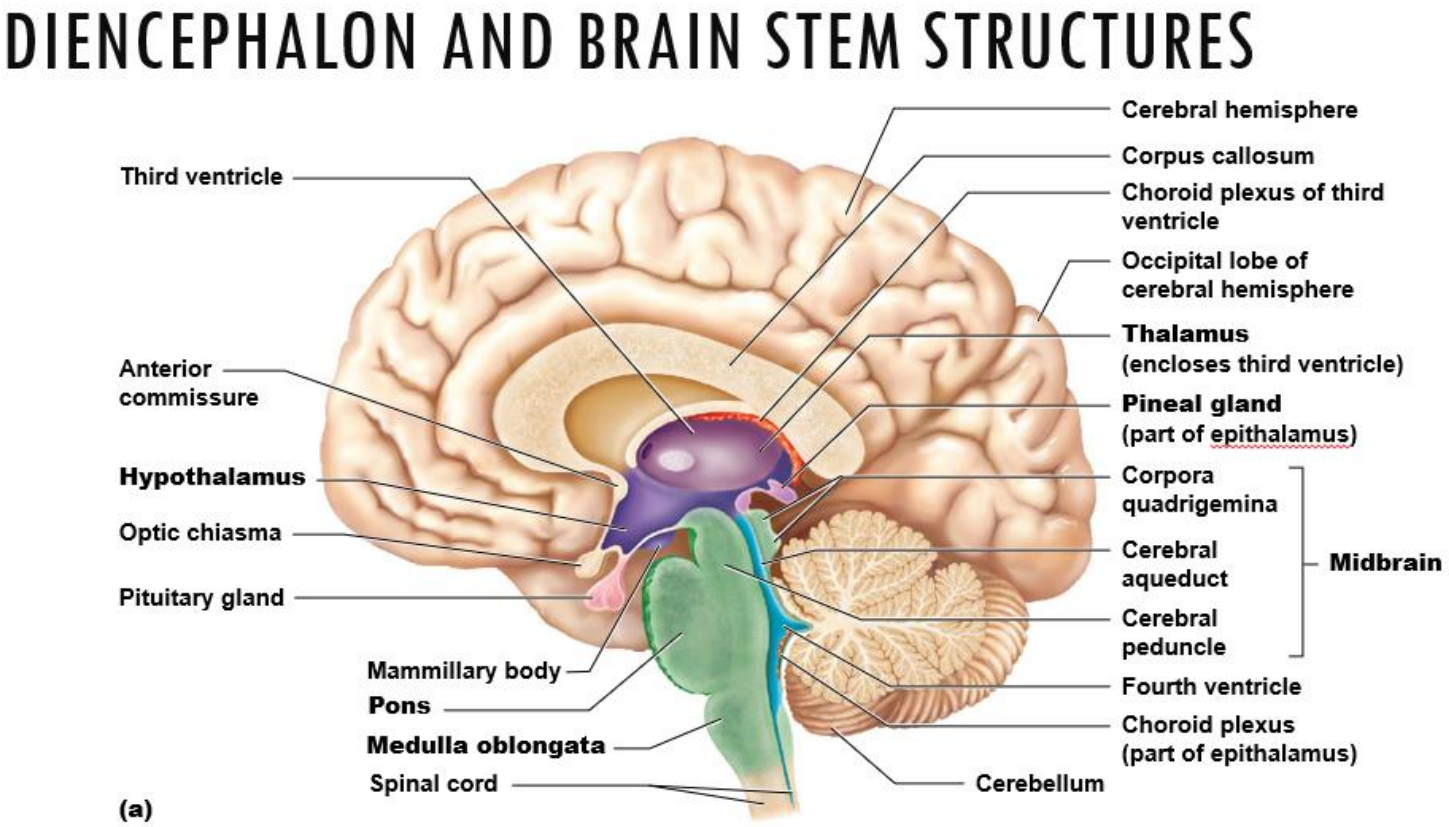


FIGURE 13.7 Diencephalon
(*a*) General overview of the right half of the diencephalon as seen in a median section. (*b*) Thalamus, showing the nuclei of the left half. (*c*) Hypothalamus, showing the nuclei and right half of the pituitary.

BRAIN
DIENCEPHALON

Main Component:

1. Thalamus
2. Hypothalamus
3. Subthalamus
4. **Epithalamus**



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BRAIN
DIENCEPHALON: EPITHALAMUS

- **Location:** Superior and posterior to the thalamus
- **Main Structures:**
  - **Habenula** – linked to sense of smell; involved in emotional and visceral responses to odors
  - **Pineal gland (pineal body)** – small, pinecone-shaped gland that modulates the **sleep-wake cycle** and **circadian rhythms** that secrete **melatonin**

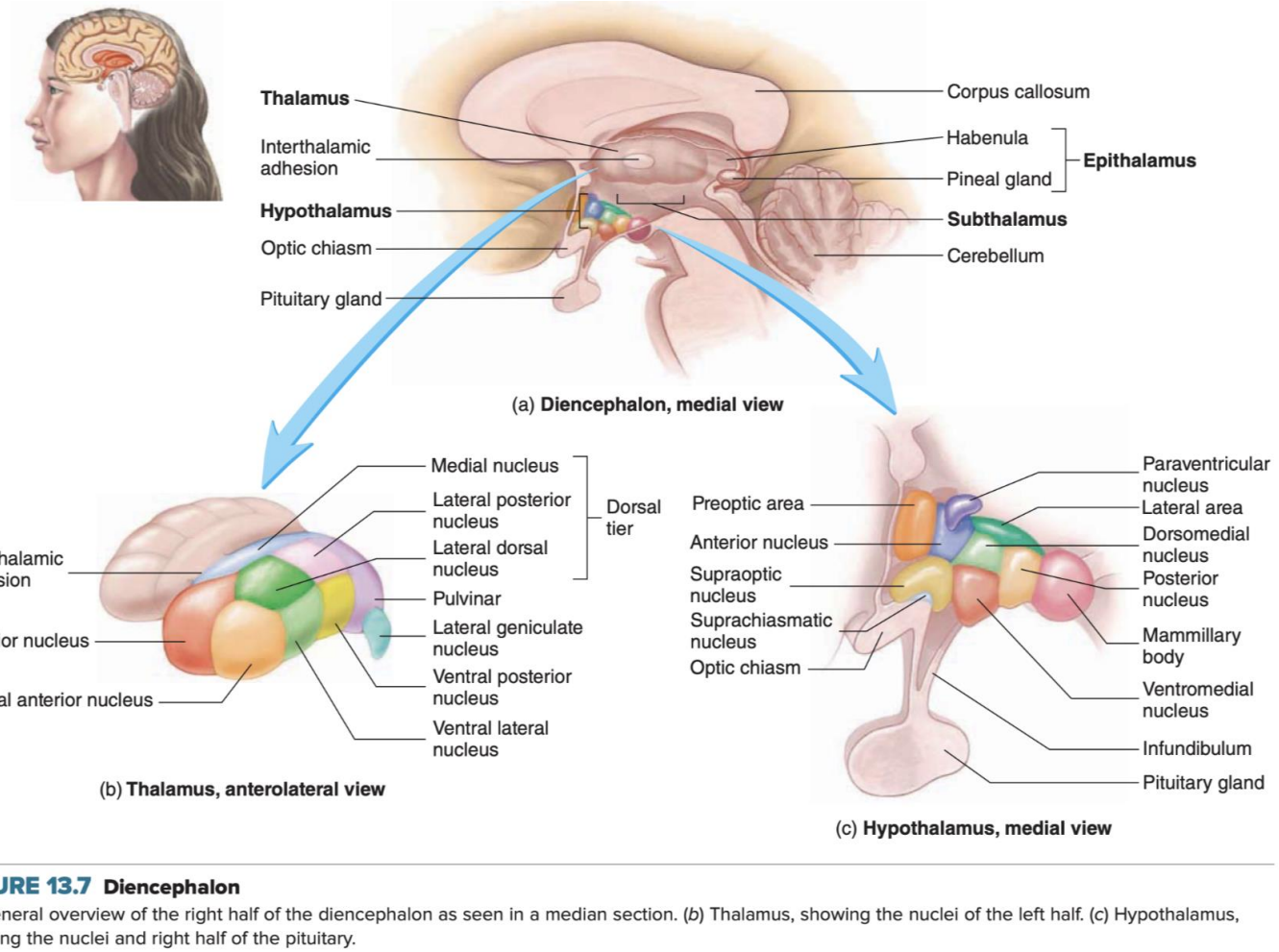


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(*a*) General overview of the right half of the diencephalon as seen in a median section. (*b*) Thalamus, showing the nuclei of the left half. (*c*) Hypothalamus, showing the nuclei and right half of the pituitary.

BRAIN
DIENCEPHALON: EPITHALAMUS

- **Functions:**

1. Influences **biorhythms** (daily physiological cycles)

2. Connects **olfactory stimuli** to emotional and visceral reactions

3. Regulates secretion of **melatonin,** a hormone involved in sleep regulation

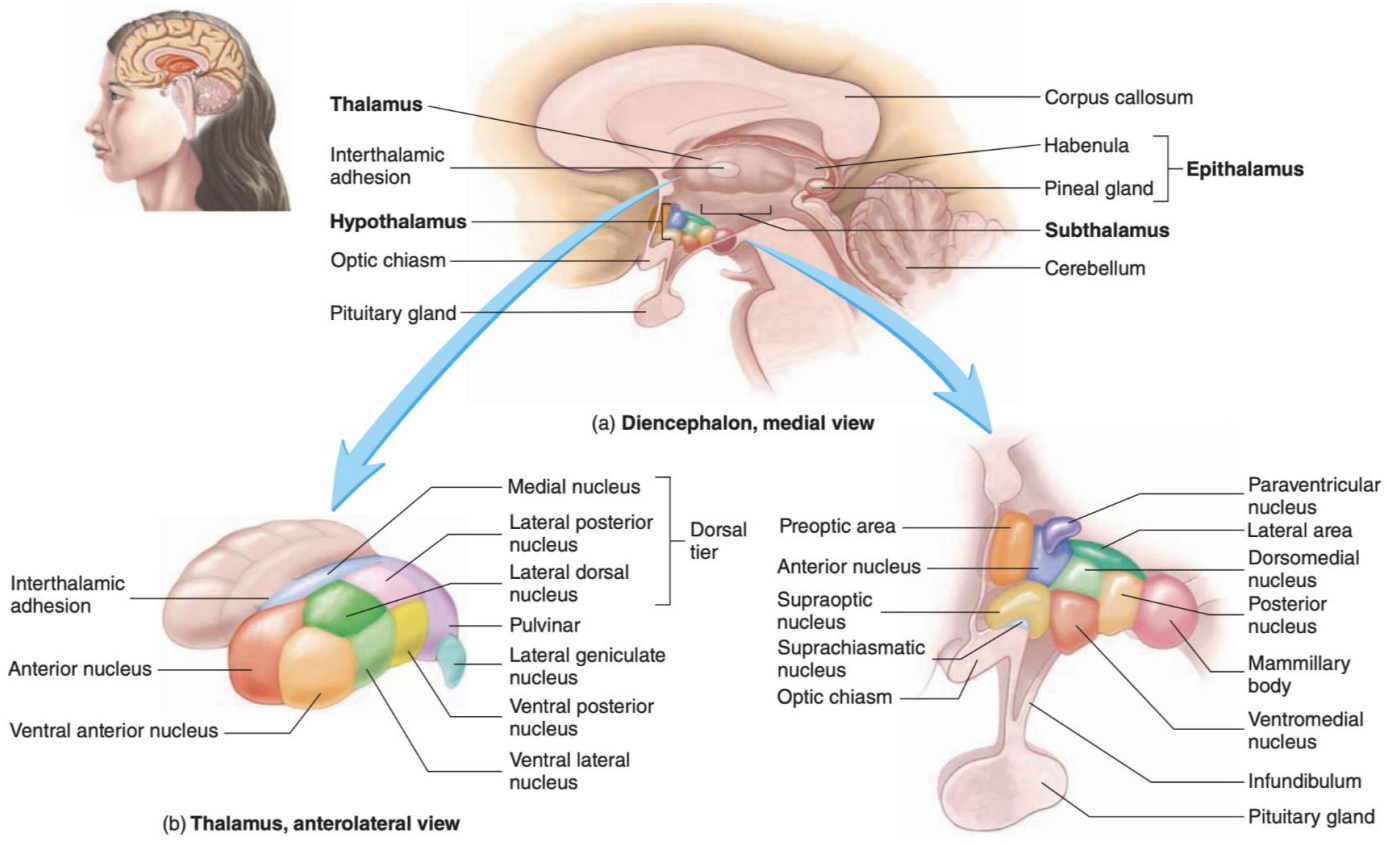
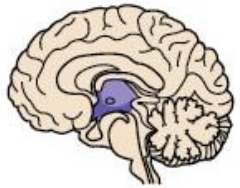


FIGURE 13.7 Diencephalon
(*a*) General overview of the right half of the diencephalon as seen in a median section. (*b*) Thalamus, showing the nuclei of the left half. (*c*) Hypothalamus, showing the nuclei and right half of the pituitary.

SUMMARY
DIENCEPHALON

Diencephalon



Thalamus:

- Relays sensory impulses to cerebral cortex
- Relays impulses between cerebral motor cortex and lower motor centers
- Involved in memory

Hypothalamus:

- Chief integration center of autonomic (involuntary) nervous system
- Regulates body temperature, food intake, water balance, and thirst
- Regulates hormonal output of anterior pituitary gland and acts as an endocrine organ (producing ADH and oxytocin)



Limbic system—A functional system:

- Includes cerebral and diencephalon structures (e.g., hypothalamus and anterior thalamic nuclei)
- Mediates emotional response; involved in memory processing

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HIGHER BRAIN FUNCTIONS

SPEECH AREAS OF THE BRAIN

Dominant Hemisphere:
In most individuals, speech centers are located in the **left cerebral hemisphere.**

Major Cortical Areas:

- **Wernicke's Area (Sensory Speech Area):** Located in the **posterior superior temporal and inferior parietal lobe;** responsible for **language comprehension** and formulation of coherent speech.
- **Broca's Area (Motor Speech Area):** Located in the **inferior frontal gyrus;** responsible for **speech production** and initiation of the motor movements for speaking.

Connection Between Areas:

- **Arcuate Fasciculus:** Bundle of association fibers that connect **Wernicke's and Broca's areas,** allowing for coordinated understanding and production of language.

SPEECH AREAS OF THE BRAIN

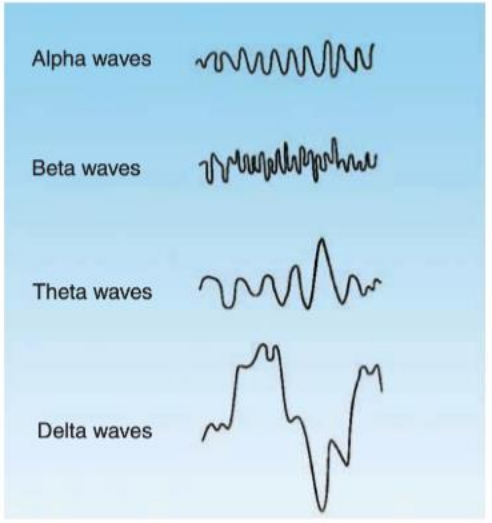
Pathway for Speech:

- **Reading Aloud (Visual to Verbal):**
  - *Primary Visual Cortex* → visual association area → *Wernicke's area* (word understood).
  - *Wernicke's area* → *Broca's area* (word formulated for speech).
  - *Broca's area* → *Premotor area* → *Primary Motor Cortex* (speech executed).
- **Repeating a Heard Word (Auditory to Verbal):**
  - *Primary Auditory Cortex* → auditory association area → *Wernicke's area* (word recognized).
  - Follows same route to *Broca's area* → *Premotor area* → *Primary Motor Cortex*.

BRAIN WAVES



(a)



(b)

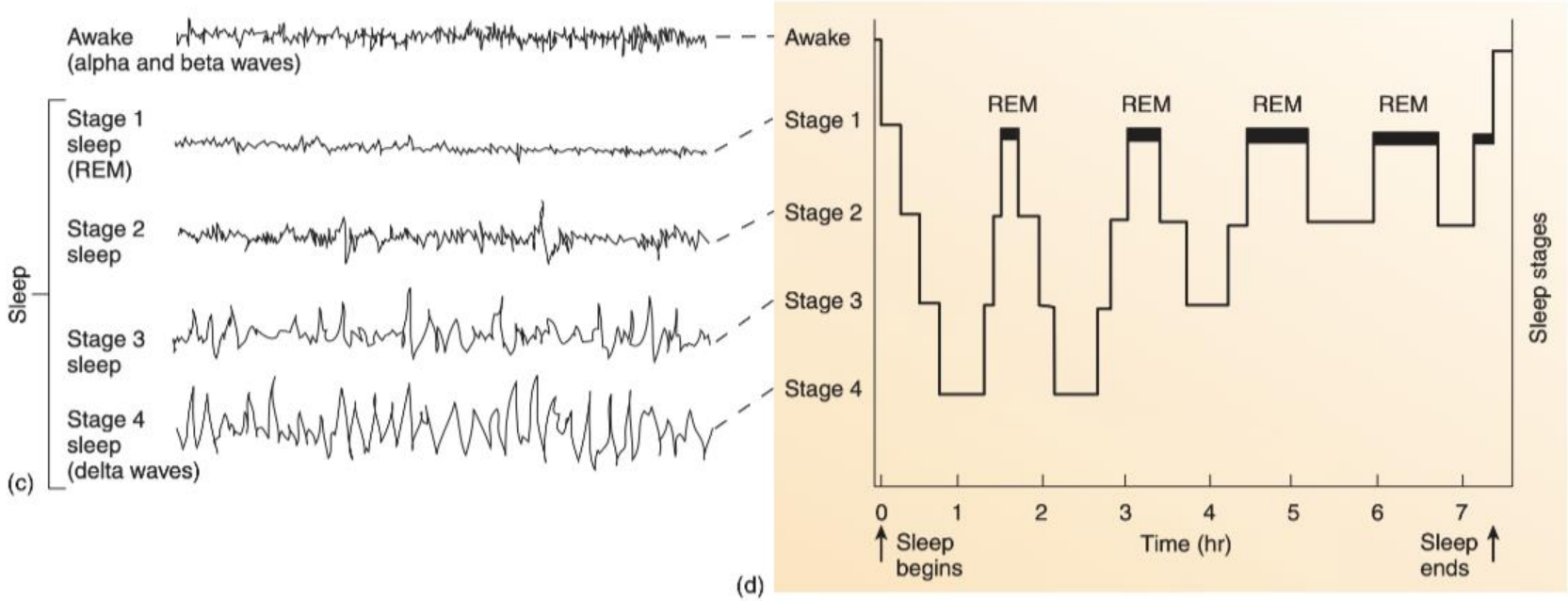
Electroencephalogram (EEG):

- Measures **electrical activity** of the brain via scalp electrodes.
- Reflects **synchronous firing** of large groups of neurons, not individual action potentials.

Types of Brain Waves:

- **Alpha (8–13 Hz):** Relaxed, awake, eyes closed.
- **Beta (14–30 Hz):** Alert, focused, active thinking.
- **Theta (4–7 Hz):** Seen in children, emotional stress, or certain brain disorders in adults.
- **Delta (≤3.5 Hz):** Deep sleep, infancy, or severe brain damage.

BRAIN WAVES AND SLEEP



Sleep Stages:

- Transition: Beta → Alpha → Delta

as sleep deepens.

Non-Rapid Eye Movement (NREM) Sleep:

- Accounts for ~75–80% of total sleep time.
- Characterized by reduced heart rate, blood pressure, and respiration.
- Divided into **4 stages:**
  - *Stage 1:* Transition from wakefulness (alpha → theta waves).
  - *Stage 2:* Light sleep; short bursts of rapid brain activity ("sleep spindles").
  - *Stage 3 & 4:* Deep sleep dominated by delta waves; restorative phase.

Rapid Eye Movement (REM) Sleep:

- Occurs about every 90 minutes.
- EEG resembles that of the awake state (beta waves).
- **Dreaming** occurs, but muscles remain **paralyzed** (except eye and diaphragm muscles).
- Vital for **emotional regulation** and **memory consolidation.**

BRAIN WAVES AND SLEEP



Transitions Between Wakefulness and Sleep

- When falling asleep, EEG patterns shift:
  Beta (alert) → Alpha (relaxed wakefulness) → Theta and Delta (deep sleep).
- A healthy adult cycles between **NREM and REM** approximately

4–6 times per night.

- Brief awakenings are normal and may not be remembered upon waking.

MEMORY

Definition: Memory is the capacity to store and retrieve information for later use.

Types of Memory Storage:

1. **Working Memory –** Temporary, task-related memory; lasts seconds to minutes.
2. **Short-Term Memory –** Lasts minutes to days; based on increased synaptic activity.
3. **Long-Term Memory –** Stable and enduring; requires new protein synthesis and synaptic growth.

Memory Types by Function:

- **Declarative Memory (Explicit):** Facts, names, events — involves the **hippocampus** and **amygdala**.
- **Procedural Memory (Implicit):** Skills and habits — involves the **cerebellum** and **premotor cortex**.

MEMORY

Memory Formation and Retention:

- **Consolidation:** Conversion of short-term to long-term memory through repeated activation and new synaptic connections.
- **Memory Engram:** Network of neurons that stores parts of a memory; retrieval involves reactivating these networks.

BRAIN
ANATOMY OF THE BRAIN

4 Major Brain Regions

1. Cerebral hemispheres/ Cerebrum
   - Cortex
   - White matter
   - Basal nuclei

2. Diencephalon
   - Thalamus
   - Hypothalamus
   - Subthalamus
   - Epithalamus

3. Brainstem
   - Midbrain
   - Pons
   - Medulla oblongata

4. Cerebellum

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BRAIN

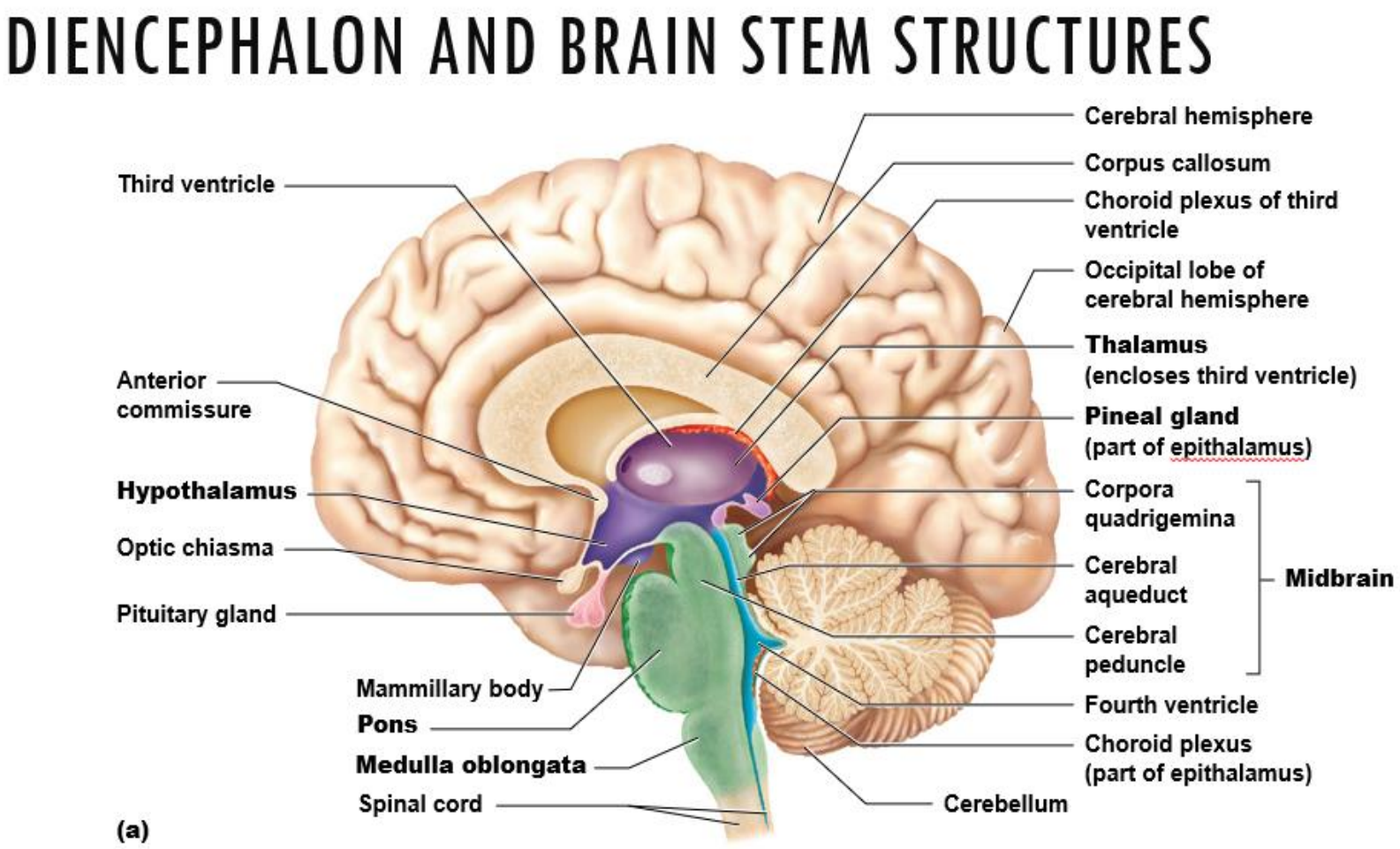
ANATOMY OF THE BRAIN

3. Brainstem

- **Midbrain:** Coordinates visual and auditory reflexes; contains motor pathways.
- **Pons:** Relays information between cerebrum and cerebellum; assists in respiratory control.
- **Medulla Oblongata:** Regulates vital autonomic functions such as heart rate, respiration, and blood pressure.

BRAIN
BRAINSTEM

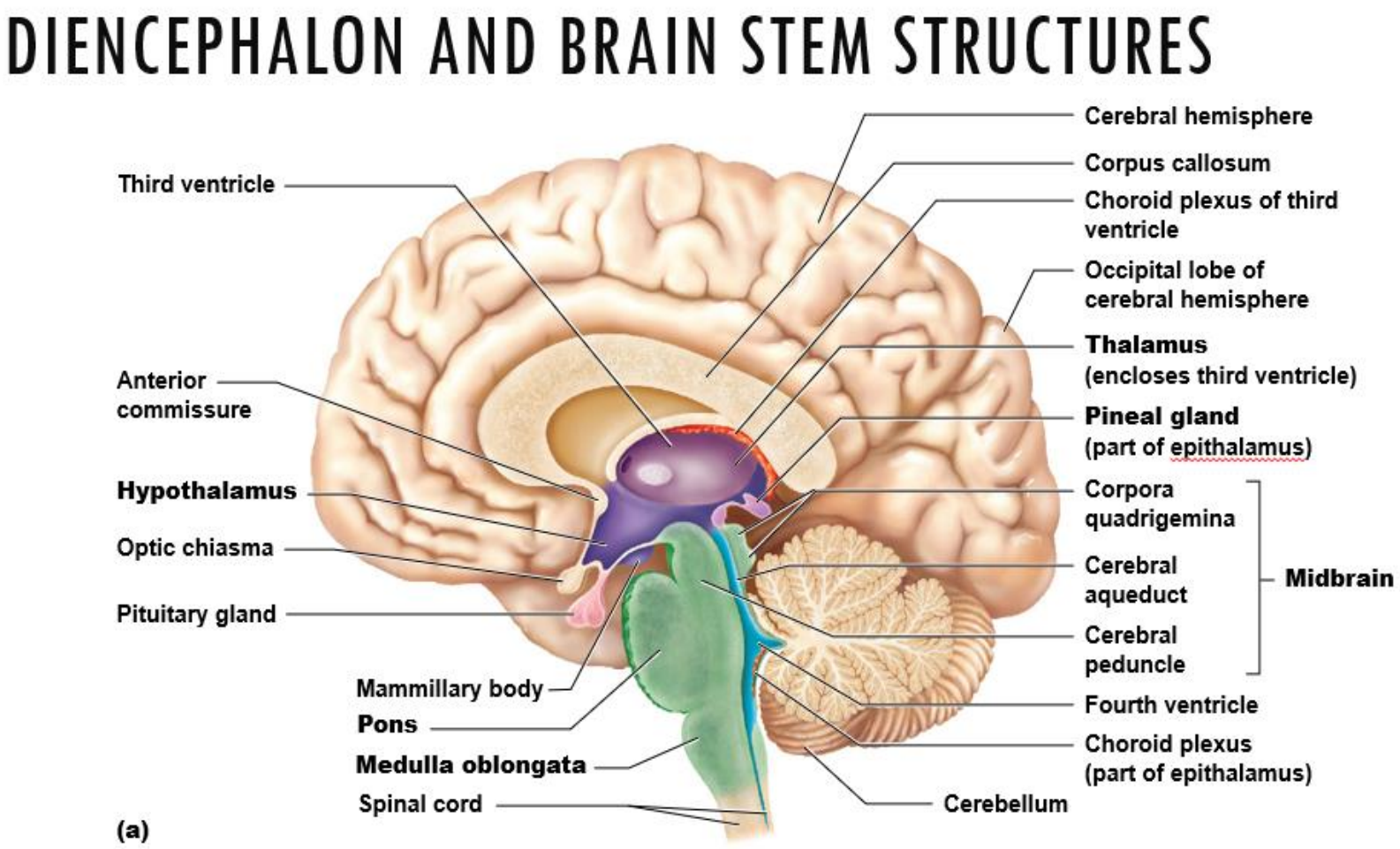
- Attaches to the spinal cord
- Parts of the brain stem
    1. Midbrain
    2. Pons
    3. Medulla oblongata



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BRAIN
BRAINSTEM

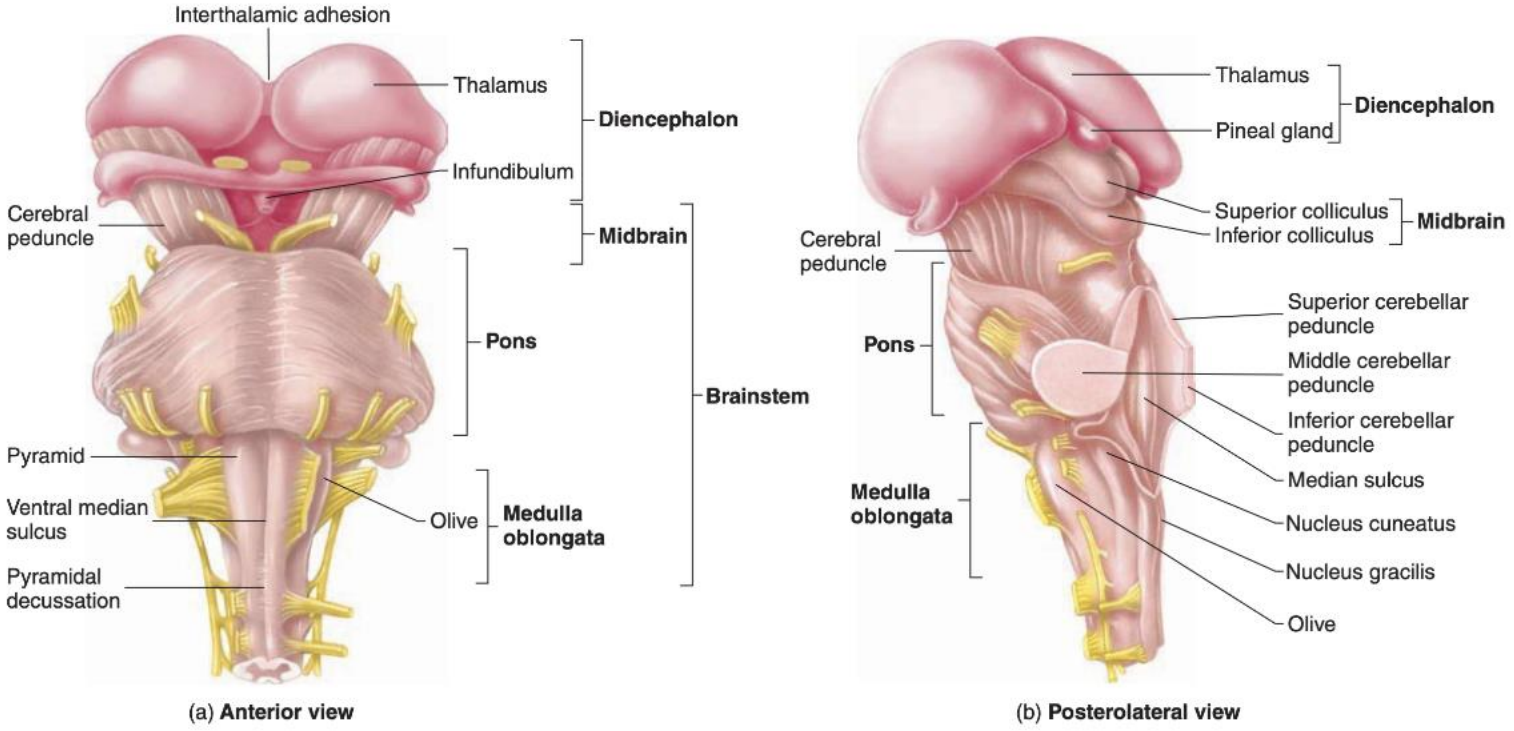
- Attaches to the spinal cord
- Parts of the brain stem
    1. Midbrain
    2. Pons
    3. Medulla oblongata



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BRAIN
BRAINSTEM: MIDBRAIN

- The *midbrain* is the smallest region of the brainstem, located **superior to the pons** and **inferior to the diencephalon.**
- **Cranial Nerves:**
  - Contains nuclei of CN III (oculomotor)
  - CN IV (trochlear)
  - Part of CN V (trigeminal).

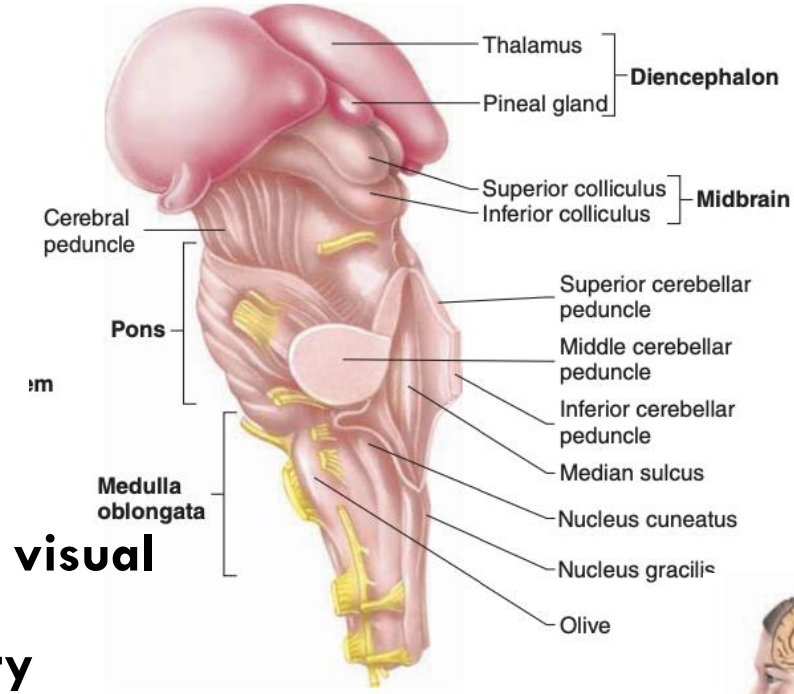


BRAIN

BRAINSTEM: MIDBRAIN

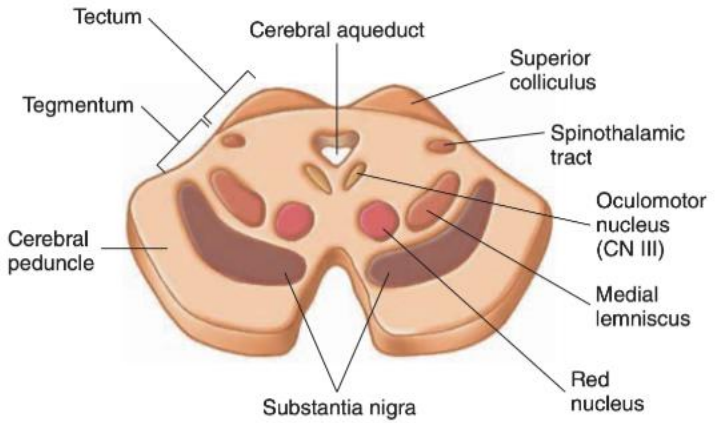
Major Structures:

- **Tectum (Corpora Quadrigemina):**
  - **Superior colliculi** – Reflex centers for **visual and tactile stimuli**.
  - **Inferior colliculi** – Part of the **auditory pathway** and initiate reflex responses to sound.
- **Tegmentum:**
  - Contains **ascending sensory tracts, red nuclei, substantia nigra,** and **cerebral peduncles.**
- **Red Nuclei:** Coordinate **unconscious motor control.**
- **Substantia Nigra:** Produces **dopamine;** helps maintain **muscle tone and movement coordination.**



(b) **Posterolateral view**





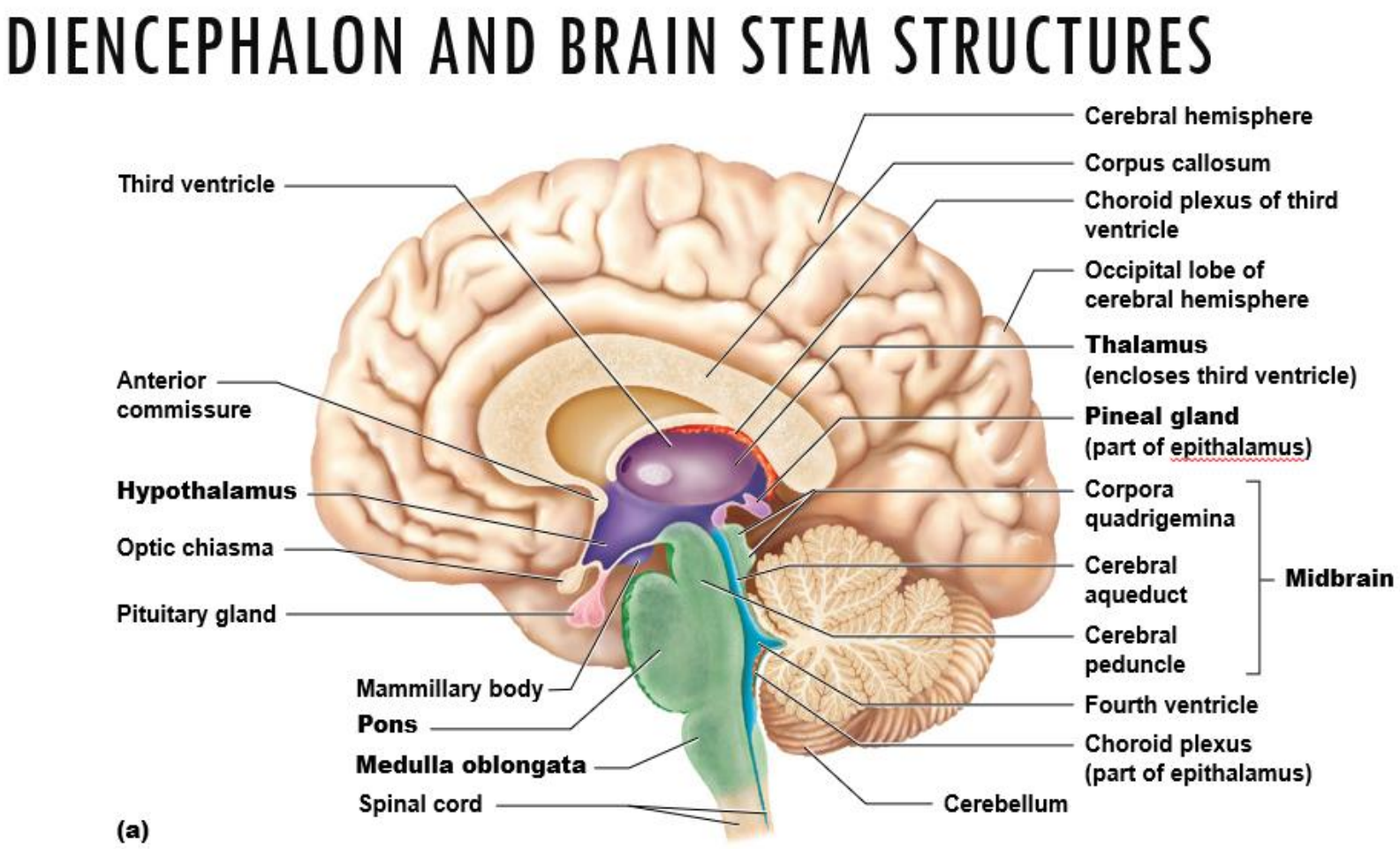
Anterosuperior view of oblique section

FIGURE 13.5 Oblique Section Through the Midbrain
Inset shows the level of section.

BRAIN

BRAIN STEM

- Attaches to the spinal cord
- Parts of the brain stem
  1. Midbrain
  2. Pons
  3. Medulla oblongata



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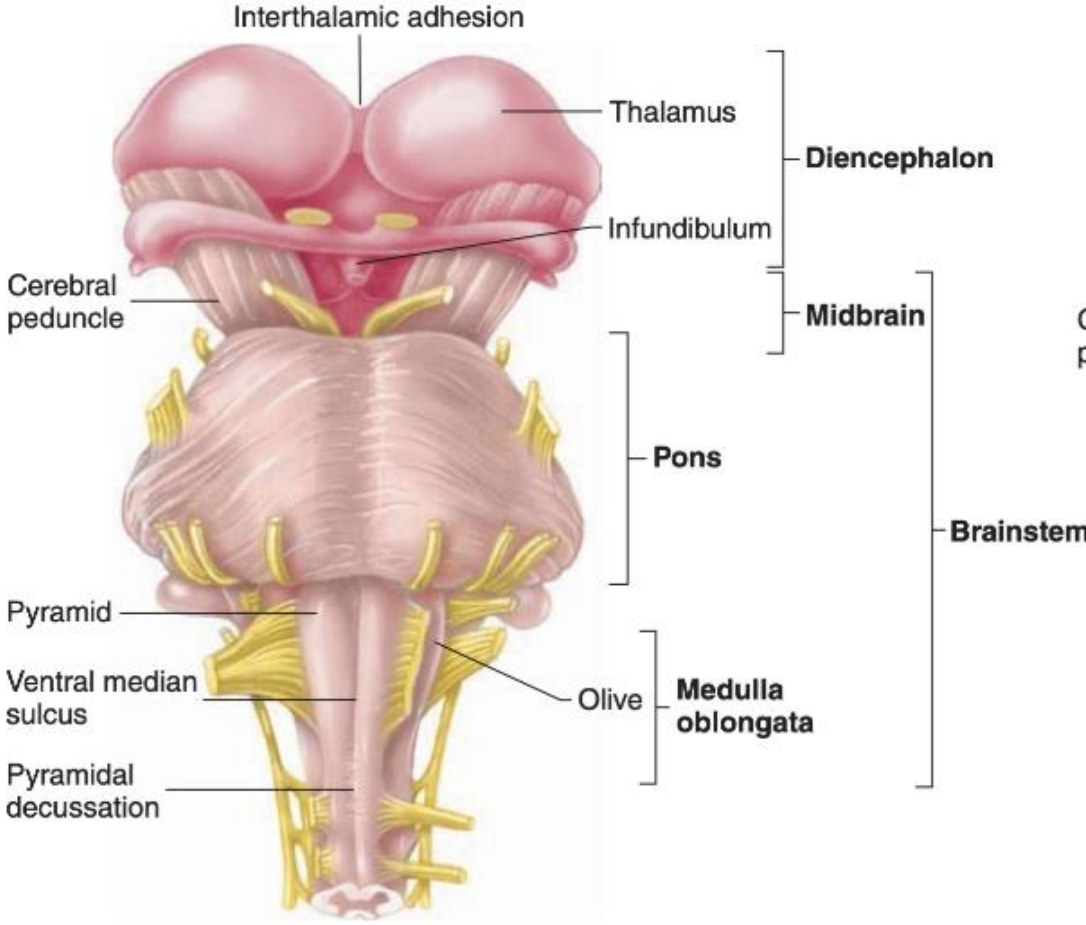
BRAIN

BRAIN STEM: PONS

The *pons* is the portion of the **brainstem located superior to the medulla oblongata** and inferior to the midbrain.

Functions:

- Acts as a **bridge** connecting the **cerebrum, cerebellum, and medulla oblongata.**
- Contains **ascending and descending tracts** for sensory and motor signal transmission.
- **Pontine nuclei** relay information between **cerebrum and cerebellum.**
- Houses nuclei for **cranial nerves V, VI, VII, and VIII.**
- Includes **pontine sleep center** (initiates REM sleep) and **pontine respiratory center** (assists in breathing regulation).

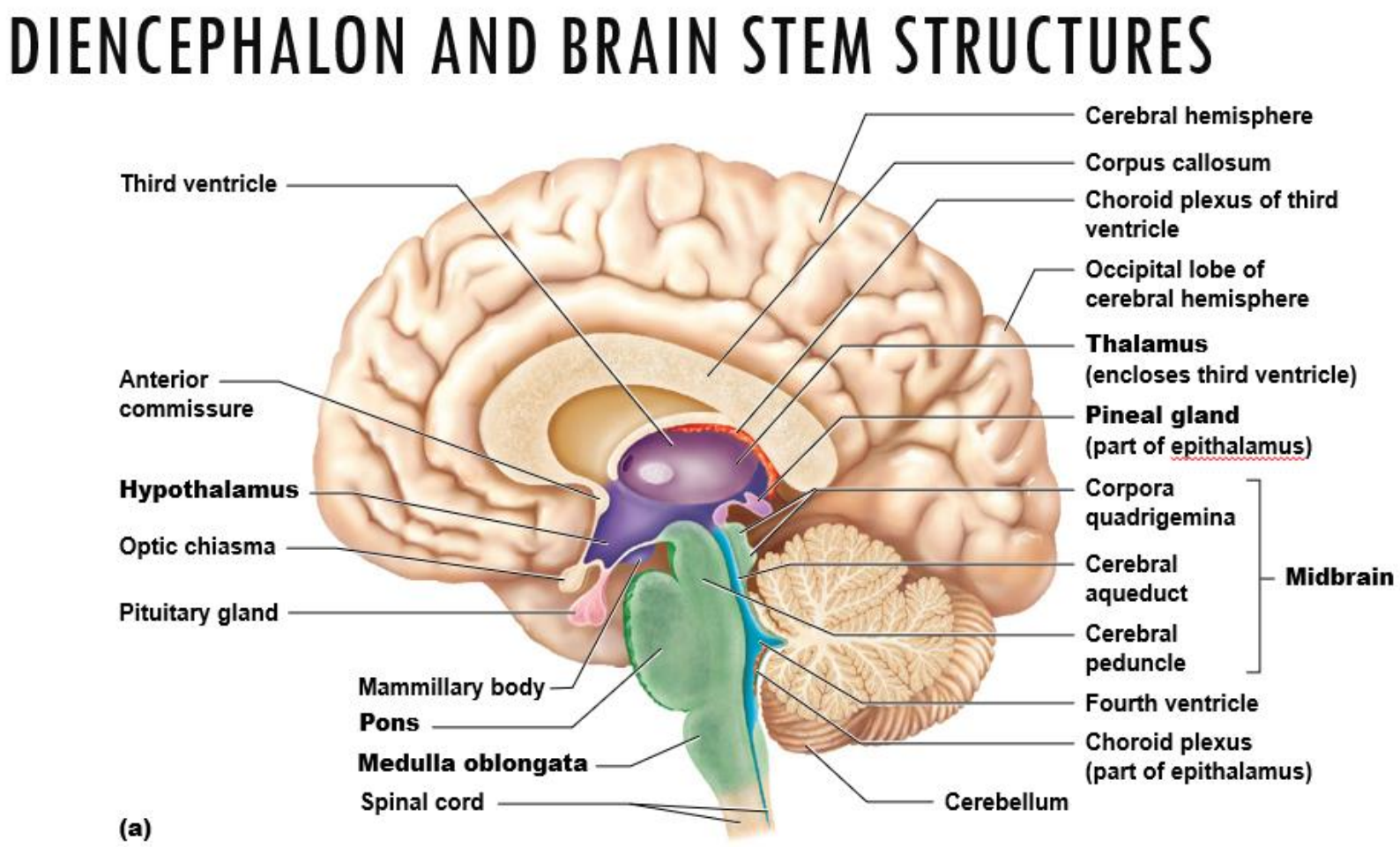


(a) **Anterior view**

BRAIN

BRAIN STEM

- Attaches to the spinal cord
- Parts of the brain stem
  1. Midbrain
  2. Pons
  3. Medulla oblongata



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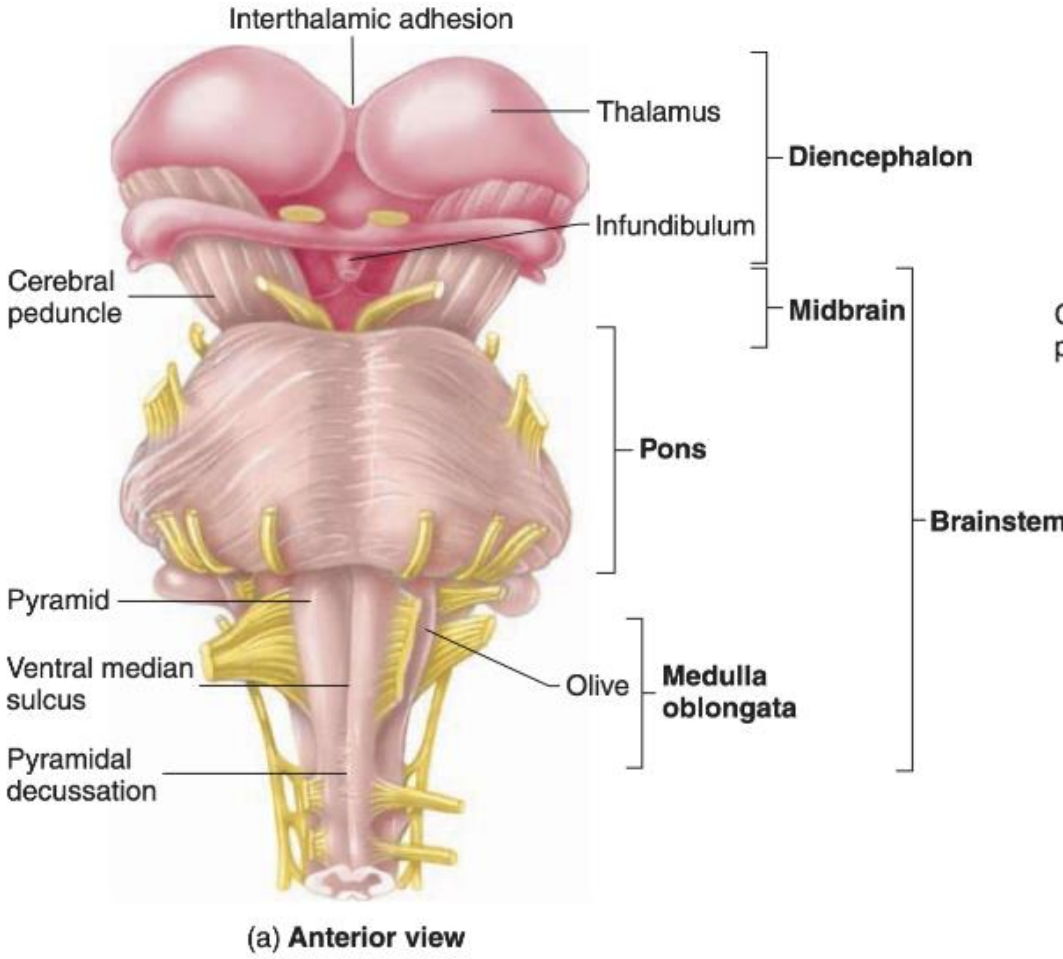
BRAIN

BRAIN STEM: MEDULLA OBLONGATA

The *medulla oblongata* is the **most inferior part of the brainstem**, continuous with the **spinal cord**, approximately **3 cm long**.

Major Features:

- Contains **sensory and motor tracts, cranial nerve nuclei,** and **vital reflex centers.**
- **Pyramids –** anterior enlargements formed by **descending motor tracts;** fibers **decussate** (cross over), causing contralateral control.
- **Olives –** oval structures involved in **balance, coordination, and auditory modulation.**
- Houses nuclei for **cranial nerves V, VII, IX, X, XI, and XII.**

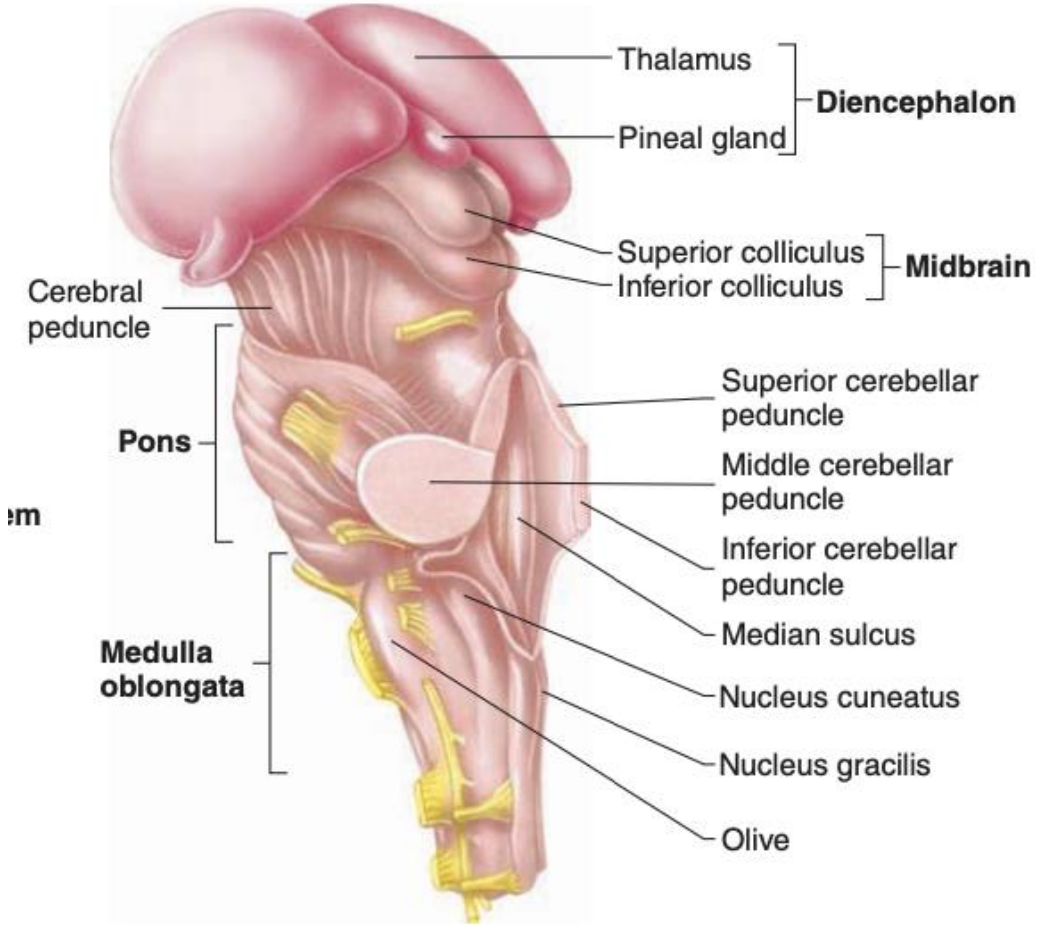


BRAIN

BRAIN STEM: MEDULLA OBLONGATA

Vital Reflex Centers:

- **Cardiac Center** – regulates heart rate.
- **Vasomotor Center** – controls blood vessel diameter and blood pressure.
- **Respiratory Rhythmicity Center** – controls rate and depth of breathing.
- Coordinates **swallowing, vomiting, coughing, sneezing, and hiccuping.**



(b) **Posterolateral view**

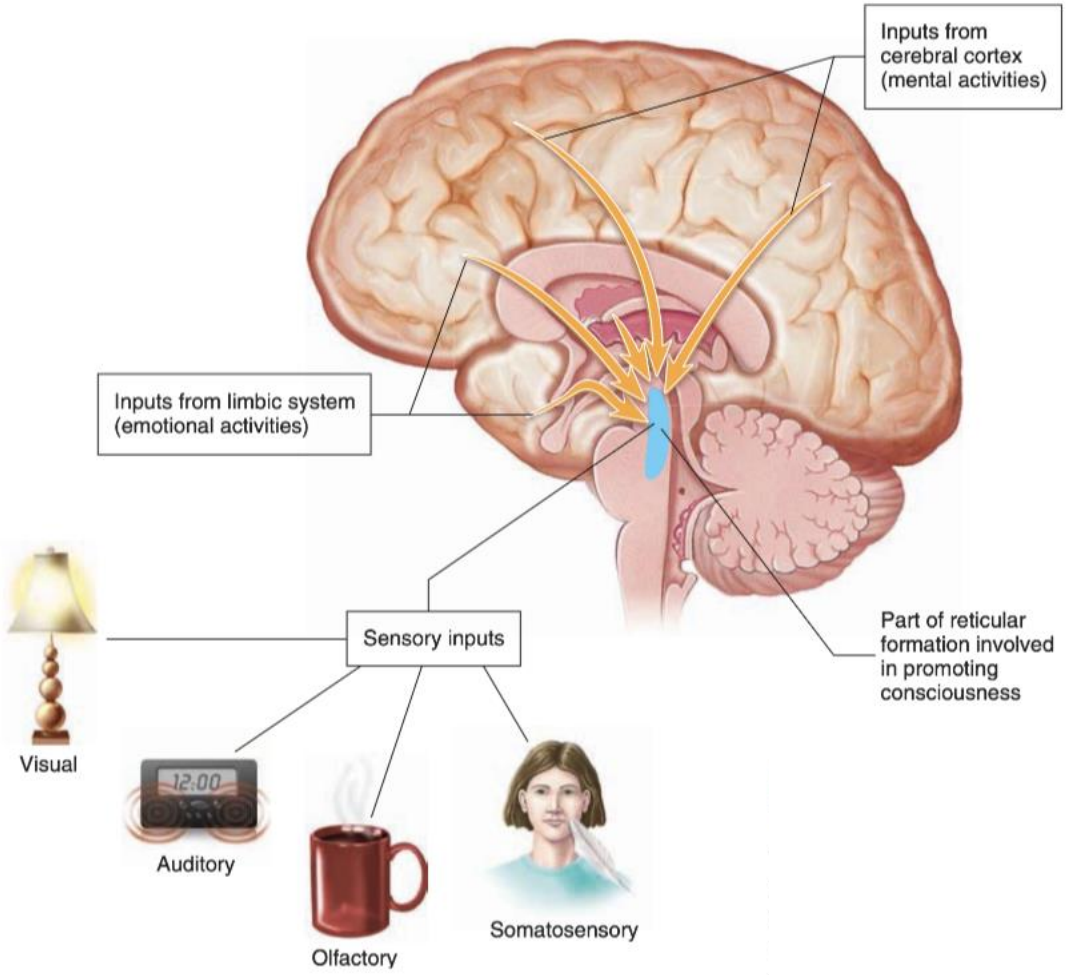
BRAIN

BRAIN STEM: THE RETICULAR FORMATION

A **diffuse network of interconnected nuclei** scattered throughout the **brainstem** (medulla, pons, and midbrain).

Functions:

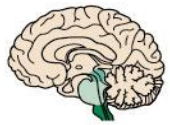
- **Arousal and Consciousness –** maintains alertness and wakefulness (part of the *Reticular Activating System*, RAS).
- **Motor Control –** regulates **posture** and **muscle tone** through spinal tracts.
- **Autonomic Regulation –** coordinates **swallowing, breathing, and heart rate.**
- **Pain Modulation –** reduces transmission of pain signals from the spinal cord.
- **Sleep-Wake Cycle –** controls transitions between **sleep and wakefulness.**



SUMMARY

BRAIN STEM

Table 7.1 **Functions of Major Brain Regions** (*continued*)

Region	Function
Cerebral hemispheres	
Brain stem	
	**Midbrain:** • Contains visual and auditory reflex centers • Contains subcortical motor centers • Contains nuclei for cranial nerves III and IV; contains projection fibers (e.g., fibers of the pyramidal tracts)
	Pons: • Relays information from the cerebrum to the cerebellum • Cooperates with the medullary centers to control respiratory rate and depth • Contains nuclei of cranial nerves V–VII; contains projection fibers
	Medulla oblongata: • Relays ascending sensory pathway impulses from skin and proprioceptors • Contains nuclei controlling heart rate, blood vessel diameter, respiratory rate, vomiting, etc. • Relays sensory information to the cerebellum • Contains nuclei of cranial nerves VIII–XII; contains projection fibers • Site of crossover of pyramids
	**Reticular formation—A functional system:** • Maintains cerebral cortical alertness; filters out repetitive stimuli • Helps regulate skeletal and visceral muscle activity

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BRAIN
ANATOMY OF THE BRAIN

4 Major Brain Regions

1. Cerebral hemispheres/ Cerebrum
   - Cortex
   - White matter
   - Basal nuclei

2. Diencephalon
   - Thalamus
   - Hypothalamus
   - Subthalamus
   - Epithalamus

3. Brain stem
   - Midbrain
   - Pons
   - Medulla oblongata

4. Cerebellum

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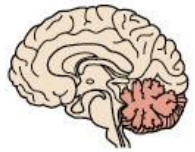
BRAIN

ANATOMY OF THE BRAIN

4. Cerebellum

- Coordinates **voluntary movements, posture, balance,** and **motor learning.**

Cerebellum



Cerebellum:

- Processes information from cerebral motor cortex, proprioceptors, and visual and equilibrium pathways
- Provides "instructions" to cerebral motor cortex and subcortical motor centers, resulting in smooth, coordinated skeletal muscle movements
- Responsible for proper balance and posture

BRAIN
CEREBELLUM

- Also known as the "little brain"

Location & Gross Anatomy

- Situated in the **posterior cranial fossa,** attached to the brainstem (dorsal to pons & medulla) via three paired cerebellar peduncles (superior, middle, inferior).
- Outer grey matter cortex, internal white matter "arbor vitae", and deep cerebellar nuclei (dentate, interposed [globose & emboliform], fastigial).
- Surface has tightly packed ridges called **folia,** grooves/fissures dividing lobes (anterior, posterior, flocculonodular)

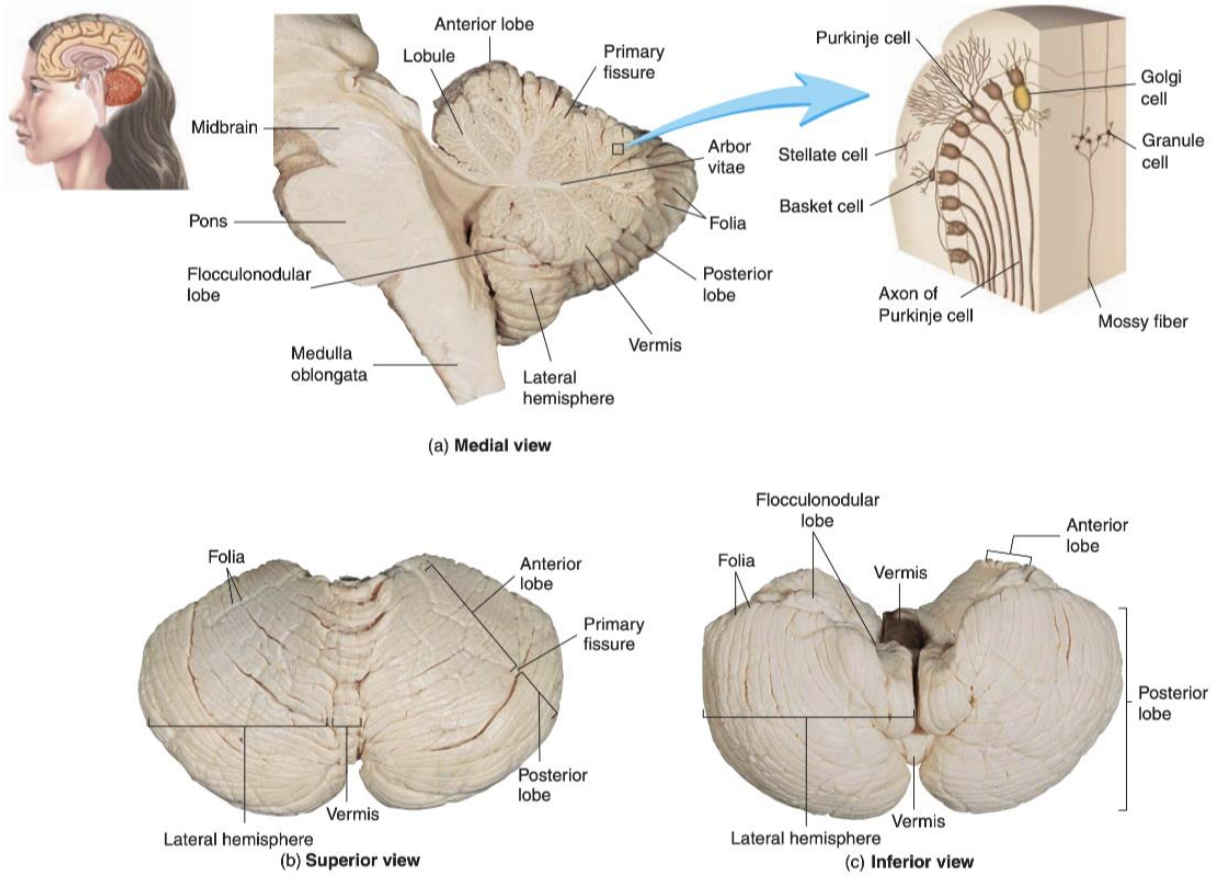


FIGURE 13.6 Cerebellum
(*a*) Right half of the cerebellum and brainstem as seen in a median section. Inset shows the histology of the cerebellum. (*b*) Superior view of the cerebellum. (*c*) Inferior view of the cerebellum. AP|R

BRAIN
CEREBELLUM

Functional & Phylogenetic Divisions

- **Vestibulocerebellum (flocculonodular lobe):** balance, eye movements.
- **Spinocerebellum (vermal & paravermal zones):** regulates posture, locomotion, limb coordination.
- **Cerebrocerebellum (lateral hemispheres):** complex movement planning, motor learning, cognitive functions

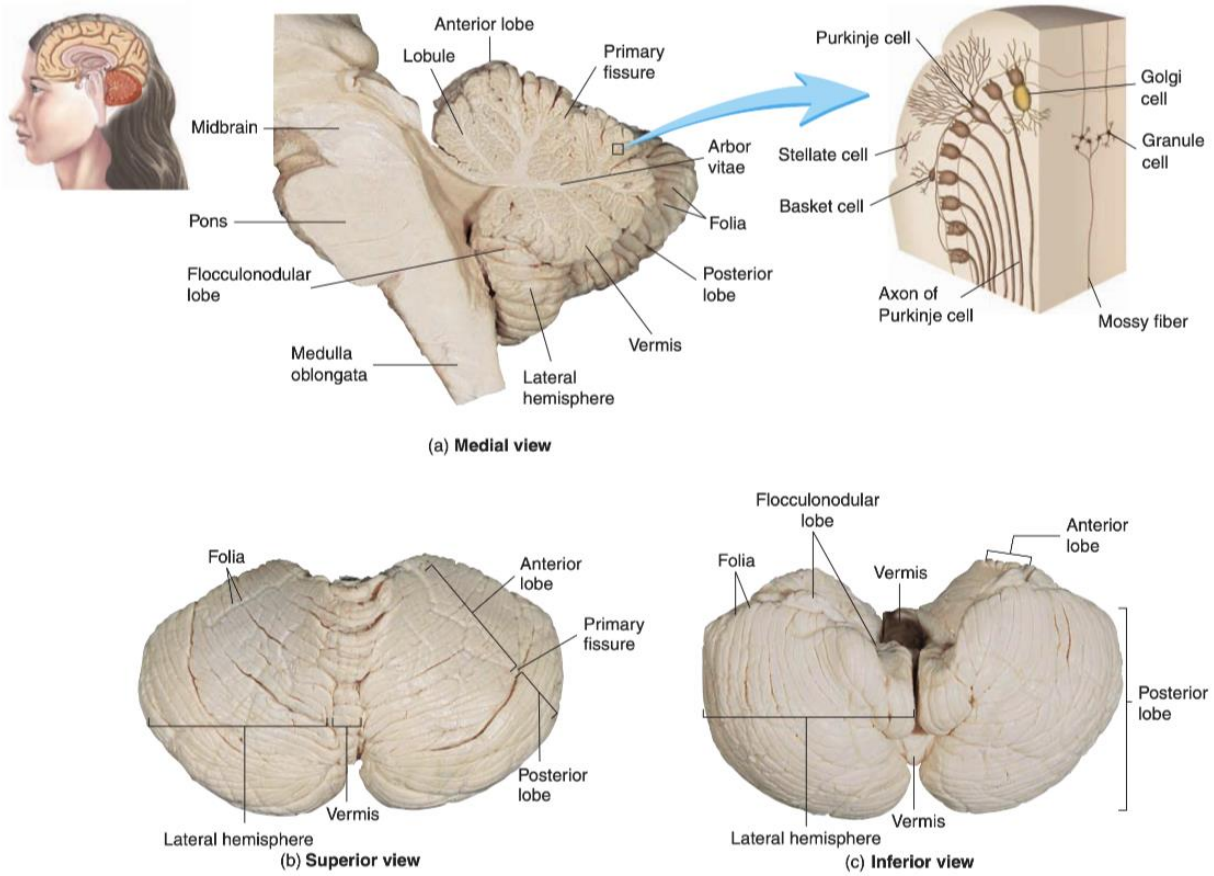


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BRAIN
CEREBELLUM

Connections & Pathways

- Afferent and efferent connections: spinal cord, vestibular nuclei, cerebral cortex (via pontine nuclei), red nucleus, thalamus etc.
- Peduncles: Superior (mainly efferent to midbrain/thalamus), Middle (afferent from pontine nuclei), Inferior (afferent vestibulocerebellar & spinocerebellar, some efferent to vestibular nuclei).

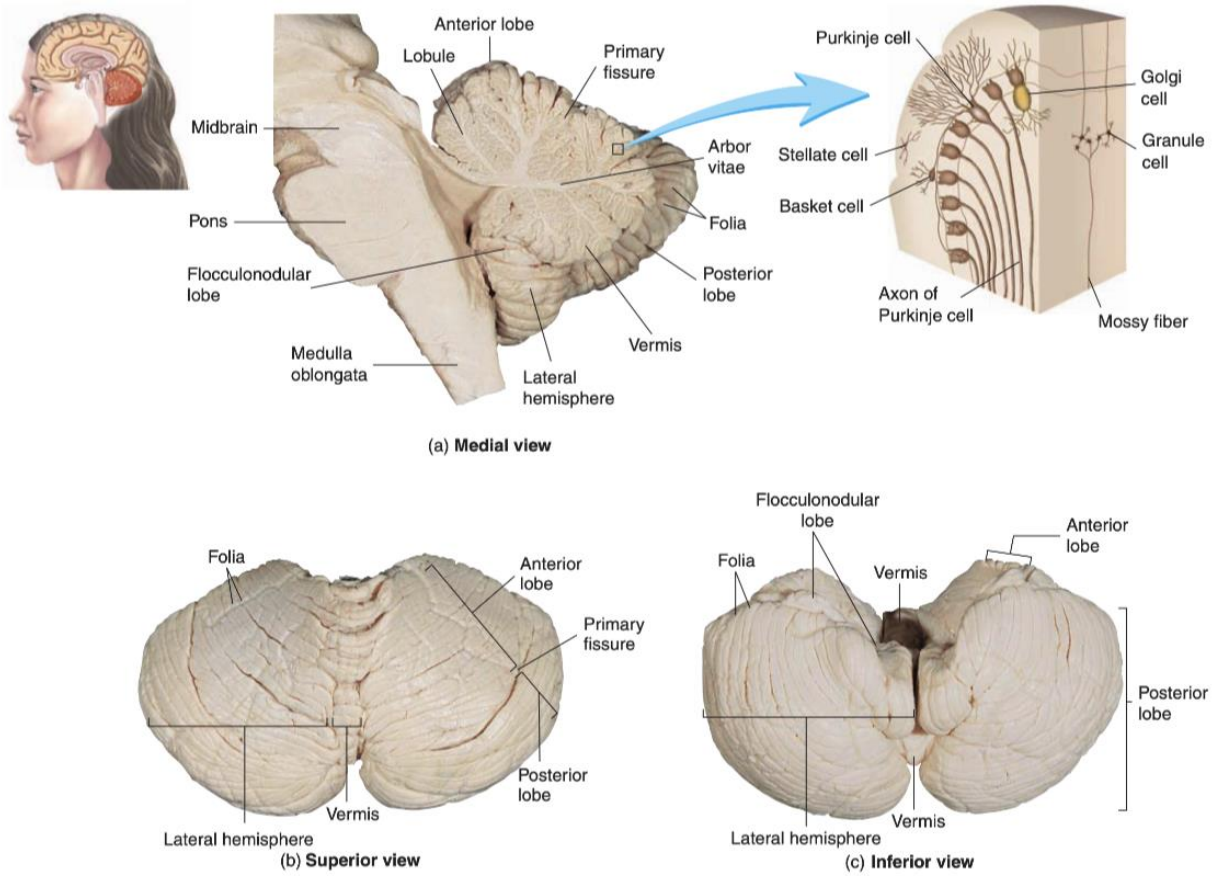


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BRAIN
CEREBELLUM

Major Functions

- Ensures **smooth, coordinated voluntary movements,** proper timing and force of muscle contractions.
- Maintains **balance, posture,** and eye-movement coordination.
- Facilitates **motor learning,** skill acquisition (e.g., playing instrument, typing, sports) and adapts movements.
- Emerging roles in **cognition, language, emotion,** and **predictive control** of behavioral responses.

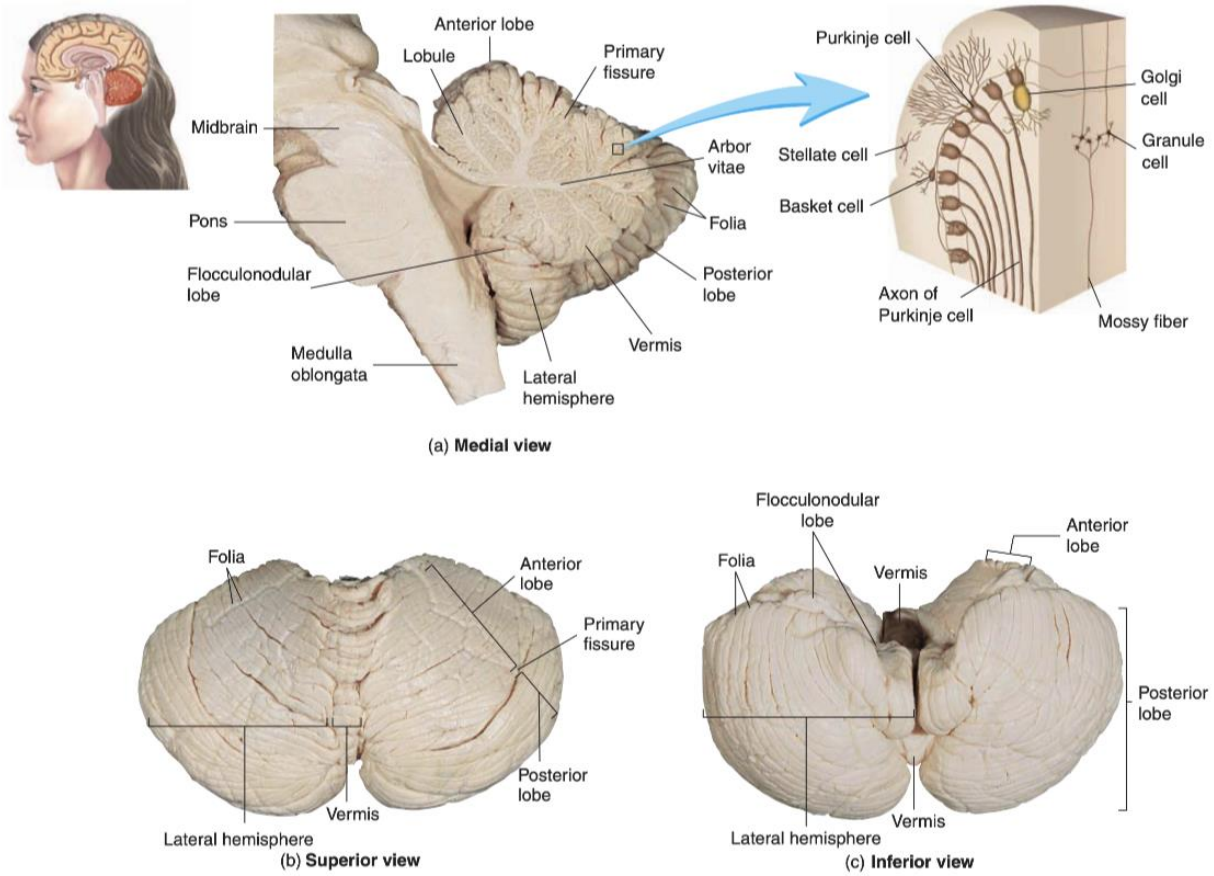


FIGURE 13.6 Cerebellum
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PROTECTION OF THE CENTRAL NERVOUS SYSTEM

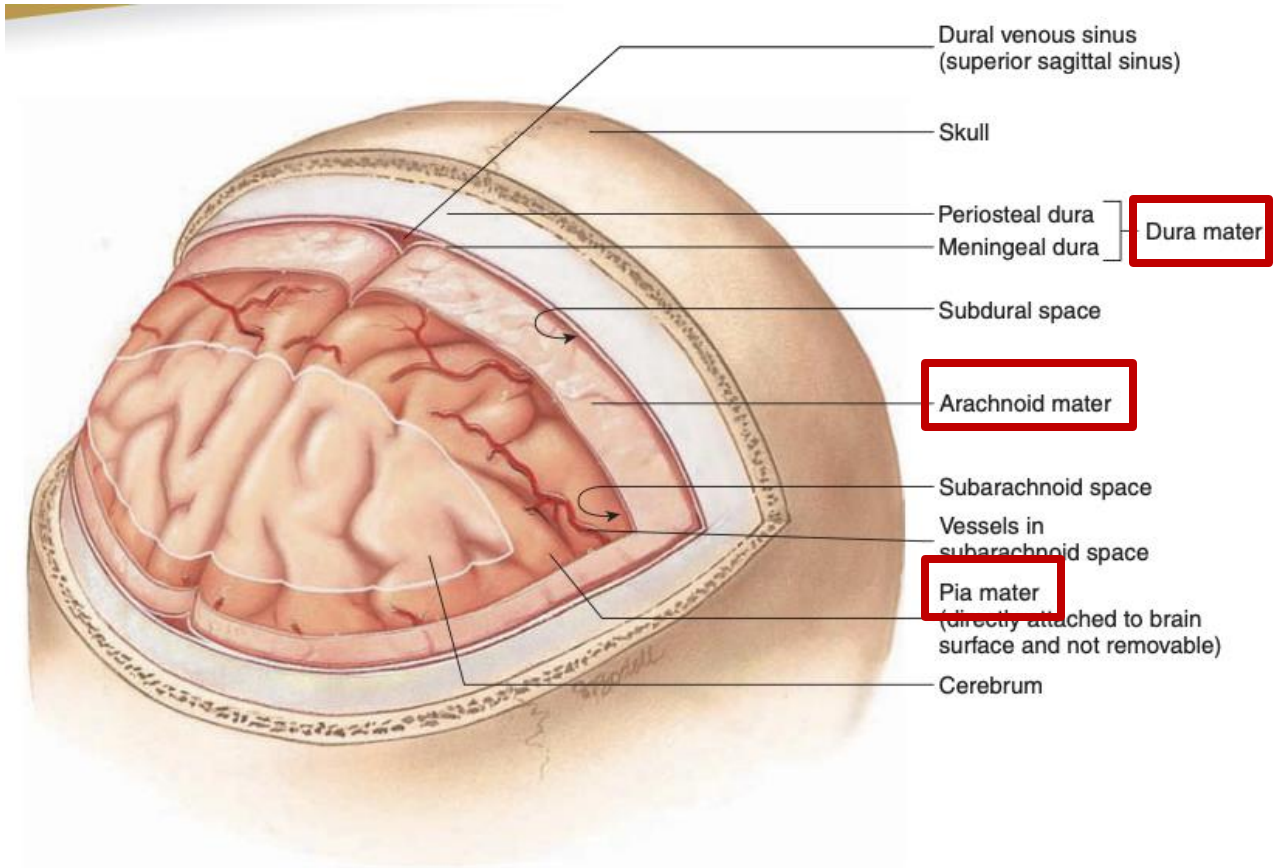
The CNS is protected by multiple structural and physiological mechanisms that ensure its stability and safety:

1. **Meninges** – Three protective connective tissue layers (dura mater, arachnoid mater, and pia mater) that encase the brain and spinal cord.
2. **Ventricles** – Interconnected cavities within the brain that produce and circulate cerebrospinal fluid (CSF).
3. **Cerebrospinal Fluid (CSF)** – A clear, colorless fluid that cushions, nourishes, and removes waste from CNS tissues.
4. **Blood-Brain Barrier (BBB)** – A selective permeability barrier that regulates the passage of substances from the bloodstream into the CNS, maintaining a stable internal environment.

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PROTECTION OF THE CENTRAL NERVOUS SYSTEM

- Meninges
- Ventricles
- Cerebrospinal fluid (CSF)
- Blood-brain barrier

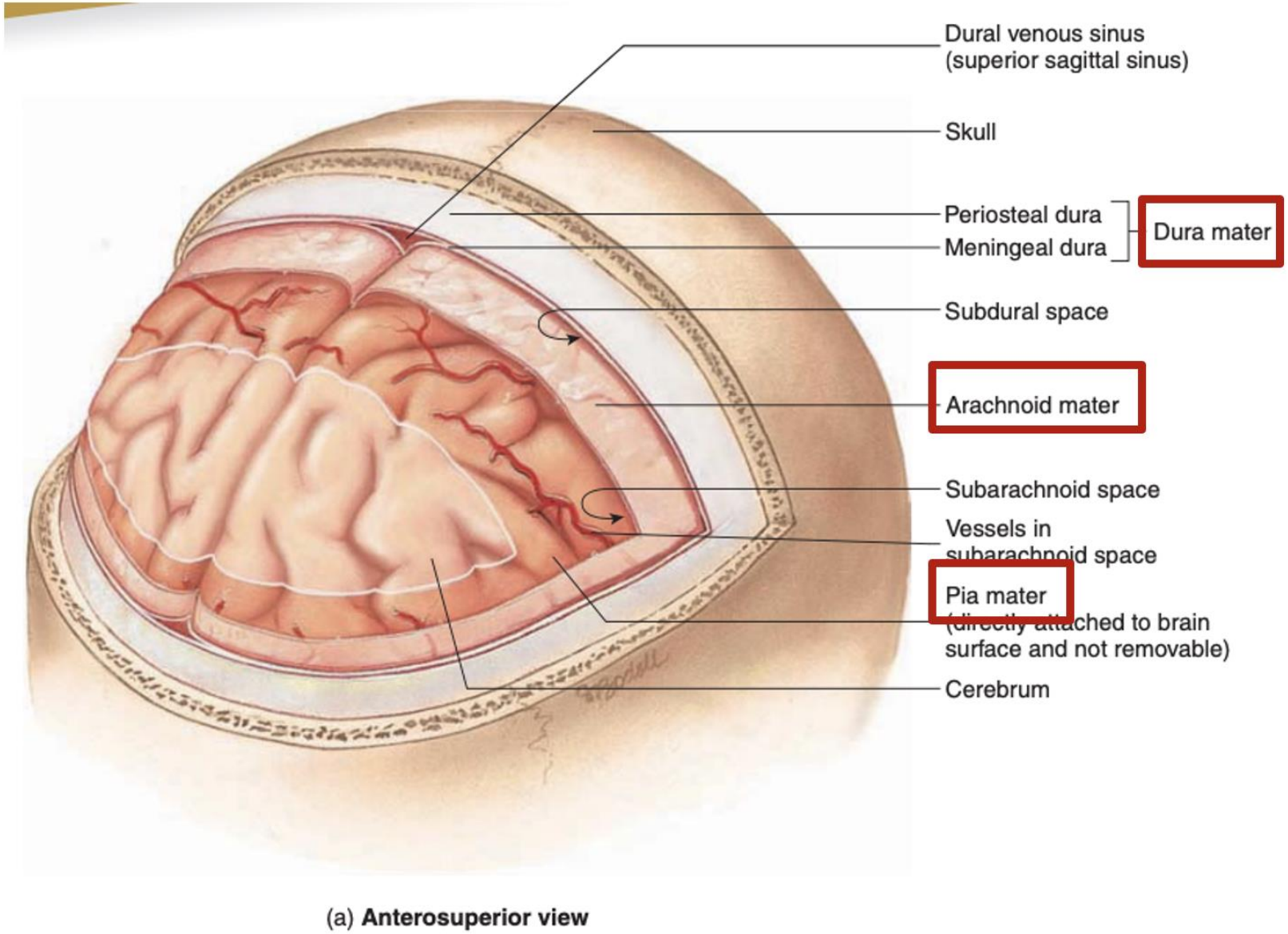


(a) **Anterosuperior view**

MENINGES

The **meninges** are three connective tissue membranes that protect and support the brain and spinal cord:

1. **Dura Mater**
2. **Arachnoid Mater**
3. **Pia Mater**



(a) **Anterosuperior view**

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MENINGES

1. Dura Mater "Tough Mother"

- Outermost and strongest meningeal layer made of **dense irregular connective tissue.**
- **Cranial dura mater** has two layers:
  1. **Periosteal dura** – adheres to the skull's inner surface.
  2. **Meningeal dura** – continuous with the spinal dura mater.
- Forms **dural folds** (e.g., *falx cerebri*, *tentorium cerebelli*, *falx cerebelli*) that stabilize the brain within the skull.
- **Dural venous sinuses** (e.g., *superior sagittal sinus*) act as venous drainage channels for brain blood and cerebrospinal fluid (CSF).

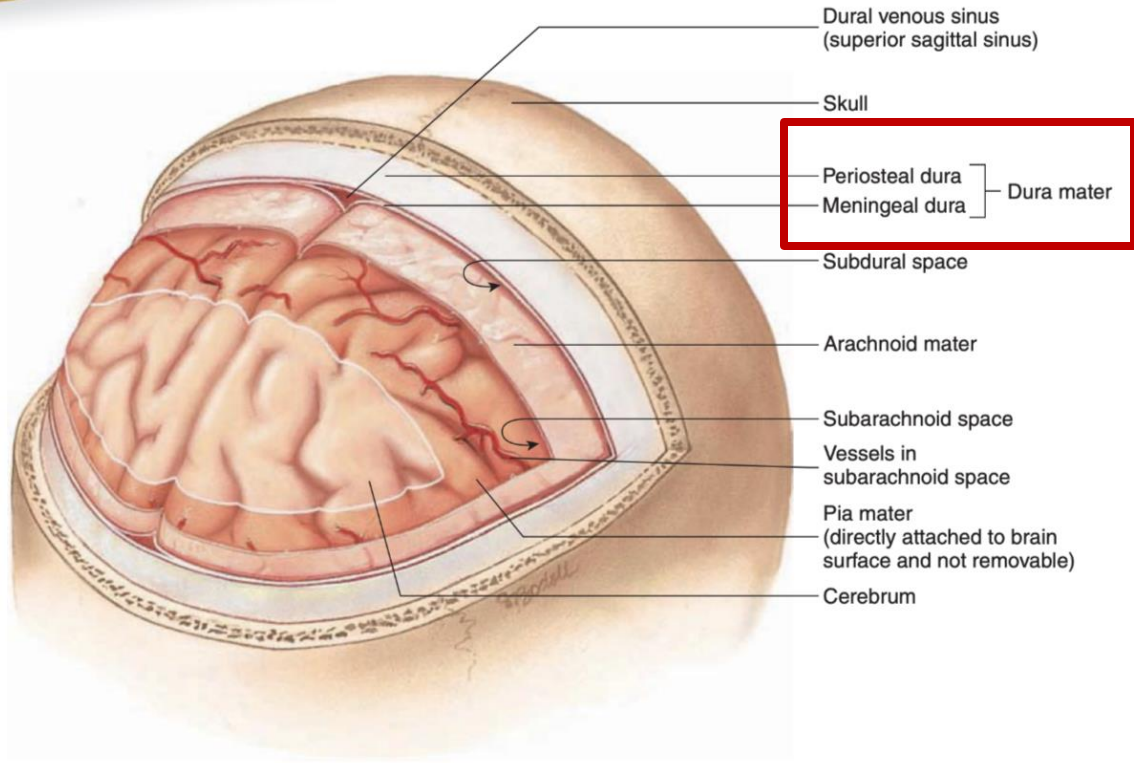


FIGURE 13.12 Meninges
(*a*) Meningeal membranes surrounding the brain. (*b*) Frontal section of the head to show the meninges.

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MENINGES

2. Arachnoid Mater "Spider-like"

- Middle, delicate, web-like layer.
- The **subdural space** (between dura and arachnoid) contains a thin film of serous fluid.

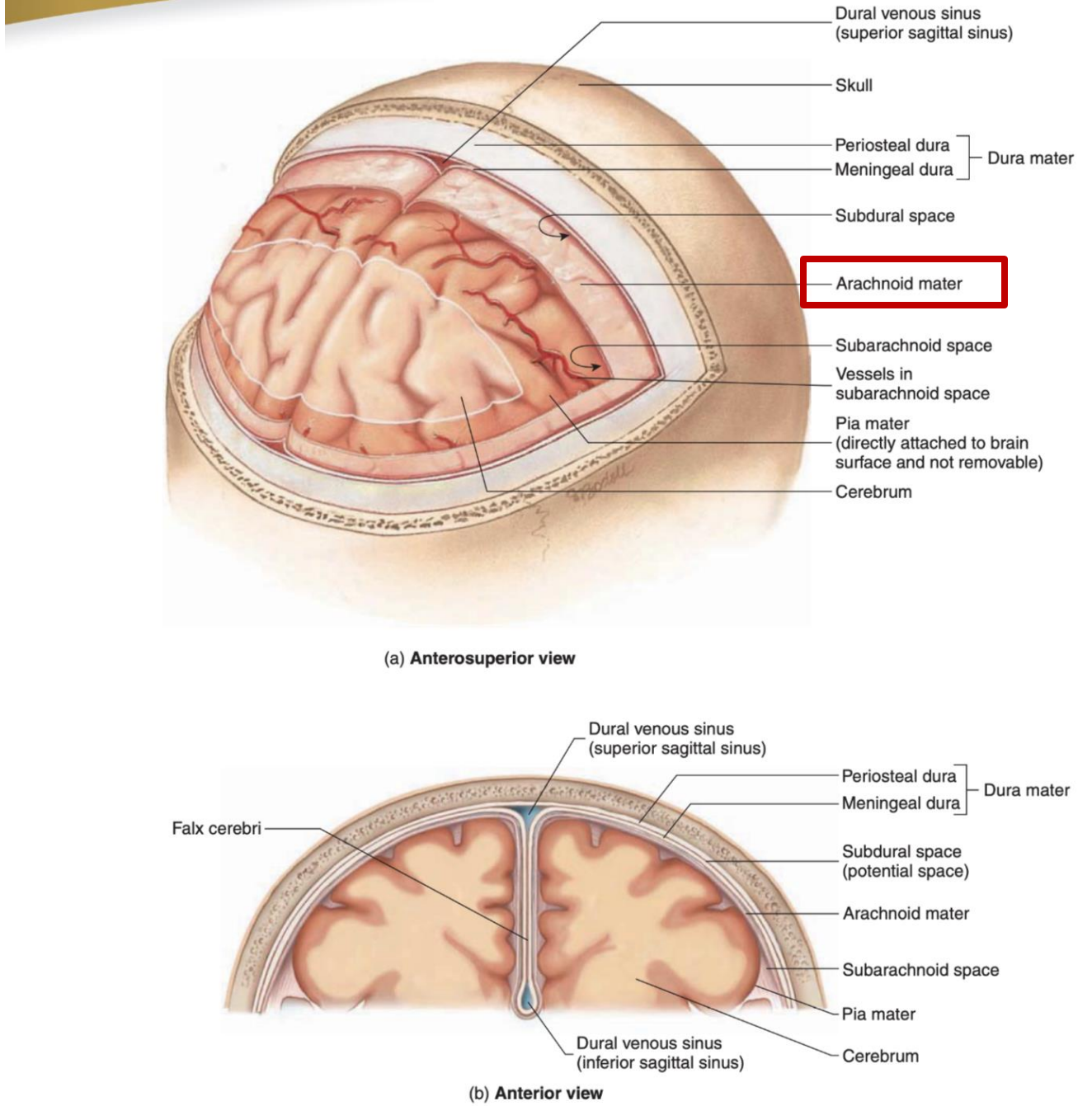


FIGURE 13.12 Meninges
(*a*) Meningeal membranes surrounding the brain. (*b*) Frontal section of the head to show the meninges. APR

MENINGES

3. Pia Mater "Tender Mother"

- Innermost, thin, highly vascular layer closely adhering to the brain's contours.
- The **subarachnoid space**, between the arachnoid and pia mater, contains **CSF** and major **cerebral blood vessels.**

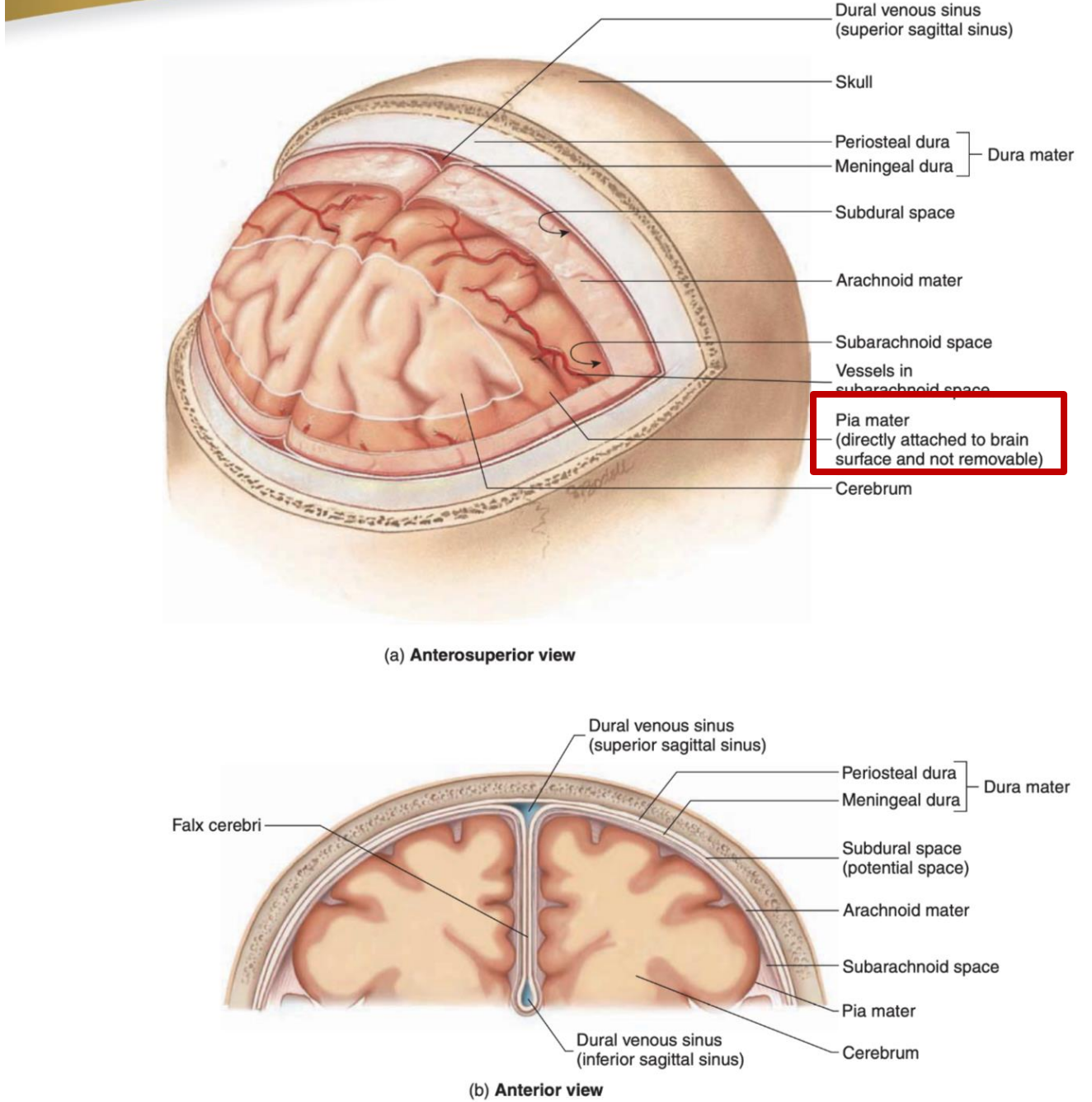


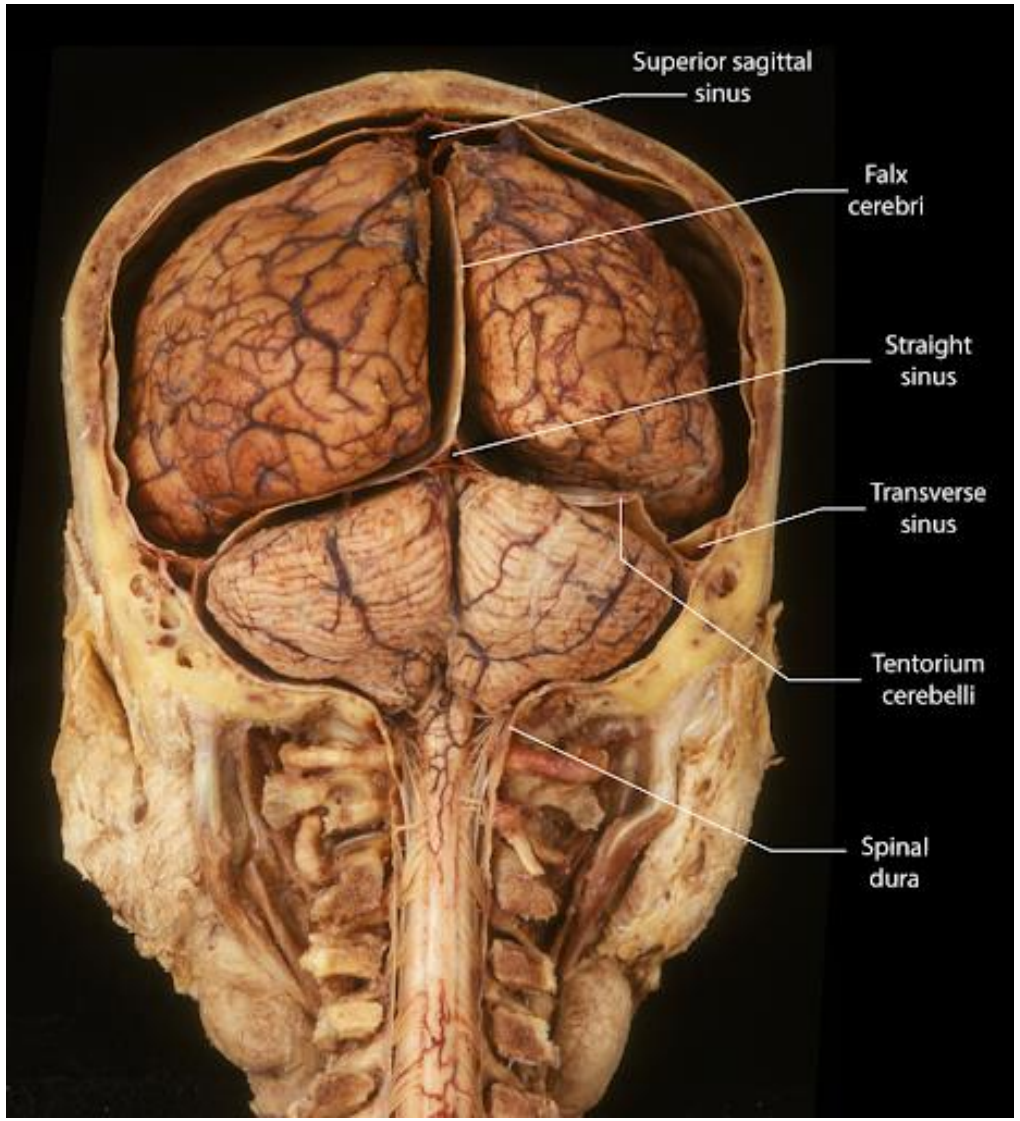
FIGURE 13.12 Meninges
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PROTECTION OF THE CENTRAL NERVOUS SYSTEM

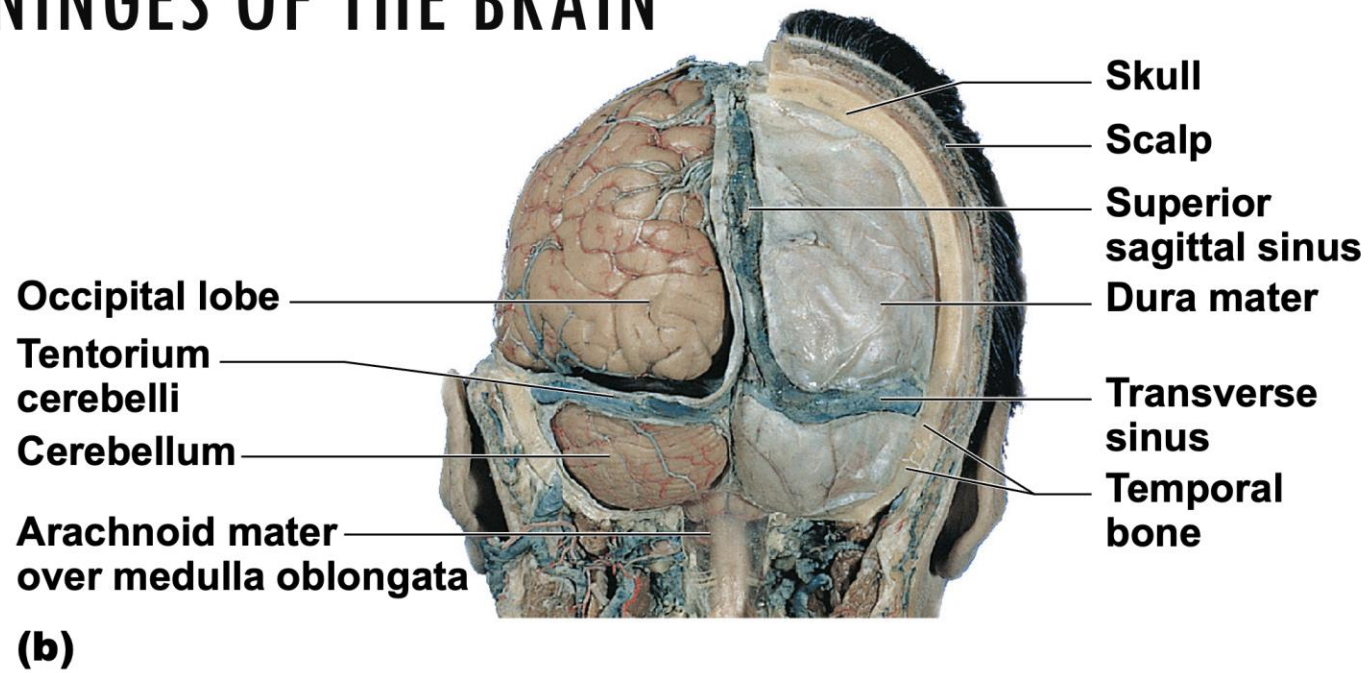
MENINGES

This cadaveric image demonstrates key meningeal and dural structures that protect and support the brain and spinal cord:

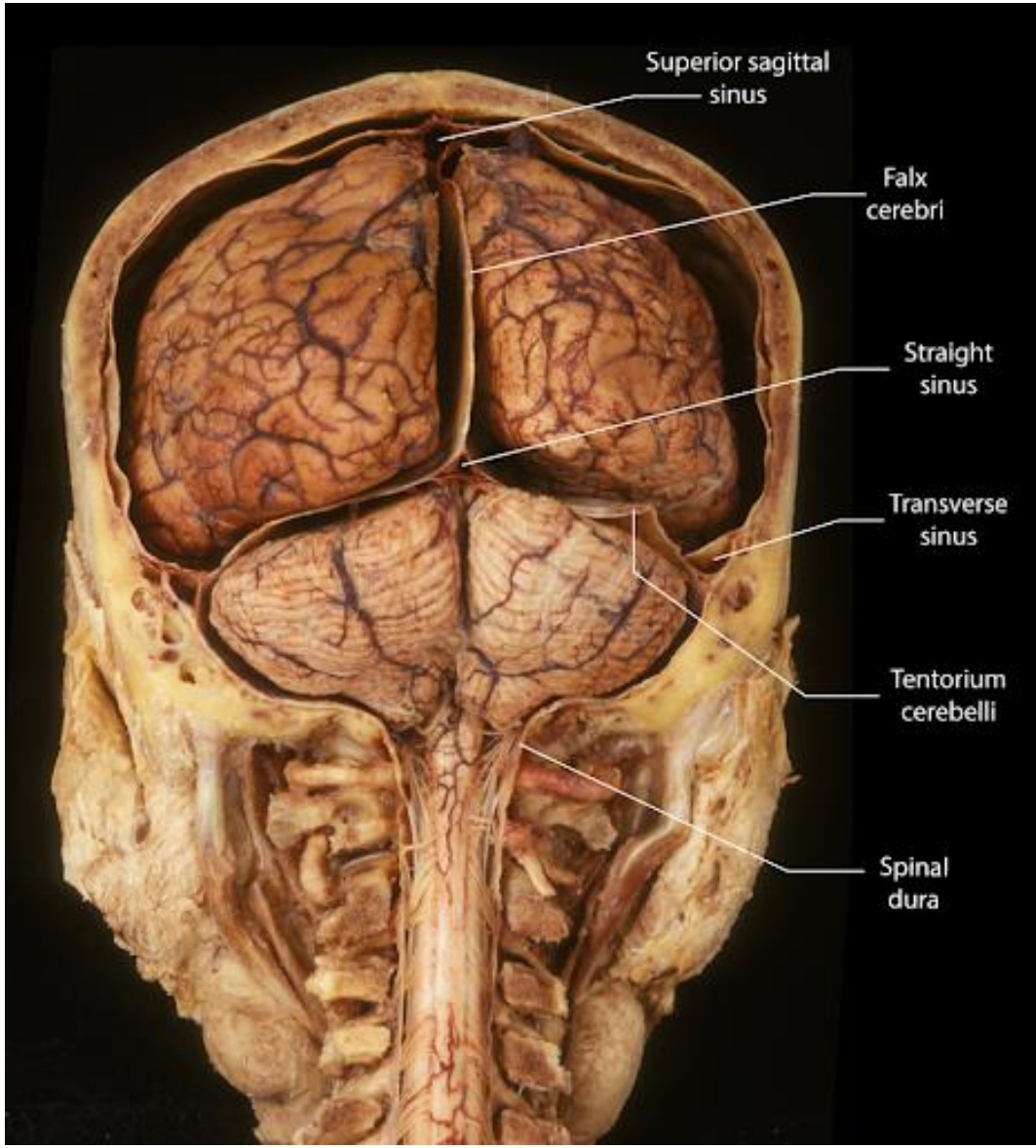
- **Superior Sagittal Sinus** – Major venous channel running along the midline between the cerebral hemispheres.
- **Falx Cerebri** – A vertical dural fold that separates the right and left cerebral hemispheres.
- **Straight Sinus** – Venous sinus located at the junction of the falx cerebri and tentorium cerebelli.
- **Transverse Sinus** – Drains blood from the straight and superior sagittal sinuses toward the internal jugular vein.
- **Tentorium Cerebelli** – A horizontal dural fold separating the cerebrum from the cerebellum.
- **Spinal Dura** – Extension of the meningeal dura mater surrounding the spinal cord.



MENINGES OF THE BRAIN



(b)



PROTECTION OF THE CENTRAL NERVOUS SYSTEM

- Meninges
- Ventricles
- Cerebrospinal fluid (CSF)
- Blood-brain barrier

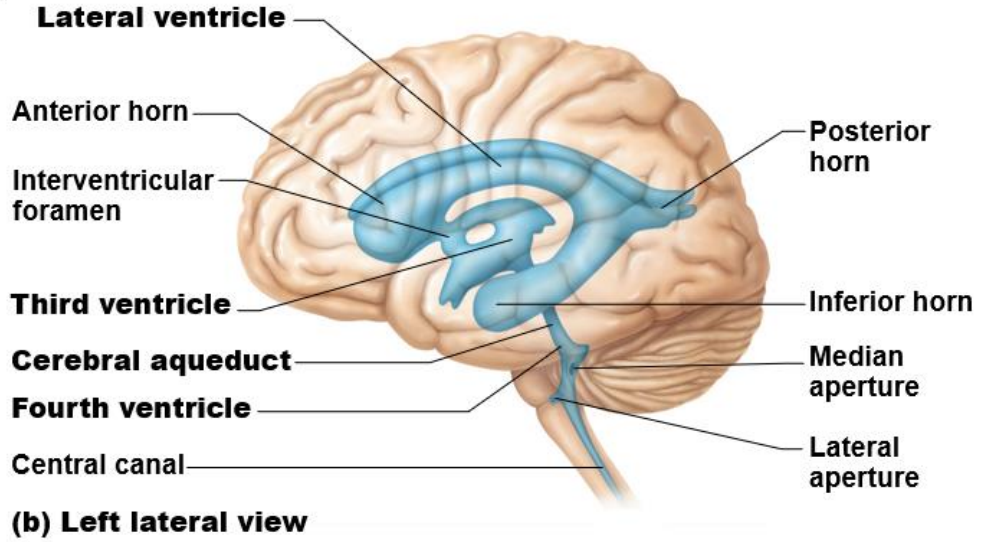
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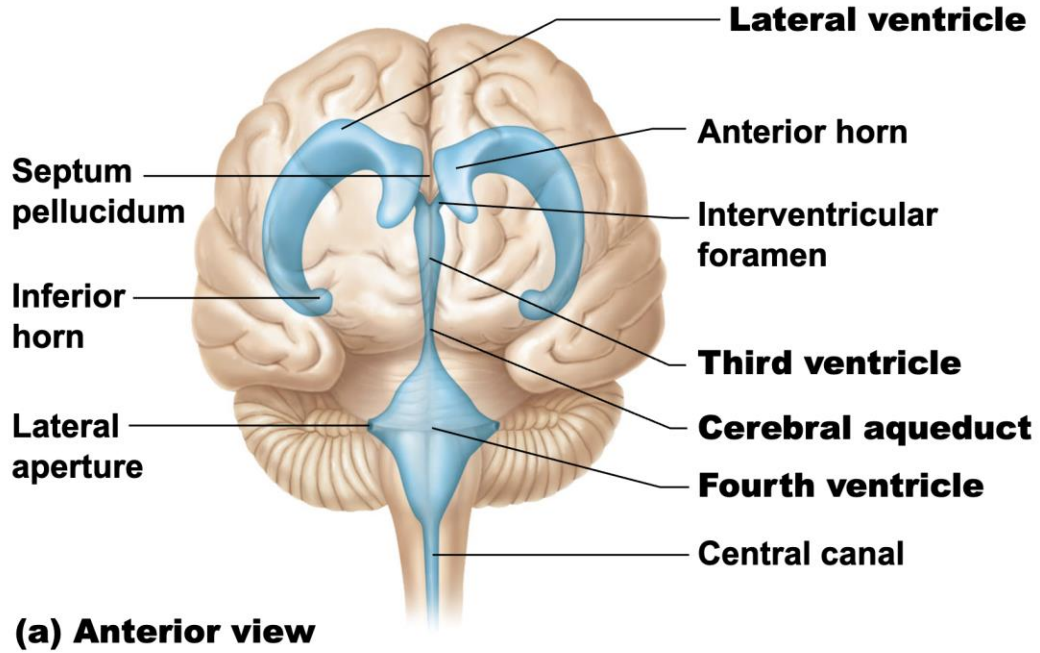
VENTRICLES

- The **ventricles** are interconnected cavities within the brain derived from the hollow neural tube.
- Lined with **ependymal cells,** which form part of the **choroid plexus—**the site of **cerebrospinal fluid (CSF)** production.

- **Lateral Ventricles (1st & 2nd):**
- **Third Ventricle**
- **Cerebral Aqueduct (Aqueduct of Sylvius)**
- **Fourth Ventricle**

VENTRICLES AND LOCATION OF THE CEREBROSPINAL FLUID



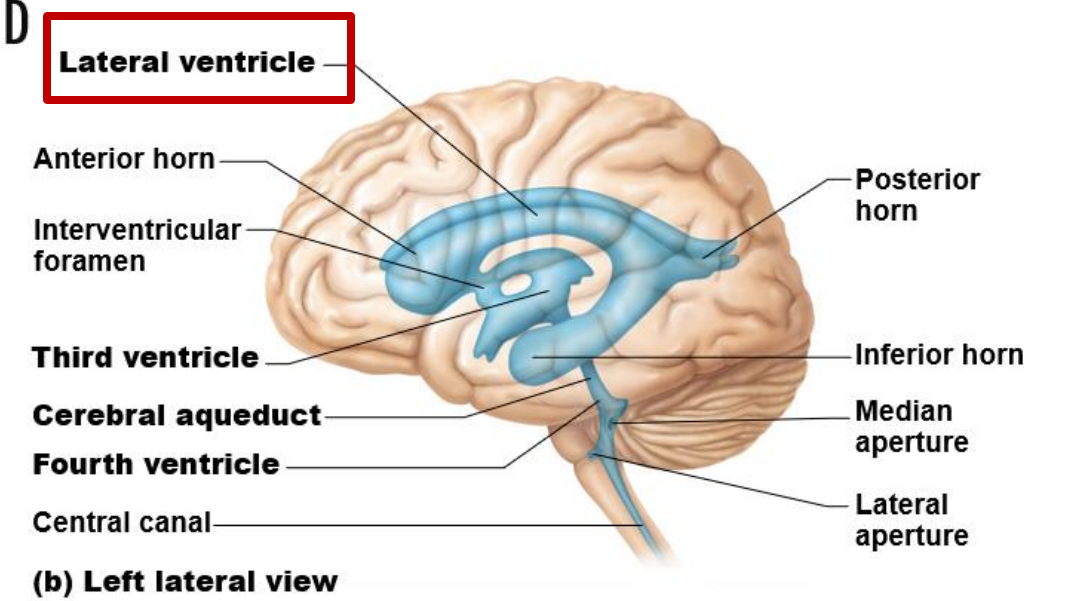


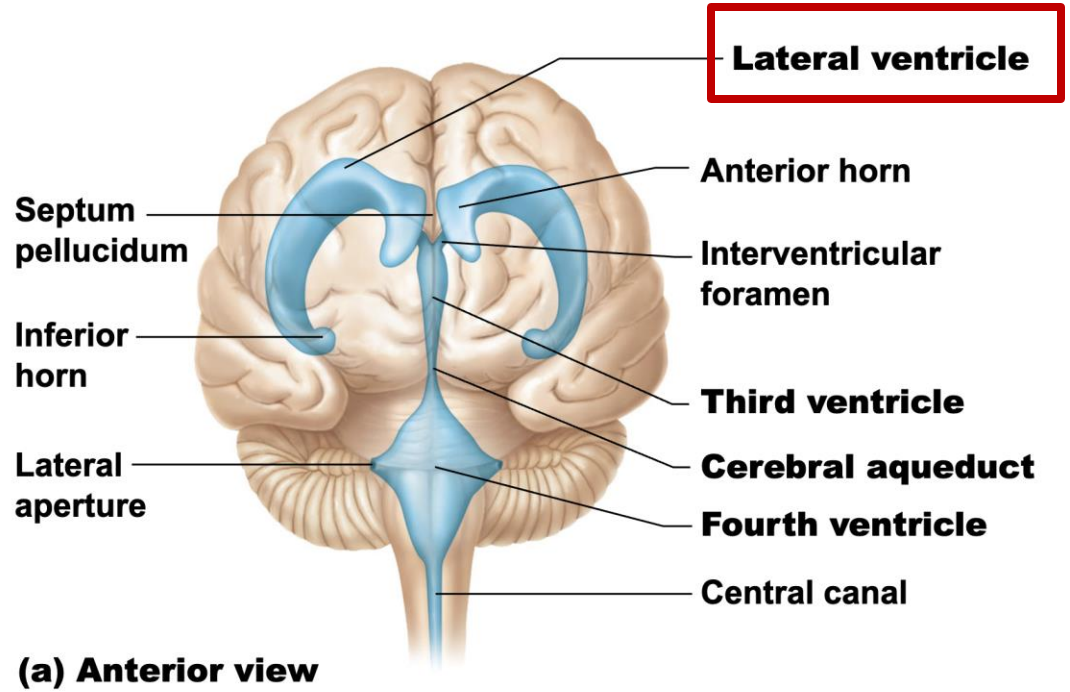
VENTRICLES

Lateral Ventricles (1st & 2nd):

- One in each cerebral hemisphere.
- Separated by the **septum pellucidum**, located beneath the **corpus callosum**.

VENTRICLES AND LOCATION OF THE CEREBROSPINAL FLUID



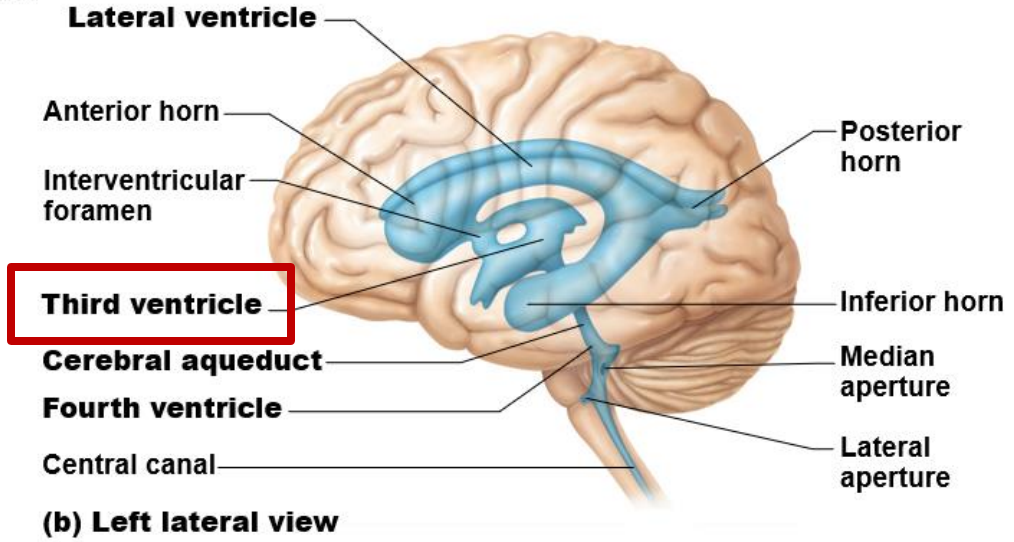


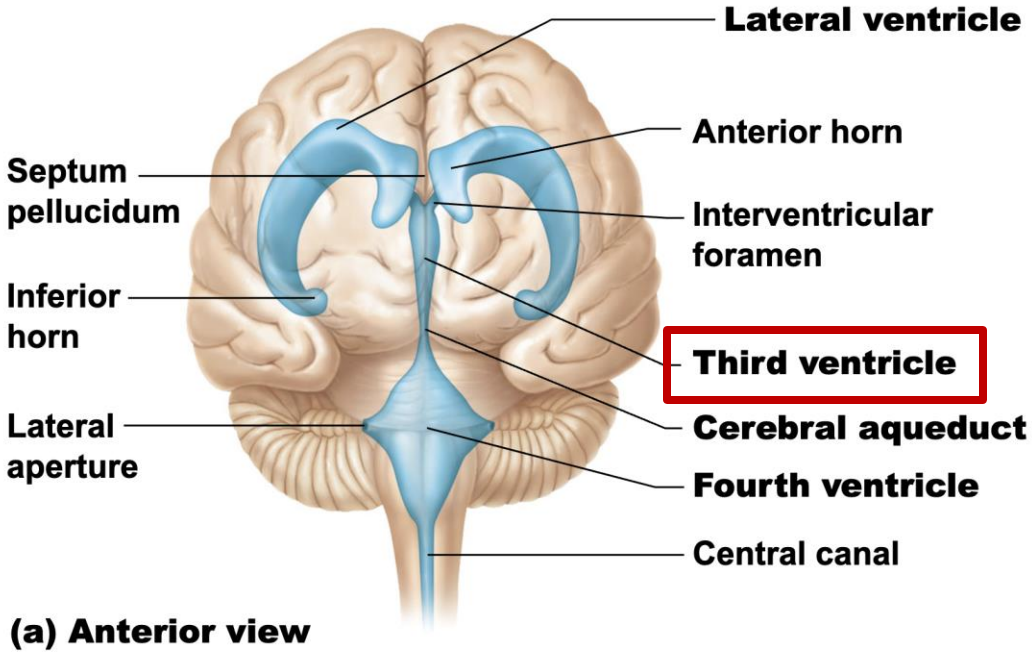
VENTRICLES

Third Ventricle:

- Narrow, midline cavity between the two halves of the **thalamus.**
- Connected to lateral ventricles via **interventricular foramina (foramina of Monro).**

VENTRICLES AND LOCATION OF THE CEREBROSPINAL FLUID

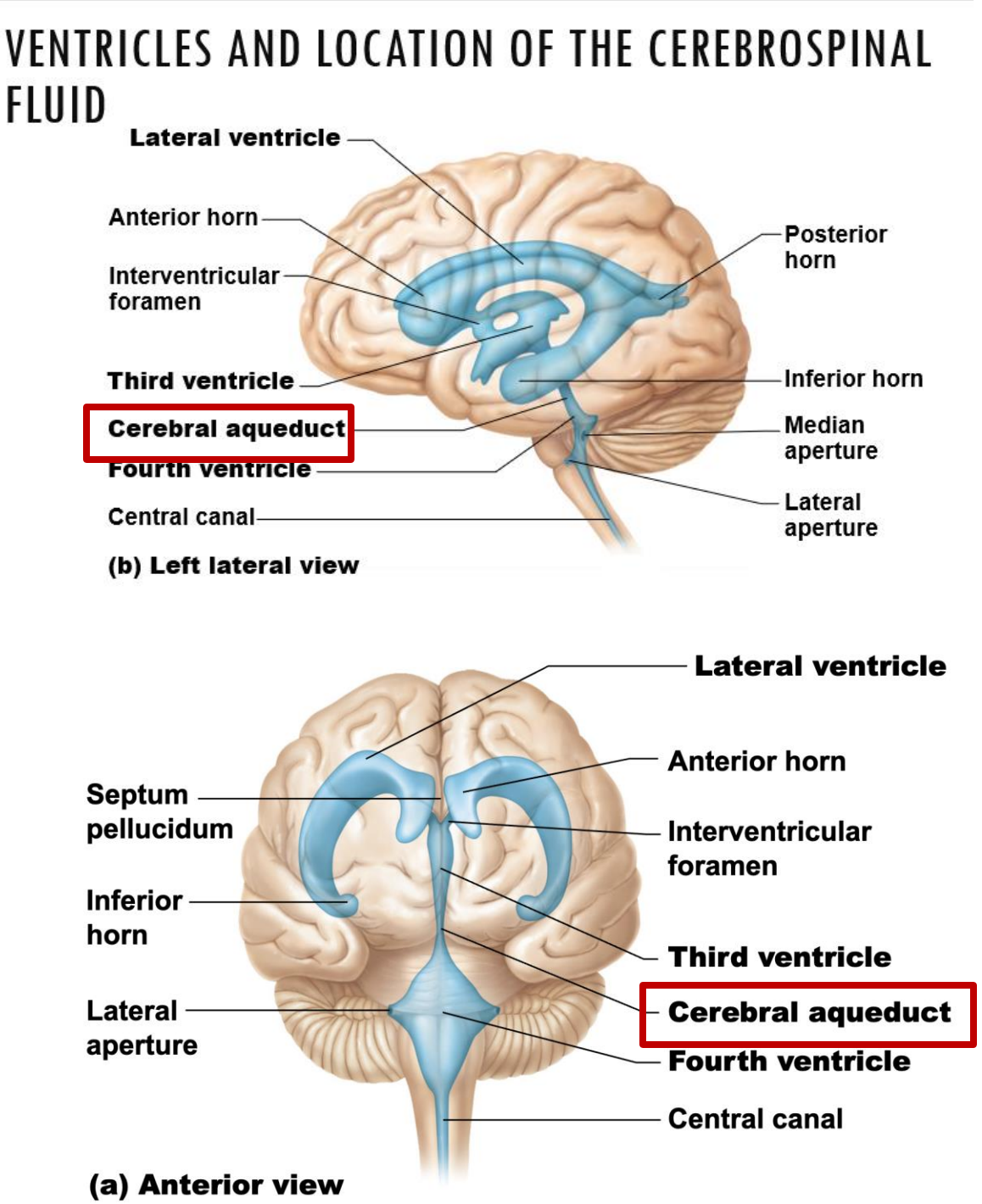




VENTRICLES

Cerebral Aqueduct (Aqueduct of Sylvius):

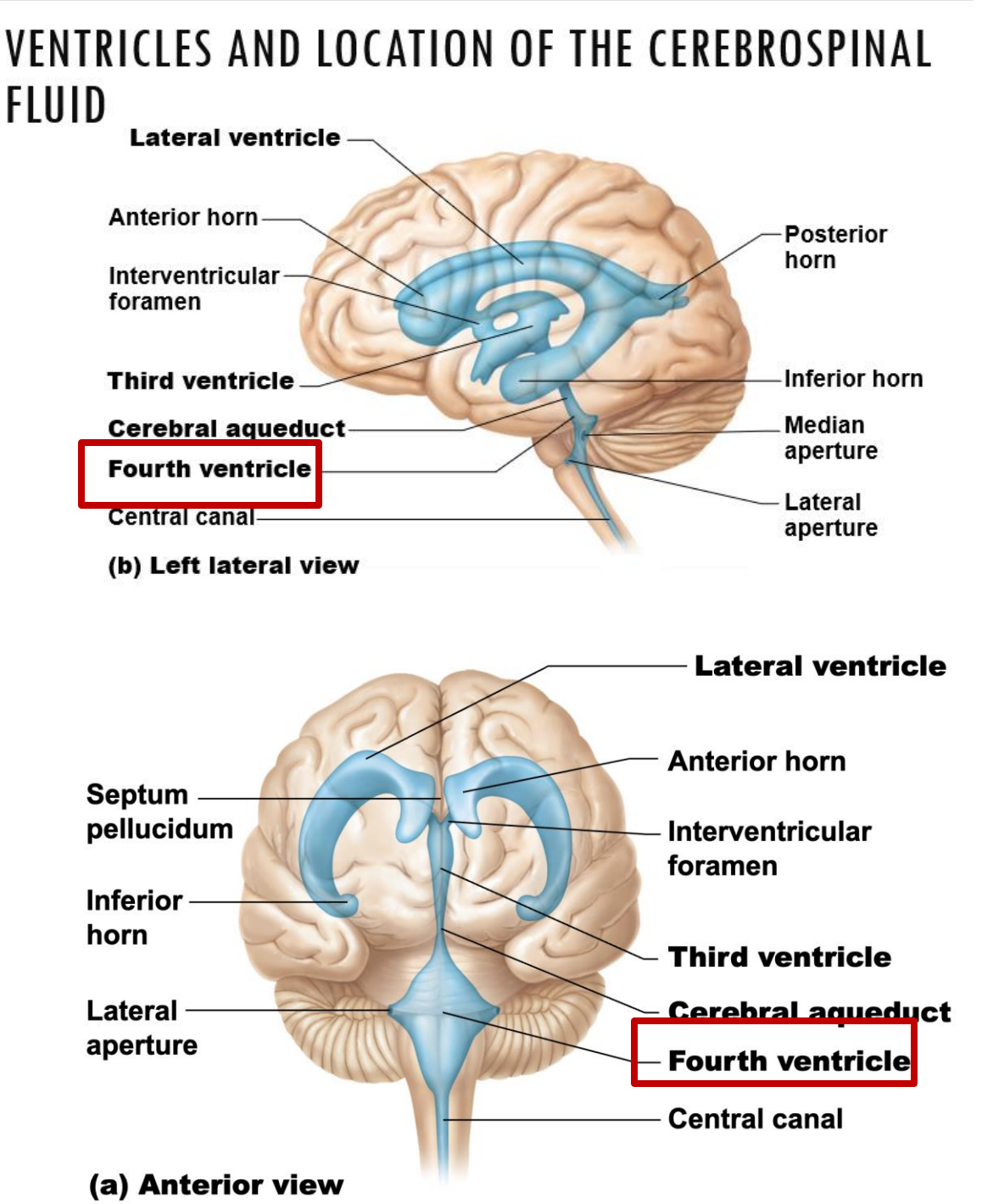
- Slender canal through the **midbrain** connecting the 3rd and 4th ventricles.



VENTRICLES

Fourth Ventricle:

- Located between the **pons and medulla oblongata (anteriorly)** and the **cerebellum (posteriorly).**
- Continuous with both the **central canal** of the spinal cord and the **subarachnoid space,** allowing CSF circulation.



PROTECTION OF THE CENTRAL NERVOUS SYSTEM

- Meninges
- Ventricles
- Cerebrospinal fluid (CSF)
- Blood-brain barrier

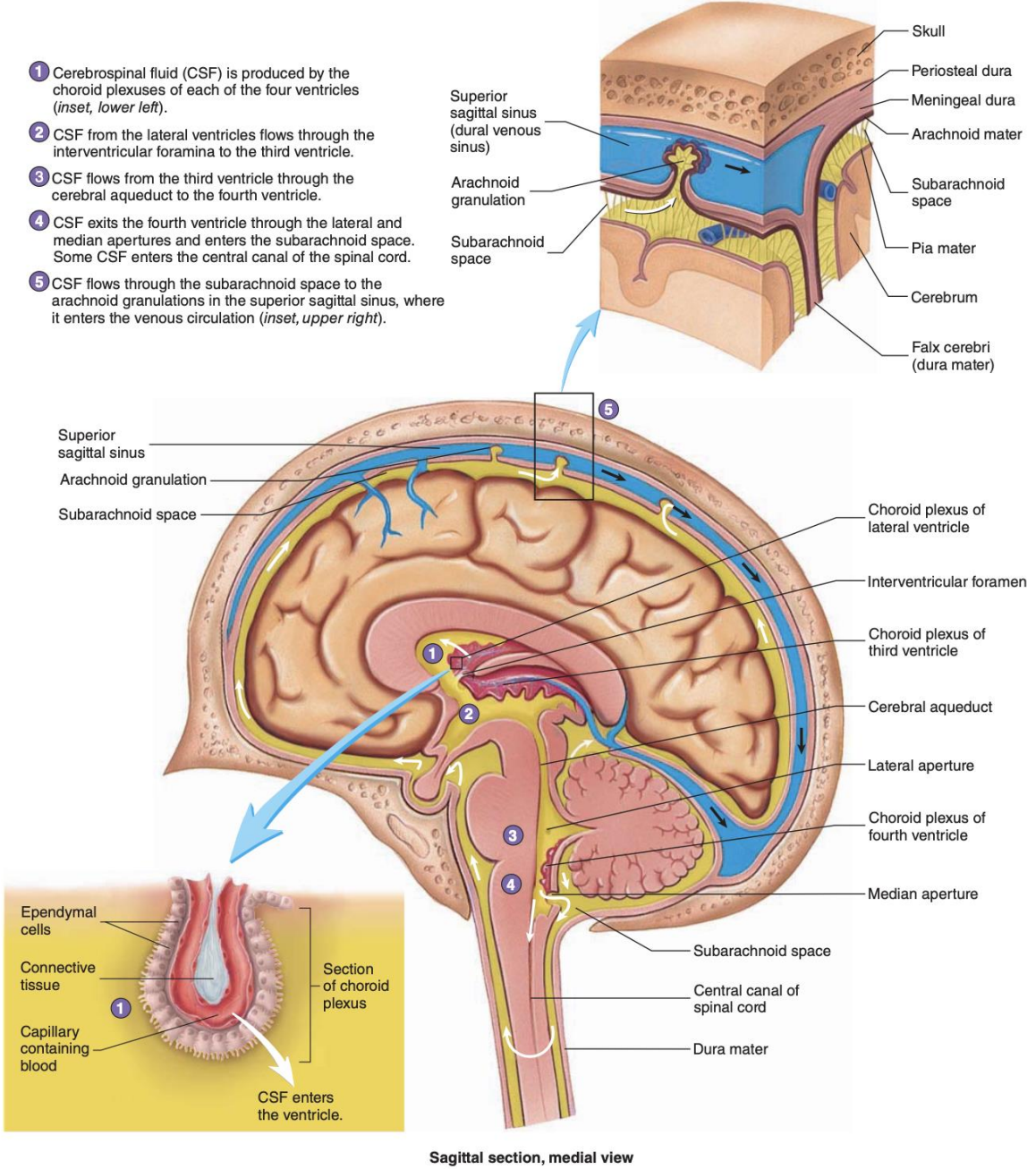
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CEREBROSPINAL FLUID

Cerebrospinal fluid (CSF) is a **clear, colorless fluid** similar to blood plasma but with **very low protein content.** It surrounds the **brain and spinal cord,** providing protection, buoyancy, and nutrient exchange.

Functions of CSF:

1. **Mechanical protection:** Cushions the CNS from trauma and allows the brain to "float," reducing pressure on neural tissues.
2. **Chemical stability:** Maintains optimal ionic composition and removes metabolic waste.
3. **Circulatory function:** Distributes nutrients (especially glucose) and hormones throughout the CNS.



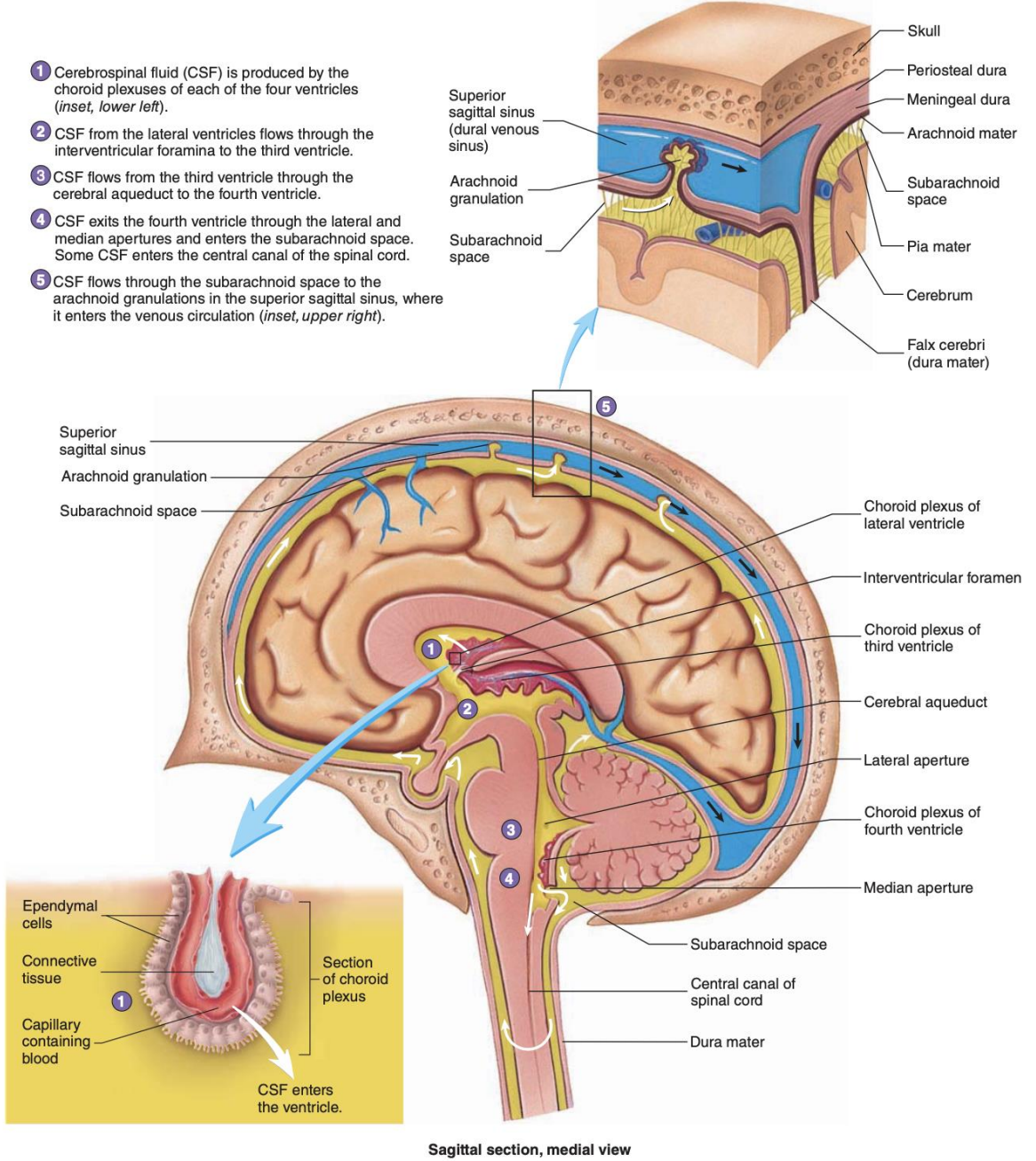
PROCESS FIGURE 13.14 Choroid Plexuses and the Flow of CSF

CSF flows through the ventricles and subarachnoid space, as shown by the *white arrows*. Arrows going through the foramina in the wall and roof of the fourth ventricle represent the CSF entering the subarachnoid space. CSF passes back into the blood through the arachnoid granulations (*white and black arrows*), which penetrate the dural sinus. The *black arrows* show the direction of blood flow in the sinuses. *Yellow* indicates CSF; *blue* indicates venous blood.

CEREBROSPINAL FLUID

Production (by the Choroid Plexus):

- Found in all ventricles—most abundant in the **lateral ventricles (80–90%).**
- Composed of **specialized ependymal cells** and **vascular pia mater.**
- CSF formation involves **active Na^+ transport → passive water movement,** and **glucose transport via GLUT1.**
- Tight junctions between endothelial cells form the **blood-CSF barrier.**



PROCESS FIGURE 13.14 Choroid Plexuses and the Flow of CSF

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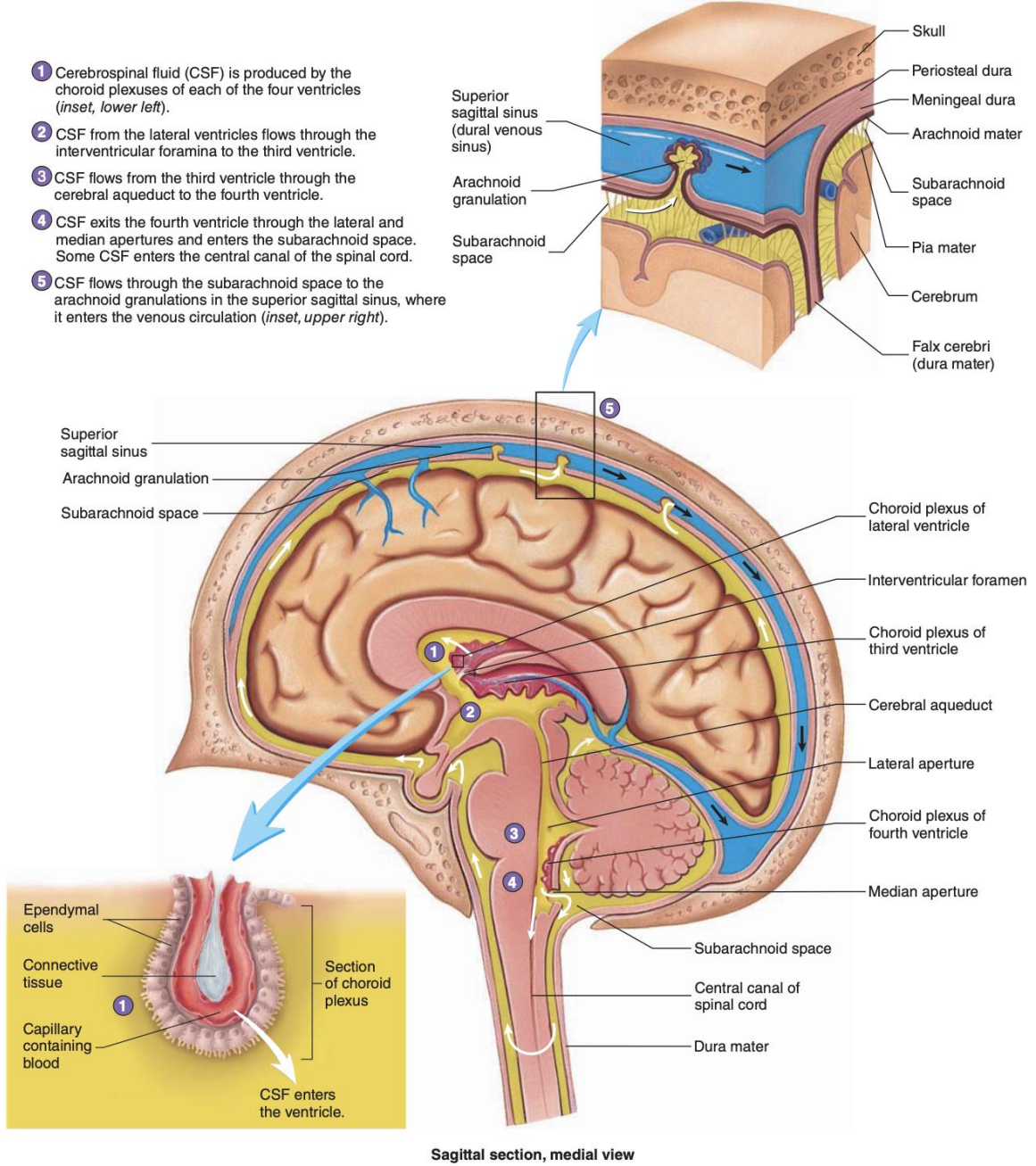
CEREBROSPINAL FLUID

Circulation Pathway:

Lateral Ventricles → Interventricular Foramina → Third Ventricle → Cerebral Aqueduct → Fourth Ventricle → Subarachnoid Space (via Median and Lateral Apertures) → Arachnoid Granulations → Dural Venous Sinuses → Internal Jugular Veins → General Circulation.

Volume and Turnover:

- Total CSF volume: ~140 mL (117 mL in subarachnoid space, 23 mL in ventricles).
- Replenished **4–5 times daily** at a rate of **~0.4 mL/min.**



PROCESS FIGURE 13.14 Choroid Plexuses and the Flow of CSF

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PROTECTION OF THE CENTRAL NERVOUS SYSTEM

- Meninges
- Ventricles
- Cerebrospinal fluid (CSF)
- **Blood-brain barrier**

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BLOOD-BRAIN BARRIER

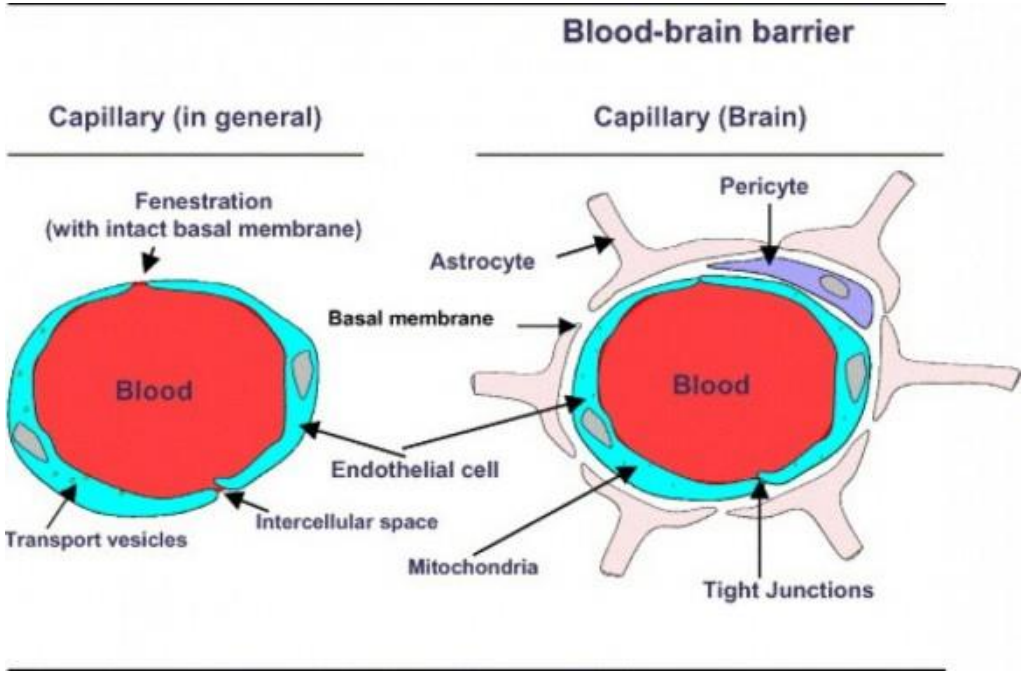
A highly selective permeability barrier formed by tight junctions between capillary endothelial cells, reinforced by astrocyte foot processes and a basement membrane.

Functions:

- Maintains CNS homeostasis by regulating movement of substances from blood to brain tissue.
- Protects the brain from toxins and pathogens.
- Allows passage of essential nutrients and gases.

Composition:

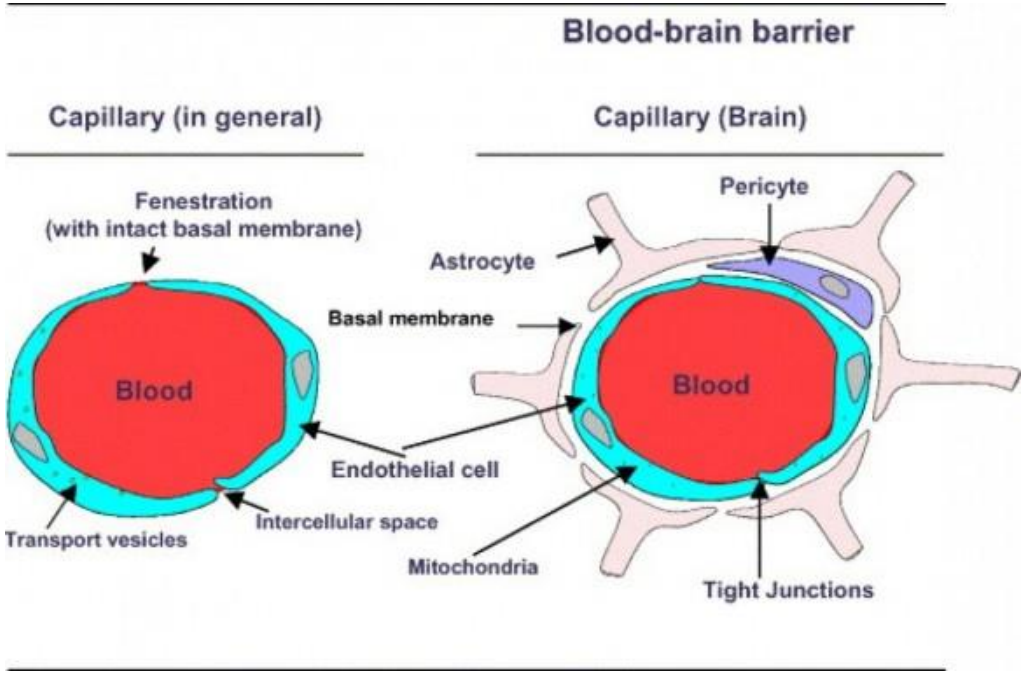
- Endothelial cells with tight junctions
- Basement membrane
- Astrocyte foot processes



BLOOD-BRAIN BARRIER

Selective Permeability:

- **Freely Pass:** O_2, CO_2, alcohol, nicotine, anesthetics
- **Transported:** Glucose, amino acids via specific transporters (e.g., GLUT1)
- **Blocked:** Proteins, toxins, most drugs



BRAIN TEASER



Can alcohol pass the blood brain barrier?

WERNICKE'S ENCEPHALOPATHY

PATHOLOGY

A serious neurological disorder caused by **thiamine (vitamin B_1) deficiency** affecting the **CNS and PNS**.

Etiology: Most commonly associated with **chronic alcohol use disorder**, malnutrition, or impaired thiamine absorption.

Pathophysiology:

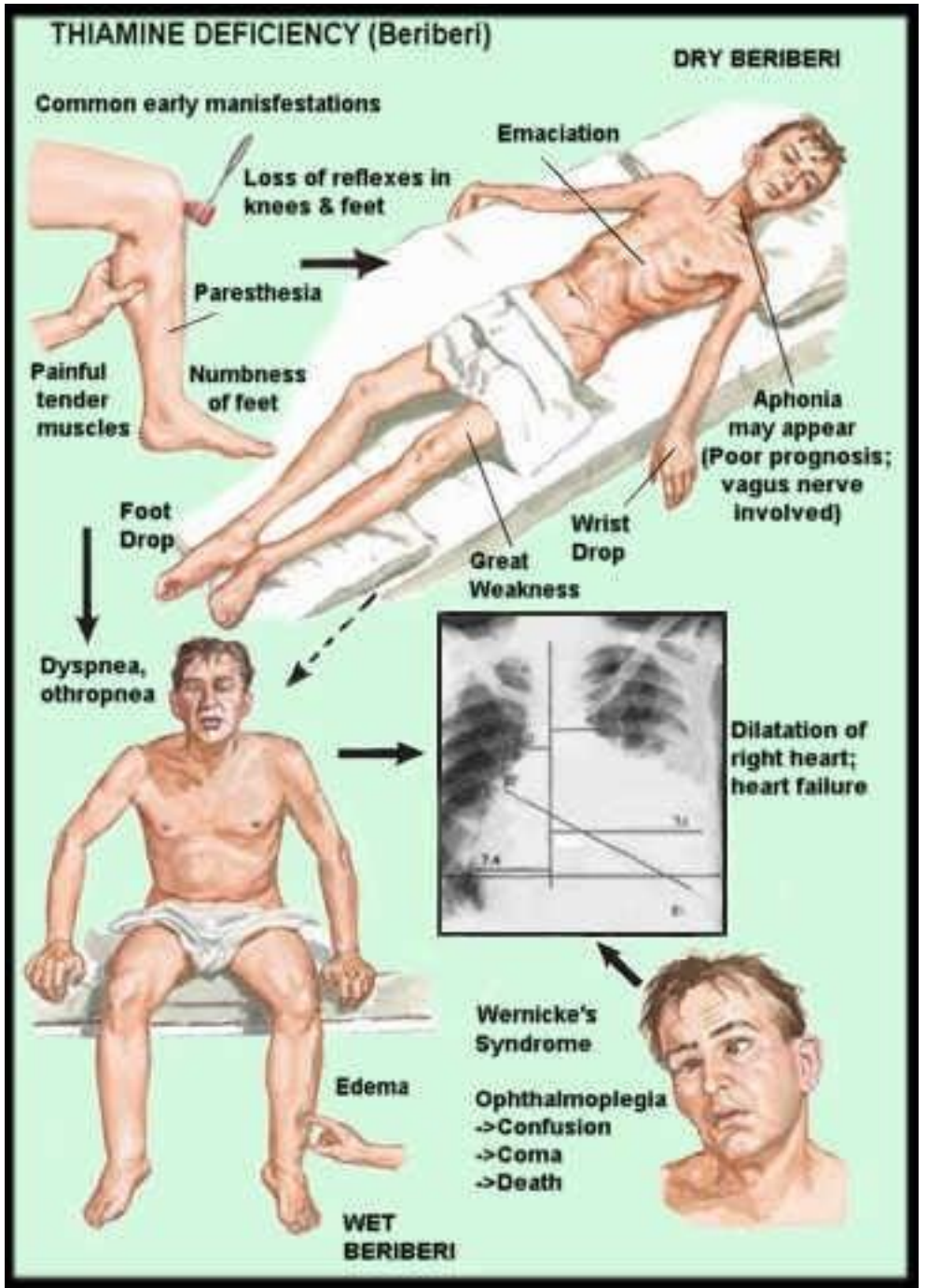
- Thiamine is a coenzyme in glucose metabolism; its deficiency impairs neuronal energy production.
- Affects regions with high metabolic demand: **mammillary bodies, thalamus, hypothalamus, midbrain, and cerebellum**.

Clinical Triad:

1. **Ocular abnormalities** (nystagmus, ophthalmoplegia)
2. **Ataxia** (poor coordination, gait imbalance)
3. **Confusion** or mental status changes

Complication: If untreated, it can progress to **Korsakoff syndrome** (chronic memory disorder).

Treatment: Prompt **IV thiamine administration** before glucose to prevent worsening neurologic damage.



BRAIN TEASER



Can caffeinated drinks, such as coffee pass the blood brain barrier?

BRAIN TEASER



Caffeine and the BBB:

- Caffeine is a **lipid-soluble molecule**, allowing it to **easily cross the blood-brain barrier (BBB)**.

Physiologic Effects:

- Stimulates the **central nervous system (CNS)**
- Increases **dopamine** and **norepinephrine** activity
- Improves **focus, reaction time,** and **mood** in moderate amounts

Safe Limit:

- **Maximum recommended intake: ≤ 400 mg/day** (≈ 4–5 cups of brewed coffee) (**Johns Hopkins Medicine, 2024**)

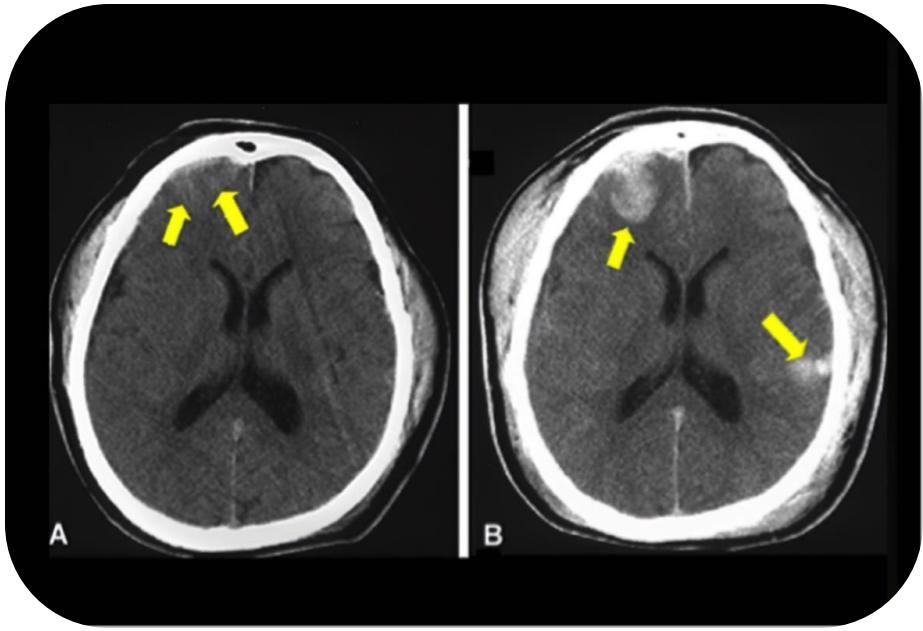
Excess Intake:

- May cause **anxiety, palpitations, tremors, insomnia,** and **elevated blood pressure**
- Chronic overuse can lead to **dependence** and **withdrawal symptoms**

BRAIN DYSFUNCTIONS

TRAUMATIC BRAIN INJURIES





Concussion

A **concussion** is a **mild traumatic brain injury** caused by a **blow or jolt to the head** that temporarily disrupts **normal brain function.**

Pathophysiology:

- Results from **sudden acceleration–deceleration** or **rotational forces.**
- Leads to **microscopic neuronal damage** and **transient chemical changes** (e.g., altered neurotransmitter release).
- Usually **no structural brain lesion** visible on imaging.

Clinical Features:

- Brief **loss of consciousness** (sometimes absent)
- **Headache, dizziness, confusion, nausea**
- **Temporary memory loss (amnesia)** or concentration difficulty

Prognosis:

- **Full recovery** is expected in most cases with **proper rest and management.**
- **Repeated concussions** can lead to **chronic traumatic encephalopathy (CTE)** or **post-concussion syndrome.**

BRAIN DYSFUNCTIONS

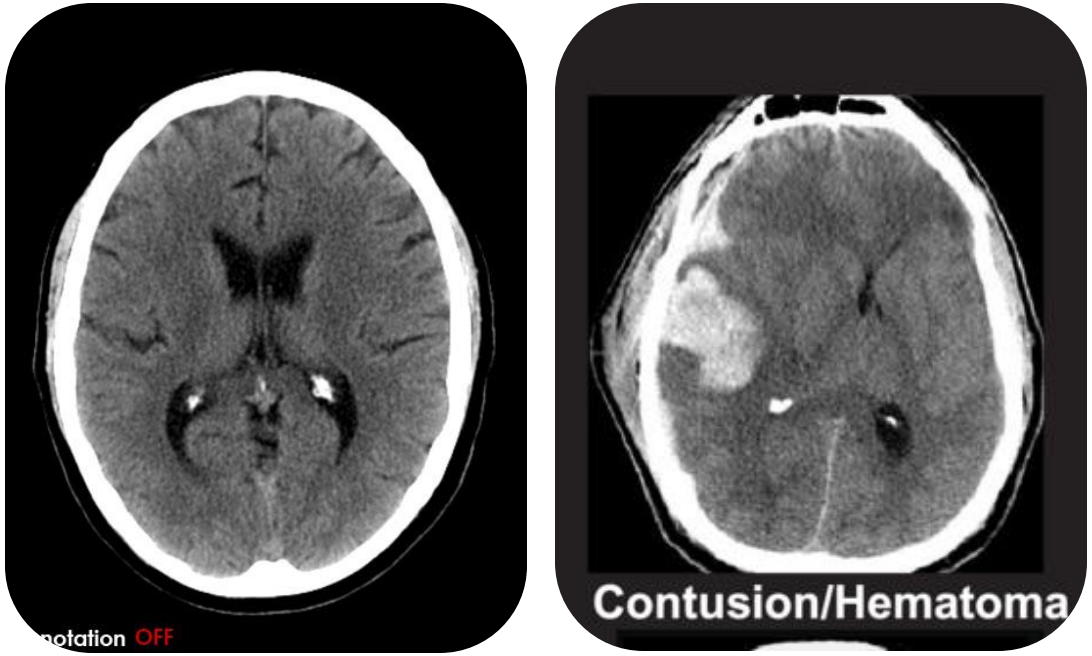
TRAUMATIC BRAIN INJURIES

Contusion

Slide Content:

- **Definition:**
  A **contusion** is a **bruising of brain tissue** caused by **direct trauma** leading to **localized bleeding** and **destruction of nervous tissue.**

- **Pathophysiology:**
  - Results from **coup–contrecoup injury** (brain strikes the skull at the point of impact and/or opposite side).
  - Causes **capillary rupture, edema,** and **neuronal necrosis.**
  - May be accompanied by **subdural or subarachnoid hemorrhage.**



- **Clinical Features:**
  - **Loss of consciousness** (can range from minutes to days).
  - **Neurological deficits** (e.g., weakness, speech impairment, confusion).
  - **Possible coma** in severe cases.

- **Prognosis:**
  - **Variable** depending on severity and location (frontal, temporal lobes often affected).
  - May result in **permanent brain damage** or **post-traumatic epilepsy.**

BRAIN DYSFUNCTIONS

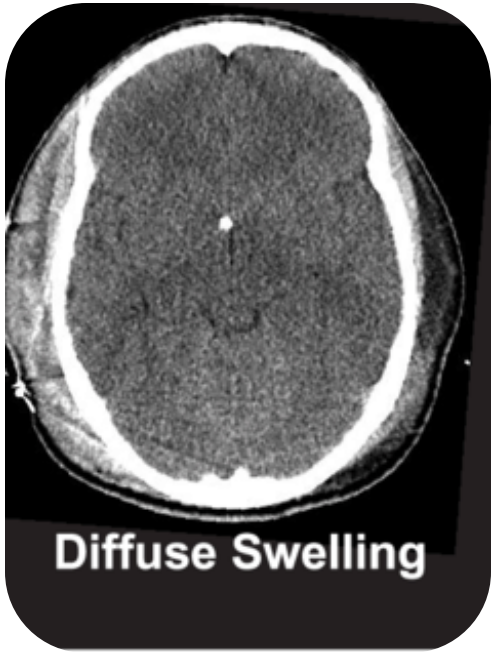
TRAUMATIC BRAIN INJURIES

Death may occur after head blows due to:

- **Intracranial Hemorrhage** – bleeding within the skull that compresses brain tissue.
- **Cerebral Edema** – swelling of the brain due to accumulation of fluid within neural tissue.



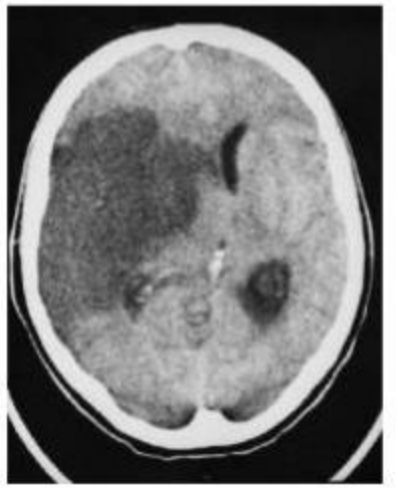


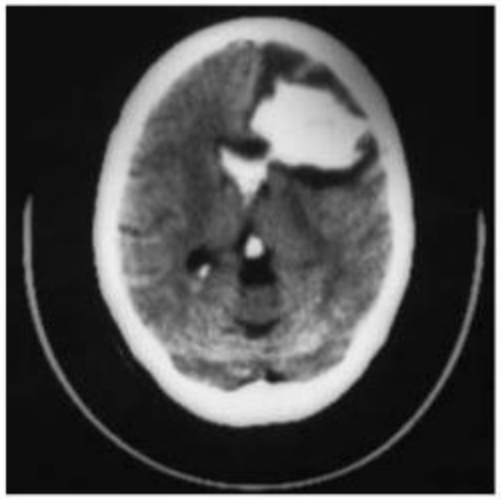


BRAIN DYSFUNCTIONS

CEREBROVASCULAR ACCIDENT (CVA) OR STROKE

- Results when **blood circulation to a brain area is blocked,** causing **ischemia and death of brain tissue.**
- **Types of Stroke:**
  - **Ischemic Stroke (Infarct)** – due to blockage of cerebral blood flow.
  - **Hemorrhagic Stroke (Bleed)** – due to rupture of a cerebral blood vessel.
- **Consequences:**
  - **Loss of function** (motor, sensory, cognitive, or speech)
  - **Hemiplegia** – paralysis of one side of the body
  - **Aphasia** – impaired speech or language comprehension (typically from left hemisphere damage)
  - Severe cases may result in **coma or death**





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BRAIN DYSFUNCTIONS

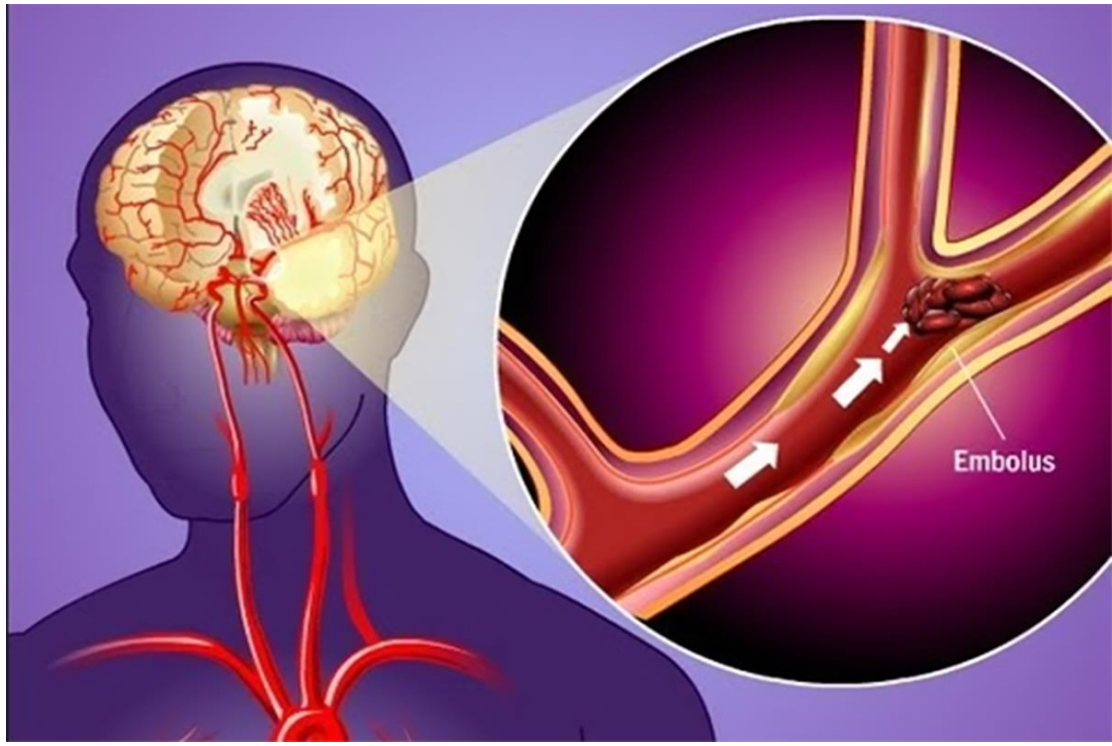
TRANSIENT ISCHEMIC ATTACK (TIA)

Temporary episode of **cerebral ischemia** (restricted blood flow to the brain).

Duration: Neurological symptoms **typically last less than 24 hours**, often **5–30 minutes.**

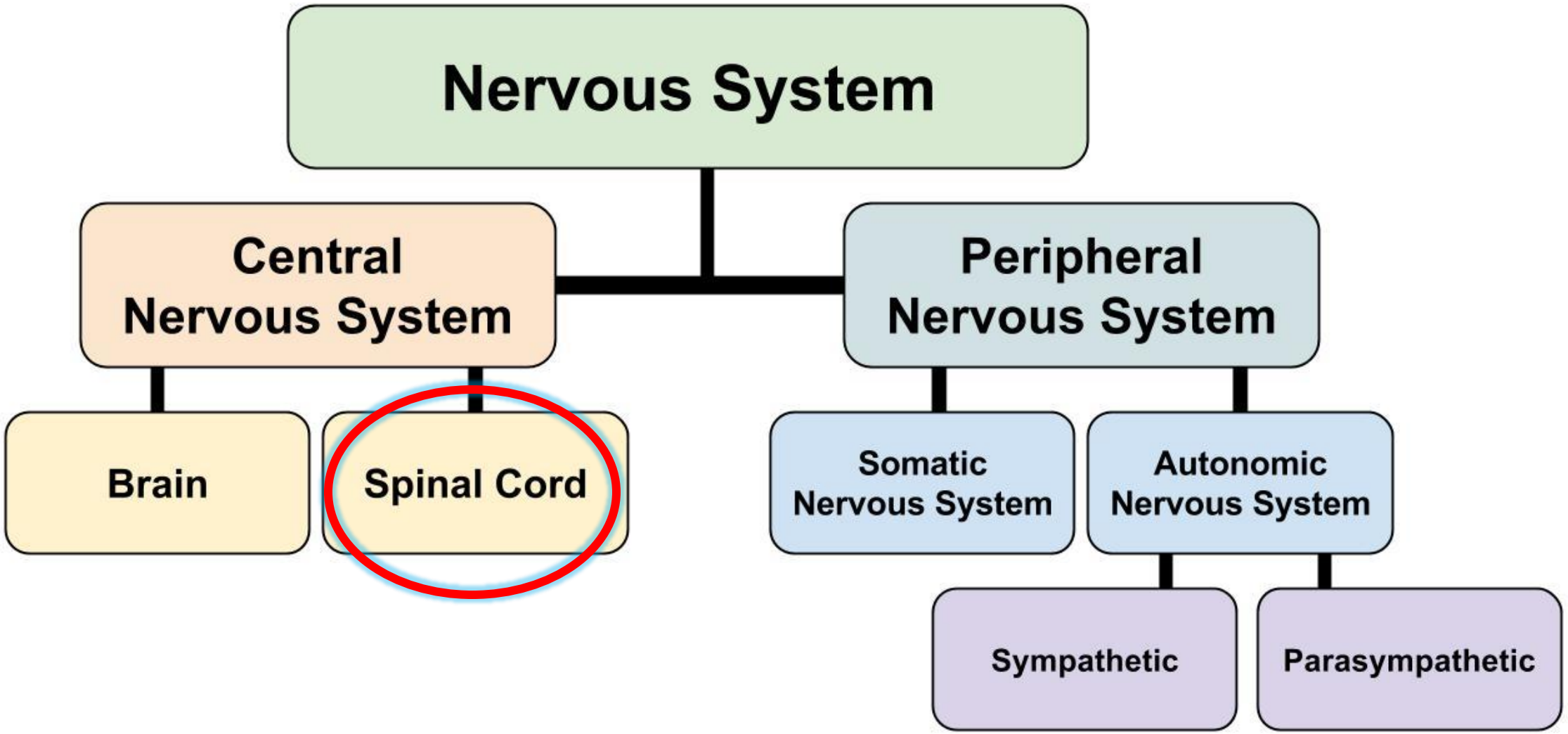
Common Symptoms:

- Sudden numbness or weakness (often on one side)
- Temporary paralysis
- Impaired or slurred speech
- Visual disturbances or dizziness



Clinical Significance:

- **Warning sign ("mini-stroke")** for impending cerebrovascular accident (CVA).
- Requires **immediate medical evaluation and preventive management**



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SPINAL CORD
STRUCTURE

Major Function:

- Acts as the **communication link** between the **brain** and the **Peripheral Nervous System (PNS)**.
- **Integrates** incoming sensory information and produces **motor responses** through **reflex mechanisms.**

General Structure:

- Extends from the **foramen magnum** to the **second lumbar vertebra (L2)**.
- **Shorter** than the vertebral column (due to slower growth during development).
- Composed of **31 segments**—each giving rise to a **pair of spinal nerves.**

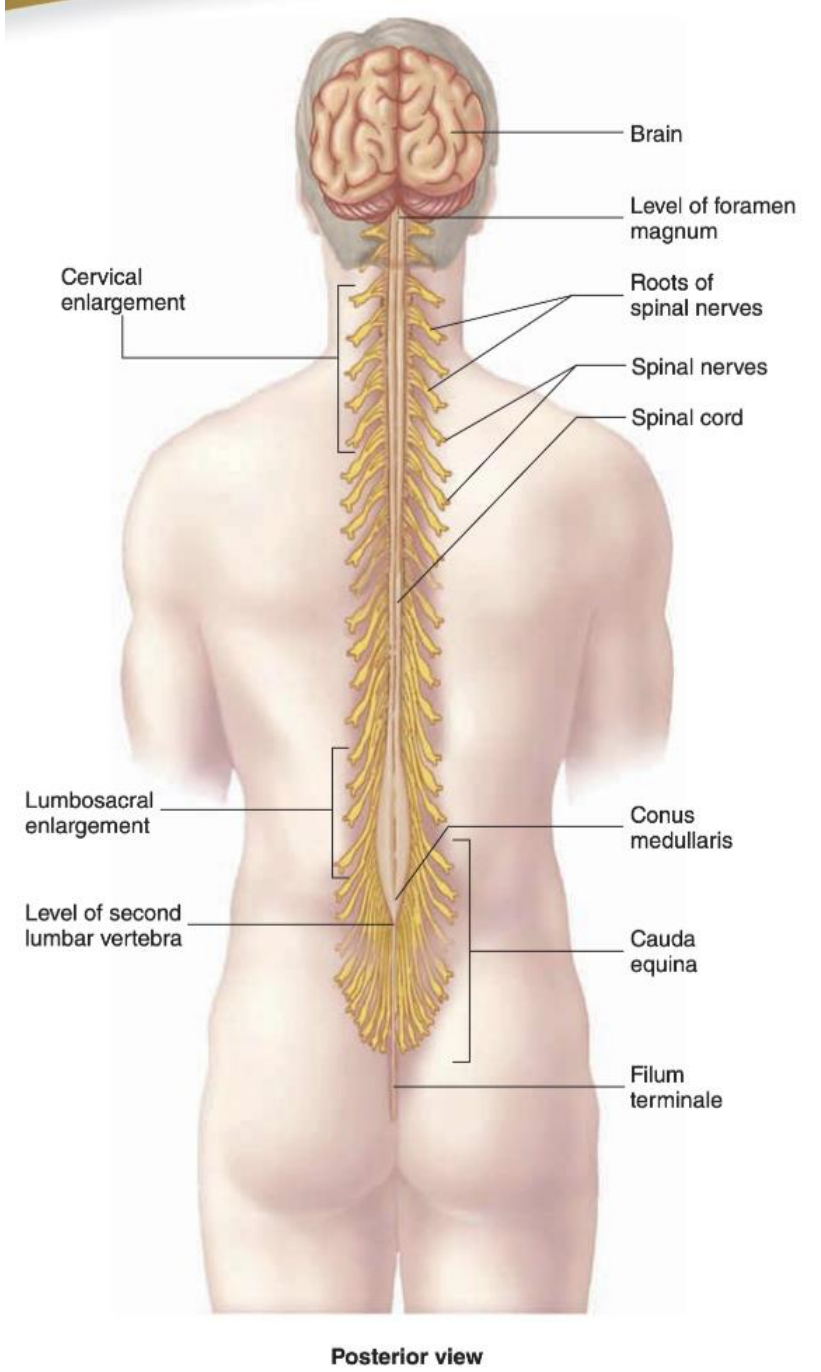
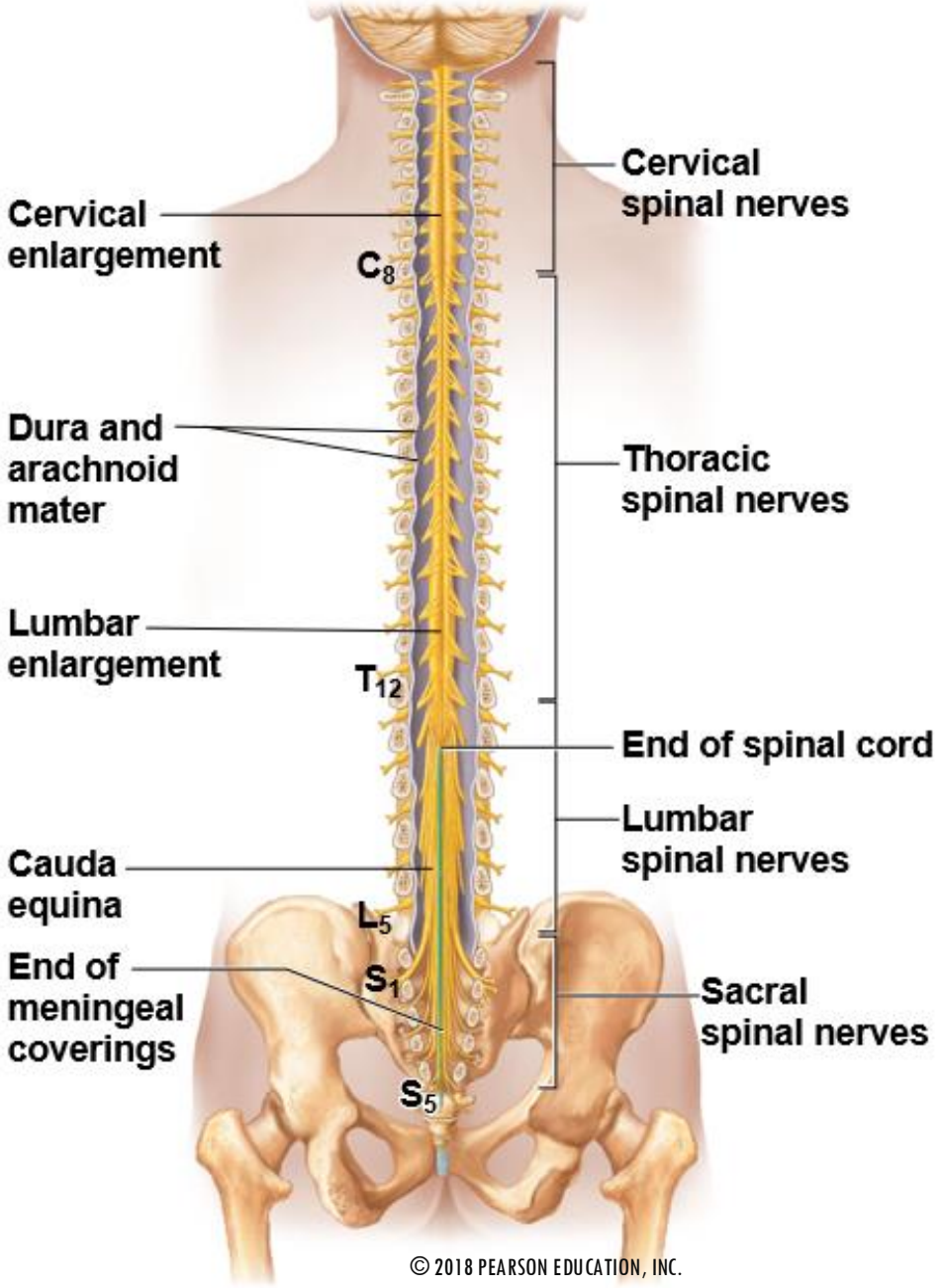


FIGURE 12.1 Spinal Cord and Spinal Nerve Roots APR

SPINAL CORD
STRUCTURE

General Structure (continuation):

- **Enlargements:**
  - **Cervical enlargement:** Nerves to upper limbs.
  - **Lumbosacral enlargement:** Nerves to lower limbs.
- **Conus medullaris:** Cone-shaped inferior end of the spinal cord.
- **Cauda equina:** "Horse's tail" – collection of nerve roots extending below the spinal cord.



MENINGES OF THE SPINAL CORD

- *Meninges* — three protective connective tissue membranes surrounding the **brain** and **spinal cord.**

Layers of the Meninges:

1. **Dura mater** – outermost, thickest layer
   - Forms the **thecal sac** around the spinal cord
   - Extends from the **foramen magnum** to the **second sacral vertebra (S2)**
   - **Epidural space** lies between dura and vertebral canal — contains fat, blood vessels, and connective tissue
   - Site for **epidural anesthesia**

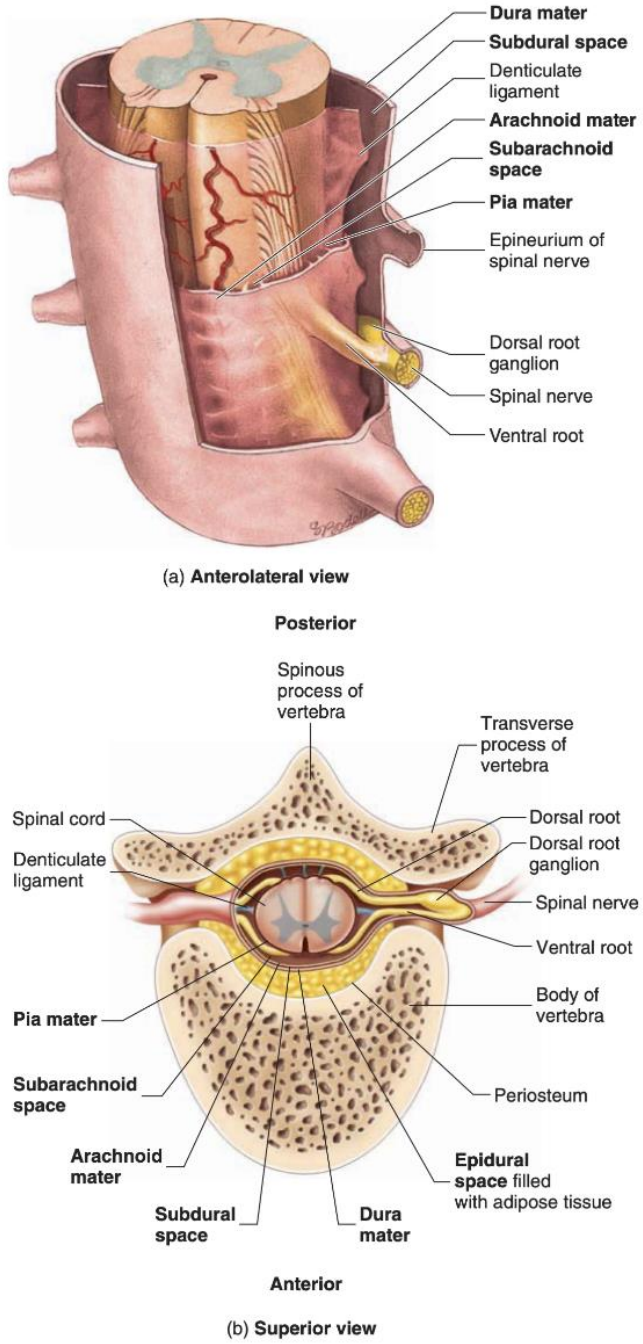


FIGURE 12.2 **Meningeal Membranes Surrounding the Spinal Cord** AP|R

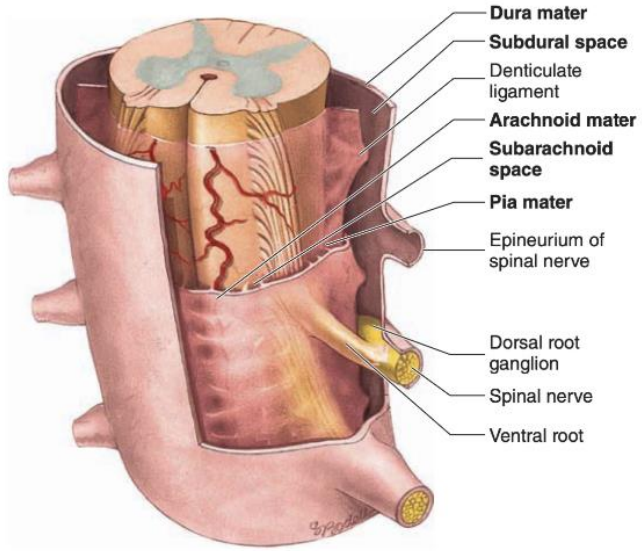
MENINGES OF THE SPINAL CORD

2. **Arachnoid mater** – middle, delicate layer
   - **Subdural space** (between dura and arachnoid): contains a small amount of serous fluid

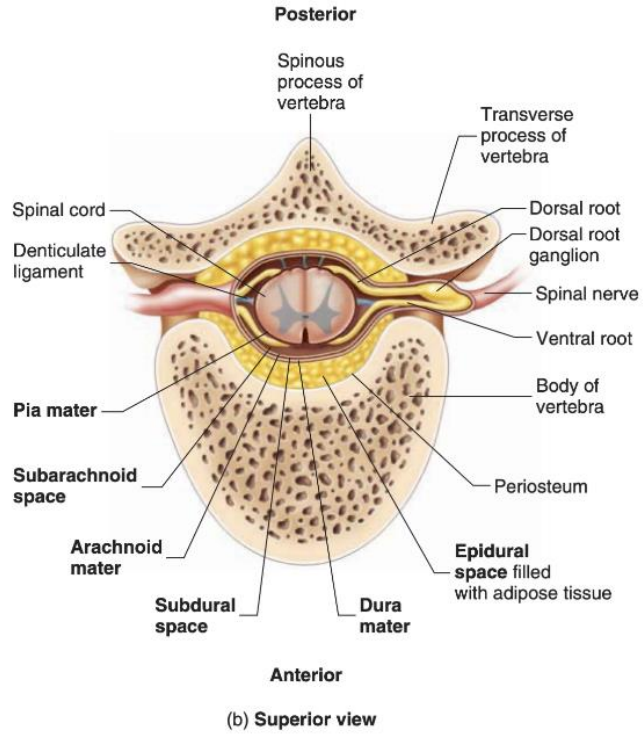
3. **Pia mater** – innermost, tightly adherent to spinal cord surface
   - **Denticulate ligaments** anchor cord laterally
   - **Filum terminale** anchors cord inferiorly to the coccyx

Subarachnoid Space:

- Between **arachnoid** and **pia mater**
- Contains **cerebrospinal fluid (CSF)**, **blood vessels**, and **web-like connective tissue**



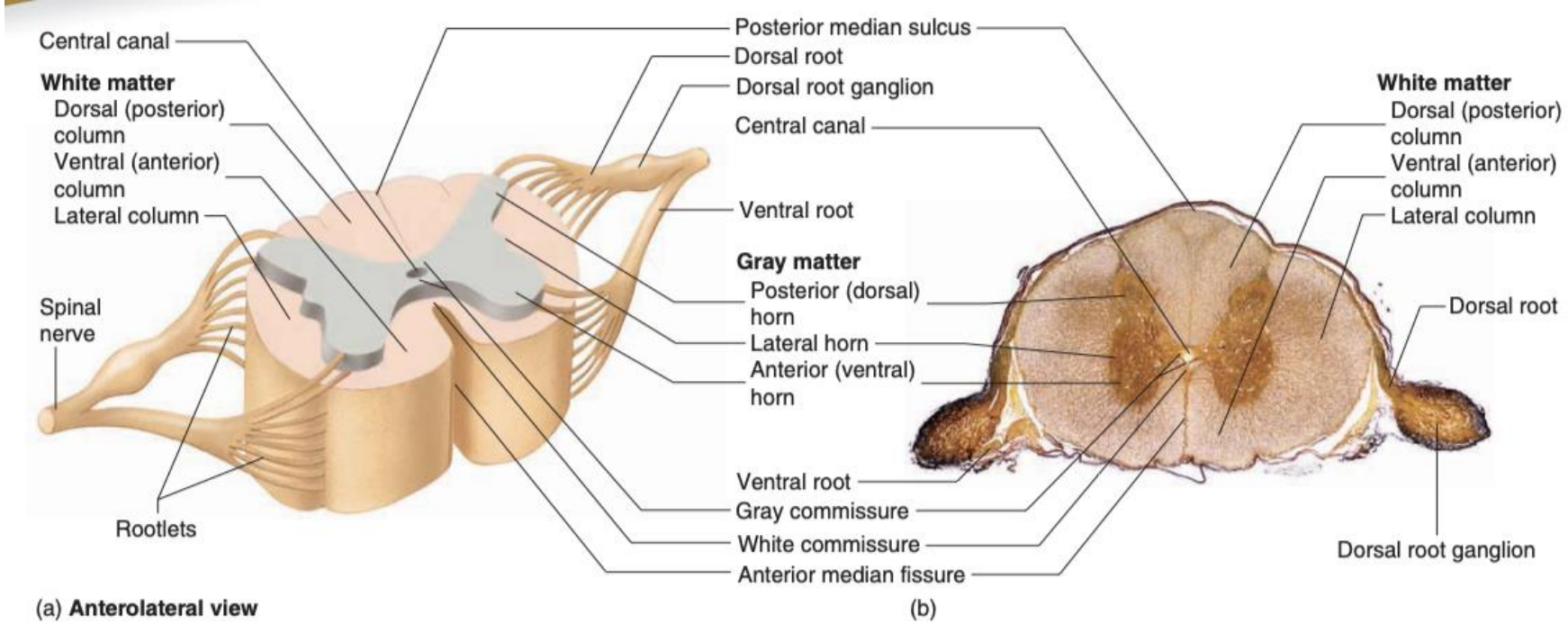
(a) **Anterolateral view**



(b) **Superior view**

FIGURE 12.2 **Meningeal Membranes Surrounding the Spinal Cord** AP|R

SPINAL CORD
CROSS SECTION



The spinal cord consists of:

- **White matter (superficial):** Myelinated axons forming nerve tracts.
- **Gray matter (deep):** Neuron cell bodies, dendrites, and unmyelinated axons.

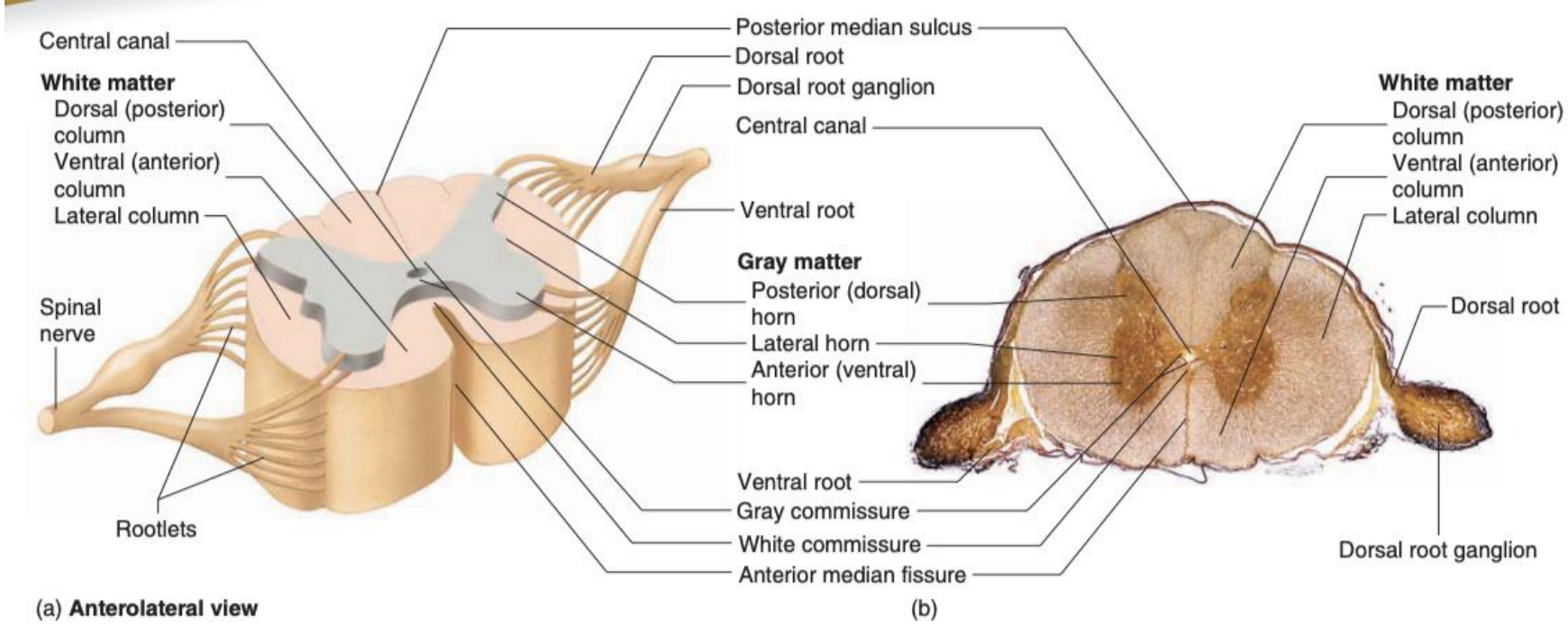
Divisions:

- White matter: dorsal, lateral, and ventral columns (funiculi).
- Gray matter: dorsal, lateral, and ventral horns.

Commissures: Gray and white commissures connect the two halves of the spinal cord.

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SPINAL CORD
CROSS SECTION

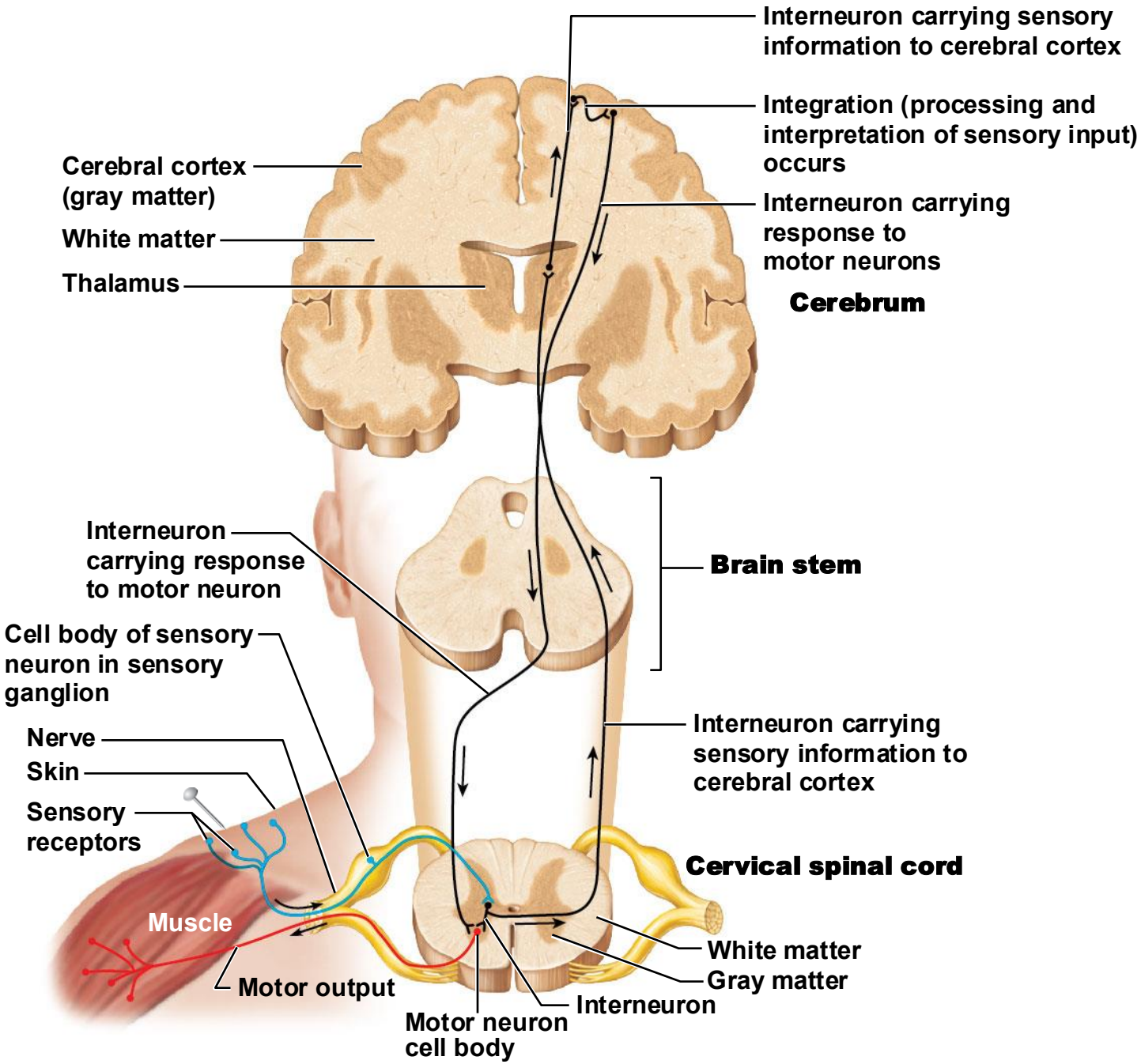


Central canal: Located in the center of the gray commissure; circulates cerebrospinal fluid (CSF).

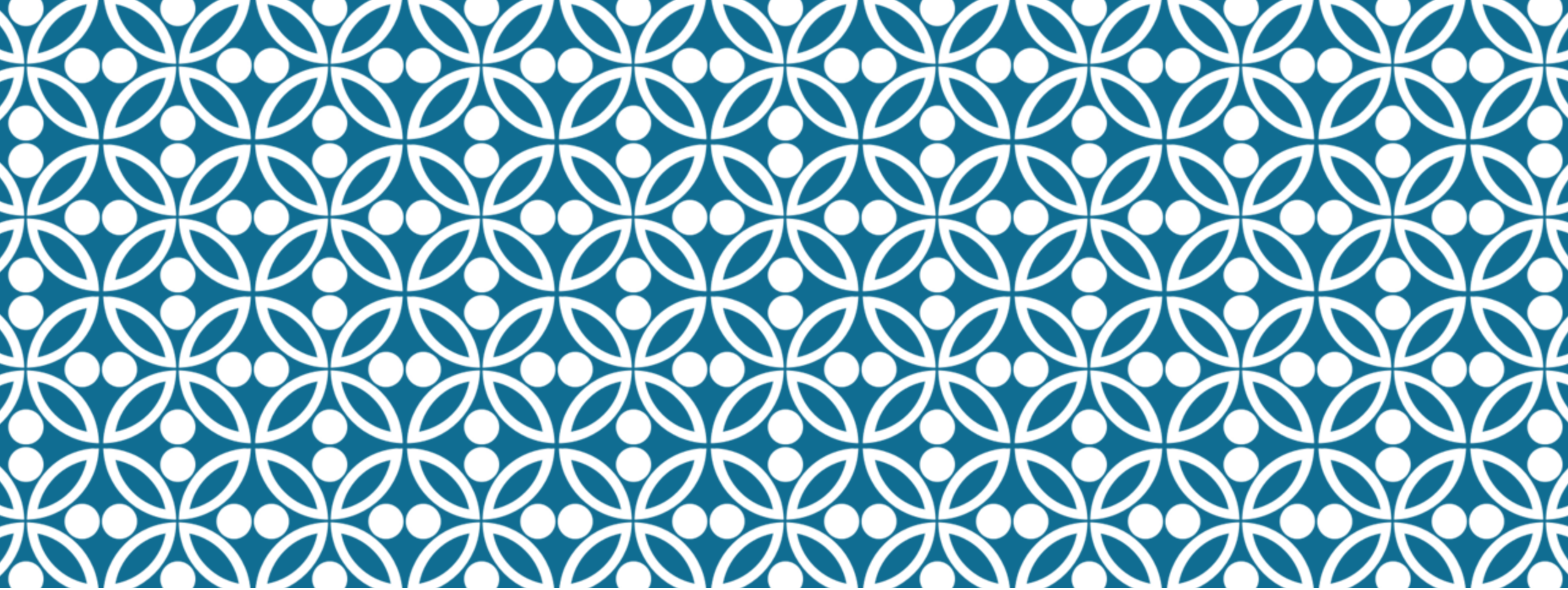
Spinal roots:

- **Dorsal roots** – contain sensory axons and a **dorsal root ganglion** with sensory neuron cell bodies.
- **Ventral roots** – contain motor axons from motor neurons in the anterior horn.
- Both unite to form a **mixed spinal nerve** (sensory + motor).

SCHEMATIC OF ASCENDING (SENSORY) AND DESCENDING (MOTOR) PATHWAYS BETWEEN THE BRAIN AND THE SPINAL CORD.



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PERIPHERAL NERVOUS SYSTEM



OBJECTIVES

At the end of the session, students are expected to:

1. Enumerate the **functions of the nervous system.**
2. Identify the **cells of the nervous system:**
    1. Neuroglia
    2. Neurons
3. Enumerate the steps in **producing an action potential** and **synaptic transmission.**
4. Define **reflex** and enumerate its **components.**
5. Identify and differentiate the **divisions of the nervous system:**
    1. **Central Nervous System (CNS):** Brain and Spinal Cord
    2. **Peripheral Nervous System (PNS):**
        1. Autonomic Nervous System (ANS)
            1. Sympathetic Nervous System
            2. Parasympathetic Nervous System
            3. Enteric Nervous System
        2. Somatic Nervous System
        3. Sensory (Afferent) Nervous System
        4. Motor (Efferent) Nervous System

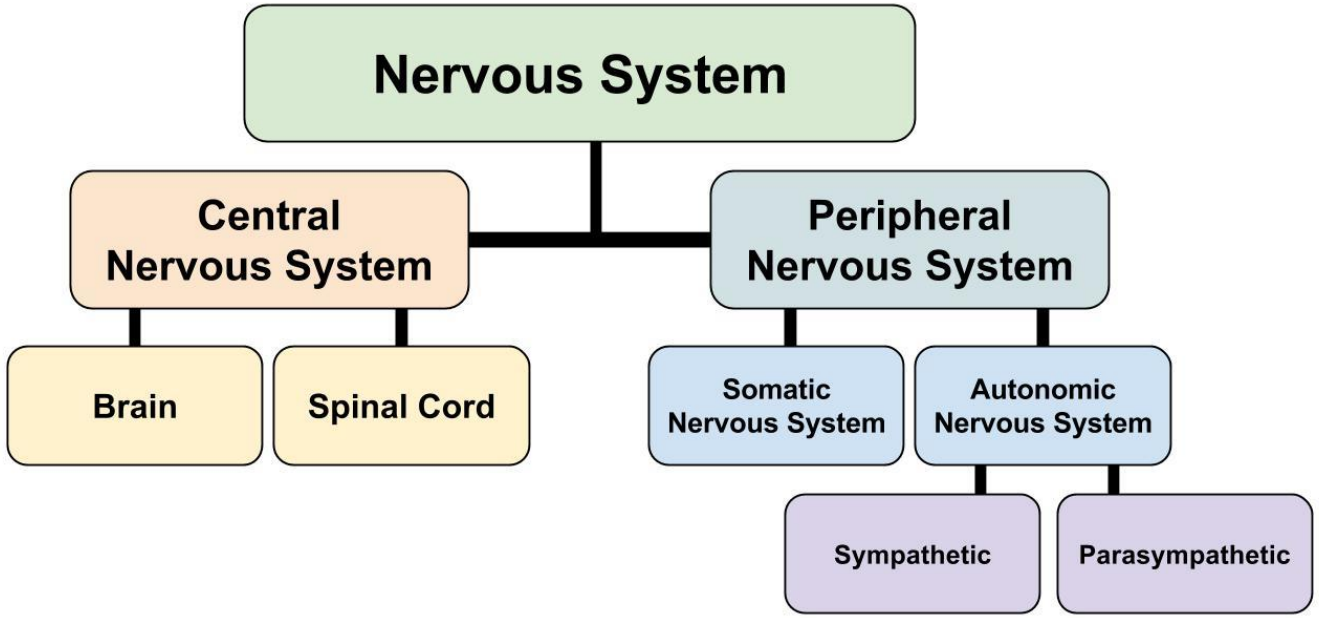
PERIPHERAL NERVOUS SYSTEM (PNS)

PNS consists of nerves and ganglia outside the CNS

Nerves

- Spinal
- Cranial

Ganglia



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PERIPHERAL NERVOUS SYSTEM (PNS)

SENSORY RECEPTORS TYPES

- *Sensory receptors* are specialized structures that detect changes (stimuli) in the internal or external environment and convert them into electrical signals (action potentials).

Types of Sensory Receptors (by Stimulus Type):

1. **Mechanoreceptors** – respond to mechanical stimuli such as touch, pressure, vibration, stretch, and sound.
2. **Thermoreceptors** – detect changes in temperature (heat or cold).
3. **Photoreceptors** – respond to light; found only in the retina of the eye.
4. **Chemoreceptors** – detect chemical stimuli such as taste, smell, and blood CO_2 or O_2 levels.
5. **Nociceptors** – respond to tissue damage or potentially harmful stimuli (pain receptors).

PERIPHERAL NERVOUS SYSTEM (PNS)

SENSORY RECEPTORS TYPES

Types (by Location):

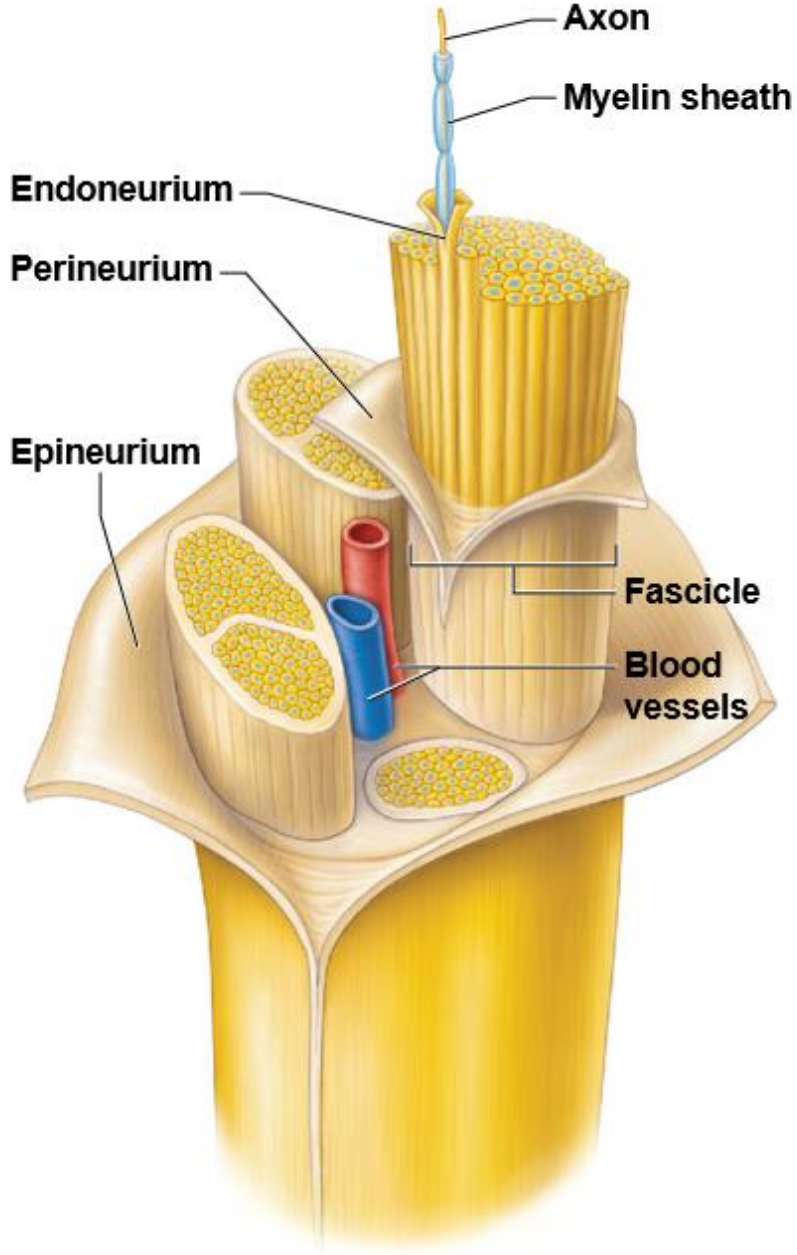
- **Exteroceptors:** Detect stimuli from the external environment (e.g., touch, temperature, vision).
- **Interoceptors (Visceroceptors):** Monitor internal organ activity (e.g., blood pressure, pH, stretch).
- **Proprioceptors:** Found in muscles, tendons, and joints; provide information about body position and movement.

SPINAL NERVE
STRUCTURE

- *Spinal nerves* are bundles of axons within the Peripheral Nervous System (PNS) that connect the Central Nervous System (CNS) to receptors, muscles, and glands.

Components of a Nerve:

- **Axons (Nerve Fibers):** Conduct electrical impulses.
- **Schwann Cells:** Produce the myelin sheath around axons in the PNS.
- **Connective Tissue Layers:**
  - **Endoneurium** – delicate connective tissue surrounding individual axons and their Schwann cells.
  - **Perineurium** – encloses groups of axons forming **nerve fascicles**.
  - **Epineurium** – outermost dense connective tissue layer binding fascicles together; continuous with the dura mater of the CNS.



SPINAL NERVE
ORGANIZATION

The **spinal nerves** connect the spinal cord to specific regions of the body through **31 pairs** of mixed (sensory + motor) nerves.

- Each nerve exits the vertebral column through **intervertebral foramina,** except the first pair (C1) and those of the sacrum.

Spinal Nerve Pairs (31 Total):

- **Cervical:** 8 pairs (C1–C8)
- **Thoracic:** 12 pairs (T1–T12)
- **Lumbar:** 5 pairs (L1–L5)
- **Sacral:** 5 pairs (S1–S5)
- **Coccygeal:** 1 pair (Co)

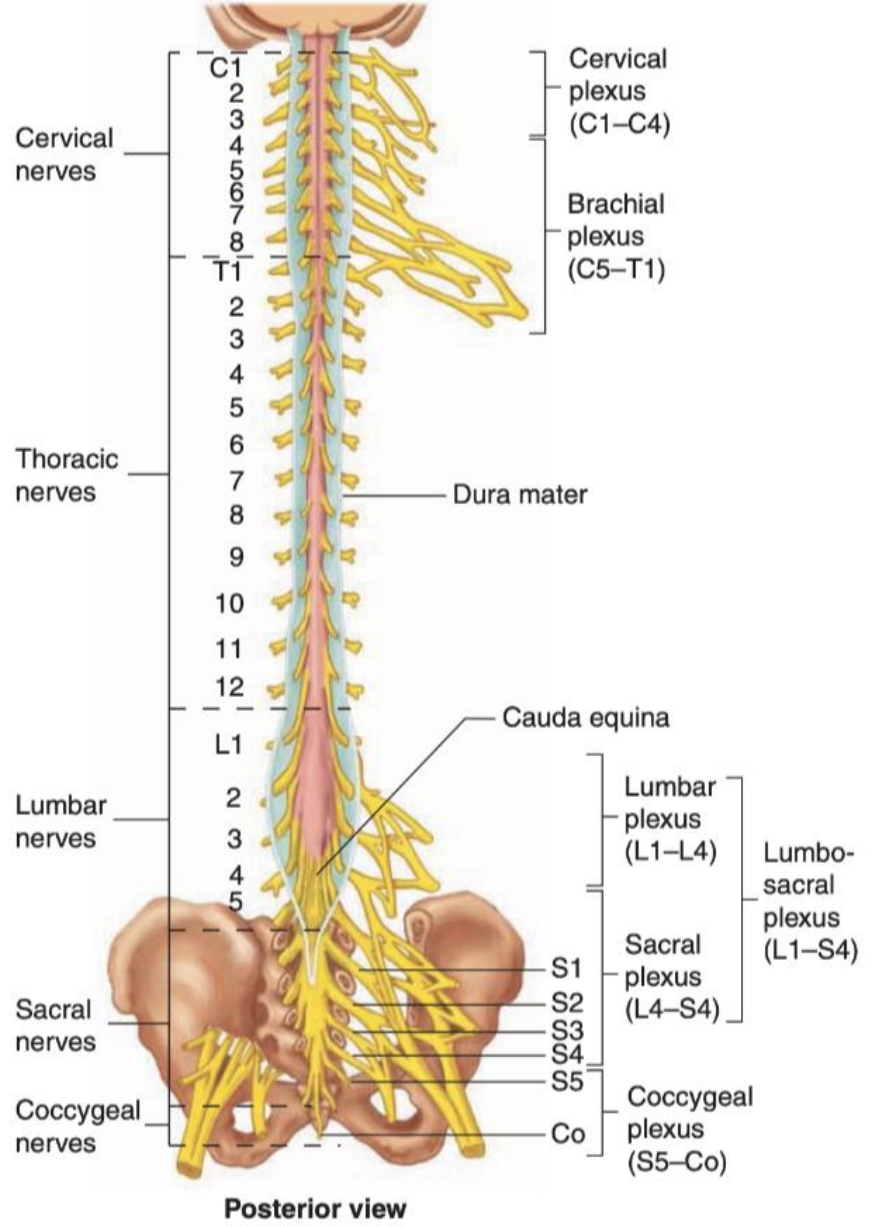
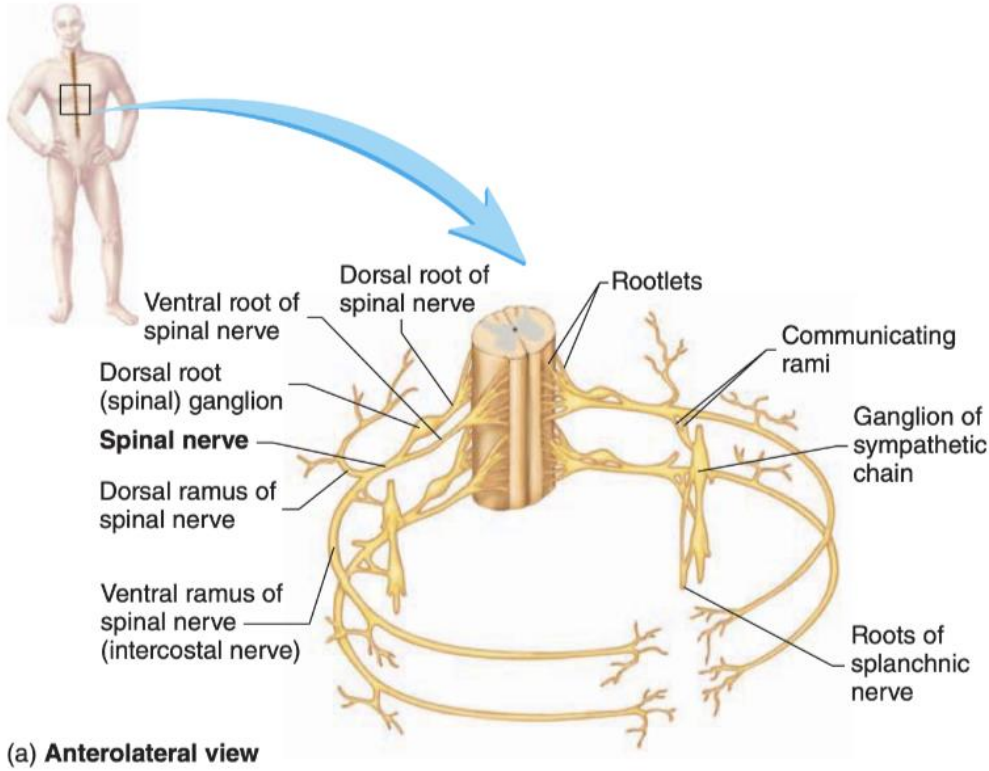


FIGURE 12.13 Spinal Nerves and Plexuses
The regional designations and the numbers of the spinal nerves are shown on the left. The plexuses formed by the spinal nerves are shown on the right. AP|R

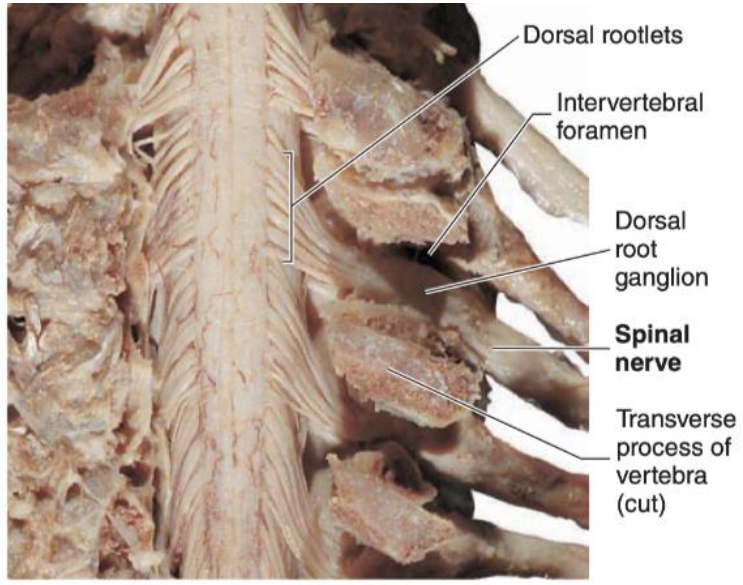
SPINAL NERVE
ORGANIZATION

Distribution and Branches:

- Each spinal nerve divides into:
  - **Dorsal ramus:** Innervates deep back muscles and skin of the posterior trunk.
  - **Ventral ramus:** Innervates anterior and lateral trunk, limbs, and forms **nerve plexuses.**
  - **Communicating rami:** Carry autonomic (sympathetic) fibers.



(a) **Anterolateral view**



(b) **Posterior view**

SPINAL NERVE ORGANIZATION

- A **dermatome** is a specific **area of skin** supplied by the **sensory fibers of a single spinal nerve.**

Key Points:

- Each spinal nerve (except **C1**) receives sensory input from a specific region of the body surface.
- **Dermatomal maps** illustrate how these sensory regions correspond to spinal nerve levels.
- Overlapping regions between adjacent dermatomes ensure that damage to one spinal nerve does not result in complete sensory loss.

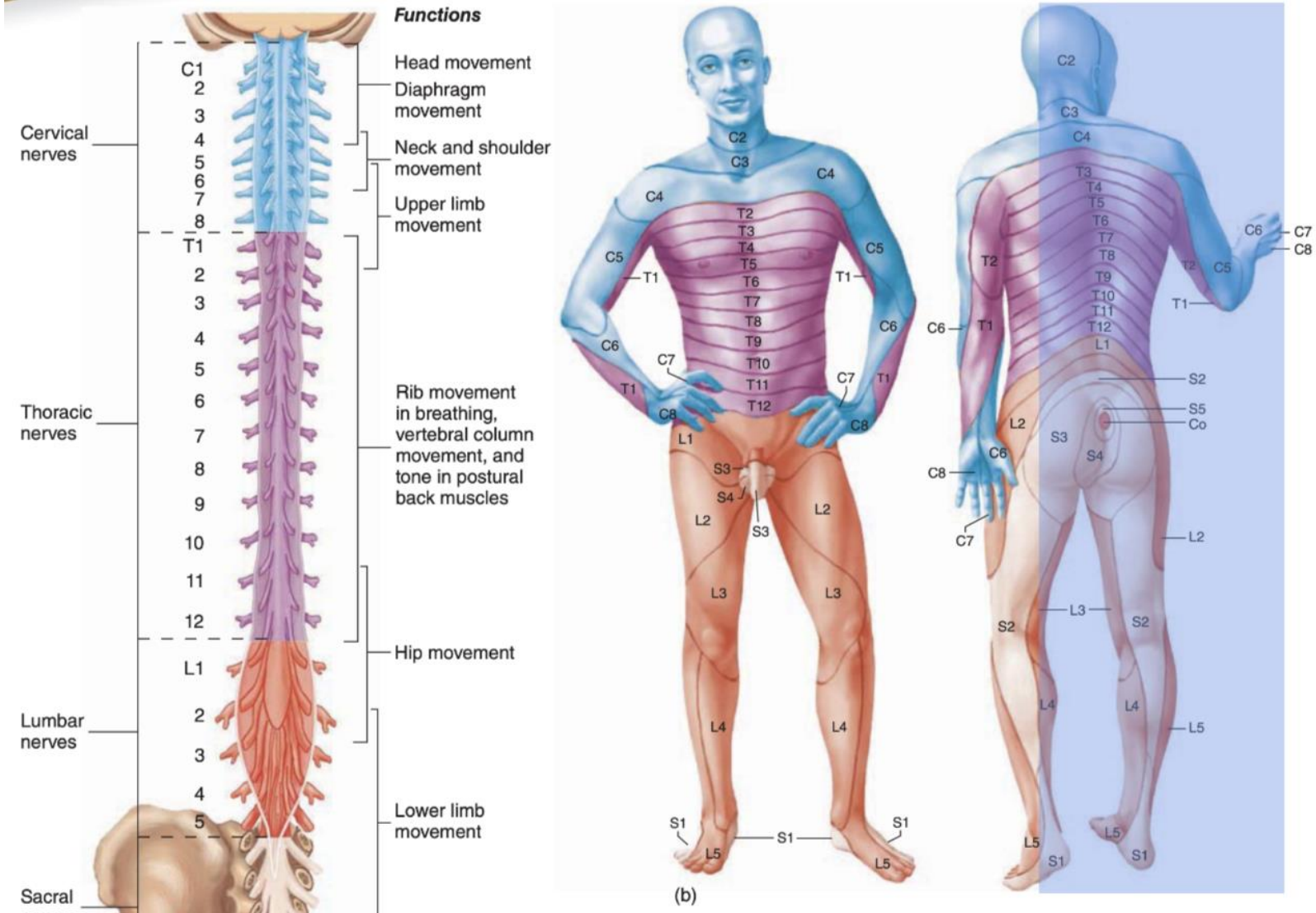


FIGURE 12.14 Spinal Cord and Dermatomal Map

(*a*) Nerves of the spinal cord and their functions. The regions are color-coded. (*b*) Letters and numbers indicate the spinal nerves innervating a given region of skin (dermatome). AP|R

SPINAL NERVE ORGANIZATION

Clinical Importance:

- **Used in neurological assessment** to determine the level of spinal cord or nerve injury.
- **Loss of sensation** in a specific dermatome indicates possible nerve root or spinal segment damage.

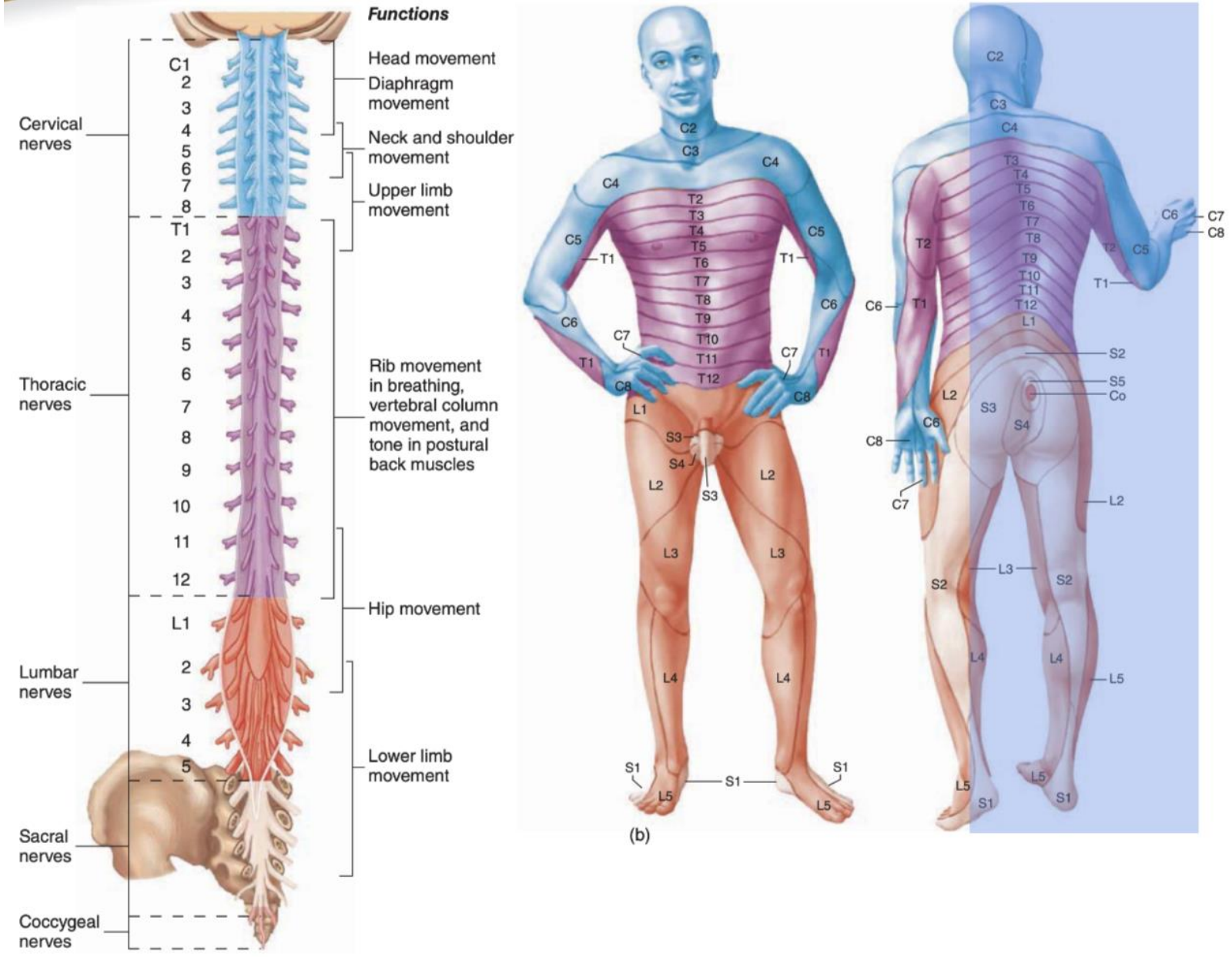


FIGURE 12.14 Spinal Cord and Dermatomal Map

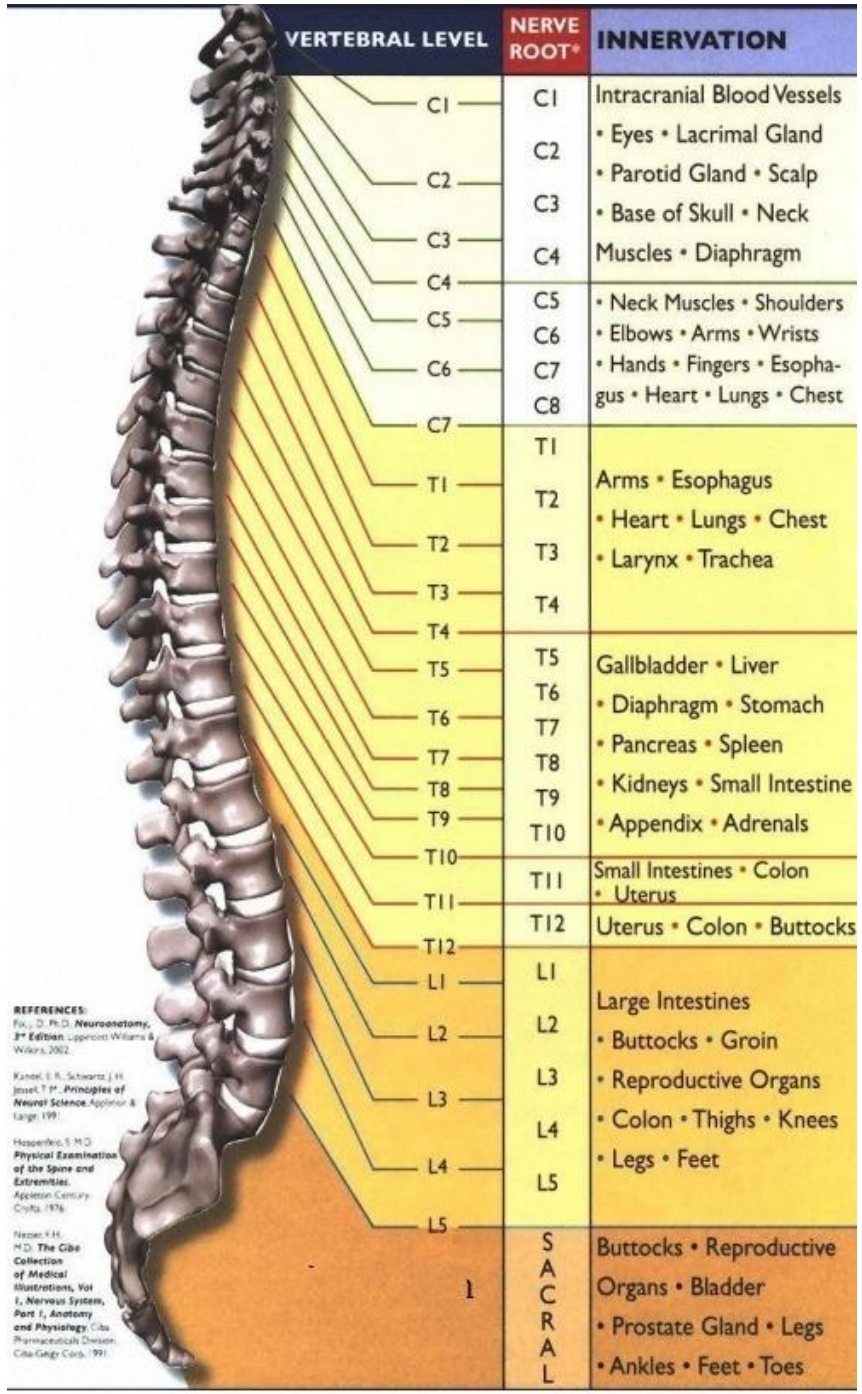
(*a*) Nerves of the spinal cord and their functions. The regions are color-coded. (*b*) Letters and numbers indicate the spinal nerves innervating a given region of skin (dermatome). AP|R

NERVE

FUNCTIONAL CLASSIFICATION

- Mixed nerves
  - Contain both sensory and motor fibers
- Sensory (afferent) nerves
  - Carry impulses toward the CNS
- Motor (efferent) nerves
  - Carry impulses away from the CNS

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VERTEBRAL LEVEL
NERVE ROOT*
INNERVATION
C1
C2
C3
C4
C5
C6
C7
T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12
L1
L2
L3
L4
L5
C1
C2
C3
C4
C5
C6
C7
C8
T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12
L1
L2
L3
L4
L5
S
A
C
R
A
L
1
Intracranial Blood Vessels • Eyes • Lacrimal Gland • Parotid Gland • Scalp • Base of Skull • Neck Muscles • Diaphragm
• Neck Muscles • Shoulders • Elbows • Arms • Wrists • Hands • Fingers • Esophagus • Heart • Lungs • Chest
Arms • Esophagus • Heart • Lungs • Chest • Larynx • Trachea
Gallbladder • Liver • Diaphragm • Stomach • Pancreas • Spleen • Kidneys • Small Intestine • Appendix • Adrenals
Small Intestines • Colon • Uterus
Uterus • Colon • Buttocks
Large Intestines • Buttocks • Groin • Reproductive Organs • Colon • Thighs • Knees • Legs • Feet
Buttocks • Reproductive Organs • Bladder • Prostate Gland • Legs • Ankles • Feet • Toes
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SPINAL NERVES
CERVICAL PLEXUS

Origin: Spinal nerves **C1–C4**

Innervation:

- **Superficial neck structures** and **skin of neck**
- **Posterior scalp region**
- **Muscles attached to the hyoid bone**

- **Ansa cervicalis (C1–C3):** Loop supplying **infrahyoid muscles**
- **Phrenic nerve (C3–C5):**
  - Supplies the **diaphragm** — essential for **breathing**
  - **"C3, C4, and C5 keep the diaphragm alive"**

Clinical note: Damage to the phrenic nerve → **severe breathing difficulty or paralysis of the diaphragm**

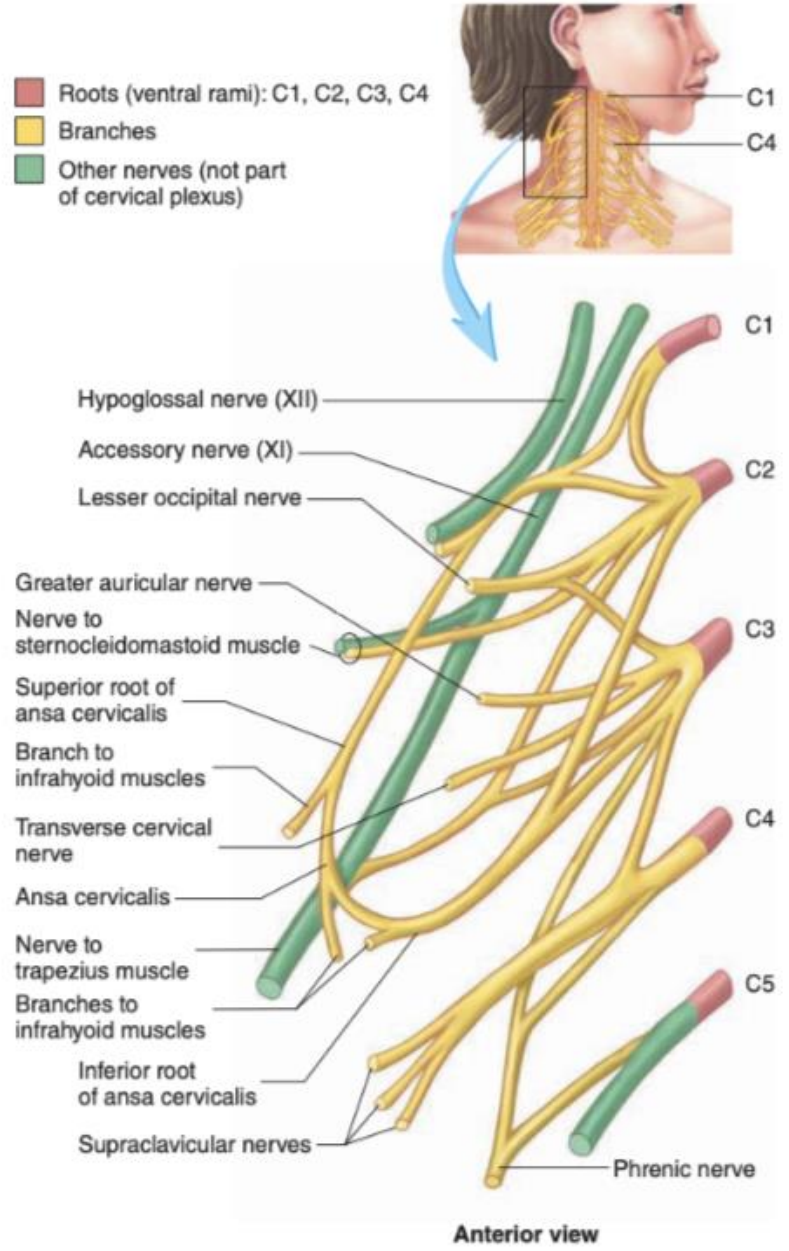


FIGURE 12.16 Cervical Plexus
The roots of the plexus are formed by the ventral rami of spinal nerves C1–C4. Branches of the cervical plexus innervate the muscles (infrahyoid) and skin of the neck. The phrenic nerve (C3–C5) innervates the diaphragm.

SPINAL NERVES

BRACHIAL PLEXUS (C5–T1)

- The **brachial plexus** is formed by the **ventral rami of spinal nerves C5–T1.**
- It supplies the **shoulder and upper limb.**
- The plexus forms a complex network consisting of:
  - **Five roots → Three trunks → Six divisions → Three cords → Five major terminal branches (nerves)**

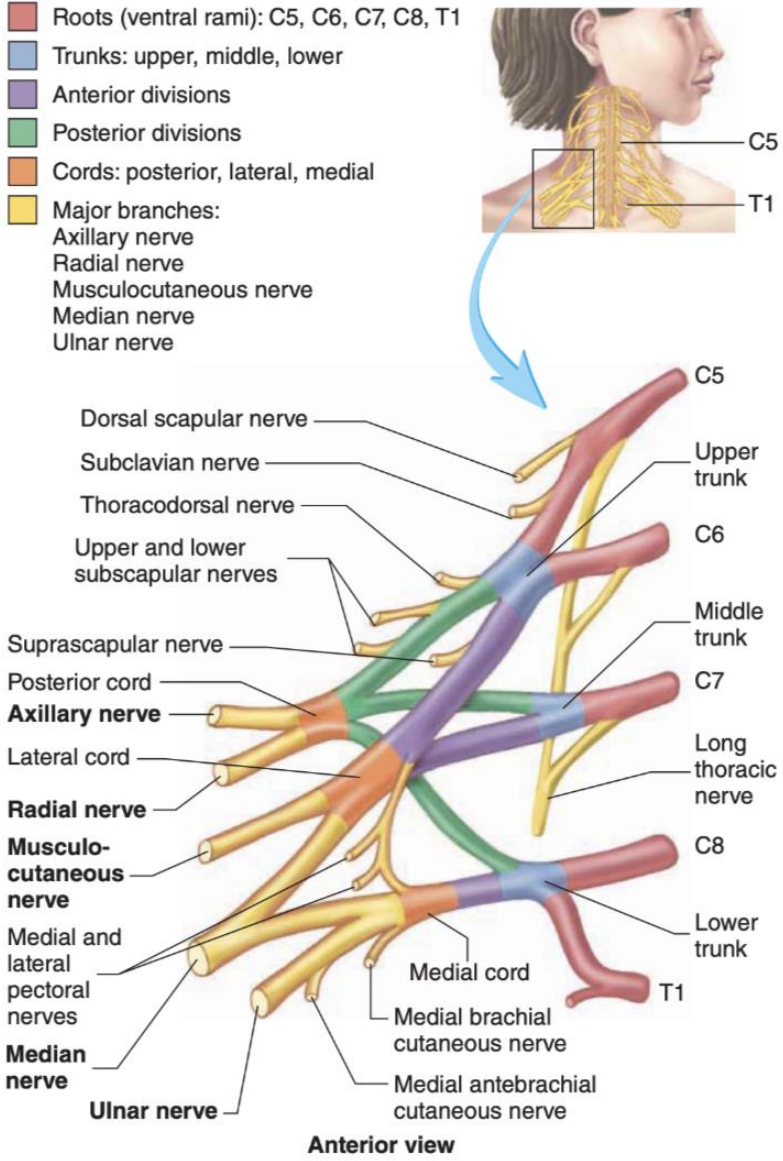


FIGURE 12.17 Brachial Plexus

The roots of the plexus are formed by the ventral rami of spinal nerves C5–T1 and join to form an upper, a middle, and a lower trunk. Each trunk divides into anterior and posterior divisions. The divisions join to form the posterior, lateral, and medial cords from which the major brachial plexus nerves arise. The major brachial plexus nerves include the axillary, radial, musculocutaneous, median, and ulnar nerves, which innervate the muscles and skin of the upper limb. AP|R

SPINAL NERVES
BRACHIAL PLEXUS (C5–T1)

Major Nerves of the Brachial Plexus:

1. **Axillary Nerve** – Innervates *deltoid* and *teres minor* muscles; provides sensory input to the shoulder region.
2. **Radial Nerve** – Innervates *extensor muscles* of arm and forearm (e.g., triceps brachii); sensory input to posterior arm and hand.
3. **Musculocutaneous Nerve** – Innervates *anterior arm muscles* (e.g., biceps brachii); provides sensation to lateral forearm.
4. **Ulnar Nerve** – Innervates *forearm flexors* and most *intrinsic hand muscles* (except those of the thumb); sensory input to medial hand and little finger.
5. **Median Nerve** – Innervates *flexor muscles of the forearm* and *thenar muscles* of the thumb; sensory input to lateral palm and fingers.

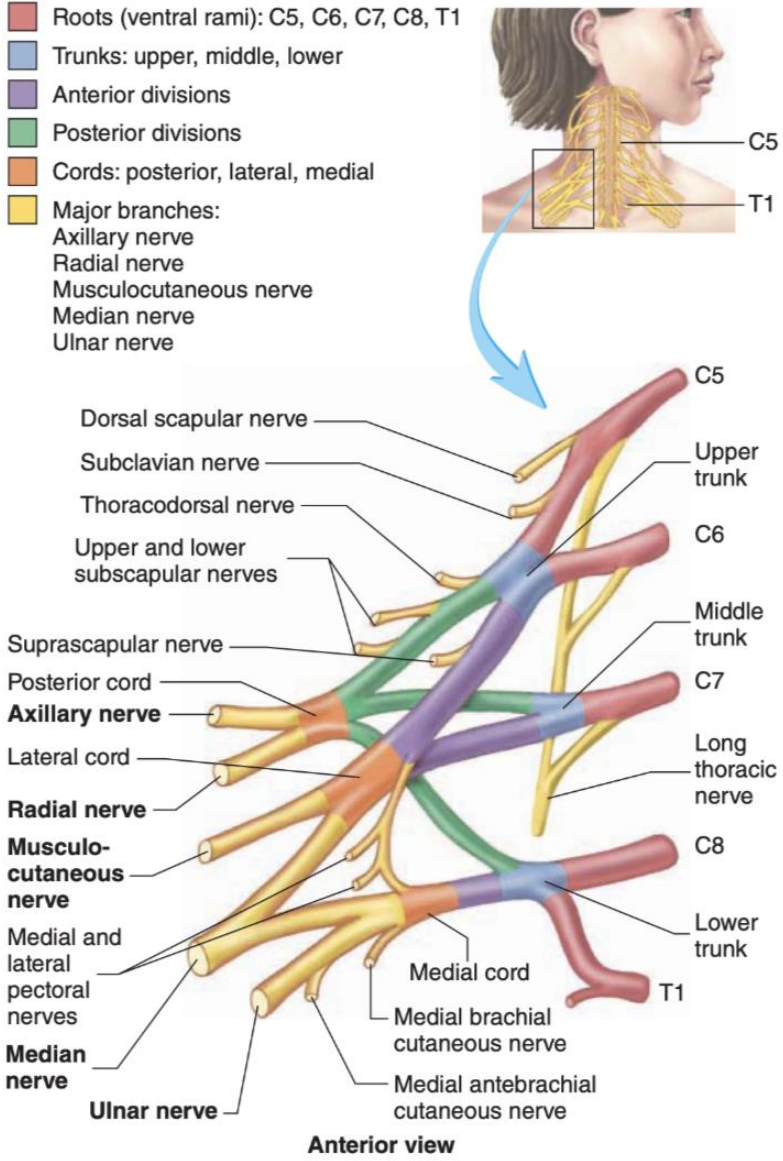


FIGURE 12.17 Brachial Plexus
The roots of the plexus are formed by the ventral rami of spinal nerves C5–T1 and join to form an upper, a middle, and a lower trunk. Each trunk divides into anterior and posterior divisions. The divisions join to form the posterior, lateral, and medial cords from which the major brachial plexus nerves arise. The major brachial plexus nerves include the axillary, radial, musculocutaneous, median, and ulnar nerves, which innervate the muscles and skin of the upper limb. AP|R

SPINAL NERVES

BRACHIAL PLEXUS (C5–T1)

Major Nerves of the Brachial Plexus:

1. Axillary Nerve – Innervates *deltoid* and *teres minor* muscles; provides sensory input to the shoulder region.

Axillary Nerve

Origin

Posterior cord of brachial plexus, C5–C6

Movements/Muscles Innervated

Laterally rotates arm
- *Teres minor*

Abducts arm
- *Deltoid*

Cutaneous (Sensory) Innervation

Inferior lateral shoulder

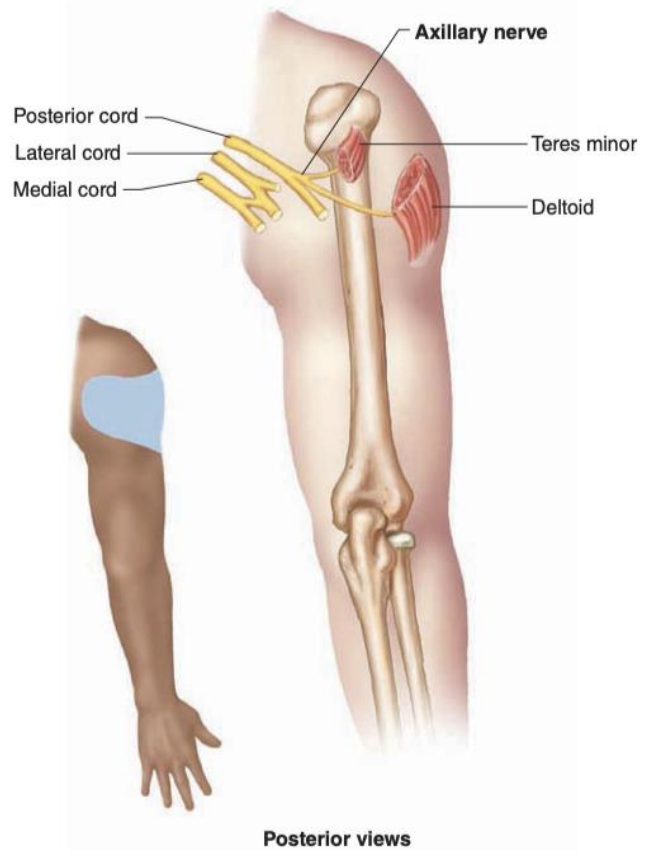


FIGURE 12.18 Axillary Nerve
The route of the axillary nerve and the muscles it innervates. The *inset* depicts the cutaneous (sensory) distribution of the nerve (*blue area*).

SPINAL NERVES

BRACHIAL PLEXUS (C5–T1)

Major Nerves of the Brachial Plexus:

2. Radial Nerve – Innervates *extensor muscles* of arm and forearm (e.g., triceps brachii); sensory input to posterior arm and hand.

Radial Nerve

Origin

Posterior cord of brachial plexus, C5–T1

Movements/Muscles Innervated

Extends elbow
- *Triceps brachii*
- *Anconeus*

Flexes elbow
- *Brachialis (part; sensory only)*
- *Brachioradialis*

Extends and abducts wrist
- *Extensor carpi radialis longus*
- *Extensor carpi radialis brevis*

Supinates forearm and hand
- *Supinator*

Extends fingers
- *Extensor digitorum*
- *Extensor digiti minimi*
- *Extensor indicis*

Extends and adducts wrist
- *Extensor carpi ulnaris*

Abducts thumb
- *Abductor pollicis longus*

Extends thumb
- *Extensor pollicis longus*
- *Extensor pollicis brevis*

Cutaneous (Sensory) Innervation

Posterior surface of arm and forearm, lateral two-thirds of dorsum of hand

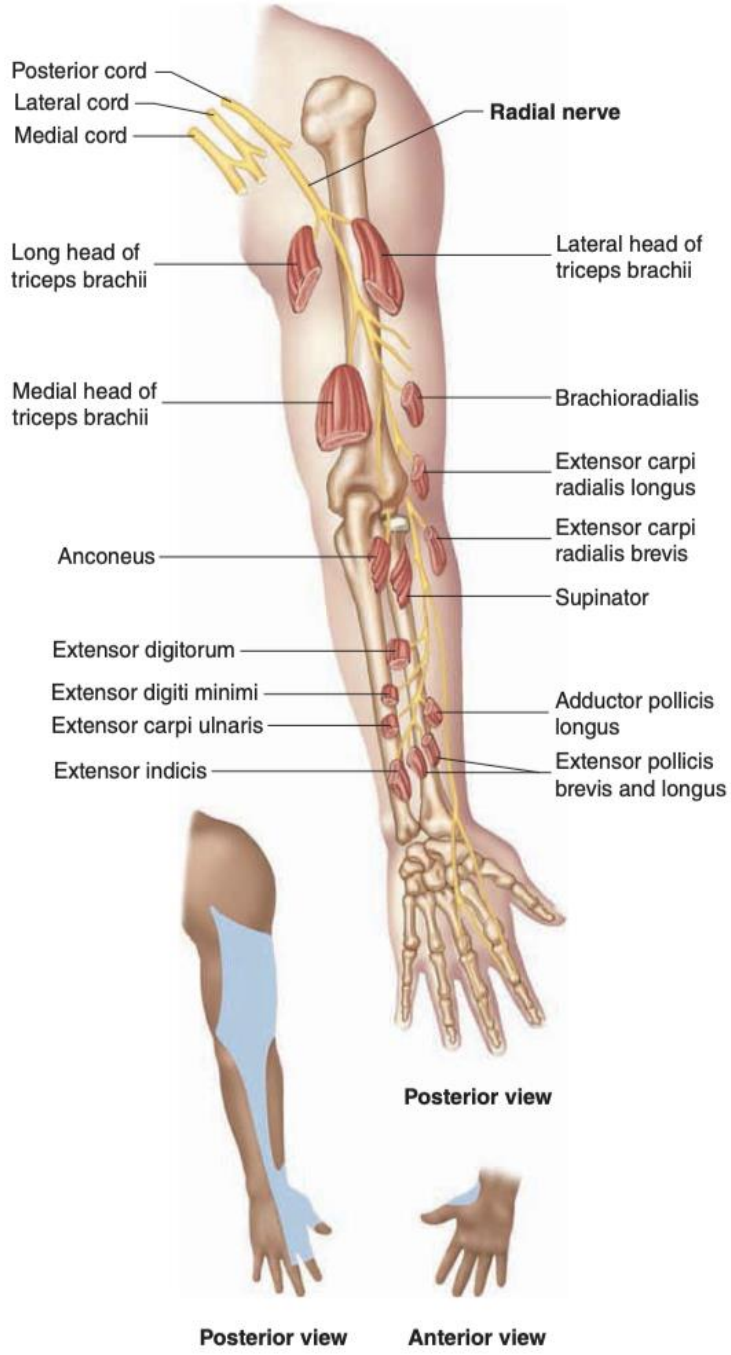


FIGURE 12.19 Radial Nerve
The route of the radial nerve and the muscles it innervates. The *insets* depict the cutaneous (sensory) distribution of the nerve (*blue areas*).

SPINAL NERVES

BRACHIAL PLEXUS (C5–T1)

Major Nerves of the Brachial Plexus:

3. Musculocutaneous Nerve – Innervates *anterior arm muscles* (e.g., biceps brachii); provides sensation to lateral forearm.

Musculocutaneous Nerve

Origin

Lateral cord of brachial plexus, C5–C7

Movements/Muscles Innervated

Flexes shoulder
- *Biceps brachii*
- *Coracobrachialis*

Flexes elbow and supinates forearm and hand
- *Biceps brachii*

Flexes elbow
- *Brachialis (also small amount of innervation from radial nerve)*

Cutaneous (Sensory) Innervation

Lateral surface of forearm

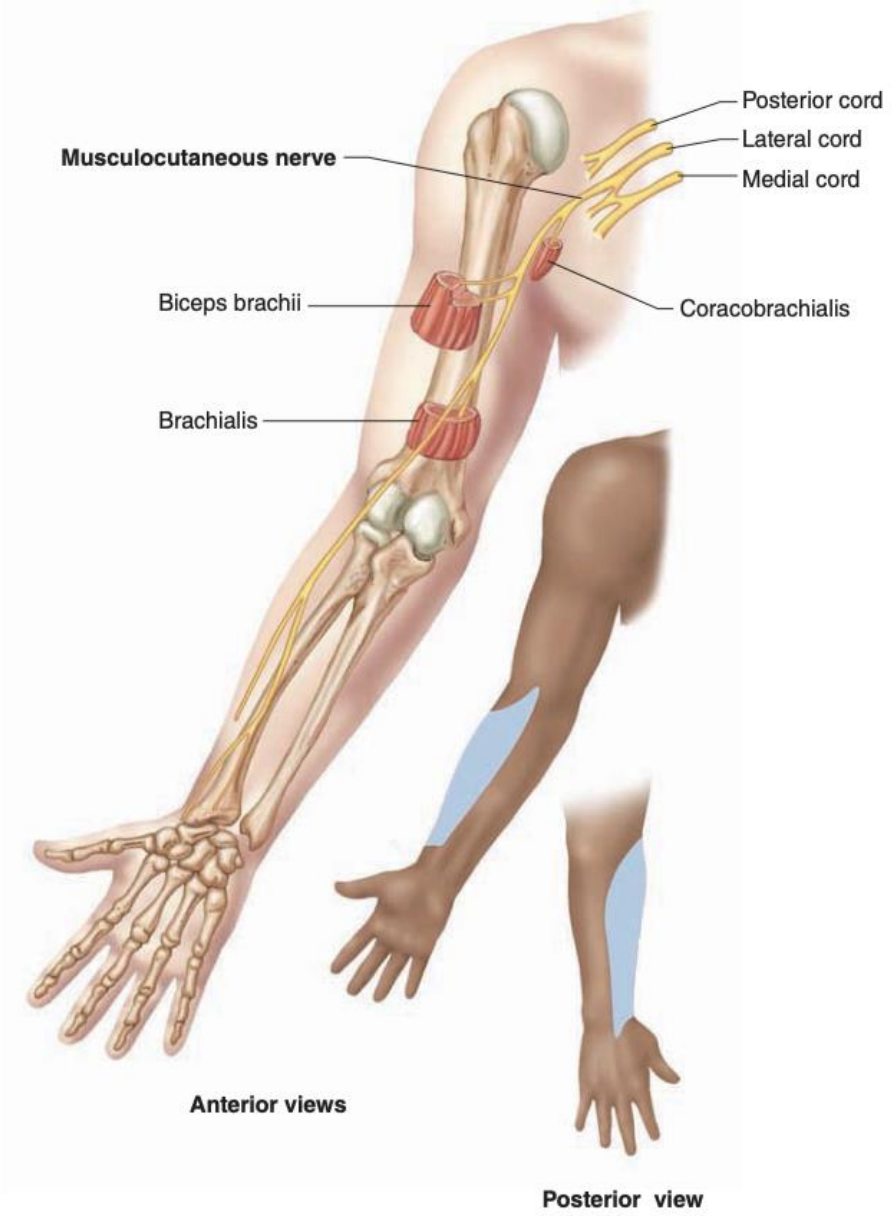


FIGURE 12.20 Musculocutaneous Nerve
The route of the musculocutaneous nerve and the muscles it innervates. The *insets* depict the cutaneous (sensory) distribution of the nerve (*blue areas*).

SPINAL NERVES

BRACHIAL PLEXUS (C5–T1)

Major Nerves of the Brachial Plexus:

4. Ulnar Nerve – Innervates *forearm flexors* and most *intrinsic hand muscles* (except those of the thumb); sensory input to medial hand and little finger.

Ulnar Nerve

Origin

Medial cord of brachial plexus, C8–T1

Movements/Muscles Innervated

Flexes and adducts wrist
- *Flexor carpi ulnaris*

Flexes fingers
- *Part of the flexor digitorum profundus controlling the distal phalanges of little and ring fingers*

Adducts thumb
- *Adductor pollicis*

Controls hypothenar muscles
- *Flexor digiti minimi brevis*
- *Abductor digiti minimi*
- *Opponens digiti minimi*

Flexes metacarpophalangeal joints and extends interphalangeal joints
- *Two medial (ulnar) lumbricales*

Abducts and adducts fingers
- *Interossei*

Cutaneous (Sensory) Innervation

Medial third of hand, little finger, and medial half of ring finger

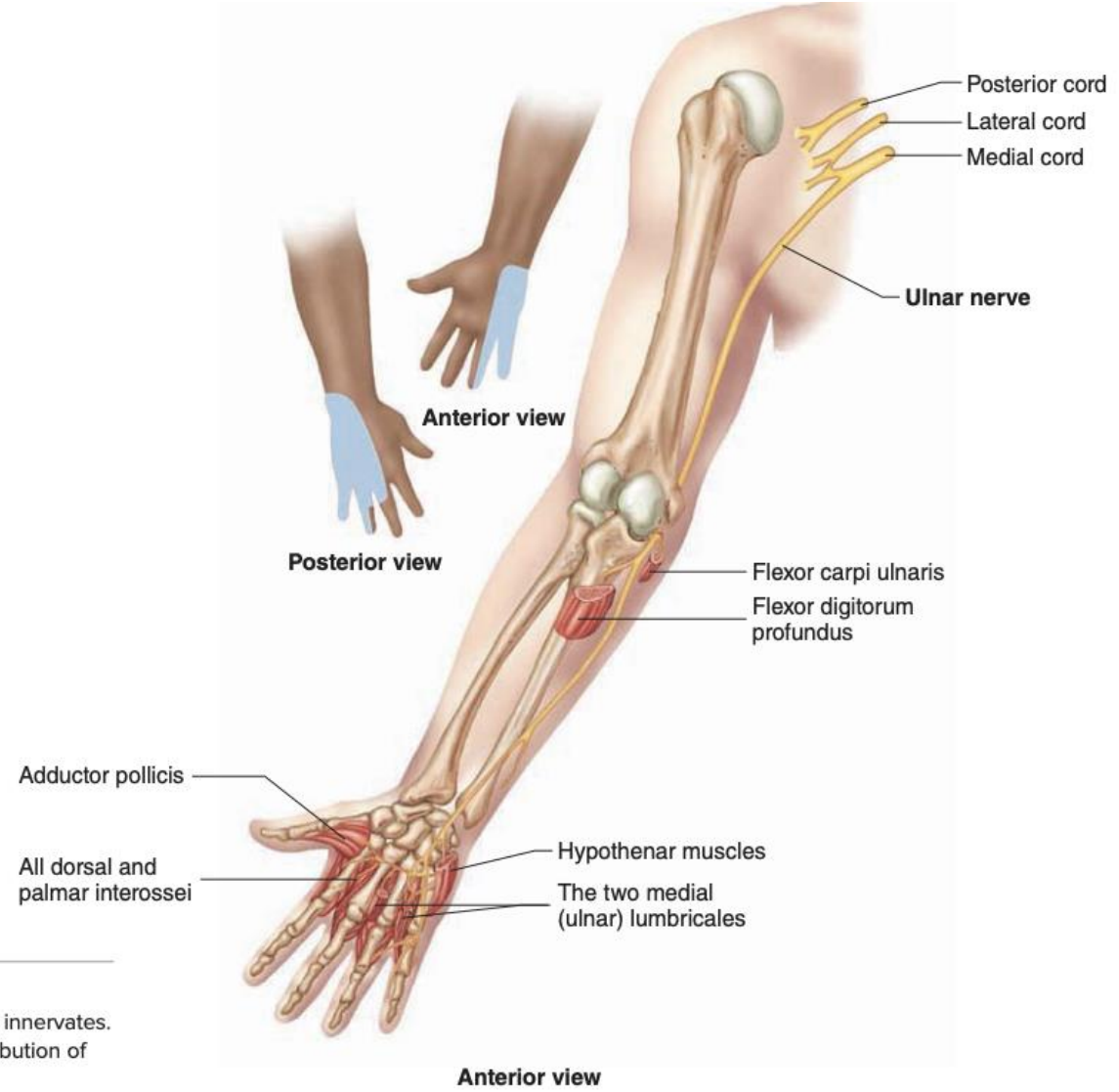


FIGURE 12.21 Ulnar Nerve
The route of the ulnar nerve and the muscles it innervates. The *insets* depict the cutaneous (sensory) distribution of the nerve (*blue areas*).

SPINAL NERVES

BRACHIAL PLEXUS (C5–T1)

Major Nerves of the Brachial Plexus:

5. Median Nerve – Innervates *flexor muscles of the forearm* and *thenar muscles* of the thumb; sensory input to lateral palm and fingers.

Median Nerve

Origin

Medial and lateral cords of brachial plexus, C5–T1

Movements/Muscles Innervated

Pronates forearm and hand
- *Pronator teres*
- *Pronator quadratus*

Flexes and abducts wrist
- *Flexor carpi radialis*

Flexes wrist
- *Palmaris longus*

Flexes fingers
- *Part of flexor digitorum profundus controlling the distal phalanx of the middle and index fingers*
- *Flexor digitorum superficialis*

Controls thumb muscle
- *Flexor pollicis longus*

Controls thenar muscles
- *Abductor pollicis brevis*
- *Opponens pollicis*
- *Flexor pollicis brevis*

Flexes metacarpophalangeal joints and extends interphalangeal joints
- *Two lateral (radial) lumbricales*

Cutaneous (Sensory) Innervation

Lateral two-thirds of palm of hand, thumb, index and middle fingers, and the lateral half of ring finger and dorsal tips of the same fingers

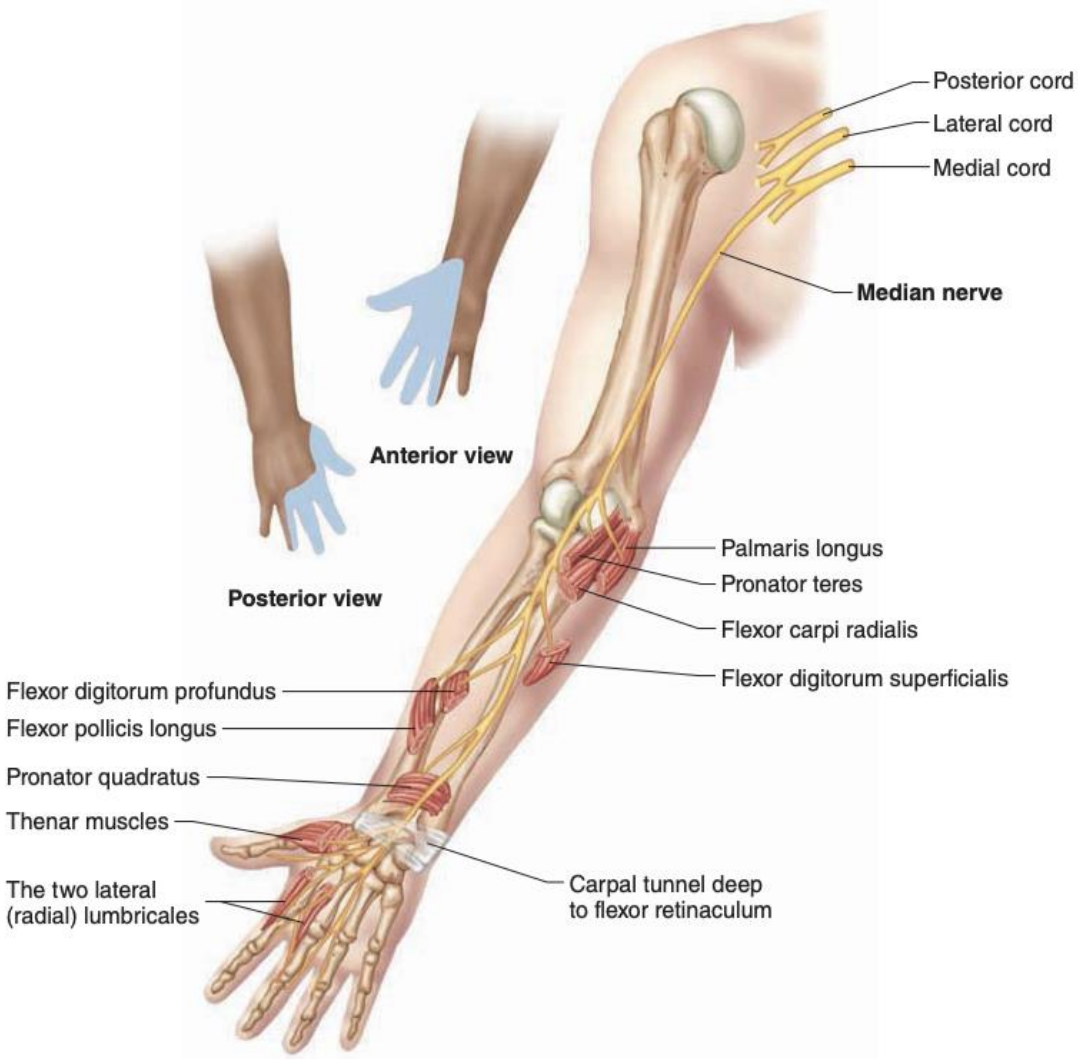


FIGURE 12.22 Median Nerve
The route of the median nerve and the muscles it innervates. The *insets* depict the cutaneous (sensory) distribution of the nerve (*blue areas*). APR

SPINAL NERVES

BRACHIAL PLEXUS (C5–T1)

Other Nerves:

- **Pectoral, long thoracic, thoracodorsal, subscapular, and suprascapular nerves** supply muscles of the scapula and upper chest.
- **Clinical Note:** The **entire upper limb** can be anesthetized by injecting near the brachial plexus (brachial anesthesia).

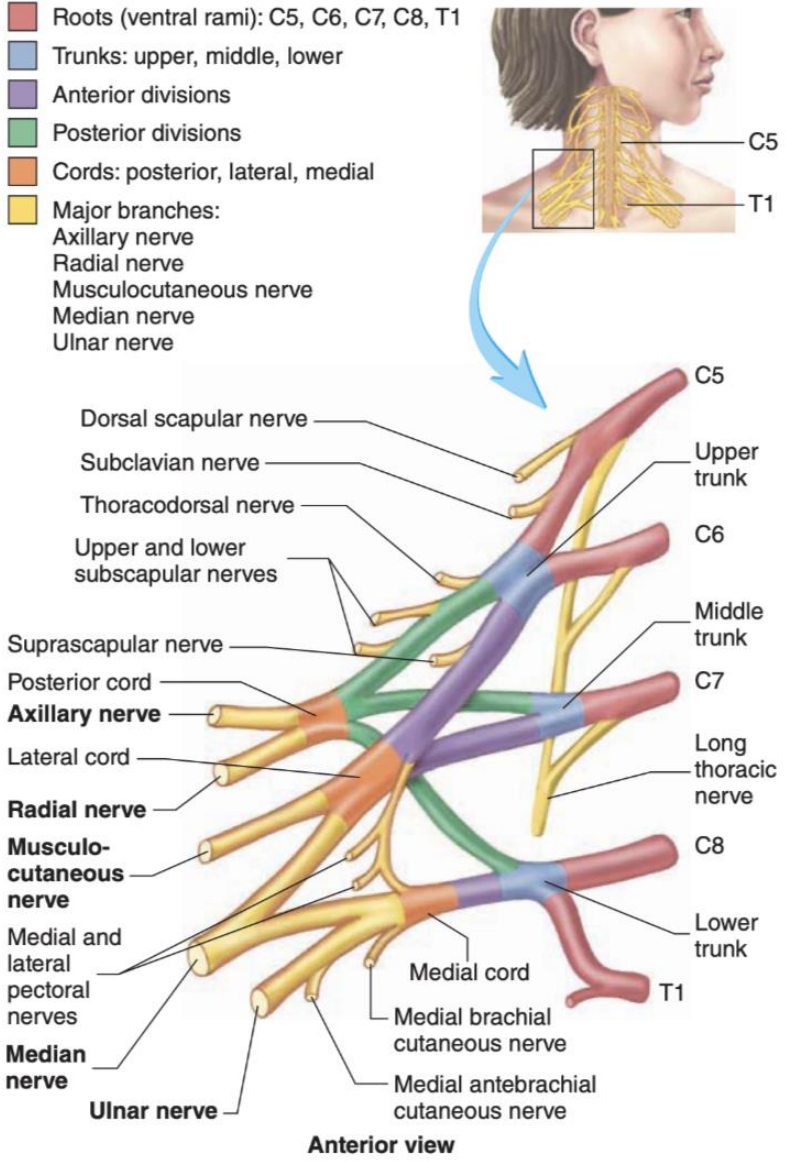


FIGURE 12.17 Brachial Plexus

The roots of the plexus are formed by the ventral rami of spinal nerves C5–T1 and join to form an upper, a middle, and a lower trunk. Each trunk divides into anterior and posterior divisions. The divisions join to form the posterior, lateral, and medial cords from which the major brachial plexus nerves arise. The major brachial plexus nerves include the axillary, radial, musculocutaneous, median, and ulnar nerves, which innervate the muscles and skin of the upper limb. AP|R

SPINAL NERVES

LUMBAR AND SACRAL PLEXUSES (L1–S4)

- The **lumbar plexus** arises from **L1–L4**, and the **sacral plexus** arises from **L4–S4**.
- Together, they are often referred to as the **lumbosacral plexus**, which supplies the **lower back, pelvis, and lower limbs**.

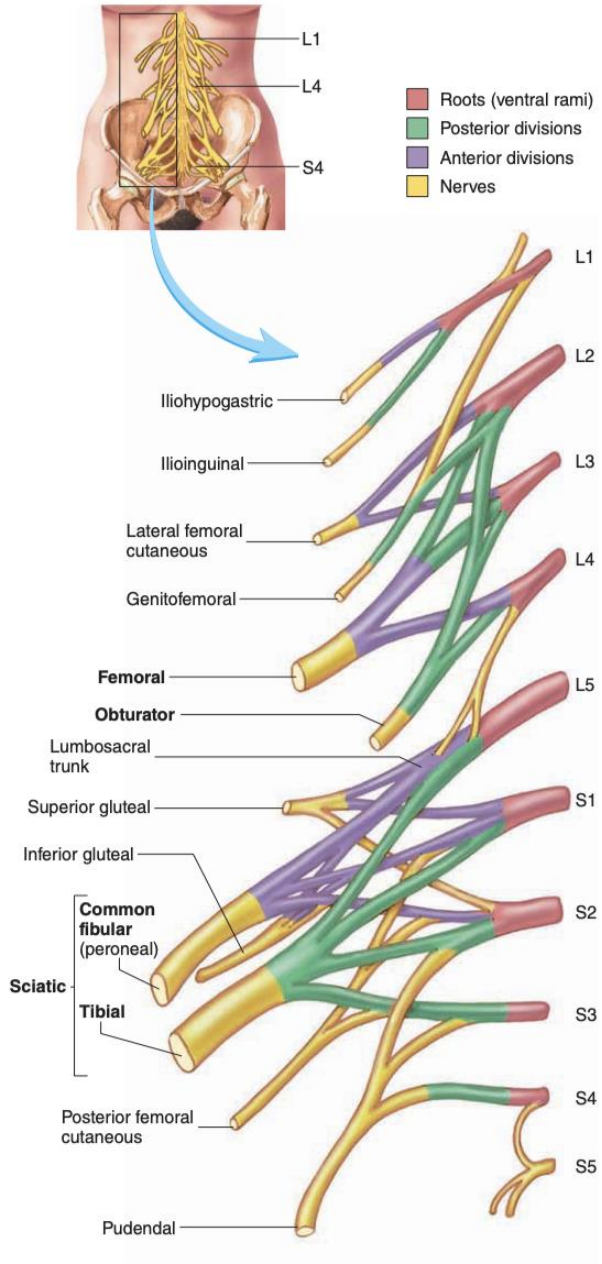


FIGURE 12.23 Lumbosacral Plexus
The roots of the plexus are formed by the ventral rami of the spinal nerves L1–S4 and form anterior and posterior divisions, which give rise to the lumbosacral nerves. The lumbosacral trunk joins the lumbar and sacral plexuses. AP|R

SPINAL NERVES

LUMBAR AND SACRAL PLEXUSES (L1–S4)

Major Nerves of the Lumbosacral Plexus:

1. **Obturator Nerve (L2–L4)**
2. **Femoral Nerve (L2–L4)**
3. **Tibial Nerve (L4–S3)**
4. **Common Fibular (Peroneal) Nerve (L4–S2)**

Other Branches:

- **Gluteal nerves**
- **Pudendal nerve, illiohypogastric, ilioinguinal, genitofemoral, and cutaneous femoral nerves**

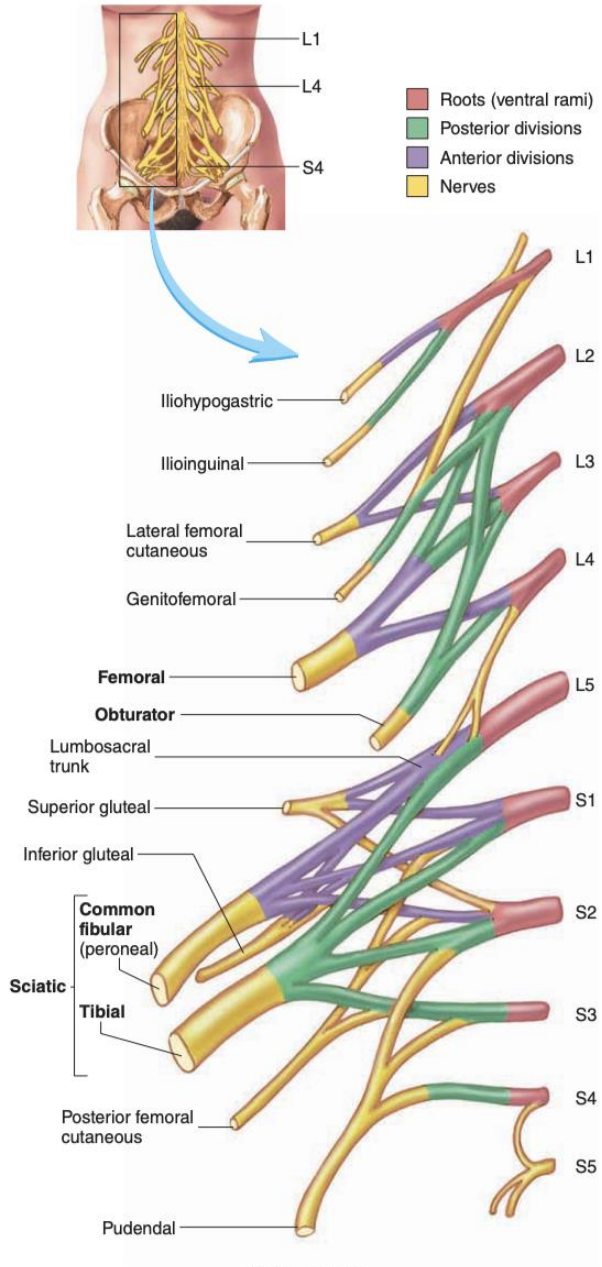


FIGURE 12.23 Lumbosacral Plexus
The roots of the plexus are formed by the ventral rami of the spinal nerves L1–S4 and form anterior and posterior divisions, which give rise to the lumbosacral nerves. The lumbosacral trunk joins the lumbar and sacral plexuses. AP|R

SPINAL NERVES

LUMBAR AND SACRAL PLEXUSES (L1–S4)

Major Nerves of the Lumbosacral Plexus:

1. Obturator Nerve (L2–L4)
– Innervates the *adductor muscles* of the thigh; provides sensation to the medial thigh.

Obturator Nerve

Origin

Lumbosacral plexus, L2–L4

Movements/Muscles Innervated

Rotates thigh laterally
- *Obturator externus*

Adducts thigh
- *Adductor magnus (adductor part)*
- *Adductor longus*
- *Adductor brevis*

Adducts thigh and flexes knee
- *Gracilis*

Cutaneous (Sensory) Innervation

Superior medial side of thigh

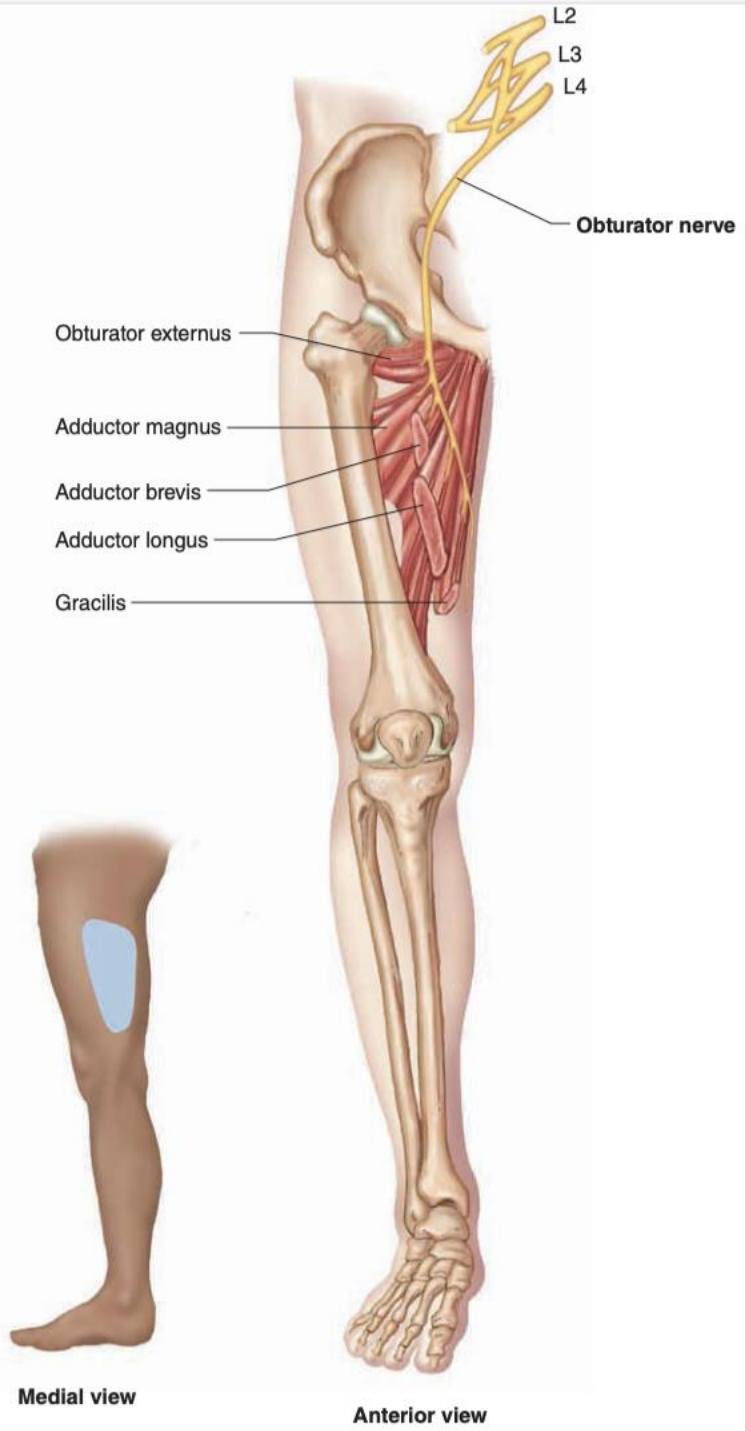


FIGURE 12.24 Obturator Nerve
The route of the obturator nerve and the muscles it innervates. The *inset* depicts the cutaneous (sensory) distribution of the nerve (*blue area*).

SPINAL NERVES

LUMBAR AND SACRAL PLEXUSES (L1–S4)

Major Nerves of the Lumbosacral Plexus:

2. Femoral Nerve (L2–L4) – Innervates *iliopsoas, sartorius,* and *quadriceps femoris* muscles; sensory distribution includes the anterior and lateral thigh and medial leg and foot.

Femoral Nerve

Origin

Lumbosacral plexus, L2–L4

Movements/Muscles Innervated

Flexes hip
- *Psoas major*
- *Iliacus*
- *Pectineus*

Flexes hip and flexes knee
- *Sartorius*

Extends knee
- *Vastus lateralis*
- *Vastus intermedius*
- *Vastus medialis*

Extends knee and flexes hip
- *Rectus femoris*

Cutaneous (Sensory) Innervation

Anterior and lateral branches supply the anterior and lateral thigh; saphenous branch supplies the medial leg and foot

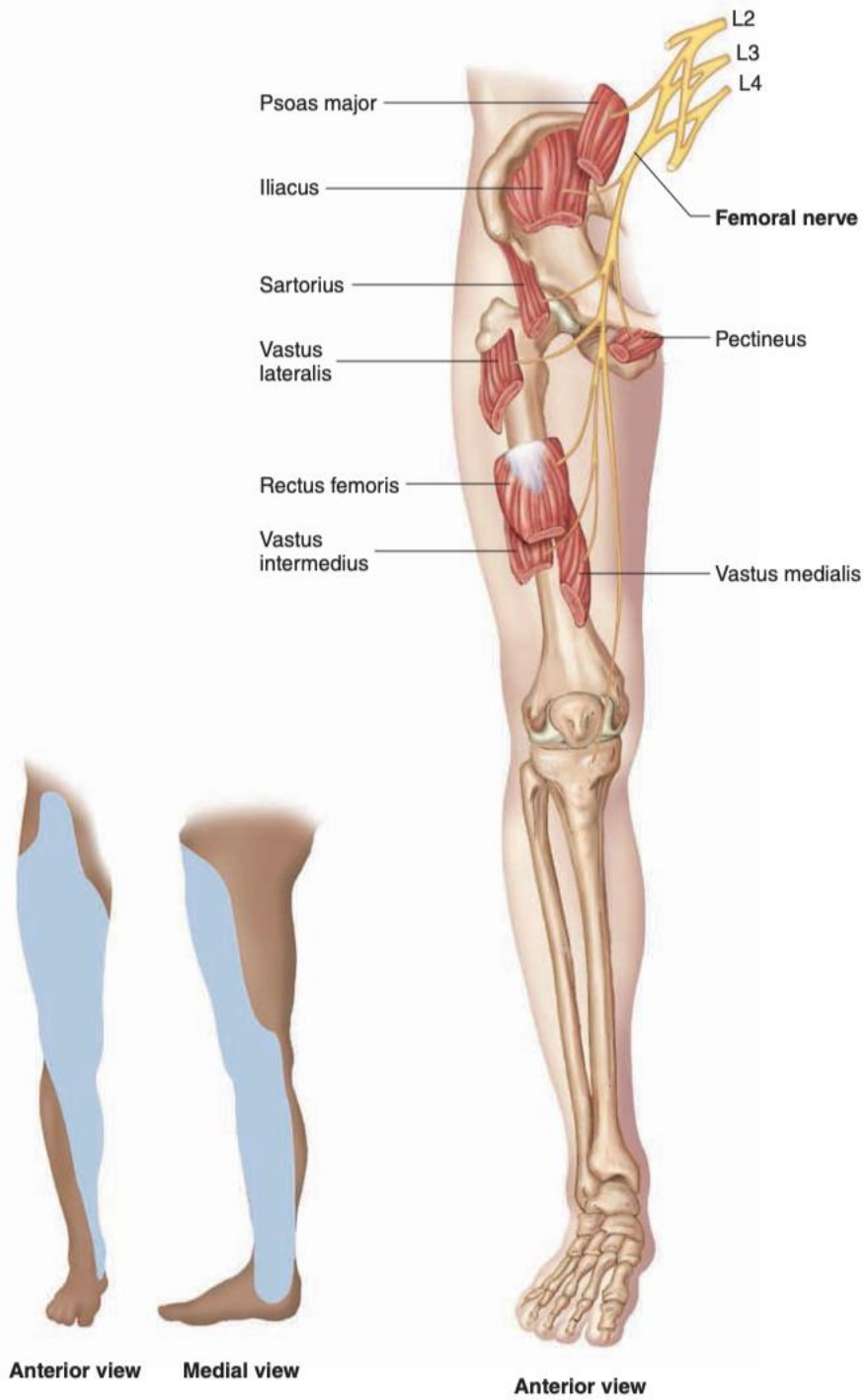


FIGURE 12.25 Femoral Nerve
The route of the femoral nerve and the muscles it innervates. The *insets* depict the cutaneous (sensory) distribution of the nerve (*blue areas*).

SPINAL NERVES

LUMBAR AND SACRAL PLEXUSES (L1–S4)

Major Nerves of the Lumbosacral Plexus:

3. Tibial Nerve (L4–S3) – Innervates *posterior thigh and leg muscles*; branches into *medial and lateral plantar nerves* in the foot; provides sensory innervation to the sole of the foot.

Tibial Nerve

Origin
Lumbosacral plexus, L4–S3

Movements/Muscles Innervated
Extends hip and flexes knee
- *Biceps femoris (long head)*
- *Semitendinosus*
- *Semimembranosus*

Extends hip and adducts thigh
- *Adductor magnus (hamstring part)*

Plantar flexes foot
- *Plantaris*
- *Gastrocnemius*
- *Soleus*
- *Tibialis posterior*

Flexes knee
- *Popliteus*

Flexes toes
- *Flexor digitorum longus*
- *Flexor hallucis longus*

Cutaneous (Sensory) Innervation
None

Medial and Lateral Plantar Nerves

Origin
Tibial nerve

Movements/Muscles Innervated
Flex and adduct toes
- *Plantar muscles of foot*

Cutaneous (Sensory) Innervation
Sole of foot

Sural Nerve (Not Shown)

Origin
Tibial nerve

Movements/Muscles Innervated
None

Cutaneous (Sensory) Innervation
Lateral and posterior one-third of leg and lateral side of foot

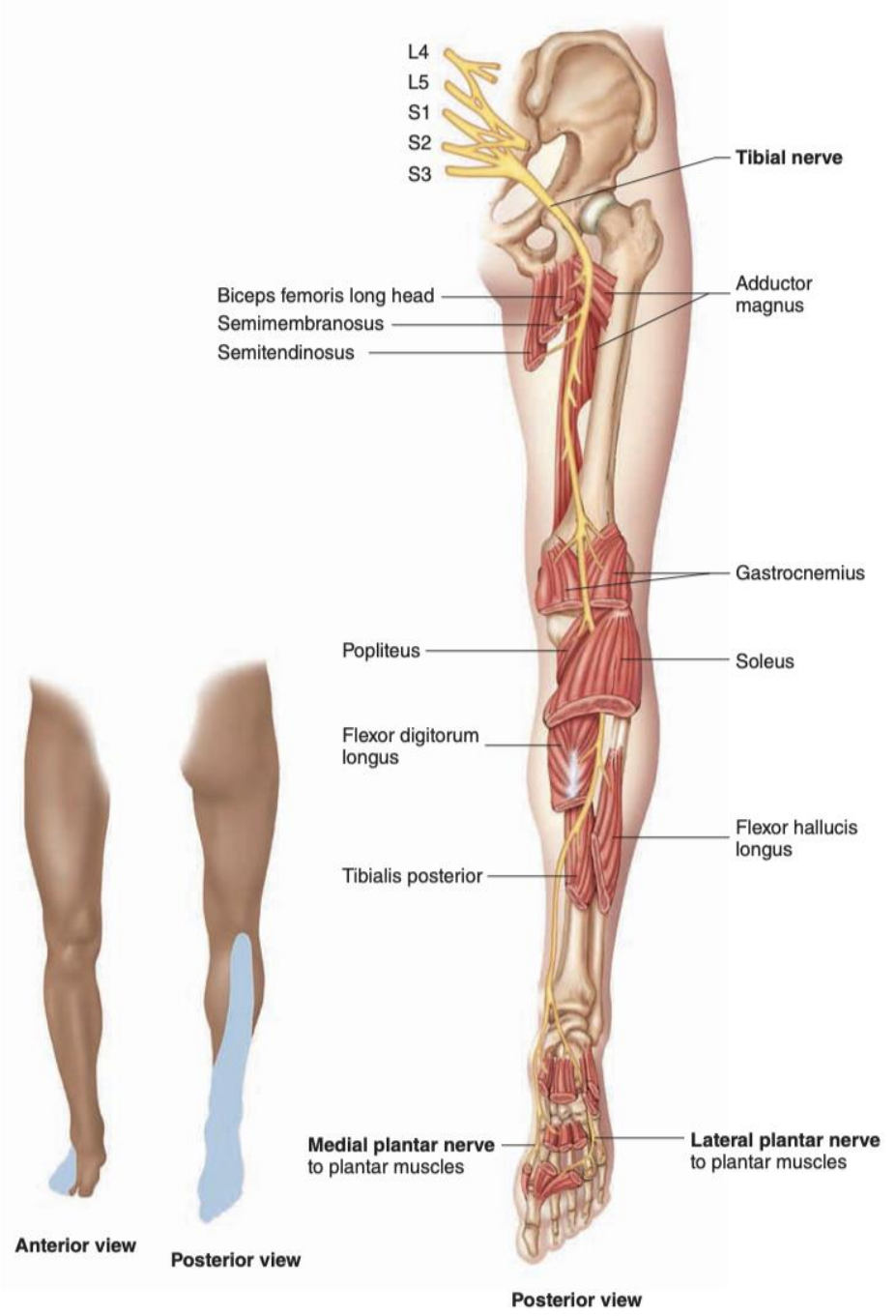


FIGURE 12.26 Tibial Nerve
The route of the tibial nerve and the muscles it innervates. The *insets* depict the cutaneous (sensory) distribution of the nerve (*blue areas*).

SPINAL NERVES

LUMBAR AND SACRAL PLEXUSES (L1–S4)

Major Nerves of the Lumbosacral Plexus:

4. Common Fibular (Peroneal) Nerve (L4–S2) – Innervates *anterior and lateral leg and foot muscles*; provides sensation to the lateral leg and dorsum of the foot.

- Together, the **tibial and common fibular nerves** form the **sciatic nerve**, the *largest peripheral nerve* in the body.

Common Fibular (Peroneal) Nerve

Origin
Lumbosacral plexus, L4–S2

Movements/Muscles Innervated
Extends hip and flexes knee
- *Biceps femoris (short head)*

Cutaneous (Sensory) Innervation
Lateral surface of knee

Deep Fibular (Peroneal) Nerve

Origin
Common fibular (peroneal) nerve

Movements/Muscles Innervated
Dorsiflexes foot
- *Tibialis anterior*
- *Fibularis tertius*

Extends toes
- *Extensor digitorum longus*
- *Extensor hallucis longus*
- *Extensor digitorum brevis*

Cutaneous (Sensory) Innervation
Great and second toes

Superficial Fibular (Peroneal) Nerve

Origin
Common fibular (peroneal) nerve

Movements/Muscles Innervated
Plantar flexes and everts foot
- *Fibularis longus*
- *Fibularis brevis*

Cutaneous (Sensory) Innervation
Dorsal anterior third of leg and dorsum of foot

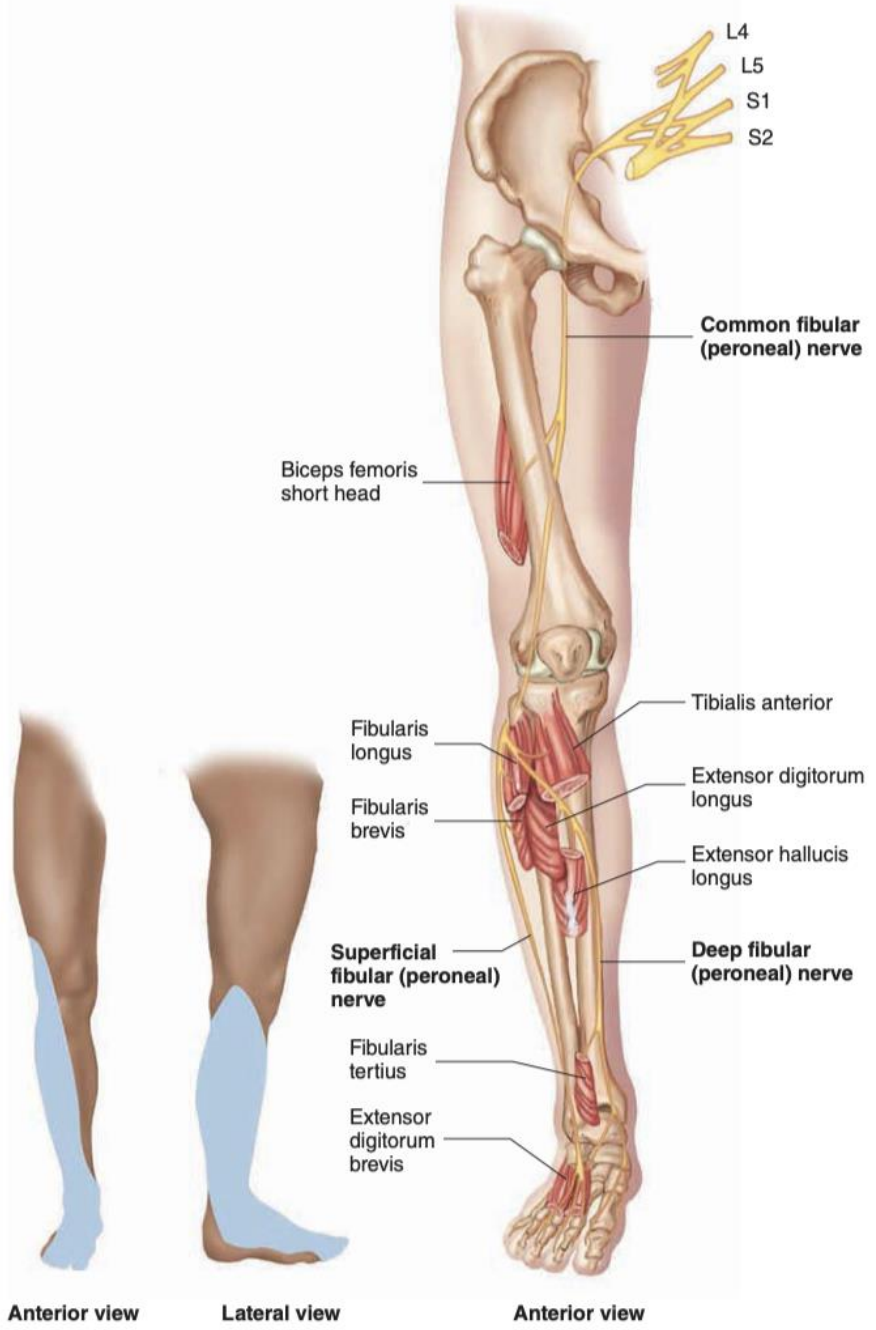


FIGURE 12.27 Fibular Nerve
The route of the common fibular (peroneal) nerve and the muscles it innervates. The *insets* depict the cutaneous (sensory) distribution of the nerve (*blue areas*).

SPINAL NERVES

LUMBAR AND SACRAL PLEXUSES (L1–S4)

Other Branches:

- **Gluteal nerves –** Supply hip muscles.
- **Pudendal nerve –** Innervates perineal muscles; plays a key role in sexual function and is anesthetized during childbirth.
- **Iliohypogastric, ilioinguinal, genitofemoral, and cutaneous femoral nerves –** Innervate the skin of the suprapubic area, external genitalia, and parts of the thigh.

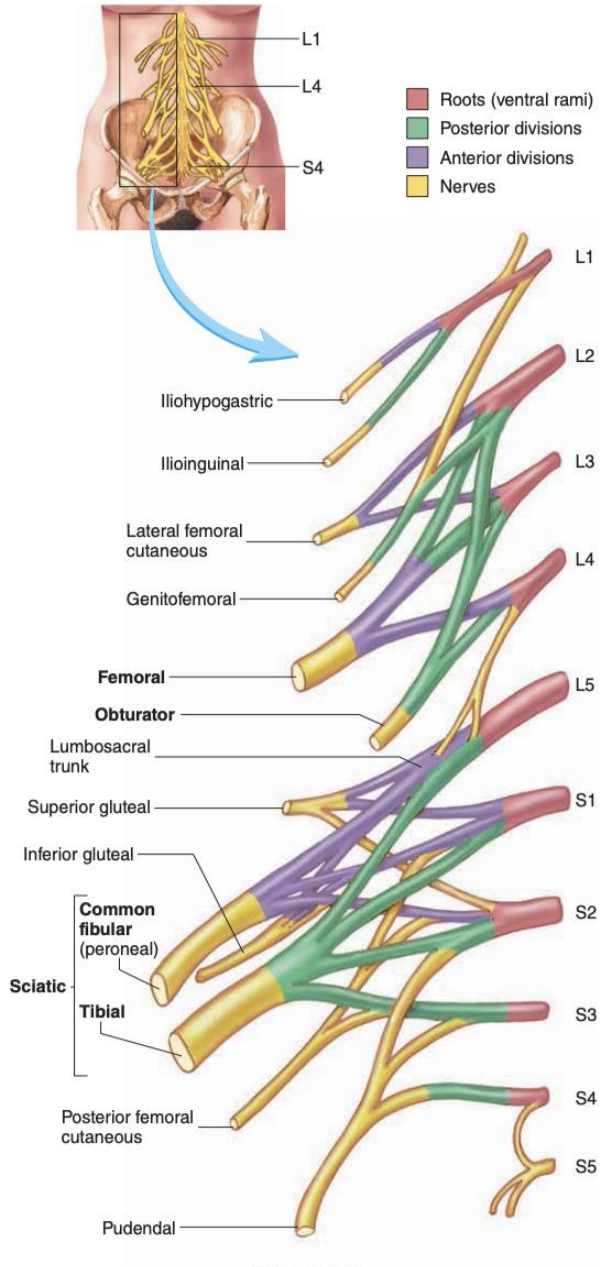


FIGURE 12.23 Lumbosacral Plexus
The roots of the plexus are formed by the ventral rami of the spinal nerves L1–S4 and form anterior and posterior divisions, which give rise to the lumbosacral nerves. The lumbosacral trunk joins the lumbar and sacral plexuses. AP|R

CRANIAL NERVES

STRUCTURAL CLASSIFICATION

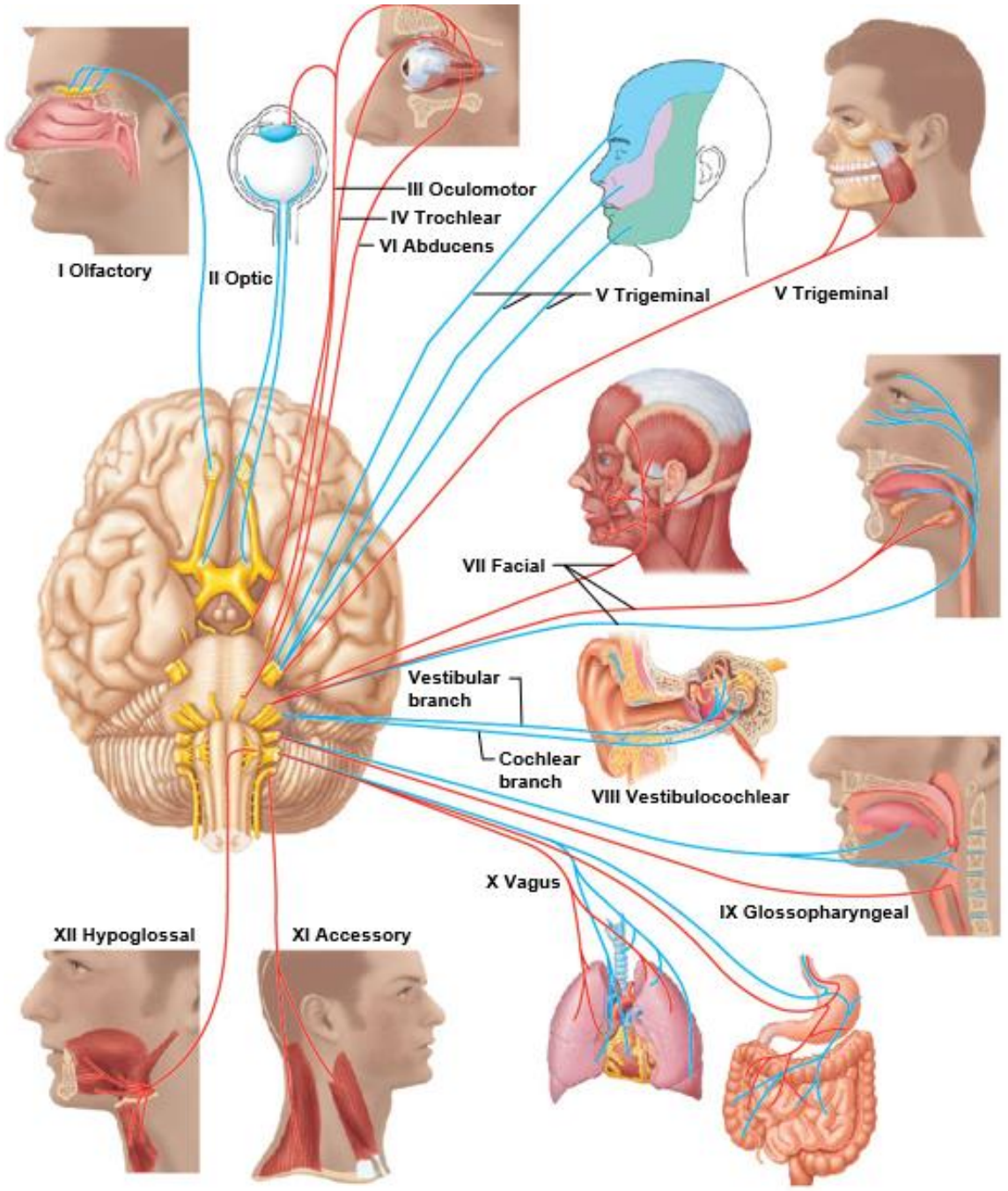
Cranial nerves are 12 pairs of nerves that transmit sensory, motor, and parasympathetic information directly to and from the **brain** (not the spinal cord).

Each cranial nerve may have one or more functions:

1. **Sensory** – transmits information from special senses (vision, smell, taste) and general senses (touch, pain).
2. **Somatic Motor** – controls skeletal muscles; also carries proprioceptive fibers for awareness of body position.
3. **Parasympathetic (Autonomic)** – regulates glands, smooth muscle, and cardiac muscle activity.

The **12 pairs** of cranial nerves are numbered **I–XII** using **Roman numerals**, arranged from **anterior to posterior**.

The **first two** connect to the **cerebrum and diencephalon**; most others arise from the **brainstem**, except the **accessory nerve (XI)**, which also has a spinal component.



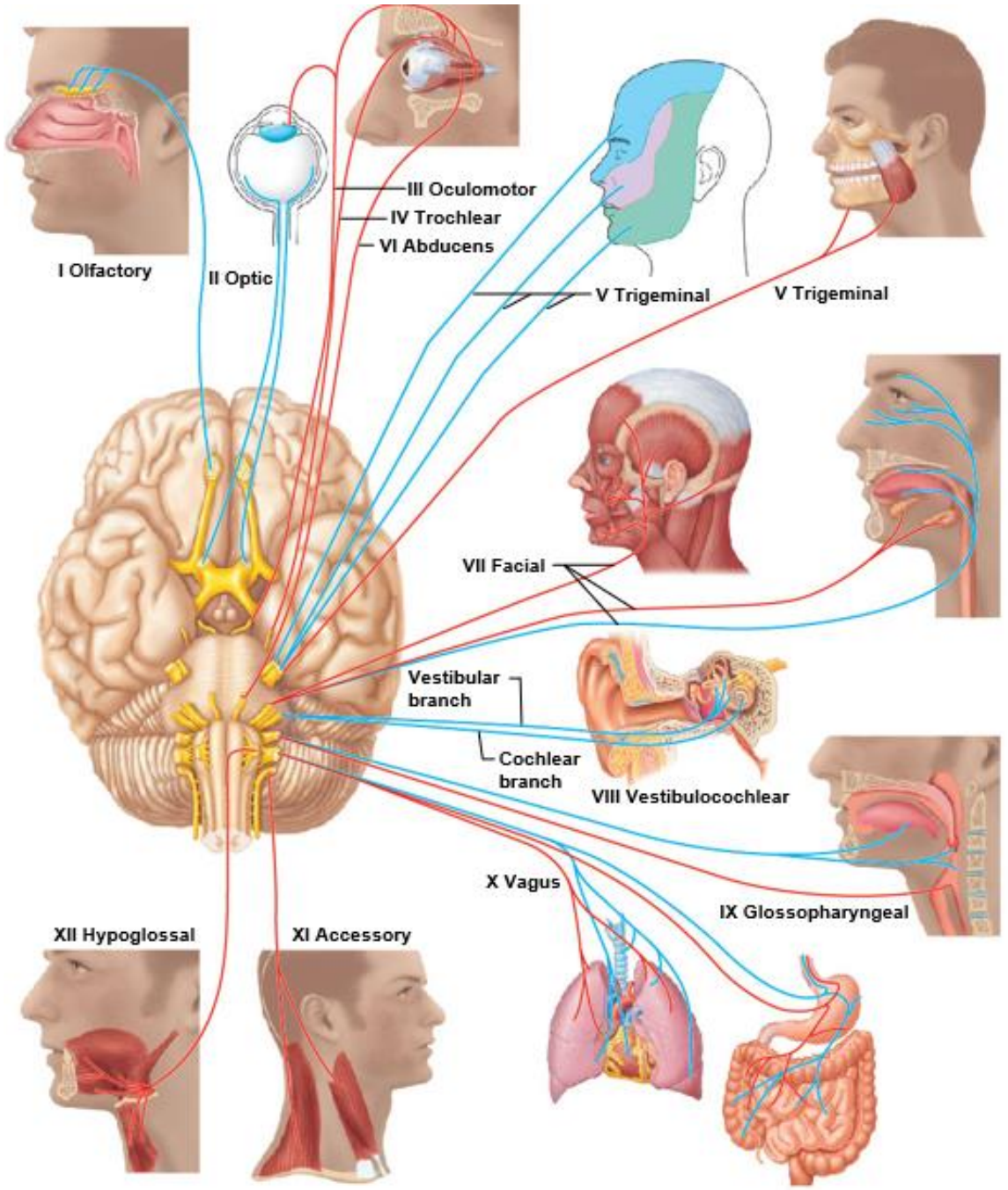
CRANIAL NERVES

STRUCTURAL CLASSIFICATION CRANIAL NERVES

Cranial nerves are **12 pairs of nerves** that emerge directly from the **brain and brainstem** (rather than the spinal cord). They primarily serve the **head and neck** regions.

General Characteristics:

- **12 pairs** numbered I–XII (Roman numerals).
- Most are **mixed nerves** (carry both sensory and motor fibers).
- **Three pairs** are **purely sensory:**
  - **Olfactory (I):** smell
  - **Optic (II):** vision
  - **Vestibulocochlear (VIII):** hearing and balance
- The **Vagus (X) nerve** is unique—it extends beyond the head and neck to the **thoracic and abdominal cavities,** innervating visceral organs.



NERVES

CRANIAL NERVES

Olfactory (I)

Optic (II)

Oculomotor (III)

Trochlear (IV)

Trigeminal (V)

Abducens (VI)

Facial (VII)

Vestibulocochlear (VIII)

Glossopharyngeal (IX)

Vagus (X)

Accessory (XI)

Hypoglossal (XII)

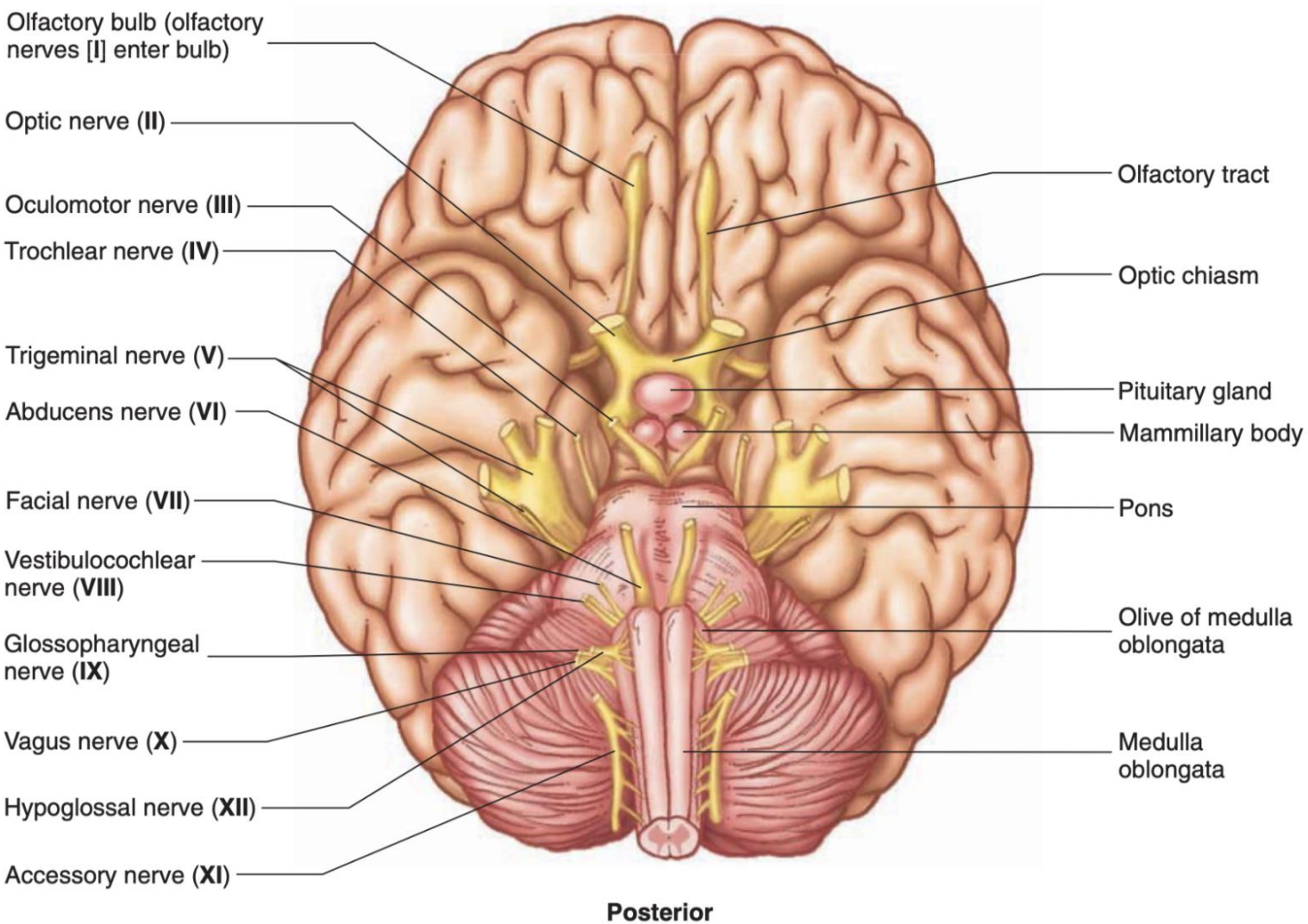


FIGURE 13.15 Inferior Surface of the Brain, Showing the Origins of the Cranial Nerves AP|R

NERVES

CRANIAL NERVES

Oh – **Olfactory (I)**

Oh – **Optic (II)**

Oh – **Oculomotor (III)**

To – **Trochlear (IV)**

Touch – **Trigeminal (V)**

And – **Abducens (VI)**

Feel – **Facial (VII)**

Very – **Vestibulocochlear (VIII)**

Good – **Glossopharyngeal (IX)**

Velvets – **Vagus (X)**

Ah – **Accessory (XI)**

Heaven – **Hypoglossal (XII)**

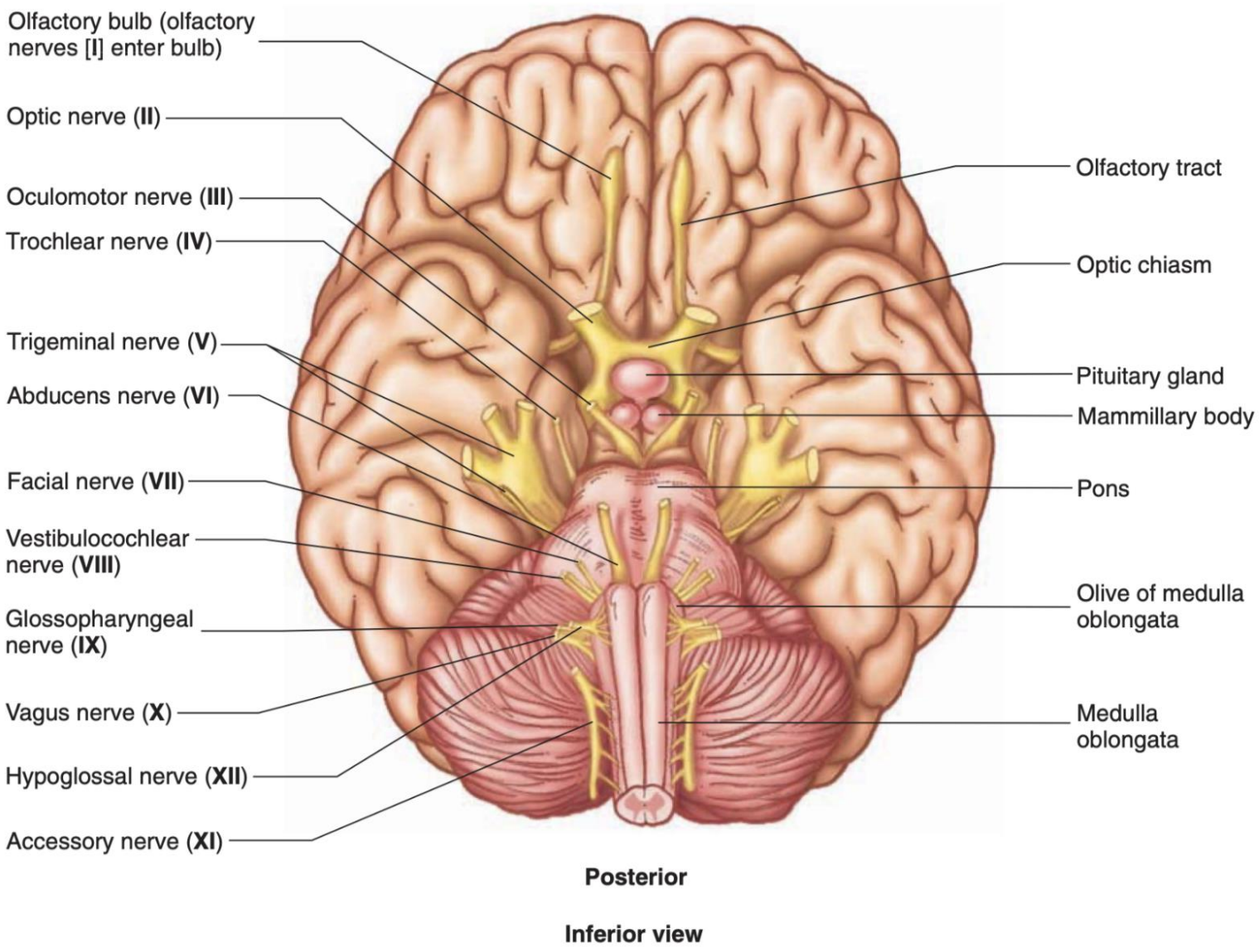
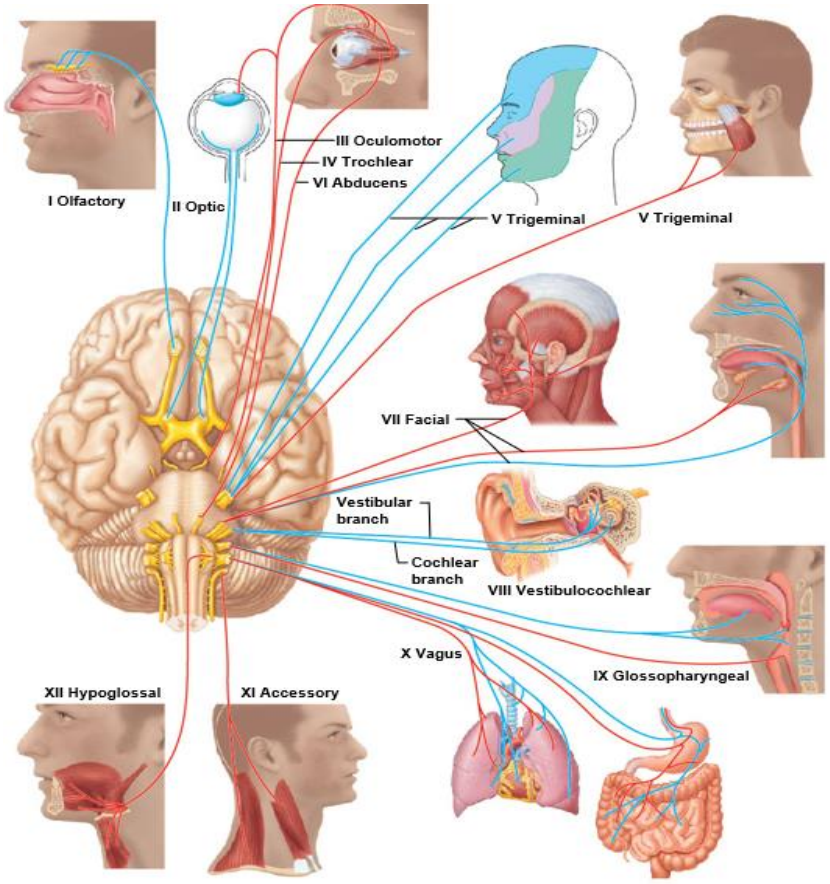


FIGURE 13.15 Inferior Surface of the Brain, Showing the Origins of the Cranial Nerves AP|R

Table **7.2** **The Cranial Nerves**

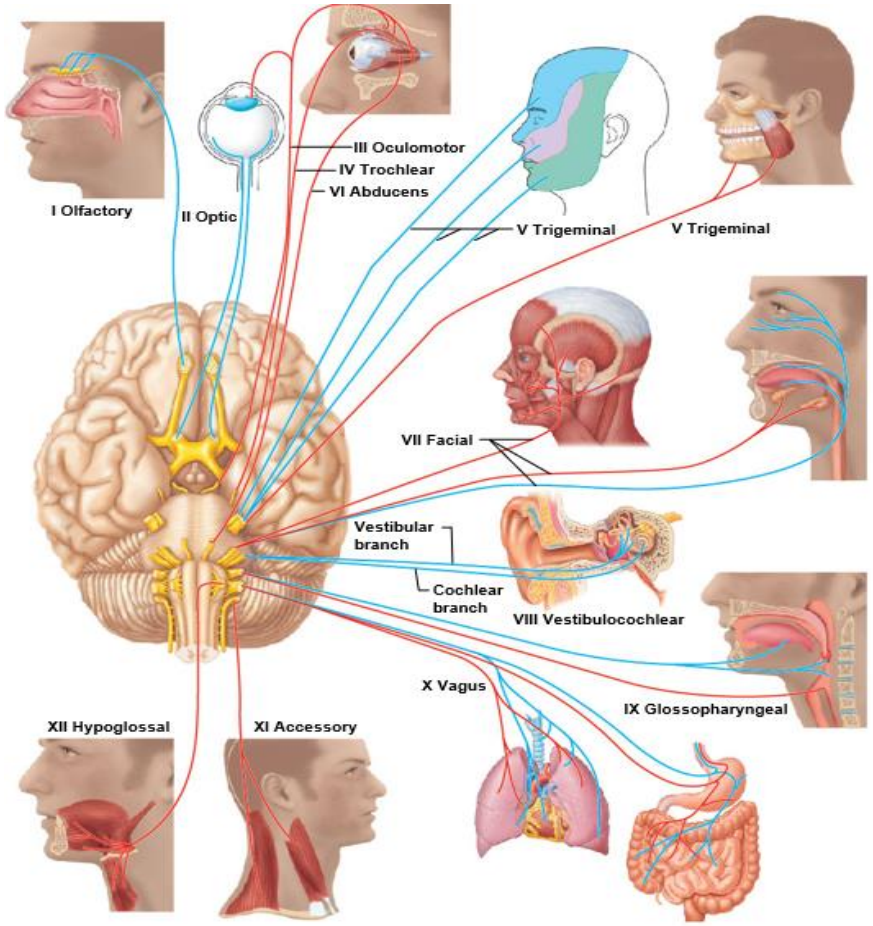
Name/number	Origin/course	Function	Test
I. Olfactory	Fibers arise from olfactory receptors in the nasal mucosa and synapse with the olfactory bulbs (which, in turn, send fibers to the olfactory cortex)	Purely sensory; carries impulses for the sense of smell	Subject is asked to sniff and identify aromatic substances, such as oil of cloves or vanilla
II. Optic	Fibers arise from the retina of the eye and form the optic nerve. The two optic nerves form the optic chiasma by partial crossover of fibers; the fibers continue to the optic cortex as the optic tracts	Purely sensory; carries impulses for vision	Vision and visual field are tested with an eye chart and by testing the point at which the subject first sees an object (finger) moving into the visual field; eye interior is viewed with an ophthalmoscope



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Table **7.2** **The Cranial Nerves** (*continued*)

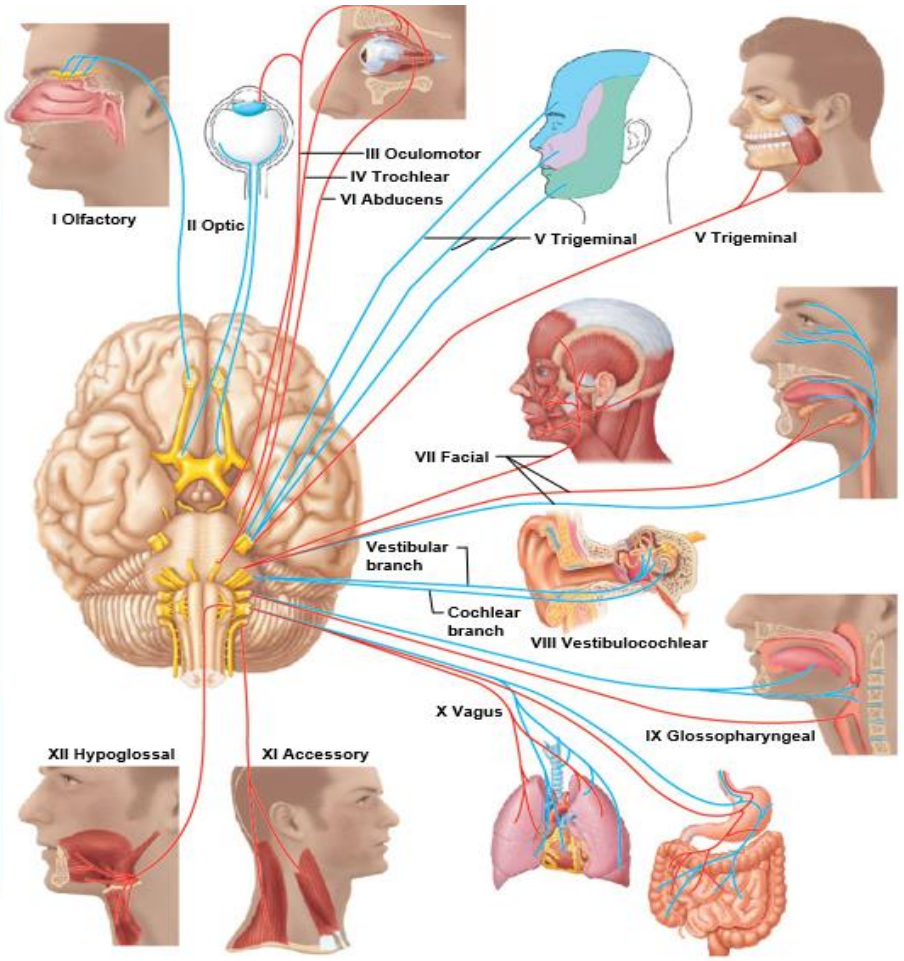
Name/number	Origin/course	Function	Test
III. Oculomotor	Fibers run from the midbrain to the eye	Supplies motor fibers to four of the six muscles (superior, inferior, and medial rectus, and inferior oblique) that direct the eyeball; to the eyelid; and to the internal eye muscles controlling lens shape and pupil size	Pupils are examined for size, shape, and size equality; pupillary reflex is tested with a penlight (pupils should constrict when illuminated); eye convergence is tested, as is the ability to follow moving objects
IV. Trochlear	Fibers run from the midbrain to the eye	Supplies motor fibers for one external eye muscle (superior oblique)	Tested in common with cranial nerve III for the ability to follow moving objects



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Table **7.2** **The Cranial Nerves** (*continued*)

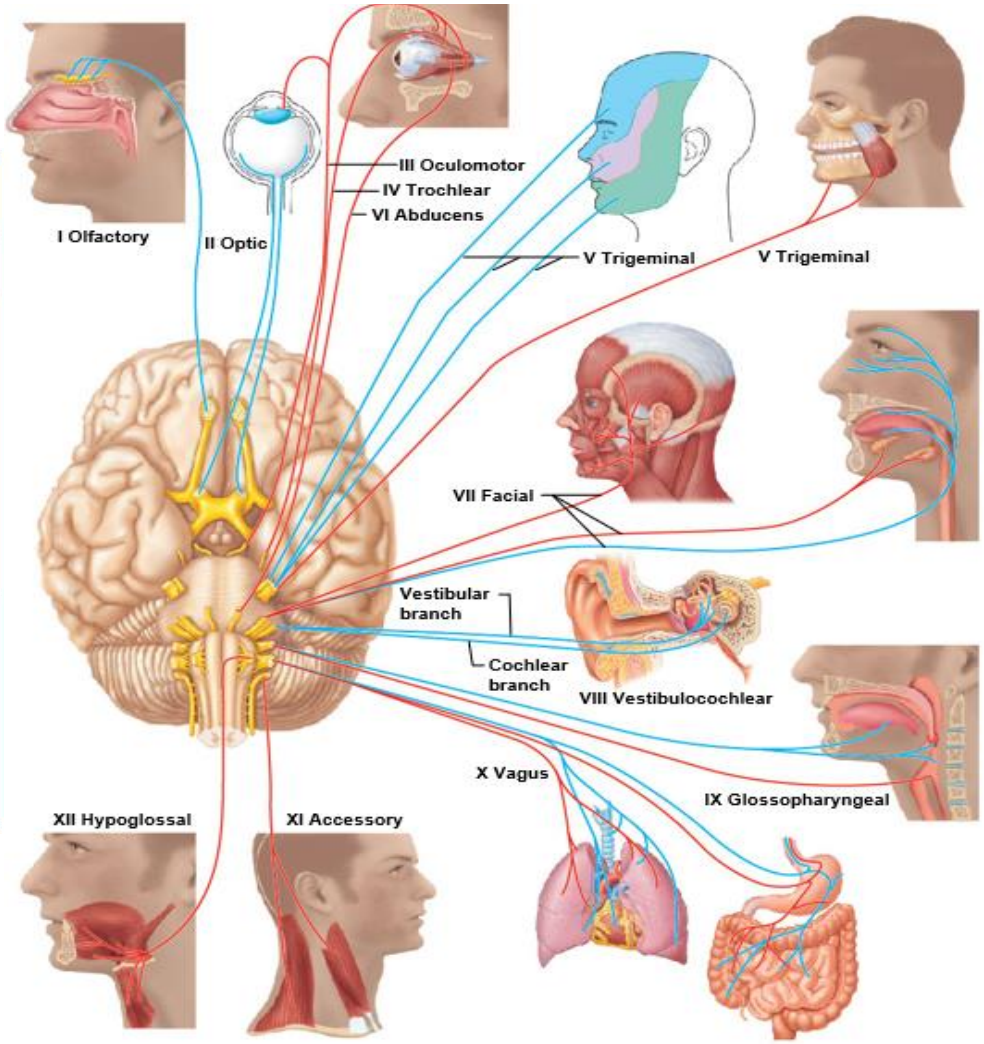
Name/number	Origin/course	Function	Test
V. Trigeminal	Fibers emerge from the pons and form three divisions that run to the face	Conducts sensory impulses from the skin of the face and mucosa of the nose and mouth; also contains motor fibers that activate the chewing muscles	Sensations of pain, touch, and temperature are tested with a safety pin and hot and cold objects; corneal reflex tested with a wisp of cotton; motor branch tested by asking the subject to open mouth against resistance and move jaw from side to side
VI. Abducens	Fibers leave the pons and run to the eye	Supplies motor fibers to the lateral rectus muscle, which rolls the eye laterally	Tested in common with cranial nerve III for the ability to move each eye laterally



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Table **7.2** **The Cranial Nerves** (*continued*)

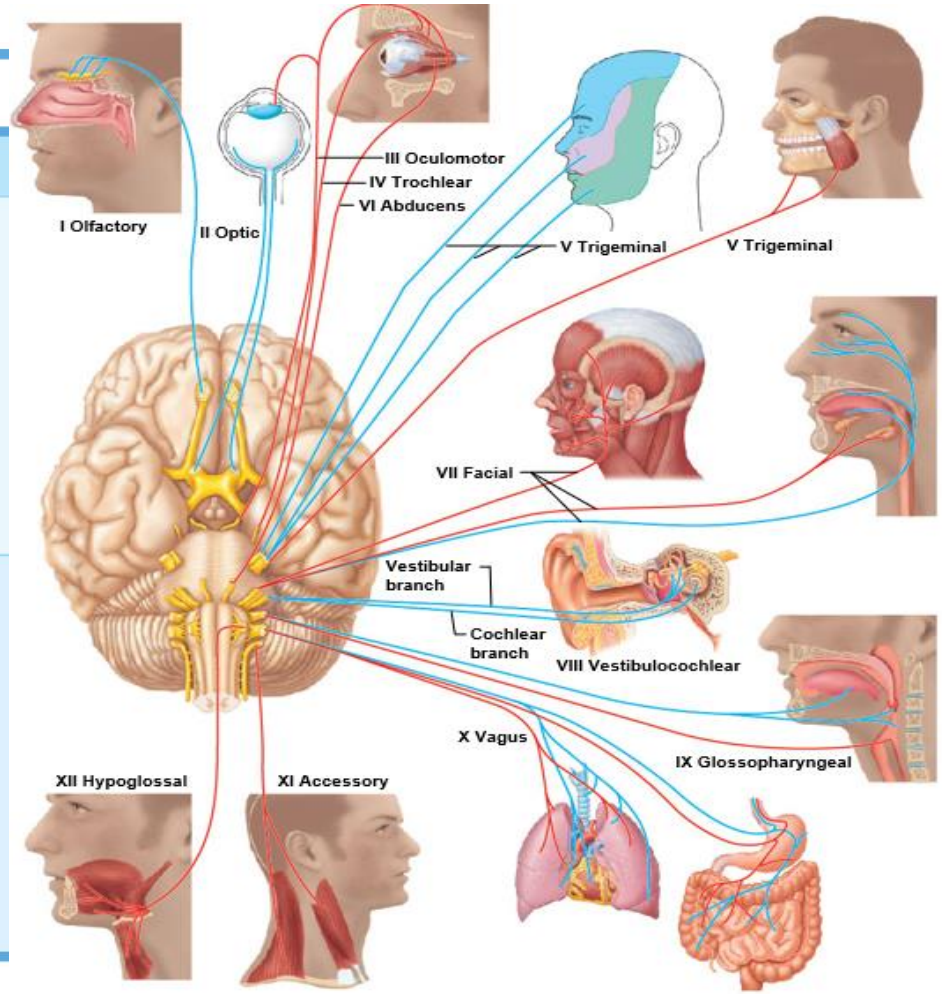
Name/number	Origin/course	Function	Test
VII. Facial	Fibers leave the pons and run to the face	Activates the muscles of facial expression and the lacrimal and salivary glands; carries sensory impulses from the taste buds of anterior tongue	Anterior two-thirds of tongue is tested for ability to taste sweet, salty, sour, and bitter substances; subject is asked to close eyes, smile, whistle, etc.; tearing of eyes is tested with ammonia fumes
VIII. Vestibulocochlear	Fibers run from the equilibrium and hearing receptors of the inner ear to the brain stem	Purely sensory; vestibular branch transmits impulses for the sense of balance, and cochlear branch transmits impulses for the sense of hearing	Hearing is checked by air and bone conduction, using a tuning fork



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Table 7.2 **The Cranial Nerves** (*continued*)

Name/number	Origin/course	Function	Test
IX. Glossopharyngeal	Fibers emerge from the medulla and run to the throat	Supplies motor fibers to the pharynx (throat) that promote swallowing and saliva production; carries sensory impulses from taste buds of the posterior tongue and from pressure receptors of the carotid artery	Gag and swallowing reflexes are checked; subject is asked to speak and cough; posterior tongue may be tested for taste
X. Vagus	Fibers emerge from the medulla and descend into the thorax and abdominal cavity	Fibers carry sensory impulses from and motor impulses to the pharynx, larynx, and the abdominal and thoracic viscera; most motor fibers are parasympathetic fibers that promote digestive activity and help regulate heart activity	Tested in common with cranial nerve IX because they both serve muscles of the throat

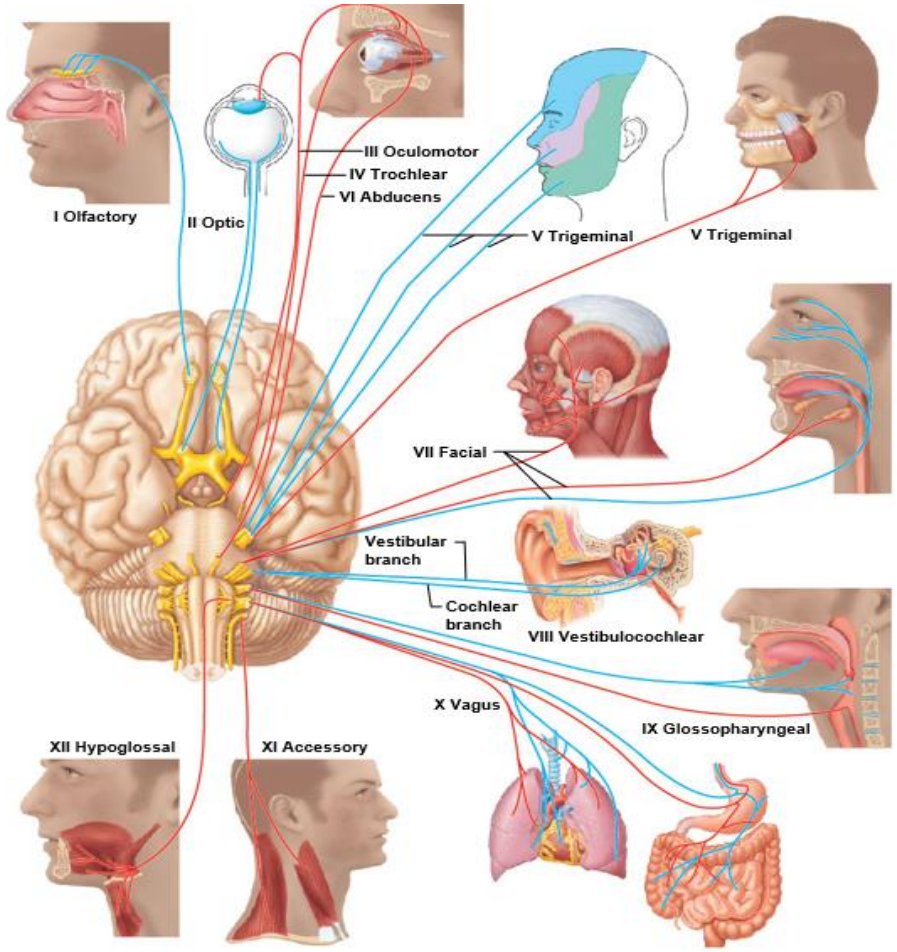


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Table **7.2** **The Cranial Nerves** (*continued*)

Name/number	Origin/course	Function	Test
XI. Accessory	Fibers arise from the superior spinal cord (C_1–C_5)* and travel to muscles of the neck and back	Mostly motor fibers that activate the sternocleidomastoid and trapezius muscles	Sternocleidomastoid and trapezius muscles are checked for strength by asking the subject to rotate head and shrug shoulders against resistance
XII. Hypoglossal	Fibers run from the medulla to the tongue	Motor fibers control tongue movements; sensory fibers carry impulses from the tongue	Subject is asked to stick out tongue, and any position abnormalities are noted

*Until recently, it was thought that the accessory nerves also received a contribution from cranial rootlets, but it has now been determined that in almost all people, these cranial rootlets are instead part of the vagus nerves. This raises an interesting question: Should the accessory nerves still be considered cranial nerves? Some anatomists say yes because they pass through the cranium. Other anatomists say no because they don't arise from the brain. Stay tuned!



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SUMMARY OF CRANIAL NERVES

Cranial Nerve	Type	Primary Functions
I – Olfactory	Sensory	Smell
II – Optic	Sensory	Vision
III – Oculomotor	Somatic Motor, Parasympathetic	Eye movement (4 of 6 muscles), eyelid elevation, pupil constriction, lens accommodation
IV – Trochlear	Somatic Motor	Eye movement (superior oblique muscle)
V – Trigeminal	Mixed (Sensory, Motor)	Sensation of face, scalp, and teeth; chewing (mastication)
VI – Abducens	Somatic Motor	Eye movement (lateral rectus muscle)
VII – Facial	Mixed (Sensory, Motor, Parasympathetic)	Facial expression, taste (anterior 2/3 tongue), salivary and lacrimal gland secretion
VIII – Vestibulocochlear	Sensory	Hearing and balance
IX – Glossopharyngeal	Mixed (Sensory, Motor, Parasympathetic)	Taste (posterior 1/3 tongue), swallowing, salivation (parotid gland), blood pressure and gas monitoring via carotid receptors
X – Vagus	Mixed (Sensory, Motor, Parasympathetic)	Taste (root of tongue), voice and swallowing muscles, regulation of thoracic and abdominal organs
XI – Accessory	Somatic Motor	Controls sternocleidomastoid and trapezius (head and shoulder movement)
XII – Hypoglossal	Somatic Motor	Tongue movement, speech, and swallowing

Cranial Nerves

Nerve	Function	Test
I. Olfactory	Smell	Identify odors with eyes closed
II. Optic	Vision	Test peripheral vision with 1 eye covered
III. Oculomotor	Eye movement & pupillary reaction	Peripheral vision, eye chart, reaction to light
IV. Trochlear	Eye movement	Test ability to depress & adduct eye
V. Trigeminal	Face sensation & mastication	Face sensation & clench teeth
VI. Abducens	Eye movement	Test ability to abduct eye past midline
VII. Facial	Facial muscles & taste	Close eyes & smile; detect various tastes—sweet, sour, salty, bitter
VIII. Vestibulocochlear (Acoustic)	Hearing & balance	Hearing; feet together, eyes open/ closed x 5 sec; test for past-pointing
IX. Glossopharyngeal	Swallow, voice, gag reflex	Swallow & say "ahh" Use tongue depressor to elicit gag reflex
X. Vagus	Swallow, voice, gag reflex	
XI. Spinal Accessory	SCM & trapezius	Rotate/SB neck; shrug shoulders
XII. Hypoglossal	Tongue mov't	Protrude tongue (watch for lateral deviation)

OBJECTIVES

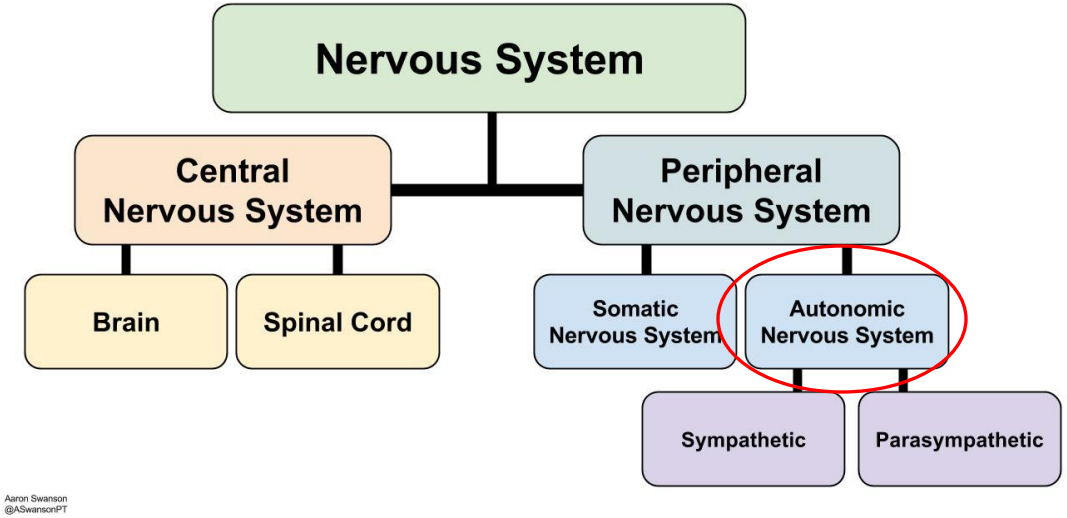


At the end of the session, students are expected to:

1. Enumerate the **functions of the nervous system.**
2. Identify the **cells of the nervous system:**
    1. Neuroglia
    2. Neurons
3. Enumerate the steps in **producing an action potential** and **synaptic transmission.**
4. Define **reflex** and enumerate its **components.**
5. Identify and differentiate the **divisions of the nervous system:**
    1. **Central Nervous System (CNS):** Brain and Spinal Cord
    2. **Peripheral Nervous System (PNS):**
        1. **Autonomic Nervous System (ANS)**
            1. **Sympathetic Nervous System**
            2. **Parasympathetic Nervous System**
            3. **Enteric Nervous System**
        2. Somatic Nervous System
        3. Sensory (Afferent) Nervous System
        4. Motor (Efferent) Nervous System

AUTONOMIC NERVOUS SYSTEM (ANS)

- A **motor subdivision** of the **peripheral nervous system (PNS)**.
- Composed **only of motor (efferent) neurons.**
- Functions **automatically**—without conscious control.
- Also called the **involuntary nervous system.**
- **Regulates the activity** of **cardiac muscle, smooth muscle**, and **glands.**
- Maintains **homeostasis** by adjusting body functions in response to internal and external changes.



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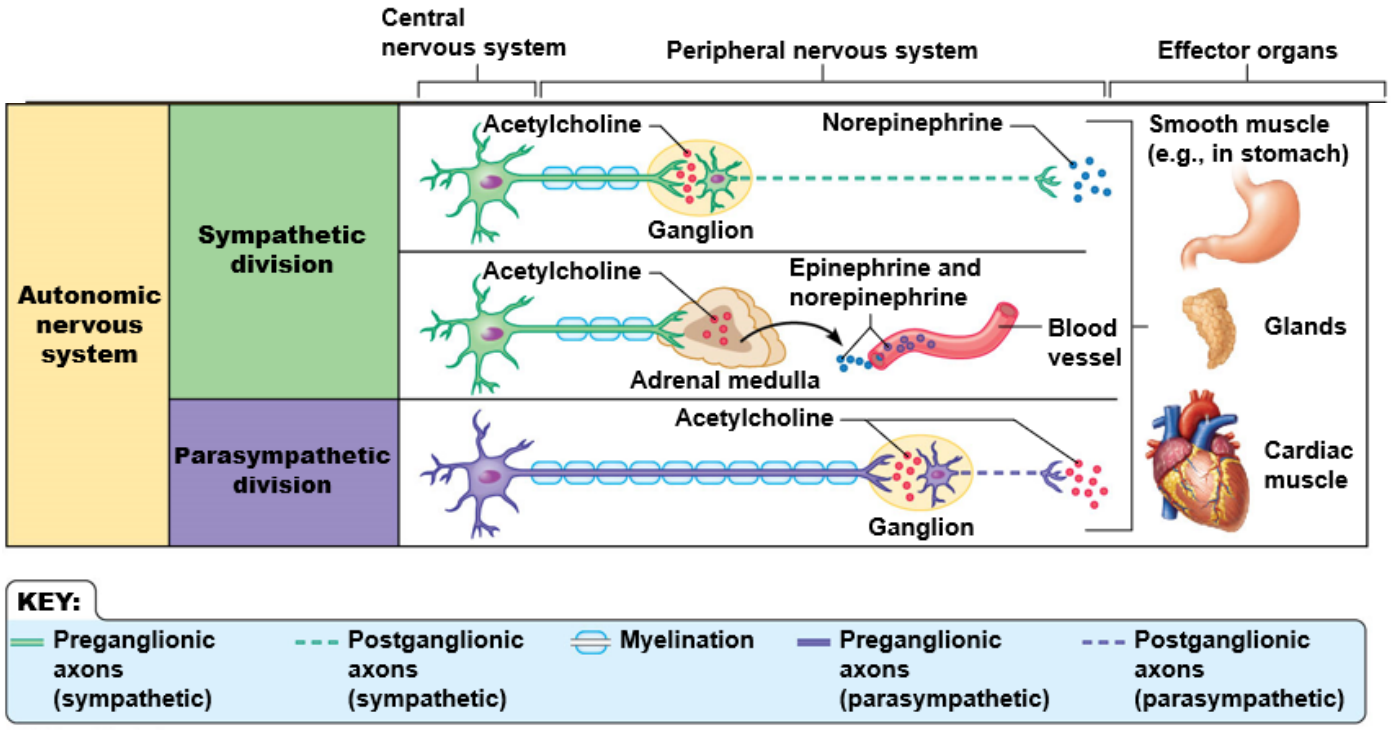
AUTONOMIC NERVOUS SYSTEMS COMPARISON

Composed of a **chain of two motor neurons:**

- **Preganglionic neuron:** Cell body located in the **brainstem or spinal cord.**
- **Postganglionic neuron:** Extends from the **autonomic ganglion** to the **effector organ** (smooth muscle, cardiac muscle, or glands).

Divided into two major functional divisions:

1. **Sympathetic Division –** "Fight or Flight"
2. **Parasympathetic Division –** "Rest and Digest"



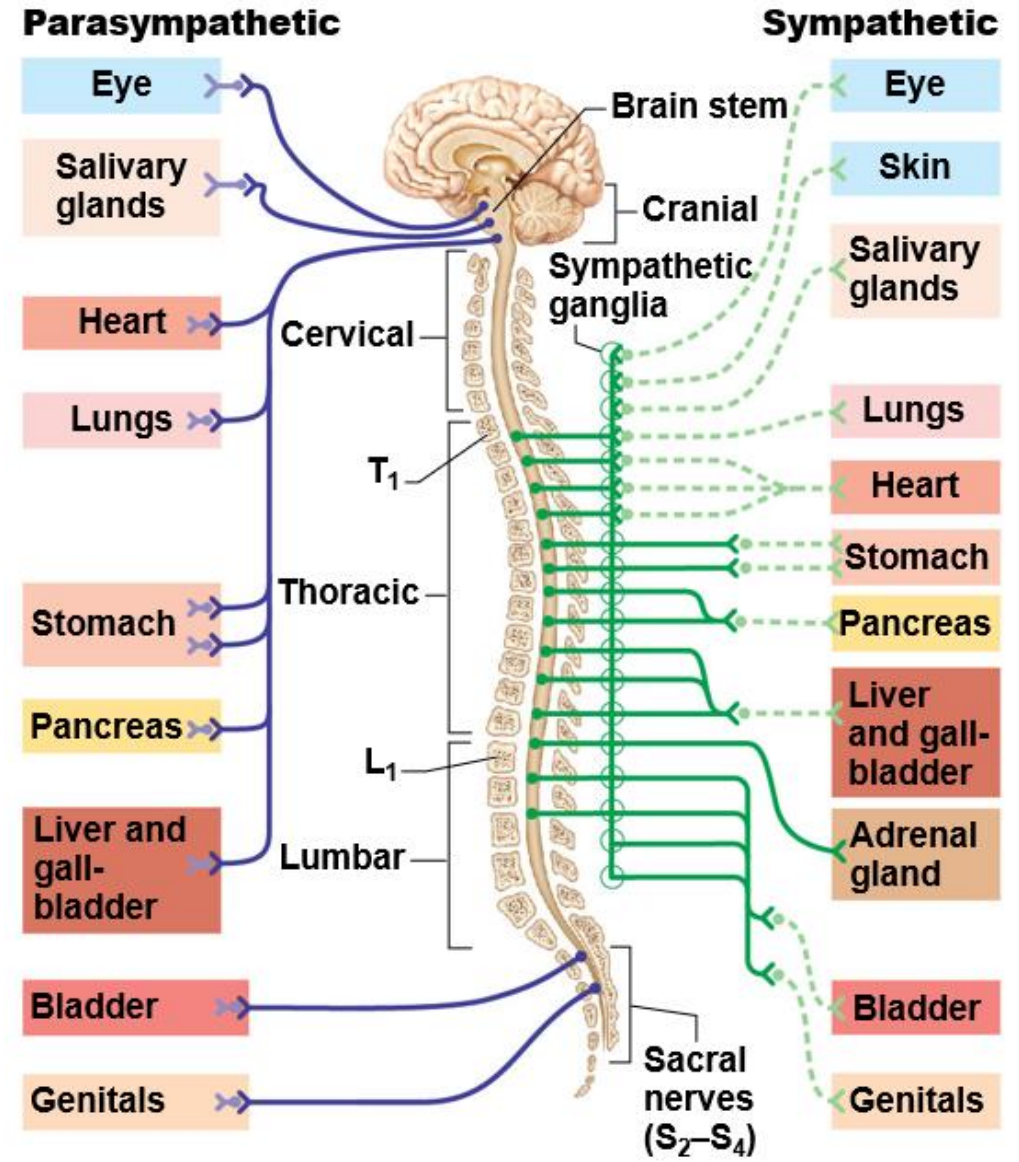
SYMPATHETIC DIVISION

ANATOMY

- Also called the **thoracolumbar division** of the ANS
- **Preganglionic neurons** originate from the **lateral horns of spinal cord segments T1–L2**
- **Preganglionic axons** pass through the **white ramus communicans** to enter a **sympathetic trunk (chain) ganglion**
- The **sympathetic trunk (chain)** lies **parallel and close to the vertebral column** on each side

Postganglionic fibers may:

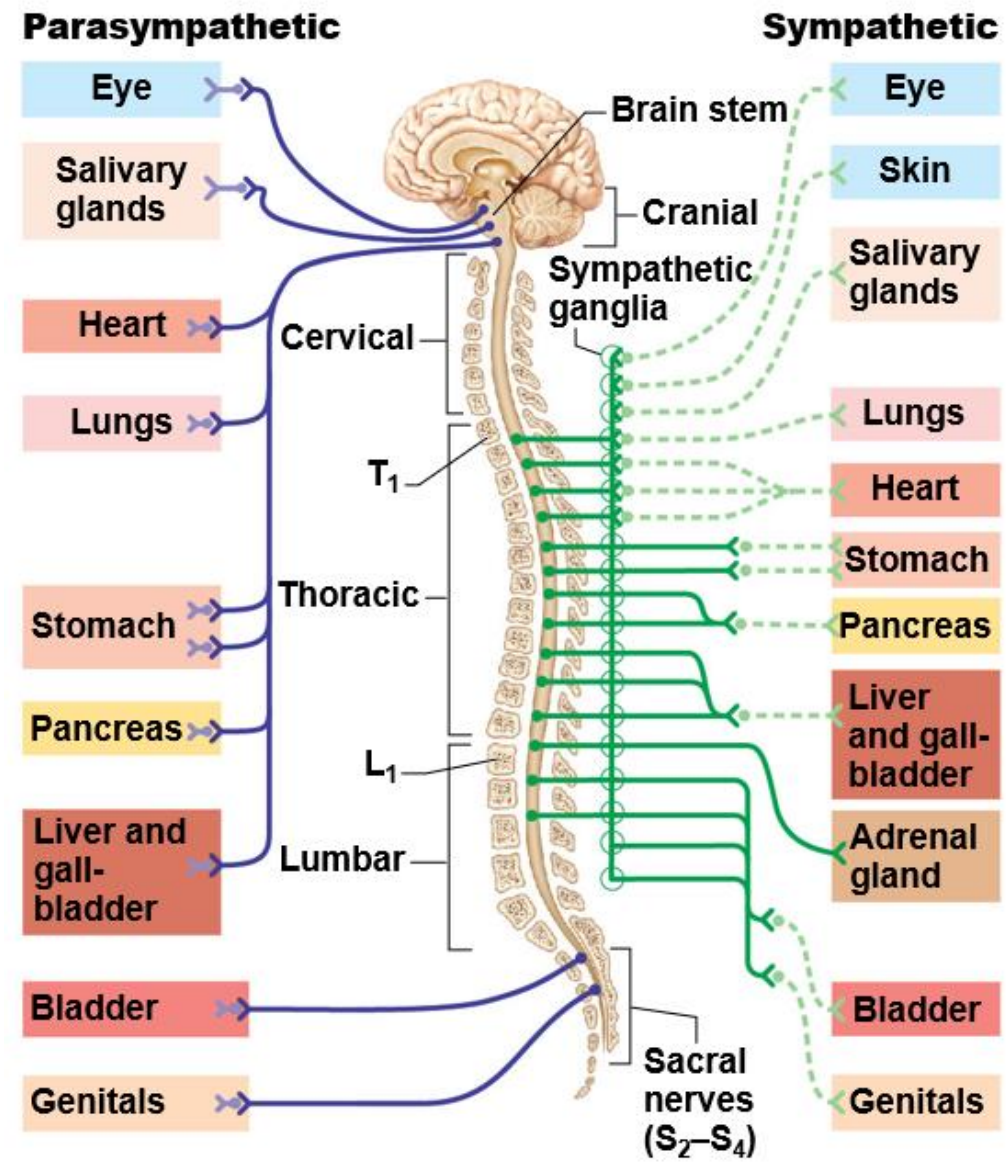
1. Synapse in the same ganglion,
2. Ascend or descend to synapse in another ganglion, or
3. Pass through to form **splanchnic nerves** that synapse in collateral ganglia



PARASYMPATHETIC DIVISION

ANATOMY

- Also called the **craniosacral division** of the ANS
- **Preganglionic neurons** originate in:
  - **Cranial nerves III (oculomotor), VII (facial), IX (glossopharyngeal), and X (vagus)**
  - **Sacral spinal cord segments S2–S4**
- **Preganglionic fibers** synapse with **terminal (intramural) ganglia** located **near or within the target organs**
- **Postganglionic axons** are **short** and extend directly to the **effector organs** (e.g., eyes, glands, heart, lungs, and digestive organs)



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AUTONOMIC FUNCTIONING

- Most **visceral organs** receive **dual innervation** from both sympathetic and parasympathetic divisions.
- **Exceptions:**
  - **Blood vessels, structures of the skin** (e.g., sweat glands, arrector pili muscles), **some glands**, and the **adrenal medulla** receive **only sympathetic innervation**.
- When both divisions serve the same organ, they produce **antagonistic effects** due to **different neurotransmitters** and **receptor types:**
  - **Parasympathetic (cholinergic) fibers** → release **acetylcholine (ACh)**
  - **Sympathetic postganglionic (adrenergic) fibers** → release **norepinephrine (NE)**
  - **Preganglionic fibers** of both divisions release **acetylcholine (ACh)**

AUTONOMIC FUNCTIONING
SYMPATHETIC

- Known as the **"fight or flight"** division
- Activated in response to **emotional or physical stress, threats,** or **emergencies**
- Functions to **prepare the body for heightened activity** by mobilizing energy resources
- Remember as the **"E Division"**:
  - **Exercise**
  - **Excitement**
  - **Emergency**
  - **Embarrassment**

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AUTONOMIC FUNCTIONING

PARASYMPATHETIC

- Known as the **"rest-and-digest"** or **"housekeeping"** system
- **Conserves energy** and promotes **maintenance of normal body functions**
- Dominates during **restful** conditions
- Supports essential **daily physiological activities** such as digestion and excretion
- Remember as the **"D Division"**:
  - **Digestion**
  - **Defecation**
  - **Diuresis**

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Table 7.4 Effects of the Sympathetic and Parasympathetic Divisions of the Autonomic Nervous System

Target organ/system	Parasympathetic effects	Sympathetic effects
Digestive system	Increases smooth muscle mobility (peristalsis) and amount of secretion by digestive system glands; relaxes sphincters	Decreases activity of digestive system and constricts digestive system sphincters (for example, anal sphincter)
Liver	No effect	Causes glucose to be released to blood
Lungs	Constricts bronchioles	Dilates bronchioles
Urinary bladder/urethra	Relaxes sphincters (allows voiding)	Constricts sphincters (prevents voiding)
Kidneys	No effect	Decreases urine output
Heart	Decreases rate; slows and steadies	Increases rate and force of heartbeat
Blood vessels	No effect on most blood vessels	Constricts blood vessels in viscera and skin (dilates those in skeletal muscle and heart); increases blood pressure
Glands—salivary, lacrimal, gastric	Stimulates; increases production of saliva, tears, and gastric juice	Inhibits; result is dry mouth and dry eyes

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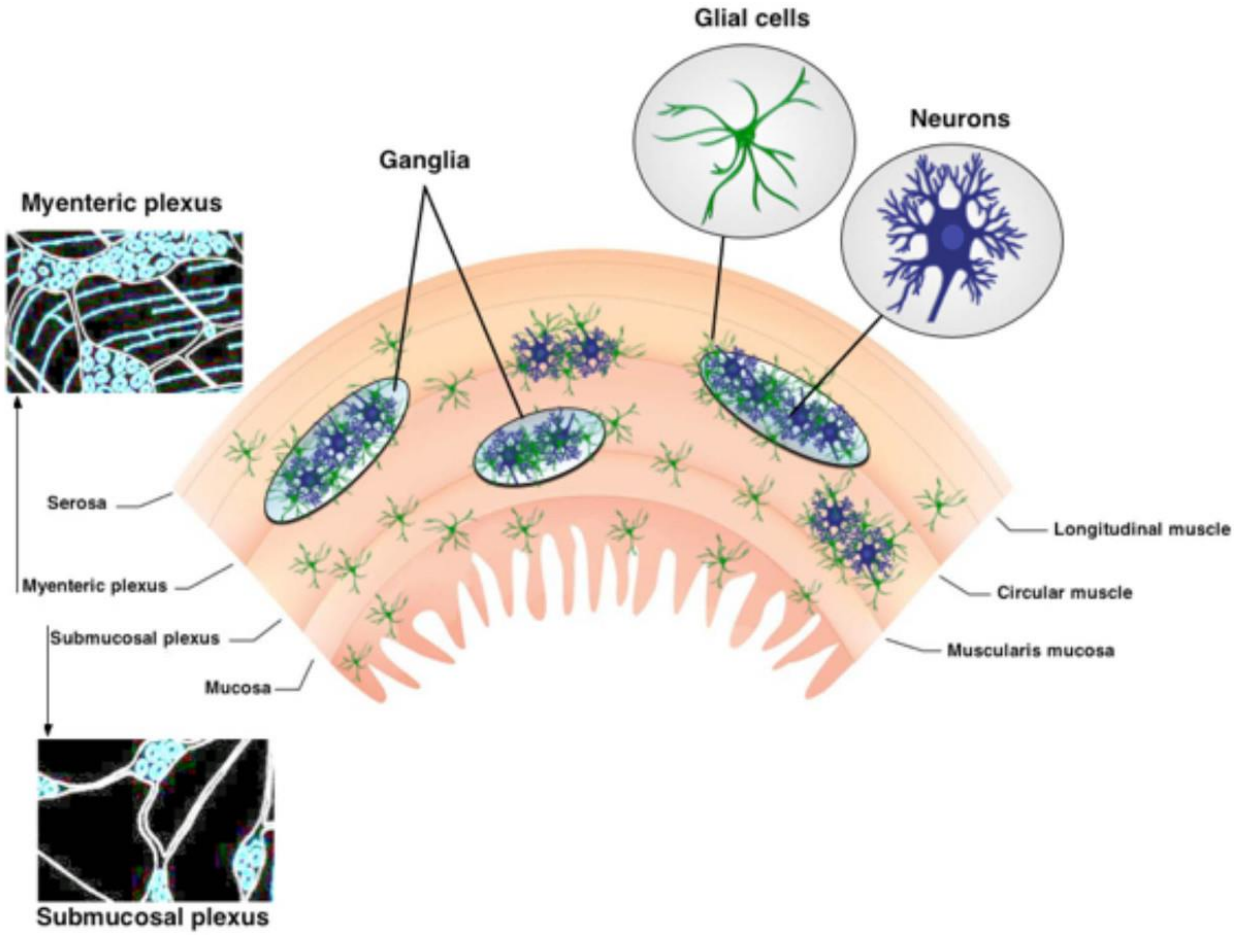
Table 7.4 Effects of the Sympathetic and Parasympathetic Divisions of the Autonomic Nervous System (*continued*)

Target organ/system	Parasympathetic effects	Sympathetic effects
Eye (iris)	Stimulates constrictor muscles; constricts pupils	Stimulates dilator muscles; dilates pupils
Eye (ciliary muscle)	Stimulates to increase bulging of lens for close vision	Inhibits; decreases bulging of lens; prepares for distant vision
Adrenal medulla	No effect	Stimulates medulla cells to secrete epinephrine and norepinephrine
Sweat glands of skin	No effect	Stimulates to produce perspiration
Arrector pili muscles attached to hair follicles	No effect	Stimulates; produces "goose bumps"
Penis	Causes erection due to vasodilation	Causes ejaculation (emission of semen)
Cellular metabolism	No effect	Increases metabolic rate; increases blood sugar levels; stimulates fat breakdown
Adipose tissue	No effect	Stimulates fat breakdown

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ENTERIC NERVOUS SYSTEM (ENS)

- Also called the **intrinsic nervous system**
- A **network of neurons** that governs the function of the **gastrointestinal (GI) tract**
- Regulates **motility, secretion,** and **blood flow** within the GI tract
- Can function **independently** of the **sympathetic and parasympathetic divisions,** but is **modulated** by both



OBJECTIVES

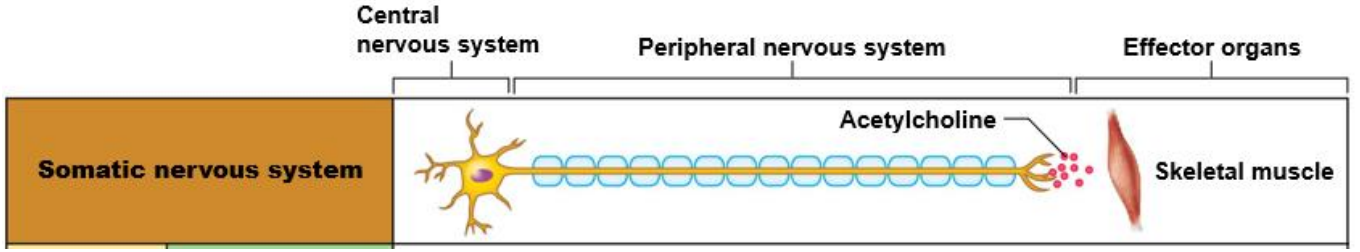


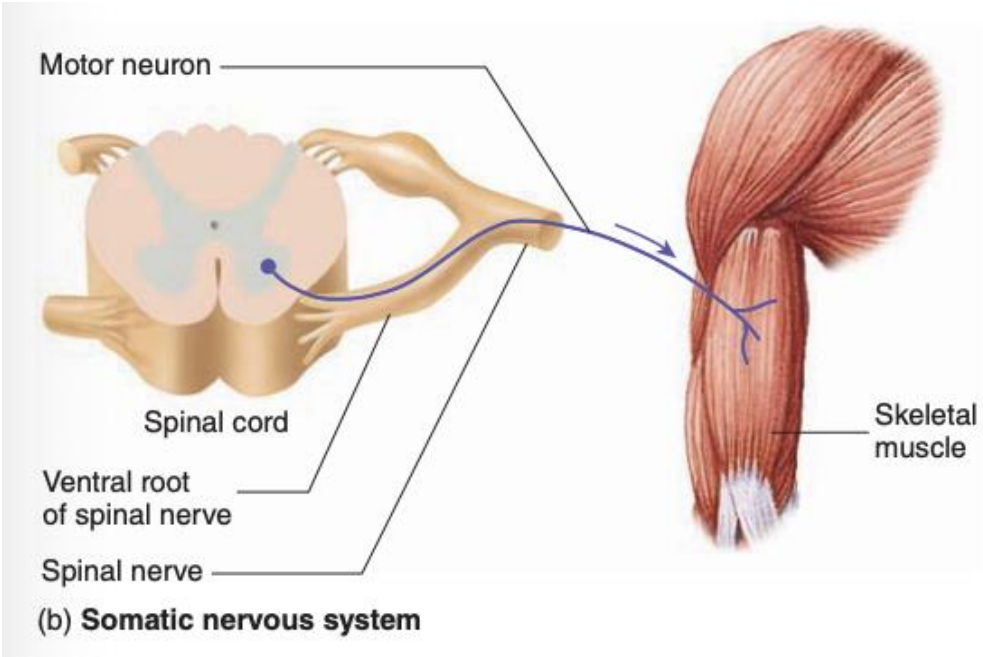
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SOMATIC NERVOUS SYSTEM

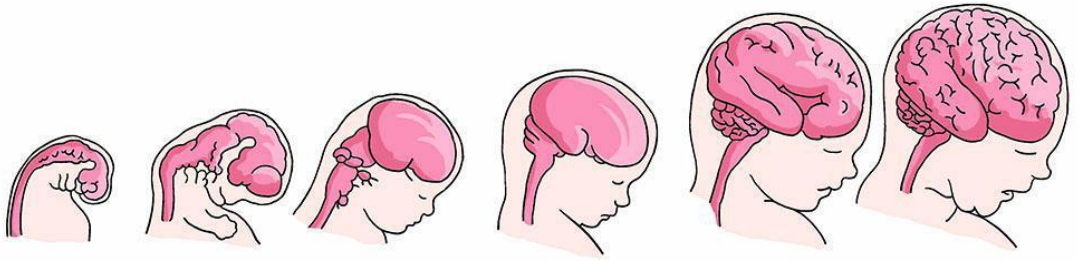
- A **voluntary (conscious)** division of the **peripheral nervous system (PNS)**.
- **Motor neuron cell bodies** are located within the **central nervous system (CNS)**.
- **Axons** extend outward from the CNS to **innervate skeletal muscles.**
- Responsible for **conscious control of skeletal muscle movements** such as posture, locomotion, and reflex actions.





DEVELOPMENTAL ASPECTS OF THE NERVOUS SYSTEM

- The **nervous system** develops from the **neural tube,** which forms the **brain and spinal cord**
- **Neural crest cells** give rise to **peripheral nerves** and **ganglia**
- The brain continues to **grow rapidly after birth,** especially during early childhood
- **Myelination** of neurons progresses after birth, improving **nerve conduction**
- **Aging** is associated with **neuronal loss, slower reflexes,** and **decline in sensory and motor functions**



DEVELOPMENTAL ASPECTS OF THE NERVOUS SYSTEM

Severe Congenital Brain Diseases

- **Cerebral Palsy (CP):**
  A group of disorders caused by brain damage before, during, or shortly after birth, leading to impaired muscle coordination and motor control.
- **Anencephaly:**
  A condition where the brain and skull fail to develop completely, resulting in absence of major brain parts; infants are usually stillborn or die shortly after birth.
- **Hydrocephalus:**
  A buildup of cerebrospinal fluid (CSF) in the brain ventricles, leading to increased intracranial pressure and an enlarged head.
- **Spina Bifida:**
  A neural tube defect where the vertebral arches do not form completely, exposing the spinal cord; severity ranges from mild to severe paralysis.

DEVELOPMENTAL ASPECTS OF THE NERVOUS SYSTEM

Severe congenital brain diseases include:

Cerebral Palsy (CP)

- A **group of neurological disorders** that affect **movement, muscle tone, or posture.**
- Caused by **damage to the developing brain—**often before, during, or shortly after birth.
- Results in **impaired voluntary muscle control** without progressive brain degeneration.



DEVELOPMENTAL ASPECTS OF THE NERVOUS SYSTEM

Severe congenital brain diseases include:

Anencephaly

- A **severe congenital disorder** where the **brain and skull bones fail to develop properly.**
- Occurs when the **neural tube fails to close** at the cranial end during early embryonic development.
- Results in **absence of major portions of the brain, skull, and scalp,** primarily affecting the **forebrain** and **cerebrum.**
- The condition is **fatal,** with most affected infants being **stillborn or dying shortly after birth.**





DEVELOPMENTAL ASPECTS OF THE NERVOUS SYSTEM

Severe congenital brain diseases include:

Hydrocephalus

- A condition characterized by **abnormal accumulation of cerebrospinal fluid (CSF)** within the **ventricles of the brain.**
- Causes **increased intracranial pressure,** leading to **enlargement of the head** and **brain tissue compression.**
- May result in **developmental, physical, and intellectual impairments** if untreated.
- Can be **congenital** (present at birth) or **acquired** (develops later due to infection, trauma, or tumor).



DEVELOPMENTAL ASPECTS OF THE NERVOUS SYSTEM

Severe congenital brain diseases include:

Spina Bifida

- A **congenital defect** in which the **vertebral column and spinal cord** fail to develop completely.
- Results from **incomplete closure of the neural tube** during early embryonic development.
- Leads to **exposure or protrusion of spinal tissues and nerves**, depending on severity.
- May cause **neurological deficits, motor weakness,** or **loss of sensation** below the level of the defect.



DEVELOPMENTAL ASPECTS OF THE NERVOUS SYSTEM

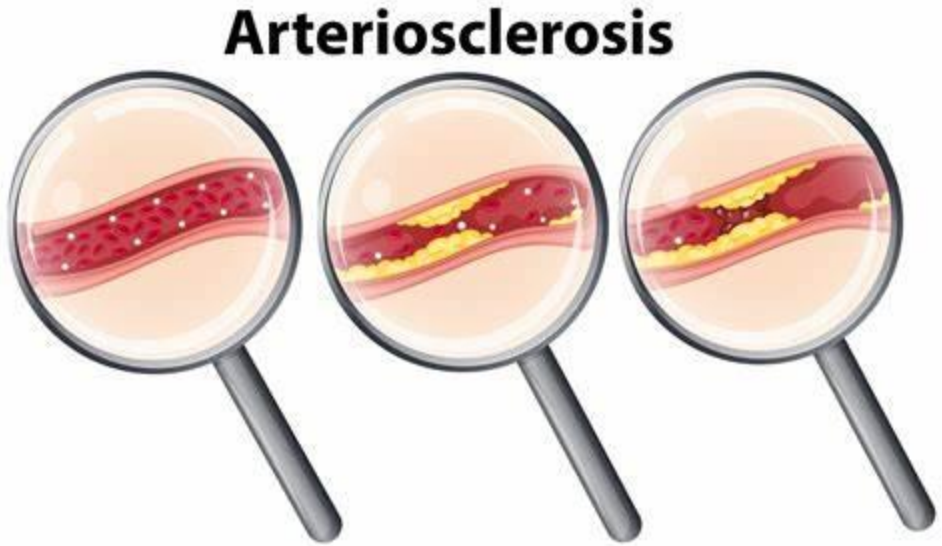
Brain Growth: Ends in young adulthood; neuronal proliferation ceases.

Neuronal Loss: Neurons die throughout life and are not replaced, leading to gradual decline in brain mass.

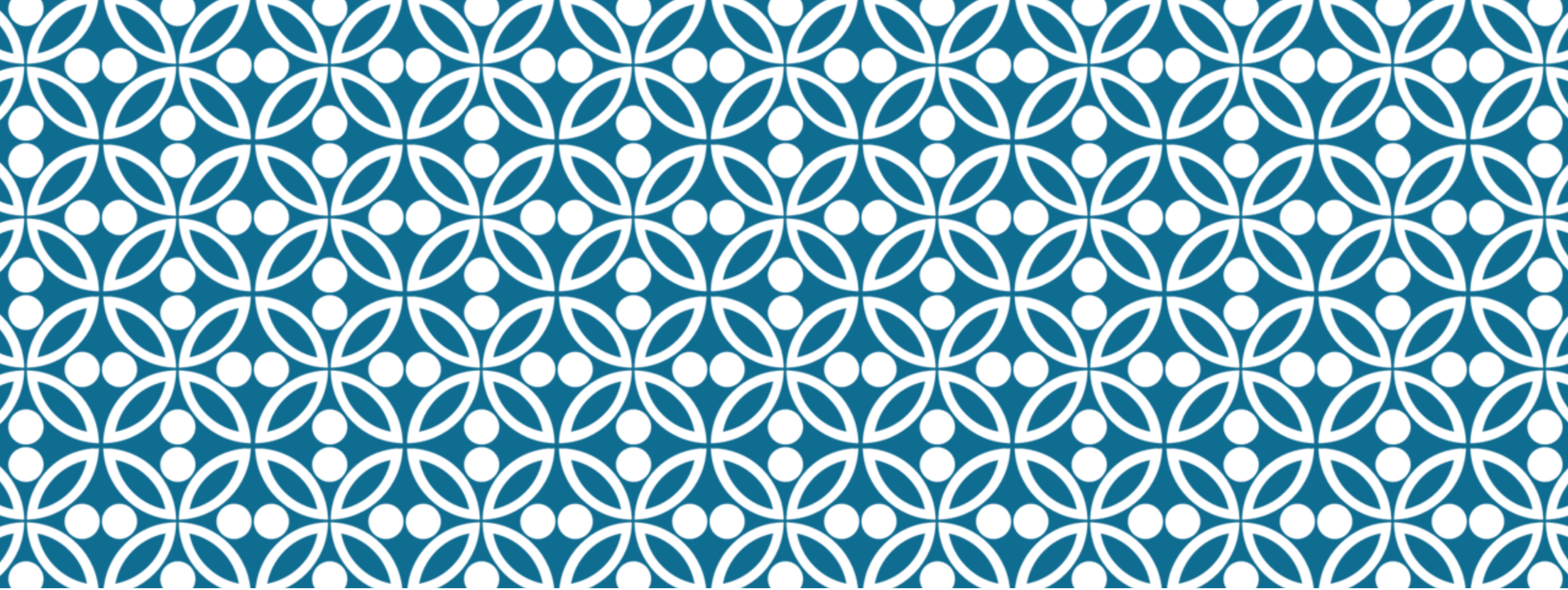
Cognitive Function in Aging: Healthy older adults typically maintain near-optimal intellectual abilities.

Impact of Disease: Cardiovascular disease is a major contributor to age-related cognitive decline.

Arteriosclerosis: Refers to decreased elasticity of blood vessels, impairing cerebral blood flow.



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THANK YOU!